Training dynamics of tree tensor networks: saddle-to-saddle, lazy and rich regimes

TTN regimes project*

Draft of October 10, 2026

Abstract

[Abstract to be written last: question, model (TTN regression with exact population loss), the regimes found and the parameters that select them, main new results with their status.]

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*Working draft; authors to be added.

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Conventions. Every claim carries a status tag: proven (proof given or cited), derived (derivation under stated assumptions), empirical (supported by experiments only) or conjecture (stated without sufficient derivation or evidence). Results we believe to be new are marked new, known results known with a reference; unmarked statements are definitions or standard background. Notation is defined in Section 2 and collected in `macros.tex`.

1 Introduction and summary of findings

[Skeleton. To be written once the results sections exist; intended content is in the TODO comments of this file.]

1.1 Question and motivation

[Which training regimes do tree tensor networks show under gradient methods, which parameters select them, and how far theory predicts them; the main focus is saddle-to-saddle dynamics.]

1.2 Summary of findings

[One table with one row per question (Q1–Q8 of the research plan): finding, status tag, new or known, and where it is shown.]

1.3 Relation to prior work

[What is known (deep linear networks, diagonal networks, tensor factorization, lazy versus rich, tensor networks for learning) and what is new here.]

1.4 Outline

[Roadmap of the report.]

2 Setup: model, data, teacher, training and diagnostics

This section fixes the model and the measurements used to classify training regimes. Every experiment uses these definitions; deviations are stated where they occur.

2.1 Tree tensor network regression

Inputs. An input x has N sites; site i carries a local feature vector $\phi_i(x) \in \mathbb{R}^{d_i}$ (default $d_i = d$). Unless stated otherwise $\phi_i(x) = x^{(i)} \sim \mathcal{N}(0, \Sigma_i)$, independent across sites, and the inputs are *whitened*, $\Sigma_i = I$. We write

$$\Phi(x) = \phi_1(x) \otimes \cdots \otimes \phi_N(x) \in \mathbb{R}^{d_1 \times \cdots \times d_N}, \qquad \Sigma = \Sigma_1 \otimes \cdots \otimes \Sigma_N.$$

Trees and cores. A tree tensor network (TTN) is specified by a rooted tree $\mathcal{T}$ with node set $\mathcal{V}$, $|\mathcal{V}| = K$, and an assignment of each site to one node. For a node v, S_v is its (possibly empty) set of sites, $\mathrm{ch}(v)$ its children and $p(v)$ its parent. Every non-root node v is joined to its parent by the edge (bond) e_v of *bond dimension* χ_v; $\chi_{\mathrm{root}} = 1$. Node v carries a *core* G_v with one *physical leg* of dimension d_i per $i \in S_v$, one leg of dimension χ_c per child c, and a parent leg of dimension χ_v (at the root, the scalar output). The dimensions of the other (*child*) legs multiply to the *fan-in*

$$F_v = \prod_{i \in S_v} d_i \prod_{c \in \mathrm{ch}(v)} \chi_c,$$

so G_v is an $F_v \times \chi_v$ matrix; the parameters are $\theta = (G_v)_{v \in \mathcal{V}}$.

Model. The tensor $W(\theta) \in \mathbb{R}^{d_1 \times \cdots \times d_N}$ is the contraction of all cores over all bonds, and the model is

$$f_\theta(x) = \langle W(\theta),\, \Phi(x) \rangle. \tag{1}$$

Equivalently, f_θ is computed by passing messages $m_v(x) \in \mathbb{R}^{\chi_v}$ from the leaves to the root,

$$m_v(x)_a = \sum_{(j_i),(b_c)} G_v[(j_i)_{i \in S_v}, (b_c)_{c \in \mathrm{ch}(v)}, a] \prod_{i \in S_v} \phi_i(x)_{j_i} \prod_{c \in \mathrm{ch}(v)} m_c(x)_{b_c}, \qquad f_\theta(x) = m_{\mathrm{root}}(x). \tag{2}$$

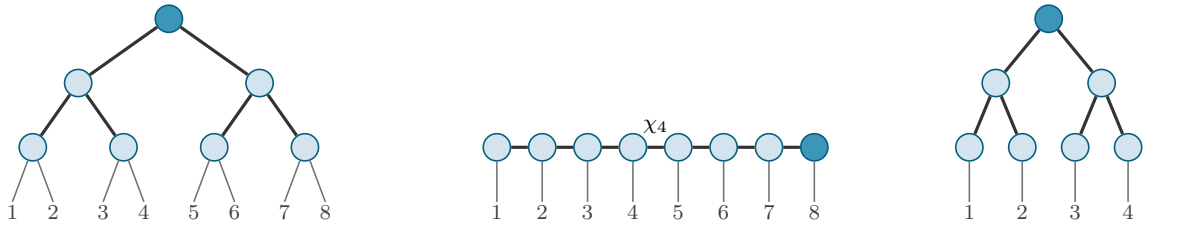
The model is multilinear in the site features, and W is multilinear in the cores (each appears once), so $\theta \mapsto W(\theta)$ is homogeneous of degree K: $G_v \mapsto c_v G_v$ rescales W by $\prod_v c_v$.

Bipartitions and bond spectra. Let A_v be the sites in the subtree rooted at v; the edge e_v separates them from the rest, and $W^{\langle e_v \rangle}$ is the matricization of W with the sites of A_v as rows. Contracting the subtree below v and the rest separately gives $W^{\langle e_v \rangle} = T_v C_v^\top$ with $T_v \in \mathbb{R}^{(\prod_{i \in A_v} d_i) \times \chi_v}$, so $\operatorname{rank} W^{\langle e_v \rangle} \le \chi_v$ (Hackbusch and Kühn, 2009; Grasedyck, 2010). The *bond spectrum* (Schmidt values) of $e = e_v$ is the list $s_1^e \ge \cdots \ge s_{\chi_v}^e \ge 0$ of the χ_v largest singular values of $W^{\langle e \rangle}$; they include all nonzero ones, so $\sum_k (s_k^e)^2 = \|W\|_F^2$ on every edge. With anisotropic inputs we use the function-space spectra of $(\Sigma^{1/2} W)^{\langle e \rangle}$, $\Sigma^{1/2} = \bigotimes_i \Sigma_i^{1/2}$. QR sweeps over the cores give them to about 10^{-15} relative accuracy without forming W.

Why this model. It is the simplest genuine TTN. (i) With at least three sites and an internal bond, W is a genuine tensor; a deep linear network is a chain with physical legs only at its two ends. (ii) Whitened inputs reduce the population loss to a Frobenius distance (Section 2.3), the setting in which Saxe et al. (2014) solved deep linear networks. (iii) The matrix product state (MPS) classifiers of Stoudenmire and Schwab (2016) compute, for each class label ℓ, $f^\ell(x) = \langle W^\ell, \bigotimes_i \phi_i(x) \rangle$ (the label index sits on one core), with non-Gaussian local features $(\cos(\pi x^{(i)}/2), \sin(\pi x^{(i)}/2))$ that are correlated across sites. (iv) Training is then a fully observed tensor factorization. On HT trees it has the mode tree of the hierarchical tensor factorization of Razin et al. (2022) (one leaf matrix per site), but there each internal node is a local CP factorization (their Eq. 3), $G_v[b_1, \ldots, b_c, a] = \sum_r \delta_{b_1 r} \cdots \delta_{b_c r} M_{ra}^{(v)}$, trained on general losses such as tensor completion; our dense cores carry the matrix gauge symmetries of Section 2.6 instead of one scalar invariant per local component.

Table 1: Built-in tree topologies for N sites (K cores; depth: edges on the longest root-to-leaf path). Binary-tree depths are for $N = 2^m$; otherwise the sites are split recursively into halves ($\lceil N/2 \rceil$, $\lfloor N/2 \rfloor$), a pair of sites becomes a bottom core and a single site joins the core above, so $K = N - 1$.

name	structure	K	depth
binary	balanced binary tree; bottom cores carry two adjacent sites, internal cores none	$N-1$	$\log_2 N - 1$
MPS	chain (caterpillar): core k carries site k, its parent is core $k+1$	N	$N-1$
HT	hierarchical Tucker: balanced binary tree over N leaf cores, one site each; internal cores none	$2N-1$	$\lceil \log_2 N \rceil$
custom	any parent array and site assignment	–	–



(a) binary, $N = 8$, $K = 7$ **(b)** MPS, $N = 8$, $K = 8$ **(c)** HT, $N = 4$, $K = 7$

Figure 1: The built-in tree topologies (Table 1). Circles are cores G_v; the shaded core is the root, whose parent leg (dimension one, the scalar output) is not drawn. Thick lines are bonds: the edge e_v between core v and its parent has dimension χ_v (one label shown in (b), where cores are numbered $1, \ldots, 8$ from the left). Thin dangling lines are physical legs, labelled by their site i (dimension d_i).

2.2 Tree topologies

Table 1 lists the built-in topologies and Figure 1 draws them. Binary trees and MPS with the same N are the extremes of tree shape at nearly equal K, the default pair for saddle-to-saddle studies. The hierarchical Tucker (HT) format adds one leaf core per site; with leaf bond dimension d this adds no expressivity but changes K and the number of cores that must move to learn a component. Unless stated otherwise every edge has the same nominal bond dimension χ, capped at the largest rank its bipartition allows,

$$\chi_v = \min\Big(\chi,\ \prod_{i \in A_v} d_i,\ \prod_{i \notin A_v} d_i\Big). \tag{3}$$

2.3 Teacher, data and loss

Teachers. The target is realizable, $f^*(x) = \langle W^*, \Phi(x) \rangle$. A *CP teacher* has

$$W^* = \sum_{r=1}^{R} s_r U_r, \qquad U_r = u_r^{(1)} \otimes \cdots \otimes u_r^{(N)}, \qquad \|u_r^{(i)}\| = 1, \qquad s_1 \geq \cdots \geq s_R > 0. \tag{4}$$

Three kinds are used:

- *Odeco* (orthogonally decomposable): $\langle u_r^{(i)}, u_{r'}^{(i)} \rangle = \delta_{rr'}$ at every site (so $R \leq \min_i d_i$); by default the factors at each site are columns of an independent Haar-random orthogonal matrix.

- *Pattern teachers*: site i has an orthonormal basis $(\beta_j^{(i)})_j$ and component r uses $u_r^{(i)} = \beta_{\pi_r(i)}^{(i)}$ for a pattern $\pi_r\colon \{1,\dots,N\} \to \{1,\dots,d\}$. With $\pi_1 = (1,1,1,1)$ and $\pi_2 = (2,2,1,1)$ the second component differs from the first only on $D = \{1,2\}$ (a *local* component).
- *Random CP*: $u_r^{(i)} = g/\|g\|$ with independent $g \sim \mathcal{N}(0, I)$; the components are generically non-orthogonal.

A *TTN teacher* gives W^* as a TTN on the student's tree or on another tree; a random TTN teacher is a fan-in initialization (Section 2.4) with $\sigma = 1$, rescaled to $\|W^*\|_F = 1$. The overlap of the student with component r of a CP teacher,

$$a_r = \langle W, U_r\rangle_\Sigma / \langle W^*, U_r\rangle_\Sigma, \tag{5}$$

is 0 at $W = 0$ and 1 at $W = W^*$; for an odeco teacher with whitened inputs, $a_r = \langle W, U_r\rangle / s_r$.

Proposition 2.1 (Teacher bond spectra)**.** proven *Let e be an edge whose two sides both carry sites (every edge of the trees of Table 1). (a) For an odeco teacher the bond spectrum of W^* on e is $(s_1,\dots,s_R)$, for every tree. (b) For a two-component pattern teacher whose components differ exactly on the site set D, the bond rank of W^* on e is 2 if e lies in the smallest subtree of $\mathcal{T}$ containing the nodes that carry the sites of D, and 1 otherwise.*

Proof. With A the sites on one side of e, $x_r = \bigotimes_{i\in A} u_r^{(i)}$ and $y_r = \bigotimes_{i\notin A} u_r^{(i)}$, $(W^*)^{\langle e\rangle} = \sum_r s_r x_r y_r^\top$. (a) Both families are orthonormal, so this is a singular value decomposition. (b) If $D \subseteq A$ or $D \cap A = \emptyset$, then $y_1 = y_2$ or $x_1 = x_2$: rank one. Otherwise the components differ (by orthogonal basis vectors) on both sides, so $x_1 \perp x_2$, $y_1 \perp y_2$: rank two. And e separates two sites of D exactly when it lies on the tree path between their nodes. □

The number of cores in that subtree is the quantity k of the locality hypothesis (Section 4.4).

Population loss. Training minimizes the population square loss

$$L(\theta) = \tfrac{1}{2}\,\mathbb{E}_x\big[(f_\theta(x) - f^*(x))^2\big] = \tfrac{1}{2}\|W(\theta) - W^*\|_\Sigma^2, \qquad \|A\|_\Sigma^2 = \langle A, \Sigma A\rangle. \tag{6}$$

The second equality holds because, by independence across sites, $\mathbb{E}[\Phi\otimes\Phi] = \bigotimes_i \mathbb{E}[\phi_i\phi_i^\top] = \Sigma$. With whitened inputs L is half the squared Frobenius distance. Only second moments enter: every local feature map with $\mathbb{E}[\phi_i\phi_i^\top] = I$ gives the same dynamics, and a non-orthonormal one is equivalent to $\Sigma_i = \mathbb{E}[\phi_i\phi_i^\top] \neq I$.

Closed form. Expanding (6),

$$L(\theta) = \tfrac{1}{2}\langle W, W\rangle_\Sigma - \langle W, W^*\rangle_\Sigma + \tfrac{1}{2}\langle W^*, W^*\rangle_\Sigma, \tag{7}$$

and every term is an exact contraction:

- For two TTNs W, W' on the same tree (bond dimensions may differ), $\langle W, W'\rangle_\Sigma = E_{\text{root}}$, with environments $E_v \in \mathbb{R}^{\chi_v\times\chi'_v}$ computed from the leaves to the root at a cost linear in K,

$$(E_v)_{aa'} = \sum G_v[(j_i),(b_c),a]\; G'_v[(j'_i),(b'_c),a'] \prod_{i\in S_v} (\Sigma_i)_{j_i j'_i} \prod_{c\in \mathrm{ch}(v)} (E_c)_{b_c b'_c}. \tag{8}$$

- For a CP teacher, $\langle W, U_r\rangle_\Sigma$ is the student evaluated at the site vectors $\Sigma_i u_r^{(i)}$, so for any tree

$$\langle W, W^*\rangle_\Sigma = \sum_r s_r\, f_\theta(\Sigma_1 u_r^{(1)},\dots,\Sigma_N u_r^{(N)}), \qquad \langle W^*, W^*\rangle_\Sigma = \sum_{r,r'} s_r s_{r'} \prod_i u_r^{(i)\top}\Sigma_i u_{r'}^{(i)}, \tag{9}$$

  where $f_\theta(z_1,\dots,z_N)$ is (2) with $\phi_i(x)$ replaced by z_i.

- A TTN teacher on another tree is glued to the student along the physical legs, and the loopy network is contracted exactly along an optimized path.

For an odeco teacher with whitened inputs, (7) reads $L = \frac{1}{2}\|W\|_F^2 - \sum_r s_r^2 a_r + \frac{1}{2}\sum_r s_r^2$. A *partial fit* $W_S = \sum_{r\in S} s_r U_r$ has loss $\frac{1}{2}\sum_{r\notin S} s_r^2$, so learning the components in decreasing order of s_r passes through the candidate plateau levels

$$L_k = \tfrac{1}{2}\sum_{r>k} s_r^2, \qquad k = 0, \dots, R. \tag{10}$$

Anisotropic and correlated inputs. Anisotropic product covariances are handled exactly by (6)–(9); the default family has power-law site spectra, $\Sigma_i = c_\beta \operatorname{diag}(1^{-\beta}, 2^{-\beta}, \dots, d^{-\beta})$ with c_β fixed by $\operatorname{tr}\Sigma_i = d$, aligned with the teacher's factors or rotated away from them. Cross-site correlations break the product structure of Σ; for small N the loss is then computed with the dense second-moment matrix (Section 6).

Finite samples. A training set $(x_j, y_j)_{j=1}^n$ has $x_j \sim \mathcal{N}(0, \Sigma)$ i.i.d. and $y_j = f^*(x_j) + \varepsilon_j$ with optional label noise $\varepsilon_j \sim \mathcal{N}(0, \varsigma^2)$. The empirical loss is $\hat{L}_n(\theta) = \frac{1}{2n}\sum_j (f_\theta(x_j) - y_j)^2$, and the test error is the population loss (6).

2.4 Initialization and its variants

Fan-in initialization. The default initialization is

$$G_v = \frac{\sigma_v}{\sqrt{F_v}} Z_v, \qquad Z_v \text{ with i.i.d.} \mathcal{N}(0,1) \text{ entries}, \tag{11}$$

with a single per-core scale $\sigma_v = \sigma$ unless stated otherwise. Each core is then approximately σ_v times an isometry (when $F_v \gg \chi_v$), so with $\sigma_v = 1$ the messages (2) keep their size up the tree.

Proposition 2.2 (Initial scales). proven *Under* (11), $\mathbb{E}\|W(\theta_0)\|_F^2 = \prod_v \sigma_v^2$ *exactly,* $\mathbb{E}\|G_v\|_F^2 = \sigma_v^2 \chi_v$, *and hence* $\mathbb{E}\|\theta_0\|^2 = \sum_v \sigma_v^2 \chi_v$.

Proof. Let T_v be the contraction of the subtree below v with its parent leg open (χ_v columns). Conditionally on the children's subtrees, the independent entries of G_v, of variance σ_v^2/F_v, give $\mathbb{E}[T_v^\top T_v] = \frac{\sigma_v^2}{F_v}\prod_{i\in S_v} d_i \prod_{c\in \mathrm{ch}(v)} \operatorname{tr}(T_c^\top T_c)\, I$. The children's subtrees are independent and $F_v = \prod_{i\in S_v} d_i \prod_c \chi_c$, so by induction from the leaves $\mathbb{E}[T_v^\top T_v] = (\prod_u \sigma_u^2) I$ over the subtree of v; at the root T_{root} is W flattened. Finally, $\mathbb{E}\|G_v\|_F^2 = F_v \chi_v\, \sigma_v^2/F_v$. □

We report both σ and the function-space scale $\alpha = \sigma^K$, the root-mean-square norm of $W(\theta_0)$: small σ makes $W(\theta_0)$ exponentially small in K, while $\|\theta_0\|^2 \approx \sigma^2 \sum_v \chi_v$ grows with the bond dimensions.

Variants.

(a) *Imbalance*: per-core scales $\sigma_v = \sigma c_v$ with a fixed profile (c_v) (for example root versus rest), compared at fixed $\prod_v \sigma_v$.

(b) *Centred model*: the target is $W^* + W(\theta_0)$, that is, $f_\theta - f_{\theta_0}$ is trained on f^* (Chizat et al., 2019, Sec. 3.2). The initial function is removed at any σ without changing the parameterization, so σ alone moves the model from small to large initialization; overlaps and bond spectra refer to $W - W(\theta_0)$. This is the main construction of the regime map (Section 3).

(c) *Two-copy model* (Chizat et al., 2019, Sec. 3.1; Woodworth et al., 2020, Sec. 4): $W(\theta^+) - W(\theta^-)$ with $\theta^+(0) = \theta^-(0) = \theta_0$, zero at initialization for every σ; a cross-check of the centred model (`notes/model.md`, `notes/decisions.md`). It is stored as one TTN with doubled bonds $2\chi_v$ and block-diagonal cores ($G_v^{\pm}$ on the block where all bond indices of v are in the "$\pm$" half; the root holds G^+_{root} and $-G^-_{\text{root}}$) and needs a gradient mask (Remark 2.3). Both differences are contracted without cancellation, as one TTN on doubled bonds in half-sums and half-differences of the two copies.

(d) *Aligned (decoupled) initialization* for CP teachers (Proposition 2.4): cores diagonal in the teacher's basis, giving exactly solvable Saxe-type dynamics.

(e) *Per-core learning-rate multipliers* η_v (15). A fixed forward multiplier c_v ($G_v = c_v H_v$, H_v trained) is equivalent to the initial scale $c_v\sigma_v$ with the multiplier $c_v^2\eta_v$; this expresses the NTK and μP parameterizations of wide networks (Yang and Hu, 2021), transposed to large bond dimension (Section 5).

Remark 2.3 (The two-copy model needs a gradient mask). proven Gradient flow on the doubled TTN does not keep its block-diagonal form: for a root with children c_1, c_2 and no sites, the root entry that pairs index b_1 of the + half of e_{c_1} with index b_2 of the − half of e_{c_2} has gradient $\langle W - W^*,\, T^+_{c_1}[:, b_1] \otimes T^-_{c_2}[:, b_2]\rangle_\Sigma$ (up to the order of sites), at initialization $-\langle W^*,\, T^+_{c_1}[:, b_1] \otimes T^-_{c_2}[:, b_2]\rangle_\Sigma$, generically nonzero; mixed blocks of other cores also receive gradient. The two-copy dynamics is therefore the masked flow $\dot\theta = -m \odot \nabla L(\theta)$, m the 0/1 indicator of the diagonal blocks, which keeps the blocks and equals gradient flow of $\frac{1}{2}\|W(\theta^+) - W(\theta^-) - W^*\|^2_\Sigma$ in (θ^+, θ^-).

Proposition 2.4 (Aligned initialization). proven known (deep linear networks: Saxe et al., 2014) *Let W^* be odeco, inputs whitened, $\chi_v \geq R$ on every edge, and let every subtree and its complement carry a site (true for the trees of Table 1). Initialize*

$$G_v = \sum_{r=1}^{R} q_{v,r} \Big(\bigotimes_{i \in S_v} u_r^{(i)} \Big) \otimes \Big(\bigotimes_{c \in \mathrm{ch}(v)} \varepsilon_r \Big) \otimes \varepsilon_{r(v)}, \tag{12}$$

with ε_r the r-th standard basis vector of the bond in question, $r(v) = r$ for $v \neq$ root and $r(\text{root}) = 1$. Under gradient flow with multipliers η_v the cores keep this form, $W = \sum_r \Pi_r U_r$ with $\Pi_r = \prod_v q_{v,r}$, $a_r = \Pi_r / s_r$, $L = \frac{1}{2}\sum_r (s_r - \Pi_r)^2$, and

$$\dot q_{v,r} = \eta_v \, (s_r - \Pi_r) \prod_{u \neq v} q_{u,r}. \tag{13}$$

With $\eta_v = 1$ the differences $q_{v,r}^2 - q_{u,r}^2$ are conserved; a balanced start $q_{v,r}(0) = q_r(0)$ stays balanced, and $\dot q_r = q_r^{K-1}(s_r - q_r^K)$.

Proof. On the family (12), contracting the bond basis vectors gives $W = \sum_r \Pi_r U_r$ with orthonormal U_r. The derivative of W with respect to an entry of G_v is a product of basis vectors on the physical legs of v, the child-subtree columns $T_c[:, b_c]$ and the environment column above v indexed by a; on the family these are multiples of $\bigotimes_{i \in A_c} u_{b_c}^{(i)}$ (and of the analogous product over the sites outside A_v), so their inner product with $W - W^* = \sum_r (\Pi_r - s_r) U_r$ vanishes unless all bond indices equal one r ($a = 1$ at the root). The gradient thus stays in the family, with component $(\Pi_r - s_r)\prod_{u \neq v} q_{u,r}$ along $q_{v,r}$; with $\eta_v = 1$, $\frac{\mathrm{d}}{\mathrm{d}t} q_{v,r}^2 = 2(s_r - \Pi_r)\Pi_r$ for every v. □

For $K \geq 3$ a balanced mode started at $q_r(0) \ll s_r^{1/K}$ reaches $a_r = \frac{1}{2}$ at

$$t_r = \frac{q_r(0)^{-(K-2)}}{(K-2)\, s_r} + O\big(s_r^{-(2-2/K)}\big), \tag{14}$$

and for $K = 2$ at $t_r = \log(s_r/q_r(0)^2)/(2s_r) + O(q_r(0)^2/s_r^2)$, by direct integration of the balanced ODE proven. With $q_r(0) \propto \sigma$ the first plateau grows like $\sigma^{-(K-2)}$, hence adaptive integration in double precision.

2.5 Training dynamics

Gradient flow is the reference dynamics,

$$\dot{\theta} = -D\,\nabla L(\theta), \tag{15}$$

where the diagonal D holds, for each parameter j of core v, the learning-rate multiplier η_v (default 1) times an optional 0/1 mask m_j (Remark 2.3; default 1). Here t denotes gradient-flow time. The default integrator is the adaptive Dormand–Prince 5(4) pair (Dormand and Prince, 1980) with relative and absolute tolerances 10^{-8} and 10^{-12} and standard step-size control (Hairer et al., 1993, Sec. II.4); steps follow the local time scale $\|\theta\|/\|\dot{\theta}\|$, so long plateaus are cheap. Past the first escape the flow is stiff, and many experiments use SciPy's LSODA with the exact Hessian as Jacobian (Section B). Diagnostics are recorded at $t = 0$ and at 200 log-spaced times from $10^{-3}\min(\tau_0, \tau_L, t_{\max})$ to $t_{\max}$, where $\tau_0 = \|\theta\|/\|\dot{\theta}\|$ and $\tau_L = L/|\dot{L}|$ at $t = 0$. This grid cannot resolve escapes, so the recorder also keeps a *fine row* (loss, overlaps, bond spectra) after each integrator step at which θ has moved by 1% of its norm, or $\ln L$ by 0.01, since the last kept row; each rise is sampled about 20 times or more (`Recorder` and `merged_rows` in `ttn/dynamics.py`). A run stops at $t_{\max}$, below a prescribed loss floor, or at step or wall-time limits. Accuracy is checked with the conservation laws (Section 2.6) and the aligned solutions (Proposition 2.4).

Gradient descent with learning rate η, $\theta_{k+1} = \theta_k - \eta D \nabla L(\theta_k)$, is recorded at $t = \eta k$. A run has diverged when L becomes non-finite or exceeds $10^6 L(\theta_0)$.

Stochastic gradient descent uses the minibatch loss $\hat{L}_B(\theta) = \frac{1}{2B}\sum_{b=1}^{B}(f_\theta(x_b) - y_b)^2$ in the same update. Online SGD draws fresh inputs $x_b \sim \mathcal{N}(0, \Sigma)$ with $y_b = f^*(x_b)$ at every step; with a fixed training set, each epoch uses a new random permutation. Optional label noise is drawn afresh each step.

Precision. All computations are in double precision: with $\sigma = 10^{-2}$ and $K = 7$, $\|W(\theta_0)\|_F \approx 10^{-14}$ and gradients are of order 10^{-12}. A dormant component whose escape signal falls below about $10^{-15}\|W\|_F$ is seeded by roundoff, not by the initialization; such runs are flagged.

2.6 Symmetries and conservation laws

For a bond $e = e_v$ joining core v to its parent $p = p(v)$, let $G_{u,(e)}$ denote core $u \in \{v, p\}$ matricized with its e-leg as rows (a $\chi_v \times F_v$ matrix for $u = v$). For every invertible $B \in \mathrm{GL}(\chi_v)$ the *gauge transformation*

$$G_{v,(e)} \mapsto B\,G_{v,(e)}, \qquad G_{p,(e)} \mapsto B^{-\top} G_{p,(e)} \tag{16}$$

leaves W, and hence any loss that depends on θ only through W, unchanged.

Proposition 2.5 (Conservation laws). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Under gradient flow $\dot{G}_u = -\eta_u \nabla_{G_u} L$ of any loss that depends on θ only through W (population or empirical), the $\chi_v \times \chi_v$ matrix*

$$Q_e = \frac{1}{\eta_v} G_{v,(e)} G_{v,(e)}^{\top} - \frac{1}{\eta_p} G_{p,(e)} G_{p,(e)}^{\top} \tag{17}$$

is constant in time, for every bond $e = e_v$ with $p = p(v)$.

Proof. With $X = G_{v,(e)}$, $Y = G_{p,(e)}$ and $B = I + \epsilon M$ in (16), $X \mapsto X + \epsilon MX$ and $Y \mapsto Y - \epsilon M^\top Y + O(\epsilon^2)$; first-order invariance of L for every M gives $C := \nabla_X L\, X^\top = Y(\nabla_Y L)^\top$, so $\frac{\mathrm{d}}{\mathrm{d}t}(XX^\top)/\eta_v = -(C + C^\top) = \frac{\mathrm{d}}{\mathrm{d}t}(YY^\top)/\eta_p$. □

This is the tree version of the balancedness of deep linear networks, an instance of symmetry-induced conservation laws (Kunin et al., 2021). Consequences:

- *Norm balance*: $\operatorname{tr} Q_e = \|G_v\|_F^2/\eta_v - \|G_p\|_F^2/\eta_p$; with equal multipliers all differences $\|G_u\|_F^2 - \|G_{u'}\|_F^2$ are conserved, so all cores grow together.
- *Balancedness at small initialization*: under (11), $\mathbb{E}\, Q_e(0) = (\sigma_v^2/\eta_v - \sigma_p^2\chi_p/(\eta_p\chi_v))I$, so at small σ the cores stay balanced up to $O(\sigma^2)$ while they grow to $O(1)$. Imbalanced initializations or multipliers set Q_e to other values, a second knob for the regime map (Kunin et al., 2024).
- The masked two-copy flow conserves the Q_e of each copy; gradient descent and SGD conserve Q_e only up to $O(\eta^2)$ per step.
- *Integrator check*: the conservation drift

$$\mathrm{drift}(t) = \frac{\max_e \|Q_e(t) - Q_e(0)\|_{\max}}{\max_e \|Q_e(0)\|_{\max} + \max_u \|G_u(t)\|_F^2/\eta_u}, \tag{18}$$

  with $\|\cdot\|_{\max}$ the largest absolute entry, measures the integration error relative to the terms that cancel in Q_e; it must stay below 10^{-6} under gradient flow.

2.7 Regime diagnostics

Regimes are assigned from measured quantities, never by eye, by the rules of `notes/diagnostics.md` (code: `ttn/diagnostics.py`; per-experiment details: Section B).

Raw quantities (Table 2) are recorded at the log times of Section 2.5. The tangent kernel uses a fixed probe set of $P = 128$ inputs $x_p \sim \mathcal{N}(0, \Sigma)$ and the Jacobian $J_t = \partial(f_\theta(x_1), \ldots, f_\theta(x_P))/\partial\theta$ at $\theta(t)$. The gauge directions at θ are the tangent vectors $(MG_{v,(e)}, -M^\top G_{p,(e)})$ of the orbits (16), for all bonds e and $M \in \mathbb{R}^{\chi_v \times \chi_v}$. With learning-rate multipliers or a mask the kernel is $J_t D J_t^\top$, the operator that drives the function-space dynamics of (15). The relative kernel change δ_Θ is computed, wherever the dense Jacobian fits in memory ($d^N \dim\theta \le 2 \cdot 10^7$, which covers the binary tree with $N = 4$ at every bond dimension used), from the tensor-space kernel $JDJ^\top$ on $\mathbb{R}^{d^N}$, $J = \partial W/\partial\theta$, whose constants the theory of Section A.8 gives; for whitened inputs the probe kernel is $\Phi(x_p)^\top(JDJ^\top)\Phi(x_{p'})$. Where the probe kernel was used instead it is named ($\delta_\Theta^{\mathrm{probe}}$), as in the regime labels of Section 3; near the lazy boundary $\delta_\Theta^{\mathrm{probe}}$ is 0.54–1.11 times δ_Θ (median 0.78).

Learning events. For CP teachers with orthogonal components (in the metric Σ), event r is dated at the last upward crossing t_r of $a_r = \frac{1}{2}$ before a_r first reaches 0.9: the first crossing unless a transient (while another component is learned) crossed and fell back, and unchanged if the learned component later dips below $\frac{1}{2}$. Its width w_r is the final rise, from the last upward crossing of 0.1 before a_r first reaches 0.9 to that time. This rule is recommended and is the one of the saddle-to-saddle (s2s-core) and regime-map experiments. The library (`runner.summary`) and the locality, optimizer and optim-samples experiments take instead the last upward crossing after which a_r stays at least $\frac{1}{2}$ until the end of the run, and the order, data-teacher (aligned covariance) and H2-pilot experiments

Table 2: Raw diagnostics recorded along every run.

symbol	definition
$L(t)$, $\ell(t)$	exact population loss (6), and $\ell = L(t)/L(0)$
$s_k^e(t)$, $s_k^{e,*}$	bond spectra of $W(t)$ and of W^* across edge e (function-space ones when $\Sigma \neq I$); $s_k^{e,*} = s_k$ for odeco teachers (Proposition 2.1)
$a_r(t)$	overlap (5) (CP teachers; of $W - W(\theta_0)$ in the centred model)
$\delta_\theta(t)$	$\|\theta(t) - \theta(0)\|/\|\theta(0)\|$, also per core
Θ_t	tangent kernel $J_t J_t^\top$ on the probe set; for small N also the tensor-space operator $\sum_v (\partial W/\partial G_v)(\partial W/\partial G_v)^\top$
$\delta_\Theta(t)$	relative kernel change $\|\Theta_t - \Theta_0\|_F/\|\Theta_0\|_F$
drift(t)	conservation drift (18)
$H(t)$, $H_\perp(t)$	Hessian of L and its gauge projection (see the saddle type)

the first crossing (code per experiment: `notes/diagnostics.md`). The first alternative differs only when a learned component later dips below $\frac{1}{2}$ (star teachers, SGD noise, rotations to a better component, a few deep-tree runs): it then dates the relearning, or gives no event. The second differs only when a transient crosses $\frac{1}{2}$ before the final rise. For other teachers one escape can raise several overlaps, so events (e, k) come from the bond spectra, by one of three rules (Section B gives each experiment's): (a) the onset, the first time s_k^e reaches $f s_k^{e,*}$ with $f = 0.05$ (sensitivity $f \in [0.03, 0.1]$), without widths (locality bond events, data-teacher v2 analyses); (b) half the teacher's value (first crossing for the optimizers, last upward crossing under correlated inputs), with final-rise widths from 0.1 to 0.9 of it; (c) the library default (`spectral_events`, used by `runner.summary`), the last upward crossing of half the run's final value, with final-rise widths from 0.1 to 0.9 of it; consecutive events whose gap is at most the larger width are then merged into one, with their mean time and largest width. Rule (a) is recommended: a half-value rule can skip intermediate saddles whose spectra stay below the final ones (Corollary A.37). Crossings are interpolated on the merged log and fine rows (or found by root finding on dense output); an event narrower than three sample intervals is unresolved, and separation ratios involving it are lower bounds. The *effective bond rank* is $\mathrm{rk}_e(t) = \#\{k : s_k^e(t) > 0.1\, s_{\min}^{e,*}\}$, with $s_{\min}^{e,*}$ the smallest nonzero teacher Schmidt value on e; the threshold-free alternative is the entropy rank $\mathrm{rk}_e^H = \exp(-\sum_k p_k \log p_k)$, $p_k = (s_k^e)^2/\sum_j (s_j^e)^2$.

Plateaus and the saddle-to-saddle signature. Number the events in time order, $t_1 < t_2 < \cdots$, with widths w_i; the *separation ratios* are

$$\rho_i = \frac{t_{i+1} - t_i}{\max(w_i, w_{i+1})}. \tag{19}$$

Between two events with $\rho_i > 3$ the loss sits on a plateau; its midpoint t_i^{mid} is the log time of minimal speed $\|\dot\theta\|$ between them. A run shows the *saddle-to-saddle (S2S) signature* when (i) there are at least two events, (ii) every $\rho_i > 3$, and (iii) across a sweep, plateau durations grow without bound as $\sigma \to 0$ (like $\log(1/\sigma)$ or a power of $1/\sigma$). Plateaus are judged in linear time: an escape that needs two cores takes a time of order $\log(1/\sigma)$, invisible on a log-time axis after a long first plateau.

Regime assignment. Each run, trained up to time T, receives one label (Table 3).

Table 3: Regime of one run trained up to time T (S2S signature: Section 2.7). Threshold sensitivity: the kernel threshold 0.1 is varied over $\{0.05, 0.1, 0.2\}$, the separation threshold 3 over $\{2, 3, 5\}$.

regime	rule
lazy	the loss converges and $\delta_\Theta(T) < 0.1$ (so δ_θ is small as well)
saddle-to-saddle	$\delta_\Theta(T) \geq 0.1$ and the S2S signature (i)–(iii)
rich, non-incremental	$\delta_\Theta(T) \geq 0.1$ and the loss converges, but some $\rho_i \leq 3$ (overlapping transitions)
stalled	$L(T) \geq 10^{-3} L(0)$ within the time budget
diverged	L non-finite or above $10^6 L(0)$ (GD and SGD; for gradient flow, an integrator failure)

Remark 2.6 (Reading of the rules). "Converges" means $L(T) < 10^{-3}L(0)$; labels are checked in the order diverged, stalled, then the converged regimes. Condition (iii) belongs to a sweep: a run is labelled S2S on (i)–(ii), and a region of a phase diagram is S2S only if (iii) holds there. A converged run with $\delta_\Theta(T) \geq 0.1$ and a single (merged) event is rich, non-incremental. The code (`classify_regime`) adds two labels: *lazy (projected fit)*, checked before stalled, for $\delta_\Theta(T) < 0.1$ and a final loss within 10% of that of the linearized fit (positive when a bond is deficient), and *undetermined* when every $\rho_i \leq 3$ involves an unresolved event. The regime map measures convergence against $\frac{1}{2}\|W^*\|_\Sigma^2$, which equals $L(0)$ when the fitted tensor vanishes at $t = 0$ (centred and two-copy models) and differs only for the plain model at large σ.

Saddle type at a plateau. At the plateau midpoint we compute H and its restriction $H_\perp$ to the orthogonal complement of the well-conditioned gauge directions: the left singular vectors of the unnormalized generator matrix with singular values above 10^{-3} times the largest (`project_out_gauge`). Normalizing the tiny generators of nearly unused bond indices, as an earlier cutoff of 10^{-10} did, removes part of the escape direction and biases $\lambda_{\min}$ near rank-deficient saddles (Section B.3); the raw $\lambda_{\min}(H)$ is always reported too. The type is read across a sweep from how $\lambda_{\min}(H_\perp)$ at the midpoint scales with σ (Proposition A.26): it tends to a negative constant at a *strict saddle* and vanishes like σ^{k-2} at a *higher-order saddle* of order $k \geq 3$ (a route through k cores, or the origin with $k = K$). A fixed threshold confuses the two: the single-σ fallback, strict if $\lambda_{\min}(H_\perp) < -10^{-3} \max_j |\lambda_j(H_\perp)|$, calls $k = 3$ plateaus strict for $\sigma \gtrsim 0.01$–0.04.

Reporting rules. Every scaling claim uses at least five initialization scales over at least one decade and five seeds per point, and reports mean ± standard error, a fitted exponent with its standard error, and the theoretical prediction. Every regime boundary comes with the threshold sensitivity check of Table 3, and every number states the dynamics that produced it (integrator tolerances, step size, batch size).

2.8 What the results may depend on

- *Population loss.* Plateaus and saddle sequences are properties of population gradient flow; implicit bias and generalization need finite samples.
- *Multilinear features, product input measure.* There is no pointwise nonlinearity, and the population loss is a tree contraction, which cross-site correlations break.
- *Placement of physical legs.* Binary trees and HT differ in K and in how many cores must change to learn a component, which sets plateau-time exponents (Section 4.4).

- *Fan-in initialization.* It fixes the relative scales of the cores; exponents should not depend on it, but prefactors and the learning order might, so imbalanced, balanced ($Q_e = 0$) and aligned initializations are also studied.

3 Map of regimes

Which regime of Table 3 does a run fall into, and which parameters decide it? This section maps the regimes of gradient flow on the centred model over the per-core initialization scale σ, the bond dimension χ, the number of cores K and the imbalance between the root and the other cores (Figure 2; Table 4 lists which parameter moves which boundary). On the binary tree with $N = 4$, training is saddle-to-saddle below $\sigma_c \approx 0.3/\sqrt{\chi}$, rich in a window of about 20 in σ, and lazy above $\sigma_{\text{lazy}} \approx 6.5/\sqrt{\chi}$, but lazy only when every bond reaches its cut dimension; below it the runs end on a floor. With more cores the saddle-to-saddle boundary rises and stalled runs invade its side, and a light root makes training lazy at a 100 times smaller function-space scale. Maps of this kind are known for linear and diagonal networks known (Woodworth et al., 2020; Tu et al., 2024; Kunin et al., 2024; Dominé et al., 2025); for tree tensor networks they are new new (for TTNs). Sections 5.1 to 5.4 look at the regimes other than saddle-to-saddle in more detail.

3.1 Protocol

The main family is the *centred* model (Section 2.4, variant (b)): its target $W^* + W_0$ removes the initial function at every σ without changing the parameterization, so σ alone moves a run from small to large initialization; the *plain* and *two-copy* models are cross-checks. Whitened Gaussian inputs, exact population loss, double precision; odeco teachers with weights $(3, 2, 1)$ at $d = 3$ or $(3, 2)$ at $d = 2$, Haar-random site bases from a seed stream independent of the initialization; fan-in initialization with one scale σ for every core unless stated. On the binary tree with $N = 4$ ($K = 3$, *binary4*) the bonds are not capped: $\chi = 9 = d^2$ is the cut dimension, larger χ over-parameterizes the bonds; other trees are capped by (3). Gradient flow (Dormand–Prince 5(4), or LSODA for long plateaus) runs until the loss falls below $10^{-6} \cdot \frac{1}{2}\|W^*\|_F^2$ or a time or step budget ends; every final rise is re-integrated on a linear grid, so the separation ratios are measured in linear time. Labels follow Table 3 at all nine threshold pairs, with events at the last upward crossing of $a_r = \frac{1}{2}$ before a_r first reaches 0.9. The lazy label uses the 128-probe kernel of Section 2.7 ($\delta_\Theta^{\text{probe}}$); comparisons with theory use the tensor-space kernel $JDJ^\top$ (δ_Θ). Near the lazy boundary the probe value is 0.54–1.11 times the tensor value (median 0.78; 43 runs at $\chi \geq 32$) empirical, so probe-kernel lazy boundaries lie about 5% lower in σ. Six seeds per point; boundaries are found per seed (saddle-to-saddle where $\min_i \rho_i$ crosses 3, or a bracket where a stalled run follows the last saddle-to-saddle run; lazy where δ_Θ crosses 0.1) and reported as geometric means $\times/$ their standard-error factor, or medians with quartiles or brackets. Details and checks: Section B.4.

3.2 Initialization scale and bond dimension

Setup. Binary4, $d = 3$, teacher $(3, 2, 1)$: $\chi = 3$–36 with 19 values of $\sigma \in [0.01, 40]$, and $\chi = 9$–128 with 22 values of $\sigma \in [0.004, 4]$ (Figure 2(a)).

Three regimes, ordered in σ. empirical new (for TTNs) At every χ the runs are saddle-to-saddle at small σ and rich (converged, $\delta_\Theta \geq 0.1$, overlapping events) at intermediate σ; at large σ they are lazy if $\chi \geq 9$ and stalled if $\chi < 9$. The saddle-to-saddle boundary falls from $0.237 \times /1.32$ at $\chi = 3$ to $0.136 \times /1.11$ at $\chi = 9$ and $0.043 \times /1.05$ at $\chi = 36$; at $\chi = 3$ it lies within the spread

of 30 plain-model seeds (median 0.27–0.31, Section A.8). Below it, $\min_i \rho_i \propto \sigma^{-p}$ per seed with $p = 1.015$–1.048 in 96 of the 101 fits (all χ of both sweeps), against $K - 2 = 1$ (derived for the first gap, Section A.8); the exceptions (1.09–1.27) are seeds with transients above $\frac{1}{2}$. The gaps grow while the widths stay at the aligned-mode values of Proposition A.48 (0.464–0.468, 0.798–0.801, 2.010 against 0.464, 0.797, 2.009 for $s_r = 3, 2, 1$; $\chi = 36$, $\sigma = 0.01$). Extrapolating $\rho \propto 1/\sigma$ from each seed's deepest run gives 0.86–1.23 times the measured boundary for 53 of 54 seeds, a consistency check, not a prediction: the χ-dependence of σ_c is not derived.

The lazy boundary. empirical confirming derived Proposition A.47 known (the criterion, a threshold on the kernel change: Geiger et al., 2020; a coherent second-order kernel change in wide networks: Bordelon and Pehlevan, 2022) new (the TTN constants) With the tensor kernel the lazy boundary is 1.13, 0.94, 0.82, 0.66 and 0.571 at $\chi = 32$, 48, 64, 96, 128 (medians of six seeds), inside the predicted brackets $[1.10, 1.20]$, $[0.92, 0.96]$, $[0.80, 0.82]$, $[0.66, 0.67]$ and 0.07% below the lower edge of $[0.571, 0.577]$: $\sigma_{\text{lazy}}\sqrt{\chi} = 6.40$–$6.56$ against the asymptote 6.50 of (60). At $\chi = 9$–16, where the isotropic constants are not expected to hold, it lies 1.5–6% above the bracket. Near the boundary δ_Θ is 0.96 (median; 0.69–1.82) times the law $\sqrt{(c_1\xi)^2 + (c_2\xi^2)^2}$ of (59): at large χ the coherent second-order change sets the boundary, and the first-order estimate holds only up to $\chi = 12$ (0.59 against the measured 1.09 at $\chi = 36$).

No lazy regime below the cut dimension. derived (Section A.8) empirical With $\chi < 9$ the centred target has more bond rank than the student can carry. All 159 stalled centred runs at $\chi = 3$, 4, 6 end on the Eckart–Young minimum of $W^* + W_0$ across the cut $\{1, 2\}|\{3, 4\}$ (to 0.04%), and every seed stalls from $\sigma = 1.0$, 1.0, 1.6, 1.7–2.2 times the small-σ estimate (61) (Section 5.1).

Both boundaries scale like $\chi^{-1/2}$. empirical known (as a scaling: the width-dependent scales $\sigma^2 w$ of Tu et al., 2024 and $\alpha\sqrt{h}$ of Geiger et al., 2020) Fits to the per-seed boundaries with full bonds give the slope -0.528 ± 0.053 against χ for the saddle-to-saddle boundary ($\chi = 16$–128, 42 values, 0.9 decade; per-seed fits -0.528 ± 0.073; -0.552 ± 0.049 over $\chi = 9$–128) and -0.504 ± 0.011 for the lazy boundary ($\chi = 32$–128; -0.553 ± 0.013 over $\chi = 16$–128, with $\chi = 16$, 24 not yet asymptotic). The measured $\sigma_c\sqrt{\chi}$ is 0.27–0.33 at $\chi = 32$–128, and the rich window $\sigma_{\text{lazy}}/\sigma_c$ spans a factor 18–23 in σ (probe kernel; 19–24 with the tensor kernel), about 10^4 in $\alpha = \sigma^3$. At large χ the map thus depends on $\sigma\sqrt{\chi}$, the per-bond Gram scale $p = \chi\sigma^2/d^2$ (the eigenvalue scale of the subtree Grams $T_v^\top T_v$): saddle-to-saddle below $p \approx 0.01$, lazy above $p \approx 4.7$. Neither σ^K (slope 0) nor the amplitude per path times the number of paths, $\sigma^3\chi/d^2$ (slope $-\frac{1}{3}$), collapses the data: $-\frac{1}{3}$ lies 2.7–2.9 standard errors from the saddle-to-saddle slope in per-seed fits ($\chi \geq 16$, $\chi \geq 32$; 3.7 in the pooled fit, which treats seeds sharing a teacher across χ as independent). Scope: one tree (a three-factor matrix product), fan-in initialization, less than one decade of χ. The saddle-to-saddle scaling is not derived; heuristically, the leaf Grams grow like p while the root is light by $\sqrt{\chi}$, so the effective per-core amplitude of the escape grows like $\sigma\sqrt{\chi}$ conjecture.

Seven cores. empirical On the binary tree with $N = 8$ ($K = 7$) the regimes come in the same order. At $d = 3$ no centred run with $\chi \leq 24$ is lazy (the top bonds have cut dimension 81; floor from $\sigma = 0.91$–1.34). At $d = 2$ (cut dimensions 4 and 16) the saddle-to-saddle boundary is 0.83 $[0.72, 0.97]$ for $\chi = 4$, 8, 16 and 0.53 $[0.46, 0.61]$ for full bonds over-parameterized twice (medians of six seeds and of their brackets), and only the latter reach a lazy regime, from $\sigma \approx 1.4$ (Section 5.1): the rich window spans a factor 2.7 in σ, against 18–23 at $K = 3$ (with another d and teacher).

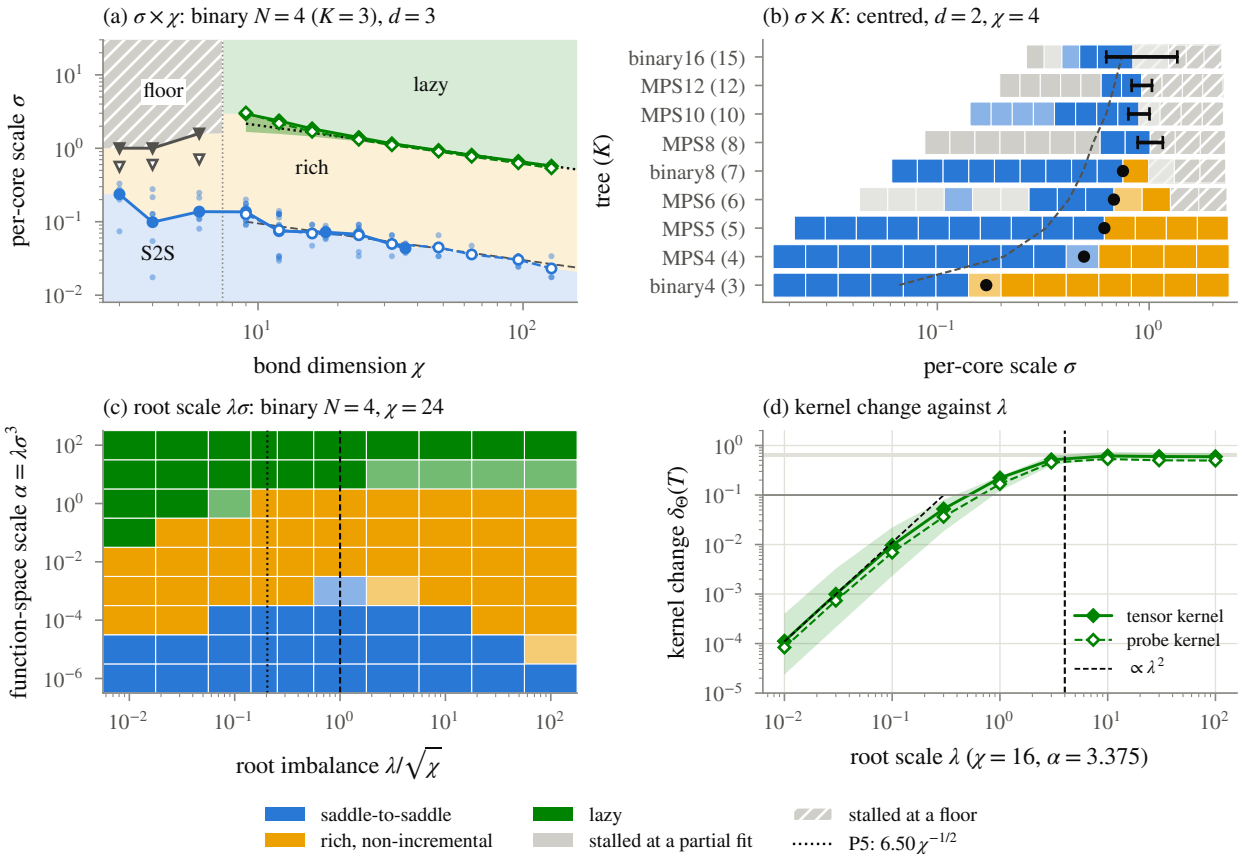


Figure 2: Map of regimes: gradient flow, centred model, fan-in initialization; regions in (a), majority labels of six seeds in (b), (c) (pale: at most three runs agree). (a) Binary $N = 4$, $d = 3$, teacher $(3, 2, 1)$; per-seed boundaries (dots) and geometric means. Blue: saddle-to-saddle boundary from the $\sigma \times \chi$ sweep (filled, $\min_i \rho_i = 3$) and the χ sweep (open, label bracket); dashed: $0.30\,\chi^{-1/2}$ (a guide). Green: lazy boundary ($\delta_\Theta = 0.1$; filled: tensor kernel, open: probe kernel), bracket of Proposition A.47 (band) and its asymptote (dotted). Triangles: σ from which every seed stalls on the floor (filled) and its small-σ estimate (61) (open); bonds are full right of the dotted vertical. (b) $d = 2$, teacher $(3, 2)$, $\chi = 4$ (full bonds only for $K \leq 5$). Dots: median per-seed boundary; bars: median bracket from the last saddle-to-saddle run to the next, stalled one; dashed: the aligned-initialization boundary $a_{0c}(K)$ (Proposition A.48). Hatched: stalled above the saddle-to-saddle runs, on the floor of the unrealizable target; plain grey: stalled below them, at a partial fit. (c) Root at scale $\lambda\sigma$, other cores at σ, fixed $\alpha = \lambda\sigma^3$, $\chi = 24$; dashed: balance ($\mathbb{E}Q_e = 0$ on the root bonds); dotted: fan-in default $\lambda = 1$. (d) δ_Θ against λ at $\chi = 16$, $\alpha = 3.375$: medians and range; dashed: $\propto \lambda^2$; grey band: predicted saturation; vertical: balance. Takeaway: on binary4 both boundaries scale like $\chi^{-1/2}$ and only full bonds give a lazy regime; with more cores the saddle-to-saddle boundary rises and stalls invade its side; a light root makes training lazy.

3.3 Number of cores

Setup. $d = 2$, teacher $(3, 2)$, $\chi = 4$ (twice the teacher rank) on every tree, so the bonds are full for $K \leq 5$ and deficient from $K = 6$ on: binary trees with $N = 4$, 8, 16 ($K = 3$, 7, 15) and MPS with $N = 4$, 5, 6, 8, 10, 12; 11–14 values of σ from $\max(10^{-8/K}, 0.02)$ to 2 (Figure 2(b)).

Resolved up to seven cores. empirical new (for TTNs) The saddle-to-saddle boundary rises with K: medians of the per-seed boundaries 0.17 (quartiles 0.11–0.20), 0.49, 0.61, 0.68 and 0.75 at $K = 3$–7, with all six seeds resolved except at $K = 6$ (four). It stays above the aligned-initialization boundary $a_{0c}(K) = 0.066$–0.49 of Proposition A.48; read as aligned initialization with per-core amplitude $\kappa_K \sigma$, $\kappa_K = a_{0c}/\sigma_c$ grows from 0.39 to 0.65, as in the runs of Section A.8. The plain model gives the same boundaries within one grid step up to $K = 7$; beyond, it must first remove W_0.

Beyond seven cores: brackets. empirical At $K = 8$, 10, 12, 15 only 1, 1, 0, 0 of six seeds are resolved: the run after the last saddle-to-saddle run is stalled on the floor of the unrealizable target, so the label rule gives brackets, with medians $[0.88, 1.16]$, $[0.80, 1.00]$, $[0.83, 1.03]$, $[0.63, 1.36]$: lower bounds, not crossings. Classified by $\min_i \rho_i$ instead, that run is saddle-to-saddle-like ($\rho = 3.9$–642) for 13 of the 18 unresolved MPS seeds. At $K = 8$ the block then ends at a crossing in 5 of 6 seeds, median 1.01; at $K = 10$ and 12 such runs persist up to $\sigma \approx 1.0$–1.3 and the next runs learn nothing within the budget, so a saddle-to-saddle-to-rich crossing is seen in 2 of 6 seeds at $K = 10$ (both at 0.90) and in none at $K = 12$. All these runs end on the floor ($L/L(0) = 0.001$–0.074), and 9 of the 13 learn the smaller component first: they do not measure the boundary of a realizable problem.

Which scale stays fixed? empirical On $K = 4$–7 both $\sigma_c^K = \text{const}$ and $\sigma_c^K/\sqrt{d^N} = \text{const}$ (a fixed per-component amplitude, Proposition A.46(b)) fit the medians ($\chi^2 = 0.79$ and 0.59, three degrees of freedom), and adding the per-seed values whose bracket ends at a crossing ($K = 8$, two seeds at $K = 10$) leaves them indistinguishable ($\chi^2 = 7.5$ and 7.3, five degrees of freedom). If $\kappa_K \to 1$, $\sigma_c \to \lim_K a_{0c}(K) = 1$, like the per-layer threshold of aligned initialization known (Gissin et al., 2020, Thm 2); a fixed per-component amplitude would give $\sigma_c \to \sqrt{d}$. With all runs at $d = 2$ and brackets beyond $K = 7$, the data distinguish neither limit conjecture.

Stalls on the saddle-to-saddle side. empirical Below the boundary stalled runs take a growing part of the map (plain grey): at the smallest σ, 0, 1, 0, 3, 0, 5, 3 and 6 of six seeds stall on binary4, MPS4, MPS5, MPS6, binary8, MPS8, MPS10 and MPS12, and 5 of 5 on binary16, each at a partial fit with one component learned ($L/L(0) = 4/13$ or $9/13$). A stalling MPS seed stalls up to a seed-dependent σ, then turns saddle-to-saddle; with χ equal to the teacher rank stalls are more frequent still (Section 5.3).

The boundary at large K. *[To be filled from the second round of the regime-map experiment (remote, running): a realizable K sweep (spare channels or full bonds) at $d = 2$ and $d = 3$ with longer budgets at $K \geq 8$, and a pre-registered comparison of $\sigma_c^K = \text{const}$ with $\sigma_c^K/\sqrt{d^N} = \text{const}$. Until then the values for $K \geq 8$ are brackets on an unrealizable target, and the limit of $\sigma_c(K)$ is a conjecture.]*

3.4 Imbalance between cores

Setup. Binary4, $d = 3$, teacher $(3, 2, 1)$. The root starts at scale $\lambda\sigma$, the other cores at σ, so $\alpha = \lambda\sigma^3$ is fixed while λ sets the conserved imbalance of the root bonds, $\mathbb{E}Q_e = \sigma^2(1 - \lambda^2/\chi)I$ (Section 2.6): balance at $\lambda = \sqrt{\chi}$; the fan-in default $\lambda = 1$ is root-light. Grids: $\chi = 24$ (full bonds)

Table 4: Which parameter moves which boundary (Section 3.7). Gradient flow, centred model, fan-in initialization, odeco teachers, six seeds per point; binary4: binary tree with $N = 4$, $d = 3$, teacher $(3, 2, 1)$. Lazy boundaries with the tensor kernel; predictions (in brackets) from Section A.8. Scope: trees with at most 16 sites; the χ scalings rest on one tree and less than one decade.

parameter	effect	measured (predicted)	status
scale σ	saddle-to-saddle, rich, then lazy or a floor	binary4: $\sigma_c = 0.24$ to 0.023, $\sigma_{\text{lazy}} = 3.0$ to 0.57 for $\chi = 3$–128	empirical
bond dimension χ, full bonds	both boundaries $\propto \chi^{-1/2}$	slopes -0.53 ± 0.05 and -0.504 ± 0.011 $(-\frac{1}{2})$; $\sigma_{\text{lazy}}\sqrt{\chi} = 6.40$–6.56 (6.50); rich window 18–23	empirical derived
χ below the cut dimension	no lazy regime, a floor	all seeds stalled from $\sigma = 1.0$, 1.0, 1.6 at $\chi = 3$, 4, 6	derived empirical
cores K $(d = 2$, $\chi = 4)$	σ_c rises; stalls below it	$\sigma_c = 0.17$, 0.49, 0.61, 0.68, 0.75 for $K = 3$–7; brackets near 1 for $K = 8$–15	empirical conjecture
root imbalance λ at fixed α	light root: lazy at up to 100× smaller α	$\delta_\Theta \propto \lambda^2$ (slope 1.85 ± 0.06; 1.8–1.9), saturation 0.59–0.61 (0.6–0.7)	empirical derived
parameterization, large χ	lazy (SP, NTK) or not (μP-TTN)	$\delta_\Theta \propto \chi^{-3.02\pm0.06}$ (-3), $\chi^{-1.01}$ for $\chi \geq 64$ (-1), $\to 6.2$ (6.2–6.3); Section 5.4	empirical derived
thresholds δ_*, ρ	boundaries move, labels robust	$\times 1.30$–1.35 for δ_*: 0.05 $\to$ 0.2 (1.27–1.36); $\times 2.14$–2.34 for ρ: 2 $\to$ 5 (2.5)	empirical

and $\chi = 3$, nine values of α from 10^{-6} to 10^2 times ten of $\lambda/\sqrt{\chi}$ from 0.01 to 100; and $\chi = 16$, $\alpha = 3.375$ $(\xi = \|W^*\|_F/\alpha = 1.11)$, $\lambda = 0.01$–100 (Figure 2(c), (d)).

A light root makes training lazy; a heavy one does not. empirical new (for TTNs) At $\chi = 24$ the runs are lazy from $\alpha = 0.1$ at $\lambda/\sqrt{\chi} = 0.01$ (4 of 6 seeds) and from $\alpha = 1$ at 0.03 (6 of 6; at 0.1 a tie, 3 of 6), against $\alpha = 10$ at balance (4 of 6): the lazy boundary moves by up to a factor 100 in α. A heavy root ($\lambda/\sqrt{\chi} \geq 3$) is lazy from $\alpha = 10$ (3 of 6; 6 of 6 at $\alpha = 100$), as at balance. The saddle-to-saddle majority reaches $\alpha = 10^{-4}$ for $0.1 \leq \lambda/\sqrt{\chi} \leq 10$ and 10^{-5} or 10^{-6} outside; on this one-decade grid no widest point near balance can be read off. With $\chi = 3$ every run stalls on the floor from $\alpha = 1$.

Kernel change at fixed α. empirical confirming derived (Section A.8) At $\chi = 16$, $\alpha = 3.375$, δ_Θ grows like λ^2 from small λ (per-seed local slopes 1.97 ± 0.01, 1.85 ± 0.06, 1.55 ± 0.11, 1.22 ± 0.09 between $\lambda = 0.01$, 0.03, 0.1, 0.3, 1; predicted 1.8–1.9 between 0.03 and 0.1) and saturates at 0.59–0.61 for $\lambda \geq 10$ (medians, seeds 0.51–0.75; at the lower edge of the predicted 0.6–0.7); the runs are lazy up to $\lambda = 0.3$ and rich from $\lambda = 3$. In the aligned mode a small root is the only core that moves, so the model is linear in it, while a large root freezes and leaves a degree-$(K-1)$ model derived. Two-layer linear networks share the first effect, a lazy phase that turns rich when the features are incomplete, but there a large output weight also gives lazy training known (Kunin et al., 2024; Dominé et al., 2025); for $K \geq 3$ the model left by a frozen root is not linear.

3.5 Threshold sensitivity and cross-checks

empirical (predictions derived) Raising the separation threshold from 2 to 5 moves the saddle-to-saddle boundary of binary4 by a factor 2.14–2.34 ($\chi = 3$–36; means of six seeds, s.e. at most 0.08), against $(5/2)^{1/(K-2)} = 2.5$: the $O(1)$ part of ρ lowers the ratio at finite σ. Raising the kernel threshold from 0.05 to 0.2 moves the lazy boundary by 1.97 ± 0.08 at $\chi = 9$, down to 1.33 ± 0.01 at $\chi = 36$ and 1.30–1.35 at $\chi = 32$–128: with $\sigma_{\text{lazy}} \propto \delta_*^{-1/(Ks)}$, the local slope $s = \mathrm{d}\ln\delta_\Theta/\mathrm{d}\ln\xi$ grows from

1 (first order, ratio 1.59) towards 2 (second order, 1.26), and the slopes 1.5–1.8 at $\chi \geq 32$ predict 1.27–1.36. Only the boundaries move: across the nine threshold pairs at most 65 of the 912 centred binary4 labels change (7.1%), each to a neighbouring regime. The two-copy model has the same tangent space as the centred one, but its kernel changes only at second order, so its lazy boundary is $2^{-1/3} = 0.794$ times lower (Proposition A.47): measured 0.79 at $\chi = 32$, 64, 128 (tensor kernel). Its saddle-to-saddle side is not comparable (merged first events, Section 5.2).

3.6 Step size

The map is that of gradient flow; Section 5.5 adds the step size. Gradient descent keeps gradient flow's saddle sequence, with event times shifted by $\eta c_e + O(\eta^2)$, up to a stability limit derived: at small initialization $\eta_c = 2/(K s_1^{2-2/K})$ (24), independent of σ (0.154 on binary4), and on the lazy side $2/\lambda_{\mathrm{GF}}$, which falls like $\sigma^{-3.965 \pm 0.014}$, close to the kernel scale $\sigma^{-2(K-1)}$ (centred binary4, $\chi = 9$) empirical. Beyond the limit GD oscillates at the edge of stability or diverges from small initialization, and catapults to flatter minima from $\sigma \geq 1$ (Figure 13(c)). Online SGD adds a mean-square limit and lengthens or traps later plateaus at fixed η/B, turning slow escapes into stalls.

3.7 Summary: which parameter moves which boundary

Table 4 collects the boundaries and what moves them; the evidence and its scope are given in Sections 3.2 to 3.5.

4 The saddle-to-saddle regime

[Skeleton. Theory and experiments for gradient flow from small initialization: plateau sequence, escape times and their dependence on each parameter, locality of later escapes, learning order, saddle types.]

4.1 Phenomenology

Gradient flow from a small fan-in initialization learns an odeco teacher one component at a time. This subsection shows such runs, where complete sequences were observed, and which saddles the plateaus are; Section 4.3 treats the first escape, and Section 4.4 teachers whose components differ on a few sites only.

Setup (details in Section B.2). Odeco teachers (Section 2.3) with $R = 3$, $d = 4$, Haar-random site bases and weights $(3, 2, 1)$ (*separated*) or $(1, 0.85, 0.7)$ (*close*); trees binary4 ($K = 3$), MPS4 ($K = 4$), binary8 and HT4 ($K = 7$), MPS8 ($K = 8$); $\chi = 3 = R$ or $\chi = 6$, capped by (3) (at $d = 4$ on MPS end bonds and HT leaf bonds). Fan-in initialization with seeds 0–9 at eight values of $\sigma \in [0.003, 0.1]$ ($K = 3, 4$) or five or six in $[0.07, 0.4]$ ($K = 7, 8$; smaller σ is out of reach, as the later plateaus grow like σ^{-5} and σ^{-6}); also the same W_0 re-gauged to $Q_e = 0$ (*balanced*). Gradient flow by LSODA with the exact Hessian as Jacobian; every event is re-integrated on a fine grid. A run is *complete* if it learns all three components; Section 4.6 treats the others.

Prediction. derived Two odeco components differ at every site, so the *route* to a new component, the set of cores that must move together (Section 4.4), is the whole tree at every saddle: $k = K$ in (22) (Proposition A.19). Hence (i) the plateaus sit at the partial fits $W_S = \sum_{r \in S} s_r U_r$, S the components learned so far, at the loss

$$L_S = \tfrac{1}{2} \sum_{r \notin S} s_r^2 \tag{20}$$

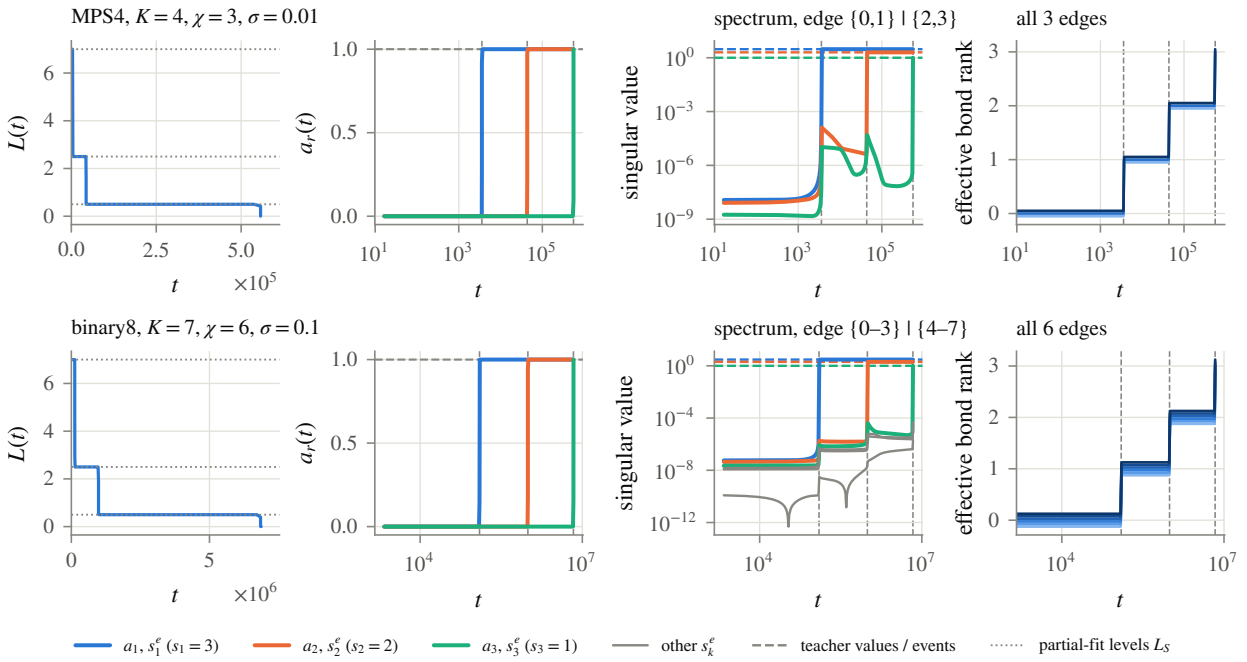


Figure 3: Two saddle-to-saddle runs: separated teacher $(3, 2, 1)$, fan-in initialization, seed 0; top: MPS4, $\chi = 3$, $\sigma = 0.01$; bottom: binary8, $\chi = 6$, $\sigma = 0.1$. Columns: loss in linear time with the partial-fit levels (20) (dotted); overlaps a_r; singular values s_k^e of the middle edge (MPS4) and the root edge (binary8), teacher values dashed; effective bond rank (Section 2.7) of every edge, offset slightly per edge. Dashed verticals: events, at $3.6 \cdot 10^3$, $4.4 \cdot 10^4$, $5.6 \cdot 10^5$ (MPS4) and $1.3 \cdot 10^5$, $9.8 \cdot 10^5$, $6.8 \cdot 10^6$ (binary8), with widths below 2 empirical. Takeaway: one component per event, in the order of the weights; every edge gains one singular value at each event, all at the same time, and the next ones stay far below the teacher's in between.

(L_k of (10) in the greedy order), with bond spectrum $(s_r)_{r \in S}$ on every edge (Proposition 2.1(a)); (ii) each event raises the rank of every edge by one; (iii) every plateau is the first-escape problem of a K-core route (Proposition A.18) and lasts $\propto \sigma^{-(K-2)}$ for a fixed initialization direction; (iv) the plateaus are higher-order saddles, $\lambda_{\min}(H_\perp) \propto \sigma^{K-2}$ (Proposition A.26). The exception to (iv) is binary4, where only two cores carry sites: a learned set that misses a larger component is a strict saddle, $\lambda_{\min} = -(s_r - s_j)\, s_j^{(K-2)/K}$ (j learned, r larger and not learned; Proposition A.22), reached only after a wrong first escape. Binary4 is a depth-3 deep linear network (Remark A.23), for which all of this is known known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024); the other four trees are genuine tensor networks.

Scope. empirical new (MPS4, binary8, HT4; known for binary4) Complete runs: binary4 320 of 320 (fan-in, both teachers, $\chi = 3, 6$) and 60 of 60 (balanced); MPS4 160/160 at $\chi = 6$, 106/160 at $\chi = 3$ and 60/60 balanced; binary8 50/50 and HT4 45/50 at $\chi = 6$ ($\sigma = 0.07$–0.3). At $\chi = 3$ few runs with $K = 7$ complete (binary8 14 of 180, HT4 12 of 150), and no MPS8 run learns its third component (0 of 284; 132 learn two; Section 4.6). Complete sequences were thus measured on binary4 and MPS4 over 1.5 decades of σ, on binary8 and HT4 at $\chi = 6$ over 0.6 decade, and on MPS8 up to the second component.

Plateaus at the rank-constrained partial fits, one rank step per event. empirical new (MPS4, binary8, HT4; known for binary4) At every plateau between resolved events ($\rho_i > 3$) the loss equals

Table 5: Plateau durations and saddle type, odeco teachers, fan-in initialization. Gaps: per-seed exponent p of $t_{i+1} - t_i \propto \sigma^{-p}$, mean ± s.e. over seeds (run sets: Section B.2), for $\chi = 3$ and 6 or, in rows "both", the separated and close teachers; brackets: range of the per-seed values (binary4, MPS4: over their four cells). $\lambda_{\min}$: mean per-seed slope of $\ln(-\lambda_{\min}(H_\perp))$ against $\ln\sigma$ at the first and second plateau midpoints ($\chi = 3$; 6, or the two teachers). Predicted: $K - 2$. Ten seeds per fit except binary4 close (9, 9; 8, 7), MPS4 gap 2 at $\chi = 3$ (6, 7), HT4 gap 2 (9), MPS8 (7; 3). σ: 0.003–0.1 (binary4, MPS4), 0.07–0.3 at $\chi = 6$, 0.03–0.4 (binary8) and 0.1 or 0.15 to 0.4 (MPS8) at $\chi = 3$; †: less than one decade (binary8 at $\chi = 3$: 1.1 decades for 11 of 20 seeds). Source: `main_exponents.csv`.

tree	teacher	χ	$t_2 - t_1$	$t_3 - t_2$	$\lambda_{\min}$ slopes
binary4	sep.	3, 6	1.003 ± 0.001, 1.007 ± 0.001	1.004 ± 0.003, 1.011 ± 0.002	1.05, 1.02; 1.08, 1.04
($K = 3$)	close	3, 6	1.004 ± 0.010, 0.999 ± 0.008 [0.95, 1.06]	1.016 ± 0.011, 1.014 ± 0.006 [0.99, 1.09]	1.17, 1.12; 1.18, 1.17
MPS4	sep.	3, 6	2.000 ± 0.000, 2.001 ± 0.001	1.990 ± 0.005, 1.998 ± 0.001	2.01, 1.99; 2.00, 2.00
($K = 4$)	close	3, 6	2.001 ± 0.001, 2.001 ± 0.001 [2.00, 2.01]	2.004 ± 0.002, 2.001 ± 0.001 [1.97, 2.01]	2.01, 2.00; 2.02, 2.00
binary8	sep.	6†	5.020 ± 0.009 [4.98, 5.05]	4.969 ± 0.022 [4.88, 5.09]	5.10, 5.07
($K = 7$)	both	3	4.93 ± 0.05, 4.92 ± 0.04 [4.65, 5.13]	–	5.03; 5.03
HT4 ($K = 7$)	sep.	6†	5.10 ± 0.08 [4.83, 5.78]	4.75 ± 0.14 [3.93, 5.33]	5.13, 4.88
MPS8	sep.	6†	5.95 ± 0.05 [5.79, 6.20]	not reached	5.99
($K = 8$)	both	3†	6.0 ± 0.3, 6.2 ± 0.6 [5.14, 7.21]	not reached	5.88; 6.07

(20) for the components learned so far, in whatever order. The largest relative deviation is $8.5 \cdot 10^{-3}$ (binary4, separated teacher, $\sigma = 0.1$), $2.0 \cdot 10^{-2}$ (binary4, close teacher, at $\sigma = 0.003$ on a plateau with $\rho = 5.1$), $2.8 \cdot 10^{-3}$ (MPS4; $8.1 \cdot 10^{-3}$ balanced), and $2.1 \cdot 10^{-5}$ and $1.5 \cdot 10^{-4}$ at $K = 7, 8$ ($\chi = 6$, 3); at $\sigma = 0.003$ it is at most $7.4 \cdot 10^{-7}$, except on binary4 with the close teacher. At these plateaus, in runs with the S2S signature, the top $|S|$ singular values of every edge equal the learned weights to a relative $7.7 \cdot 10^{-3}$ (binary4, close), $9.0 \cdot 10^{-4}$ (binary4, separated; $1.7 \cdot 10^{-9}$ at $\sigma = 0.003$) and $3.5 \cdot 10^{-4}$ (other trees), and the next one is below 0.07 times the smallest weight: the plateaus are the critical points W_S of the problem with all bond ranks fixed at $|S|$. In 497 of the 531 complete runs that recorded spectra in their event windows, every event raised by exactly one, on every edge, the number of singular values above fs_r (r the component being learned; $f = \frac{1}{2}$ or a nearby value that separates s_r from the learned weights); the exceptions are 28 binary4 runs with the close teacher at $\sigma \geq 0.05$, whose events overlap, and 6 HT4 runs.

Plateaus grow as $\sigma^{-(K-2)}$. empirical new (MPS4, binary8, HT4, MPS8; known for binary4) Every separation ratio exceeds 3 in all complete runs on MPS4 (326), binary8 (83) and HT4 (74), and in 218 of 220 on binary4 with the separated teacher; with the close teacher on binary4 in 5 of 20 runs at $\sigma = 0.1$, rising to 20 of 20 at $\sigma = 0.003$, as close weights make escapes overlap unless σ is small. Per-seed fits of $\ln(t_{i+1} - t_i)$ against $\ln\sigma$ (Table 5; run sets in Section B.2) give $K - 2$ to within 0.1 per seed on binary4 and 0.04 on MPS4, over 1.5 decades. At $K = 7, 8$ the means agree with $K - 2$ (binary8 5.02 ± 0.01 and 4.97 ± 0.02, MPS8 5.95 ± 0.05 for its only gap) but over 0.6 decade, below the reporting rule of Section 2.7; binary8 at $\chi = 3$, over up to 1.1 decades, gives 4.93 ± 0.05, and HT4 is not a clean confirmation (per seed 3.93–5.78).

Saddle type. empirical new (MPS4, binary8, HT4, MPS8; known for binary4) At the plateau midpoints $\lambda_{\min}(H_\perp)$ vanishes like σ^{K-2}: the cell means of the per-seed slopes are 1.02–1.18 (binary4), 1.99–2.02 (MPS4), 5.07–5.10 (binary8), 4.88–5.13 (HT4) and 5.99 (MPS8), with $\chi = 6$ for $K \geq 7$ (Table 5; the raw Hessian agrees to 0.004). In cell medians the magnitude is 0.87–1.15 times the balanced-route

value of (46) at $K = 3, 4$ and 1.36–3.15 times it at $K = 7, 8$, where the routes are not balanced; per seed the ratio reaches 20. The strict exception appears as predicted known (deep linear networks: Achour et al., 2024, Thm 7; Jacot et al., 2022, Lemma 15). With the close teacher on binary4 ($\chi = 3$), seed 8 learns in the order 1-3-2 at seven of eight values of σ (0.005–0.1), and seed 5 at two. At the plateau where 1 and 0.7 are learned and 0.85 is not, $\lambda_{\min}$ is -0.106 to -0.146 at the nine midpoints (raw: -0.113 to -0.149) and flat in σ (seed 8: slope 0.03), against $-(0.85 - 0.7)\,0.7^{1/3} = -0.133$; the next event follows within about one width ($\rho = 0.76$–1.25), and this gap does not grow as $\sigma \to 0$ (seed 8: $\propto \sigma^{0.08}$). At $\sigma = 0.003$ seed 8 learns greedily: the wrong order, and with it the strict saddle, becomes rare as $\sigma \to 0$ (Proposition A.15). Otherwise the full order on binary4 is greedy (160 of 160 complete runs with the separated teacher; 89% and 95% at $\chi = 3$, 6 with the close one); on the other trees it varies from seed to seed (Section 4.3).

4.2 An exactly solvable case: aligned initialization

If every core starts diagonal in the teacher's basis (Proposition 2.4), the components decouple exactly: component r is a product of K per-core amplitudes obeying (13), and a balanced start stays balanced, $\dot{q}_r = q_r^{K-1}(s_r - q_r^K)$, the mode equation of deep linear networks known (Saxe et al., 2014, eq. 15), here on any tree proven. Integrating it gives (14), $t_r \simeq q_r(0)^{-(K-2)}/((K-2)s_r)$ known (from eq. 15 of Saxe et al., 2014; symmetric matrix factorization from αI: Li et al., 2021, Thm 6.2): with one initial amplitude for all components they are learned in the order of their weights, at times in the ratio $1/s_1 : 1/s_2 : 1/s_3$, and every gap grows like $\sigma^{-(K-2)}$. This is the case $k = K$ of Proposition A.25, where a route through every core never sees the components already learned. With unequal amplitudes the escape time is set by the cores that start smallest: m of them at γ, far below the others, give a time that stays $O(1)$, grows like $\ln(1/\gamma)$, or grows like $\gamma^{-(m-2)}$ for $m = 1$, 2, ≥ 3 derived known (the trichotomy: Rawal and DeWeese, 2026, Thm 4.2; new: the TTN reduction, Proposition A.13). Random initialization is of this unbalanced kind, which is why its escape times are not those of a balanced start of the same norm (Section 4.3).

Test (Figure 4). empirical Separated teacher, $d = 4$, $\chi = 3$, balanced aligned initialization with per-core amplitude $q_0 = 0.01$ (binary4), 0.07 (MPS4), 0.29 (binary8, HT4) and 0.34 (MPS8), so that the events fall near $t = 33$, 50, 100 on every tree. Gradient flow on the full TTN (Dormand–Prince 5(4), tolerances 10^{-10} and 10^{-14}) follows the exact solutions: the largest deviation of any overlap is $2.0 \cdot 10^{-10}$ (binary4), $3.2 \cdot 10^{-10}$ (MPS4), $2.3 \cdot 10^{-9}$ (binary8), $2.6 \cdot 10^{-9}$ (HT4) and $2.7 \cdot 10^{-9}$ (MPS8); event times agree on the same output grid to $1.3 \cdot 10^{-11}$ relative; the conservation drift (18) stays below $3 \cdot 10^{-14}$. LSODA, the integrator of all other runs of this section, deviates by at most $7.1 \cdot 10^{-7}$. The test checks contractions, gradients, integrators and event detection on all five trees.

4.3 Escape from the origin

How long does the first plateau last, and which component ends it? The research plan's first guess used only the size of the initialization, $t_1 \propto \|\theta_0\|^{-(K-2)}$. The answer is a blow-up time computed for each initialization, with no free parameter; the norm gives only a lower bound, low by factors from 3 to $3 \cdot 10^7$.

Prediction: one blow-up time per initialization. On the first plateau $\|W\|_F$ is far below the teacher's weights, and gradient flow is to leading order the flow $\dot{\theta} = \nabla g$ of the correlation $g = \langle W, W^* \rangle$ (Section A.2): its direction follows the spherical gradient flow of g, independent of σ, and its norm blows up at the time (29). As g is homogeneous of degree K, the initialization $\theta_0 = \sigma Z$

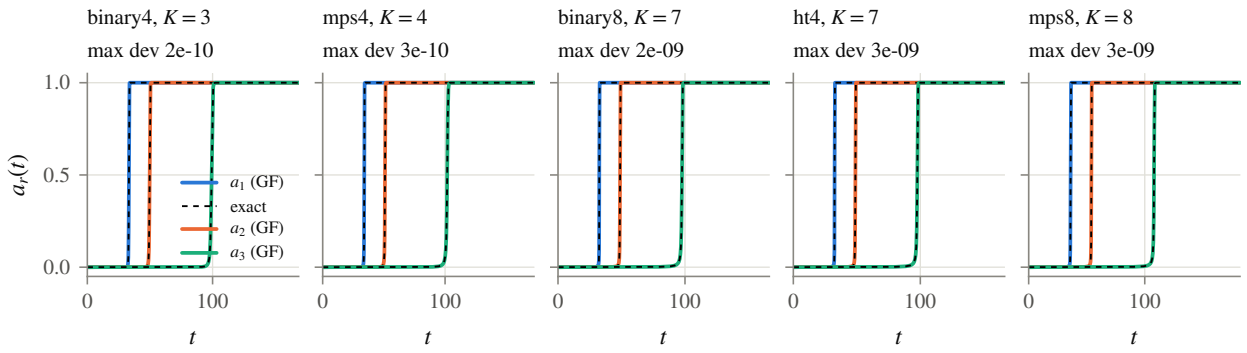


Figure 4: Aligned balanced initialization: overlaps from gradient flow on the full TTN (solid, Dormand–Prince, tolerances 10^{-10} and 10^{-14}) and the exact mode solutions (dashed); separated teacher $(3, 2, 1)$, $d = 4$, $\chi = 3$. Panel titles: largest deviation over time and components. Takeaway: with aligned initialization every tree learns the components in the order of their weights, at times $\propto 1/s_r$, as the decoupled modes of a deep linear network do.

(Z the initialization at $\sigma = 1$) blows up at $\sigma^{-(K-2)}T(Z)$, with $T(Z) = \hat{t}_1(Z)/s_1$ of Corollary A.5. The direction tends to the balanced rank-one configuration of one component r_1, which then escapes as a balanced mode (Proposition A.6):

$$t_1 = \sigma^{-(K-2)}\, T(Z) + \delta_K\, s_{r_1}^{-(2-2/K)} + o(1), \tag{21}$$

with δ_K from (30) ($\delta_3 = -0.850$, $\delta_4 = -0.266$, $\delta_7 = 0.148$, $\delta_8 = 0.201$). $T(Z)$ and r_1 come from integrating the leading-order flow from Z (Remark A.7), so (21) predicts the exponent, the prefactor and the first component of every run. Status: the scaling and (29) are proven, the offset derived (the direction must have converged to r_1 by the time $\|W\|_F$ is comparable to s_{r_1}). Known: the exponent known (deep linear networks: Jacot et al., 2022; homogeneous models, from directional convergence: Kumar and Haupt, 2025a, Thm 1), and (29) is in substance the activation law of alternating gradient flows known (Kunin et al., 2025, Eqs. 2–3); new new (TTN form): its evaluation on trees and the tests below. The escape direction is exactly balanced, with core norms $1/K$ (Proposition A.9) proven known (deep linear networks: Kumar and Haupt, 2025a, Lemma 5; also from the proof of Furman et al., 2026, Prop. F.10); at the end of the leading-order flow every core holds $1/K$ of $\|\theta\|^2$ to within 1.3% relative in all 1,200 initializations of the d sweep below empirical.

Test, seed by seed (Figure 5). empirical For 231 pairs of initialization and teacher we computed $T(Z)$ and r_1 and ran gradient flow to the first event at eight values of σ (1,848 runs): 0.002–0.1 on binary4 and MPS4 ($\chi = 3, 6$, both teachers, fan-in and balanced) and 0.01–0.4 at $K = 7, 8$ (mostly separated teacher, fan-in; Table 14). At the smallest σ the prediction (21) matches every run to a relative $8.5 \cdot 10^{-6}$ (binary4, separated teacher), $2.2 \cdot 10^{-3}$ (binary4, close teacher, whose direction has not converged at $\sigma = 0.002$), $6 \cdot 10^{-5}$ (MPS4) and $4 \cdot 10^{-8}$ (binary8, HT4, MPS8 at $\sigma = 0.01$). With the offset removed the per-seed exponents are 0.9996–0.9997 (± 0.0001; binary4, separated), 0.990–0.996 (binary4, close: 2.6–3.8 s.e. below 1, as an unconverged offset makes them), 1.9999–2.0000 (MPS4), 4.996–4.999 (binary8, HT4) and 5.9995 ± 0.0002 (MPS8). An independent integration of the spherical flow reproduces $T(Z)$ for the 160 binary4 and MPS4 jobs to $2 \cdot 10^{-10}$ (verification review).

The size of the initialization gives only a lower bound. From an initialization along a maximizer of g the escape takes $t_{\mathrm{KH}} = K^{K/2-1}\|\theta_0\|^{-(K-2)}/((K-2)s_1)$ known (Kumar and Haupt,

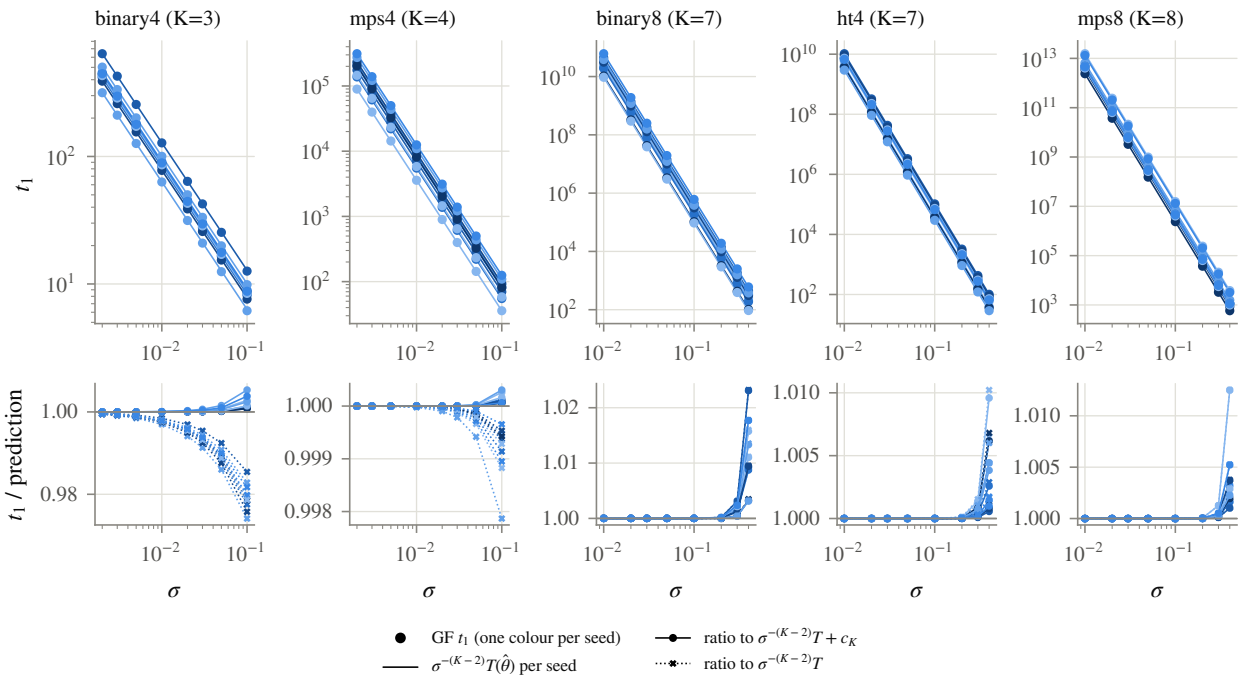


Figure 5: First escape against σ: $\chi = 3$, separated teacher, fan-in initialization, ten seeds (one colour per seed). Top: t_1 from gradient flow (dots) and each seed's prediction $\sigma^{-(K-2)}T(Z)$ (lines), with $T(Z)$ computed from that seed's initialization before training. Bottom: t_1 over the prediction (21), with the offset (solid) and without it (dotted); in the legend $\hat{\theta} = Z$ and c_K is the offset. Takeaway: one number per initialization, computed before training, predicts the first escape of every run, to within 10^{-5} at the smallest σ in every panel; the deviations at large σ are the offset and, at $K = 7, 8$, a direction that has not converged.

2025b, text before Thm 11), and every initialization of the same norm takes longer (Corollary A.12) proven. The median T/t_{KH} is 3.1–4.8 (binary4), 7.9–13.7 (MPS4), 95–210 (HT4), 184–2650 (binary8) and 1460 (MPS8) over the cells of the first-escape sweep empirical new (the bound on trees and the ratios). Most of $\|\theta_0\|$ sits in core slices orthogonal to the teacher's factors, which do not move at leading order (Proposition A.11), and the moving slices form an unbalanced product path (Proposition A.13): the norm law is a bound, not a prediction.

Dependence on d, χ and K (Figure 6). The leading-order flow gives $T = \lim_{\sigma\to 0} \sigma^{K-2} t_1$ per initialization cheaply, so these were measured on many initializations ($d = 4$, one component of weight 1, unless stated). *Physical dimension* empirical new (the closed form): at full bond $\chi = d^2$ (200 initializations per point) the binary4 medians are 1.544 $[1.468, 1.594]$, 1.569 $[1.525, 1.615]$ and 1.563 $[1.525, 1.587]$ (95% bootstrap intervals) for $d = 2, 3, 4$, independent of d and equal to $\pi/2$, the full-bond value of the closed form of Proposition A.13 (derived given end directions that do not rotate), while t_{KH} falls by a factor 1.9; $\mathrm{sd} \ln T = 0.218 \pm 0.010$, 0.157 ± 0.009, 0.135 ± 0.006 against the closed form's $1/(2\sqrt{\chi}) = 0.250, 0.167, 0.125$ (± 0.02). MPS4 gives 1.39, 1.41, 1.50 (theory note, other initializations: 1.30, 1.35, 1.46) while t_{KH} falls by 2.9. *Bond dimension* empirical new: the median falls slowly, with log-log slopes -0.335 ± 0.028 (binary4) and -0.264 ± 0.038 (MPS4) over $\chi = 4$–16 and -0.76 ± 0.08 (binary8, $\chi \le 12$), -0.69 ± 0.08 (HT4) and -0.72 ± 0.06 (MPS8) over $\chi = 2$–16, against -0.47 and -1.5 to -2.5 for t_{KH}, as the theory note found (-0.20 ± 0.05 for MPS4, -0.36 to -0.38 for binary4): χ acts only through the moving slices. *Number of cores* conjecture (heuristic) empirical (agreement) new: over 17 trees with $K = 3$–15 ($\chi = 3$, weights $(3, 2, 1)$,

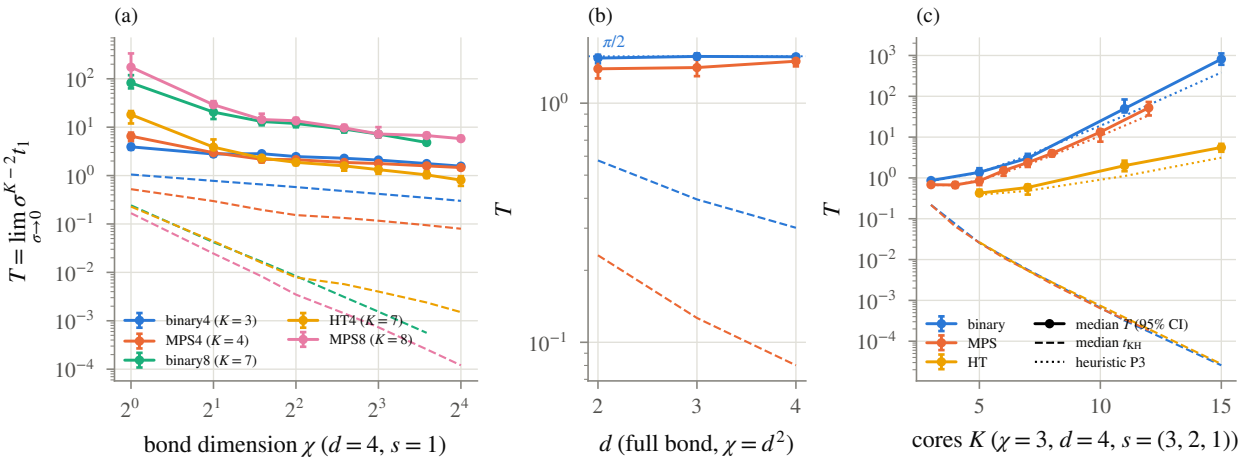


Figure 6: Prefactor $T = \lim_{\sigma\to 0} \sigma^{K-2} t_1$ from the leading-order flow, medians over initializations with 95% bootstrap intervals. (a) Against χ: one component of weight 1, $d = 4$, 100 initializations per point; dashed: median Kumar–Haupt time t_{KH}, same colours. (b) Against d at full bond $\chi = d^2$, binary4 and MPS4 (colours as in (a)), 200 initializations; dotted: $\pi/2$. (c) Against K, 17 trees, $\chi = 3$, $d = 4$, weights $(3, 2, 1)$, 50 initializations; dotted: the RMS-overlap heuristic P_3; dashed: t_{KH}. Takeaway: weak χ-dependence, none on d at full bond, P_3 follows the growth with K, and t_{KH}, which uses only the norm of the initialization, is low by factors that grow with d, χ and K.

50 initializations each) the median grows from 0.42 (HT3) to 814 (binary16). The RMS-overlap heuristic $P_3 = d^{N(K-2)/(2K)}/((K-2)s_1)$, the escape time of a balanced aligned mode started at the root-mean-square overlap $\sigma^K d^{-N/2}$ of a fan-in W_0 with a unit rank-one tensor, tracks it within a factor 0.87–2.13 with no free parameter, while the median of T/t_{KH} grows from 3.1 to $2.9 \cdot 10^7$. *Gauge balance* empirical new (minor): re-gauging the same W_0 to $Q_e = 0$ changes T per seed by a factor 0.88–1.92 (cell medians 0.99–1.18), keeps r_1 in 88 of 94 seeds and leaves the exponent unchanged.

Which component escapes first. empirical new (tensor-like trees; binary4 known, below) In 1,847 of the 1,848 runs the first component learned is the leading-order flow's r_1, and in all 231 at the smallest σ; the exception (binary4, $\chi = 3$, close teacher, balanced, seed 6, $\sigma = 0.1$) learns the weakest component first, a wrong first escape at finite σ (Proposition A.15). On binary4 only the largest odeco component is a local maximum of g on the sphere (a matrix-like tree, Proposition A.10), and it is learned first in 639 of 640 first-escape runs and in all 380 full runs, which have 20 independent initialization directions known (deep linear networks: Jacot et al., 2022, Thm 2 and Lemma 15). On the tensor-like trees every component is a local maximum, and the first component varies from seed to seed (largest first in 400 of 640 runs on MPS4, 264/328 on binary8, 104/160 on HT4, 64/80 on MPS8) but is the same at every σ for a given seed (10 of 10 seeds in every cell of the full runs). The race of independent per-component flows (Proposition A.14) names it in all 640 MPS4 and 80 MPS8 runs, where it is exact at leading order, in 328 of 328 binary8 and 152 of 160 HT4 runs, where cores without sites couple the components, and in only 537 of 640 binary4 runs, where that coupling decides. Statistics of the order: Section 4.5.

4.4 Saddle sequence and locality

After the first escape, gradient flow rests near a saddle that represents what has been learned. Which component comes next, and how long is the next plateau? In a TTN the answer depends on where the next component sits in the tree: one that differs from what is learned only on sites

carried by two adjacent cores needs those two cores to move, one spread over the tree needs many. We state the resulting rule, derived in Section A.3, and test it.

Prediction: the route rule. Take a two-component pattern teacher (Section 2.3), $W^* = s_1U_1 + s_2U_2$ with $s_1 > s_2$, whose components differ exactly on the site set D (sites numbered from 0, as in Table 13). Let $N(D)$ be the set of cores that carry a site of D, $\mathrm{St}(N(D))$ the smallest subtree containing them (their Steiner subtree), and

$$k = |\mathrm{St}(N(D))|. \tag{22}$$

The bonds whose teacher rank is two are exactly the bonds of $\mathrm{St}(N(D))$ (Proposition 2.1(b)). Once U_1 is learned, the flow sits near the balanced saddle θ_1 that represents s_1U_1, and new parameters can lower the loss only through *routes*, connected sets of cores whose new entries combine into a term proportional to the residual at no Gauss–Newton cost (Lemma A.16). The cheapest route to U_2 is $\mathrm{St}(N(D))$, with coupling $c_2 = s_2 s_1^{(K-k)/K}$ (Proposition A.19), and the escape is the first-escape problem of a k-core TTN (Proposition A.18). With $\Delta t_2 = t_2 - t_1$ the time between the two learning events,

$$\Delta t_2 \simeq \begin{cases} O(1) \text{ (no plateau)}, & k = 1, \\ \dfrac{1 - s_2/s_1}{\lambda_2} \ln \dfrac{1}{\sigma}, \qquad \lambda_2 = s_2\, s_1^{(K-2)/K}, & k = 2, \\ A\, \sigma^{-(k-2)}, & k \geq 3, \end{cases} \tag{23}$$

(Proposition A.25). For $k = 2$ the saddle is strict, with $\lambda_{\min}(H_\perp) = -\lambda_2$ (Proposition A.21); for $k \geq 3$ its Hessian has no negative eigenvalue (a higher-order saddle), and on the plateau $\lambda_{\min} \propto \sigma^{k-2} \to 0$ (Proposition A.26). The factor $1 - s_2/s_1$ is a *head start*: the route grows during the first plateau, so from the saddle itself the $k = 2$ slope is $1/\lambda_2$. The prefactor A is set by the route amplitudes when the plateau starts and is not predicted from σ alone for random initialization. Status: the Steiner identity, the upper bound k on the saddle order and, for two-component teachers learned in this (greedy) order, the matching lower bound, so that the order is exactly k on every tree (Corollary A.32), are proven; (23) is derived (leading order, small σ).

With several learned components "the sites where the next component differs" is ambiguous, and saddles need not be partial sums of components. The *route rule* (Remark A.27) works at the saddle actually reached, the best approximation of W^* with the current bond-rank profile: if the learned structure is stable (no wrong order, no star, Proposition A.22; from $k^* = 5$ on also no generalized star, see the next paragraph), the next escape takes the smallest route, of size k^*, with a nonzero local residual (Definition A.17), raises the bond rank on its internal edges, and the flow moves to the best approximation with the new profile. We evaluate it numerically (rank-constrained optima by L-BFGS with restarts); it predicts every plateau level and its k^*, and $k^* = k$ for two components. Status: derived at leading order for each step, conjecture as an algorithm.

When the saddle order is k^*. proven new (Section A.4) The route always gives a descent of order k^*; that nothing descends faster is decided, at a saddle with one learned component j, by a *generalized piggyback operator* (Definition A.29). It couples the parts of the residual that live on small sets of cores: single cores (the piggyback graph of Proposition A.22) and, from $k^* = 5$ on, also sets whose Steiner subtree has fewer than $k^*/2$ cores. If its largest eigenvalue is below s_j, the order is exactly k^* (Theorem A.30). This holds always for $k^* = 2$; for $k^* = 3, 4$ when the piggyback graph has $\lambda_{\max}(\Gamma) < s_j$ (no wrong order, no star; above s_j the saddle is strict); and for two-component teachers in the correct order on every tree (Corollary A.32). It fails from $k^* = 5$ on. On MPS6 the

teacher 000000 (1), 110001, 110010 (0.8 each) has $k^* = 5$ and a positive semidefinite Hessian at the saddle of the first component, yet new entries on cores 0, 1 that point at the non-residual pattern 110000, together with old rotations of cores 4 and 5, descend at order 4 ($-0.0350\,t^4$, as predicted); gradient flow escapes in $\propto \sigma^{-2}$ instead of σ^{-3} and then forgets the first component derived (checked on this teacher; from small initialization only some seeds visit the saddle). Which of our experiments satisfy the condition: every two-component row of Table 6 (the fits use the runs that learn the larger component first), and every saddle with one learned component of the multi-component teachers below (there the operator reduces to the piggyback graph, with $\lambda_{\max}(\Gamma) \le 0.7 < s_j$). Saddles with two or more learned components are not covered by the theorem. The condition fails for the runs on HT4 with $D = \{0, 1, 2, 3\}$ that learn the smaller component first (order 6, not $k = 7$; excluded from the fits): in 6 of the 7 such runs with $\chi = 2$ ($\sigma \ge 0.4$) the smaller component is forgotten when the larger one is learned and then relearned, as the cluster mechanism predicts empirical.

What is known and what is new. Known in general: dormant weights of lowest homogeneity degree L stay $O(\delta)$ for at least a time $\propto \delta^{-(L-2)}$, δ their initial size, if the active weights start within δ of a local minimum of the loss restricted to them (Łojasiewicz) known (Kumar and Haupt, 2026, Thm 8); escapes that need 1, 2 or $r \ge 3$ small layers take a time of order 1, $\ln(1/\epsilon)$ or $\epsilon^{-(r-2)}$ known (Rawal and DeWeese, 2026, Thm 4.2) (first escape; the first r layers at scale ϵ, the others $O(1)$); in our model a min-norm critical point has a descent curve of order N, the number of cores touching a connected component of rank-deficient bonds, and with several such components the smallest gives the least suppressed escape known (Furman et al., 2026, Lemmas D.8–D.9, Remark D.10 and Fig. 4); for $N = 2$ the curve is a direction of negative curvature, so the saddle is strict (a one-line consequence, not stated there); in deep linear networks with at least three weight matrices, which include the binary tree with $N = 4$ (Remark A.23), every saddle visited from an exactly balanced initialization that selects the top singular values is non-strict known (Achour et al., 2024, Thm 7 and the remark after Prop. 8; full-rank targets with distinct singular values); and the $k = 2$ head start known (weaker directions are $O(\epsilon^{1-s_{r+1}/s_1})$ when the top ones are learned, two-layer linear networks: Zhang et al., 2026, Thm 4; dormant norms carried across phases: Kunin et al., 2025, Sec. 2.2). New for TTNs new: (i) the dormant block is the Steiner route of the residual, which can be a proper subtree of a connected rank-deficient component, so $k^* = |\mathrm{St}(N(D))|$ can be far below Furman et al.'s N ($N = K$ at a rank-one saddle with $\chi \ge 2$); (ii) the coupling c_2 and the rate λ_2 (Equations (39) and (40)); (iii) the head start for $k \ge 3$ in closed form, an effective initial route amplitude $\sigma(1 - s_2/s_1)^{-1/(k-2)}$ (the carry-over of dormant amplitudes is that of Kunin et al., 2025); (iv) the piggyback-graph criterion for learned saddles that are strict although $k^* \ge 3$ (Proposition A.22); (v) the conservation-law trap (Proposition A.28); (vi) the measured dependence on the topology, below; (vii) the matching lower bound on the saddle order and its failure from $k^* = 5$ on (Theorems A.30 and A.31; lower bounds were known for deep linear networks, order 2 or more, Achour et al., 2024, Thm 7, and for one parity saddle, Furman et al., 2026, Prop. 5.3).

Setup (details in Section B.3). Scope of every statement below: orthogonal pattern teachers with $d = 2$ and weights $(1, 0.5)$ unless stated, $\chi = 2$ (3 for the three-component teachers B4, N4, E8), whitened inputs, gradient flow on the exact population loss. Nine trees: MPS with $N = 4, 6, 8$ (MPS4, MPS6, MPS8), binary4, binary8, HT4, HT8, and two custom trees with $N = 6$, *star3* (a root without sites, three children with two sites each) and *midroot6* (the MPS6 chain rooted at a middle core). Protocols: *small* (fan-in initialization (11) at scale σ, first plateau included), *saddle* (the exact balanced saddle $s_1 U_1$ plus a fan-in perturbation of scale σ; $\Delta t_2 = t_2$ times the escape alone) and *balanced* (the fan-in W_0 re-gauged to $Q_e = 0$). Event r is the last upward crossing of $a_r = \frac{1}{2}$.

Table 6: Plateau law after the first component, two-component pattern teachers ($d = \chi = 2$, weights $(1, 0.5)$): prediction (23) against measurement. p is the exponent of $\Delta t_2 \propto \sigma^{-p}$, fitted per seed; for $k = 2$ the slope of Δt_2 against $\ln(1/\sigma)$. Ranges are over rows (one row = tree, D, protocol); single rows give mean ± s.e. over seeds (median). Protocols as in the text; ρ is the separation ratio (19). †: less than one decade of σ, below the reporting rule of Section 2.7. Sets D number the sites from 0.

k	rows	prediction	measured
1	binary4, binary8, star3 (4 cases)	no plateau	small: $\Delta t_2 \propto \sigma$, at most $8 \cdot 10^{-4}$, $\rho < 5 \cdot 10^{-4}$; saddle: $\Delta t_2 = 0.63$–0.65, independent of σ, $\rho = 0.34$–0.35
2	MPS, midroot6 (7 cases)	log, slope 1 (small), 2 (saddle)	0.980–1.009 (small), 2.002–2.071 (saddle); log law preferred in 14 of 14 rows
3	all nine trees (22 cases)	1	0.98–1.04 (42 rows)
4	MPS, star3, midroot6 (8 cases)	2	small 1.95–2.05, balanced 1.94–2.02, $\chi = 4$: 1.99–2.03; saddle 1.48–2.39 (1.81–2.19 on never-stalling seeds)
5	binary8, HT4 (6 cases)	3	2.89–3.13, all protocols
5	HT8 $\{1, 2\}$	3	saddle 2.74 ± 0.37 (2.96); 3.10 ± 0.11 on never-stalling seeds
5	MPS6 $\{1, 3, 5\}$, MPS8 $\{1, 5\}$	3	$\sigma \in [0.03, 0.3]$: 2.86–3.76. $\sigma \in [0.012, 0.03]^\dagger$: MPS6 3.05 ± 0.06 (saddle), 3.15 ± 0.23 (balanced); MPS8 3.13 ± 0.25 (balanced), $\mathbf{3.67 \pm 0.20}$ (saddle, 4 seeds). MPS8 small, never-stalling seeds: 2.97 ± 0.05. $\chi = 4$: 3.19 ± 0.15, 2.97 ± 0.04
6	binary8 $\{0, 2, 4\}$, HT4 $\{0, 1, 2\}$	4	balanced, one decade, 8 and 7 seeds: 4.007 ± 0.021, 4.061 ± 0.048 (fan-in†: 3.68 ± 0.20, 4.23 ± 0.10)
6	MPS8 $\{1, 6\}$	4	balanced, one decade: 4.51 ± 0.19
7	HT4, all sites	5	balanced, one decade, 8 seeds: 4.957 ± 0.061 (5.01)
6, 7	other rows (6 and 3 cases)	4, 5	fan-in and saddle†, 1–5 fitted seeds: 3.4–5.1 ($k = 6$), 5.3–6.3 ($k = 7$; HT8 $\{3, 4\}$: 5.74 ± 1.06, median 5.01)
8	MPS8 $\{0, 7\}$	6	not measured (≤ 2 fitted seeds per protocol†)

Per seed (seeds 0–7), over the (at least three) σ at which it escapes, we fit $\ln \Delta t_2$ against $\ln(1/\sigma)$ (exponent p), or Δt_2 against $\ln(1/\sigma)$ (the $k = 2$ slope), and report mean ± s.e. over seeds (median); log and power laws are compared by the residual of $\ln \Delta t_2$ on the same data ("fair comparison"). Fits use the runs that learn U_1 first, without rotations (an overlap crossing $\frac{1}{2}$ more than once). About 11,600 runs in all.

Two components: the plateau exponent is $k - 2$. empirical new (Table 6 and Figure 7) There is no plateau for $k = 1$, a logarithmic one for $k = 2$, and the exponent is $k - 2$ for $k = 3$ (all nine trees), 4 and 5, on trees with K from 3 to 15: it follows the Steiner count, not K or the tree family. Named exceptions: the saddle-start $k = 4$ rows and HT8 $\{1, 2\}$ are carried by seeds that stall at some σ (Section 4.6); the MPS rows with $k = 5$ exceed 3 for $\sigma \in [0.03, 0.3]$ but give 2.97–3.15 at $\sigma \leq 0.03$ or on never-stalling seeds, except MPS8 $\{1, 5\}$ from the saddle, 3.67 ± 0.20. For $k = 6$, 7 the exponent is $k - 2$ on binary8 and HT4 *with balanced initialization only*, over one decade; MPS8 with $k = 6$ gives 4.51 ± 0.19, and the other $k \geq 6$ rows show only that the exponent grows with k. $k = 8$ is not measured. Coverage is uneven: $k = 2$ occurs only on chains (MPS, midroot6), $k = 4$ on chains and star3, $k = 7$ only on HT trees.

$k = 2$: a logarithmic plateau with a head start. empirical The fair comparison prefers the logarithmic law in all 14 rows with $k = 2$ (1.1–3 decades of σ; MPS8 $\{3, 4\}$ from small initialization

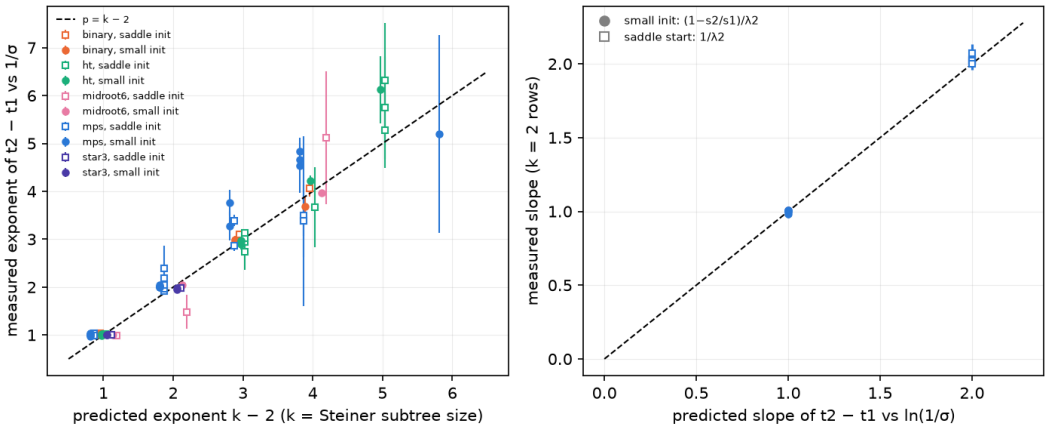


Figure 7: Plateau law after the first component: two-component pattern teachers, $d = \chi = 2$, weights $(1, 0.5)$, all rows of the main sweep. Left: per-seed exponent p of $\Delta t_2 \propto \sigma^{-p}$ (mean ± s.e. over seeds) against the prediction $k - 2$, with $k = |\mathrm{St}(N(D))|$, for every row with $k \geq 3$; filled circles: small initialization, open squares: saddle start; colour: tree family; points are offset horizontally by family and protocol. Right: slope of Δt_2 against $\ln(1/\sigma)$ for the $k = 2$ rows, against $(1 - s_2/s_1)/\lambda_2 = 1$ (small initialization) and $1/\lambda_2 = 2$ (saddle start). Dashed: equality. The points at $k - 2 \geq 4$ span 0.5–0.8 decades of σ with 1–5 fitted seeds and are not measurements of the exponent (one-decade values in Table 6). Takeaway: up to $k = 5$ the exponent is set by the number of cores of the Steiner subtree, on every tree; for $k = 2$ the head start halves the slope.

nearly ties, residuals 0.004 and 0.005). Its slope carries the head start: 0.980–1.009 from small initialization against $(1 - s_2/s_1)/\lambda_2 = 1$, and 2.002–2.071 from the saddle against $1/\lambda_2 = 2$ (Figure 7, right); without a head start both would be 2. On MPS4, $s_2 = 0.25$ gives 3.023 ± 0.010 (small) and 4.04 ± 0.05 (saddle) against 3 and 4, and $s_2 = 0.75$ gives 0.367 ± 0.025 and 1.345 ± 0.016 against 1/3 and 4/3. With weights $(2, 1)$ from the saddle on MPS4, MPS6, MPS8 (8 seeds each), $\lambda_{\min} = -1.4142$, -1.5874, -1.6818 against $\lambda_2 = 2^{(K-2)/K} = 1.4142$, 1.5874, 1.6818, and the slopes are 0.712 ± 0.007, 0.649 ± 0.016, 0.595 ± 0.003 against $1/\lambda_2 = 0.707$, 0.630, 0.595. The $k = 2$ head start is known in other models known (two-layer linear networks: Zhang et al., 2026, Thm 4; dormant norms carried across phases: Kunin et al., 2025, Sec. 2.2); the TTN rate λ_2, including its dependence on K, is new new.

Saddle type: strict for $k = 2$, higher order for $k \geq 3$. empirical new (for TTNs; deep linear networks: Achour et al., 2024) At the plateau midpoint (minimum speed between the events) we compute $\lambda_{\min}$ of $H_\perp$ and of the raw Hessian; near rank-deficient states the gauge projection removes part of the escape direction (Section B.3), so raw values are quoted where the two differ. The type is read from the scaling of $\lambda_{\min}$ with σ (Proposition A.26); the fixed threshold of Section 2.7 would call $k = 3$ plateaus strict for $\sigma \gtrsim 0.01$–0.04. For $k = 2$, $\lambda_{\min} = -0.5000 = -\lambda_2$ from the saddle at every σ, projected and raw (-0.2500 and -0.7500 for $s_2 = 0.25$, 0.75), with one negative eigenvalue per active route, also at $\chi = 4$, 8. From small initialization the raw value is -0.498 to -0.499 for $\sigma \leq 10^{-3}$ and departs from $-\lambda_2$ where the plateau is too short to reach the saddle (-0.437 for MPS8 $\{3, 4\}$ at $\sigma = 0.008$; -0.581 against -0.75 for $s_2 = 0.75$). For $k \geq 3$, $\lambda_{\min} \propto \sigma^x \to 0$ with x growing with k (Figure 8); full ranges over rows: $x = 0.81$–1.06 for $k = 3$ (raw 0.92–1.06), 1.64–2.16 for $k = 4$, 2.82–3.76 for $k = 5$, 4.00 and 4.04 for $k = 6$ (balanced binary8 and HT4, one decade; 4.27 on MPS8; 3.19–5.23 in the short rows), and 5.01 (balanced HT4) up to 6.5 (short rows) for $k = 7$. x is within 0.2 of $k - 2$ in 126 of 135 rows with $k = 3$–5, and the raw x is within 0.2 of the row's duration exponent p in 95% of 169 rows with $k \geq 3$ (median $|x - p| = 0.02$): where the plateau law

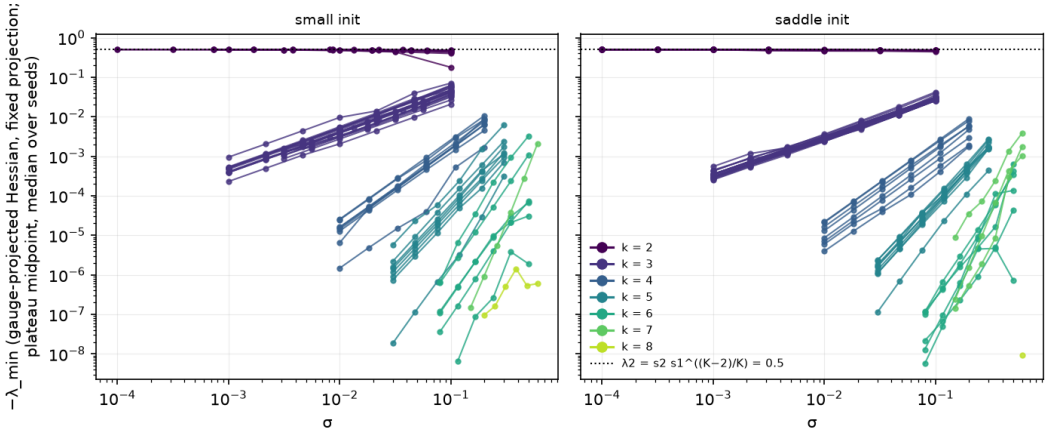


Figure 8: Saddle type from the Hessian at the plateau midpoint, same sweep as Figure 7. Plotted: $-\lambda_{\min}$ of the gauge-projected Hessian $H_\perp$ (median over seeds) against σ, one line per (tree, D), coloured by k; left small initialization, right saddle start; dotted: $\lambda_2 = s_2 s_1^{(K-2)/K} = 0.5$. For $k = 2$, $\lambda_{\min} = -\lambda_2$ at every σ: a strict saddle (the small-initialization curves drop at the largest σ, where the plateau is too short to reach the saddle). For $k \geq 3$, $\lambda_{\min} \to 0$ as $\sigma \to 0$, roughly as σ^{k-2}: higher-order saddles. Lines for $k \geq 6$ come from the short rows of Figure 7. Takeaway: negative curvature that survives $\sigma \to 0$ exists only for two-core routes.

deviates, the curvature deviates the same way. Raw and projected exponents differ by up to 0.11. No plateau with $k \geq 3$ has σ-independent negative curvature, as predicted for saddles visited in the greedy order; wrong-order and star saddles, strict with $k^* \geq 3$ by Proposition A.22, were not tested.

One teacher, different trees. empirical new Since k depends on the tree, one teacher has different plateau laws on different trees. For $D = \{3, 4\}$ and $N = 8$ the route has $k = 2$ on MPS8, 5 on binary8 and 7 on HT8 (Table 13). MPS8 shows a logarithmic plateau (saddle slope 2.002 ± 0.010 against 2; residuals 0.003 for the log law, 0.064 for a power law), binary8 a σ^{-3} plateau (2.998 ± 0.078 small, 3.106 ± 0.057 saddle), and HT8 $p = 5.74 \pm 1.06$ (median 5.01; saddle start, 0.6 decades, one seed with a pinned route), consistent with 5 but unable to separate $k = 6$, 7, 8. The order of learning changes too. Teacher B4 has patterns 0000, 0110, 1001 with weights 1, 0.3, 0.6 ($\chi = 3$): on MPS4 the 0.3 component needs $k = 2$ and the 0.6 component $k = 4$, and every escaping run learns the smaller one first (*locality beats weight*), then the larger after a σ^{-2} plateau ($p = 2.02$); on binary4 both need $k = 3$ and come after two σ^{-1} plateaus (1.05, 1.03). Figure 9 shows teacher TA on three trees.

Several components: the route rule, not the literal rule. empirical (predictions derived at leading order) For seven teachers with three or four components (TA, TB, N4, B4 with $N = 4$; E8, TC, TD with $N = 8$; Section B.3) we detect the plateaus of $L(t)$ (separation ratio above 3, linear time) and match them to the levels predicted by the route rule (within 5%). Every predicted level with $k^* \geq 3$ is detected in 88–100% of the runs (35–65% for a few $k^* = 3$ levels within 10% of the next one), with a duration exponent within 0.2 of $k^* - 2$ (1.02–1.19, 2.02–2.08, 2.97–3.17, 4.05 for $k^* = 3, 4, 5, 6$), except B4's second level on HT4 with fan-in initialization, 4.73 ± 0.90 (4 fitted seeds, 23 of 48 runs stalled; 3.17 ± 0.10 with balanced initialization). Three teachers discriminate between rules. The literal reading of H2 (Steiner count of the sites where the next component differs from the closest learned one) predicts no plateau before the third component of TA on any tree; there is a σ^{-1} plateau on binary4 (1.02) and a σ^{-2} one on HT4 (2.08), as the route rule predicts. For TD

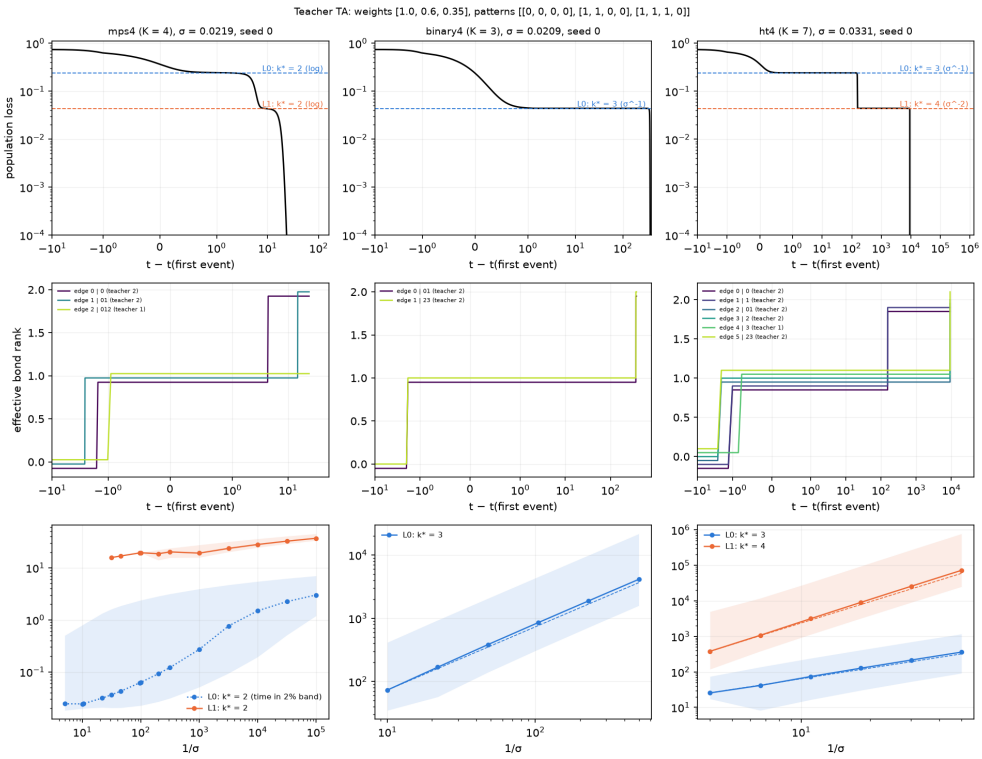


Figure 9: One teacher on three trees: teacher TA (patterns 0000, 1100, 1110, weights 1, 0.6, 0.35), $d = \chi = 2$, small fan-in initialization. Columns: MPS4 ($K = 4$), binary4 ($K = 3$), HT4 ($K = 7$). Top: loss of one run (seed 0, σ in the panel title) against time since the first event (symmetric-log axis), with the plateau levels predicted by the route rule (dashed, labelled with k^* and the predicted law). Middle: effective bond rank of every edge in that run (threshold 0.1 times the smallest nonzero teacher singular value of the edge; edges offset vertically by 0.05). Bottom: duration of each predicted level against $1/\sigma$, median over 8 seeds with the range shaded; solid: detected plateaus, dotted: time spent within 2% of a level too short to be detected; dashed: slope $k^* - 2$. Measured: MPS4 two short levels (the second logarithmic in σ; for the first the fair comparison favours a power below 1, so its law is undetermined); binary4 one σ^{-1} plateau ($p = 1.02$), where the first escape already fits U_1, U_2 and part of U_3; HT4 σ^{-1} and σ^{-2} plateaus (1.09, 2.08). Takeaway: the tree, not the teacher alone, sets the number of plateaus and their laws.

on MPS8 (patterns 0^8, $0^3 1^2 0^3$, $1^2 0^6$, weights 1, 0.6, 0.35, from the saddle) the last plateau has $k = 2$ by the literal rule, $k = 4$ by the exact bond-rank rule (cores whose bond rank must grow to add the component exactly, Lemma A.20) and $k^* = 3$ by the route rule, which first learns part of the component through a two-core route: measured 1.012 ± 0.010 (8 seeds). For TC on MPS8 (patterns 0^8, $1^2 0^6$, $1^4 0^4$, 1^8, weights 1, 0.6, 0.35, 0.2, from the saddle) the route rule predicts a $k^* = 4$ plateau where the bond-rank rule predicts $k = 5$: measured 2.018 ± 0.017. The $k^* = 2$ levels are logarithmic where measured over three decades from the saddle (four levels of TC, the first of TD); elsewhere (TD's second level, the first levels on MPS8 from small initialization) a power law below 1 fits better and the law is not determined. At the predicted saddles the measured bond-rank profile equals the predicted one in 93.9% of 3,114 resolved (run, level) pairs (99.6–100% for $k^* = 3$, 92–100% for $k^* = 4$–6, 86–97% for $k^* = 2$).

Bond spectra: the new component enters through the Steiner edges only. empirical new The second component raises the teacher's bond rank exactly on the edges of $\mathrm{St}(N(D))$

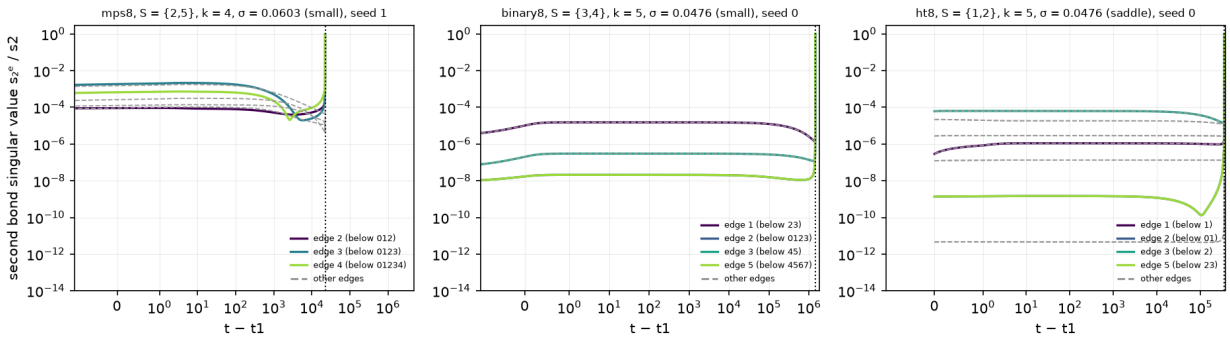


Figure 10: Bond spectra through the second event, two-component teachers, $d = \chi = 2$, weights $(1, 0.5)$. Second singular value s_2^e of every edge, relative to s_2, against the time since the first event (symmetric-log axis); coloured: the edges of the Steiner subtree $\mathrm{St}(N(D))$ (legend: the sites below each edge), grey dashed: all other edges; dotted vertical line: the second event. Left: MPS8, $D = \{2, 5\}$, $k = 4$, small initialization; middle: binary8, $D = \{3, 4\}$, $k = 5$, small initialization; right: HT8, $D = \{1, 2\}$, $k = 5$, saddle start (one run each, σ and seed in the panel titles). Before the event every edge sits at the initialization-noise level, Steiner edges included; at the event exactly the Steiner edges rise, together. Takeaway: the escape uses the cores of the Steiner subtree and no others.

(Proposition 2.1(b)), so the bond spectra show which cores the escape uses (Figure 10). Through all 5,131 two-component events the second singular value of every edge outside the Steiner subtree stays at the initialization-noise level, from the plateau midpoint to the end of the run: $s_2^e \le 4.3\,\sigma^2 s_2$, scaling as $\sigma^{2.07}$ (below $3 \cdot 10^{-3} s_2$ for $\sigma \le 0.03$). This *leakage bound* is the substantive result: the new component enters through the Steiner edges only, up to $O(\sigma^2)$. The rest is *single-route consistency*: the edges whose effective rank (number of s_k^e above 0.05 times the teacher's value) is higher at $a_2 = 0.9$ than at the plateau midpoint are the Steiner edges in 99.75% of the 4,448 events with a resolved plateau (99.6%, 99.9% with thresholds 0.03, 0.1). Given the leakage bound a non-Steiner edge cannot cross the threshold, so this number checks only that every Steiner edge rises; it is not a prediction success rate. The Steiner edges also cross together (median spread of crossing times $5 \cdot 10^{-5}$ event widths, 95th percentile 0.25, growing as $\sigma^{1.94}$), which tests that they rise as one route, not the timing of the event.

Reading k from a single run. On a balanced route the new overlap grows as $\dot{a}_2 \propto a_2^{2-2/k}$ at small a_2, so the local growth exponent $\gamma = \mathrm{d}\ln \dot{a}_2 / \mathrm{d}\ln a_2$ gives $k = 2/(2-\gamma)$ (Equation (45)) derived known (the exponent $2 - 2/k$ of a balanced product of k factors: Arora et al., 2019b; Razin et al., 2022). Fitted over $a_2 \in [10^{-4}, 10^{-2}]$, the medians per row give $k = 2.00$, 3.0, 4.0, 5.1–5.3, 6.0, 7.0–7.7 for $k = 2$ (saddle start) to 7 empirical new (as a meter of k). Validity range: from the saddle, or from small initialization if the plateau is long enough for the window to lie after the escape transient. The estimate fails for small-initialization $k = 2$ on MPS6 and MPS8 (2.4–3.3; plateau shorter than the transient), is biased high for $k \ge 5$ (HT8: 5.3 for $k = 5$, 7.7 for $k = 7$), and cannot separate $k = 7$ from 8, since $\mathrm{d}k/\mathrm{d}\gamma = k^2/2$ (per-run 5–95% ranges ± 1 at $k = 7$; MPS8 $\{0, 7\}$, $k = 8$: 7.9 from small initialization, 3 runs, 6.1 from the saddle, 2 runs).

Independent cross-check. empirical A pilot with separate code (Section B.3: $d = \chi = 2$, weights $(1, 0.5)$, MPS4, MPS6, binary4, binary8, fan-in initialization; mean $\pm$ s.d. of per-seed fits over 3–8 seeds) gives the first escape $t_1 \propto \sigma^{-(K-2)}$ to within 0.003 in the exponent for every seed; no plateau for $k = 1$; a logarithmic $k = 2$ plateau with slope 1.01 ± 0.01 on MPS4 ($\sigma = 10^{-3}$–0.1; predicted 1),

rising to 2.3 ± 0.4 from the saddle (predicted 2); and $p = 1.00$–1.01 for $k = 3$, 2.00 (MPS4) and 2.05 ± 0.06 (MPS6) for $k = 4$, 3.00 ± 0.11 (binary8) for $k = 5$ and 4.97 ± 0.05 (binary8, under one decade) for $k = 7$. It also met the failure modes of Section 4.6: 95 of 332 runs with $k \geq 4$ (none with $k \leq 3$) never learn the second component, and on MPS6 the exponents for $k = 5$ and 6 are seed-dependent and up to two units steeper than $k - 2$ (3.75 ± 0.24 for $k = 5$, with 3 of 6 seeds escaping).

4.5 Order of learning

[Which component is learned first, how random the order is, and how a wrong order is corrected.]

4.6 Failure modes

Two things end a saddle-to-saddle sequence early: a route dies, its end cores decaying to roundoff under the conservation laws (the trap of Proposition A.28), or a plateau outlasts the budget. Both appear in the odeco runs of Section 4.1, whose routes span the tree, and in the pattern-teacher runs of Section 4.4. An odeco run is *trapped* if a component not yet learned has a route-end slice below $10^{-8}\sigma$ and *stalled* if its budget ended with every such slice above it (Section B.2); a pattern-teacher run is flagged *stalled* when its gradient stays below 10^{-12}, and *dead* when a longer run confirms route ends at roundoff. The regime maps meet the same trap at χ equal to the teacher rank, as Section A.8 predicts (empirical; Section 5.3): on binary8 ($K = 7$) and MPS8 ($K = 8$) most runs never learn the last component, whose route runs through all K cores, in the plain as in the centred model; from fan-in initialization that route ends dead (28 of 32 stalled plain runs), and balanced initialization removes the dead routes but not the stalls. Spare channels remove the stalls where every route end has one (binary8 at $\chi = 6$), not where the end bonds are capped at the teacher rank (MPS8 at $d = 3$).

Odeco teachers: the trap grows with K (Figure 11). empirical new No run traps at $K = 3$ (0 of 380). On MPS4 at $\chi = 3$ the separated teacher's outcome is fixed by the initialization direction: the same 4 of 10 seeds trap at all eight values of $\sigma \in [0.003, 0.1]$ (32 of 80 runs); with the close teacher 14 of 80 runs trap and 8 stall, not fixed by the direction (one seed traps at 6 of 8 values). The dormant slices at both route ends decay at a rate $\propto \sigma^2$, constant per seed to a few percent (at most 2.4% in the committed fits, 5.3% in an independent refit), equal at the two ends and seed-dependent (0.26 to $1.06\,\sigma^2$): the signature of the mechanism, whose rate is $\propto |Q_{\nu\nu}| = O(\sigma^2)$ (its magnitude is not tested). Decay time $\propto \sigma^{-2}$ against escape time $\propto \sigma^{-(K-2)}$ explains the pattern derived: at $K = 3$ the route escapes first, at $K = 4$ both are of the same order, at $K \geq 5$ the ends are decided long before the route could escape. At $K = 7, 8$ with $\chi = 3$ (fan-in, $\sigma = 0.15$–0.3) at least 9 of 30 runs trap on binary8, 24/30 on HT4 and 14/30 on MPS8 (lower bounds: some stalled runs end with slices still decaying). With $\chi = 6$ none traps on binary8 (0 of 50), but 5 of 50 do on HT4 (one seed, at every σ) and 7 of 50 on MPS8 (34 more stall), whose route ends sit on bonds capped at $d = 4$, one channel above the teacher rank. The classes are unchanged at a 100 times smaller absolute tolerance (9 of 9 runs). The optimizer experiment met the same trap on its odeco MPS8: fan-in gradient flow converged in none of 12 runs, with dead route ends in 11 (Section B.7).

Balanced initialization; the escapes in between. empirical new Re-gauged to $Q_e = 0$, the same initializations never trap where fan-in ones do (0 of 120 runs, against 12 of 30 and 47 of 90), but 69 of the 90 balanced runs at $K = 7, 8$ stall; 17 of the 47 fan-in-trapped runs complete when re-gauged, and 5 that complete from fan-in initialization stall when re-gauged. The routes die on the plateaus,

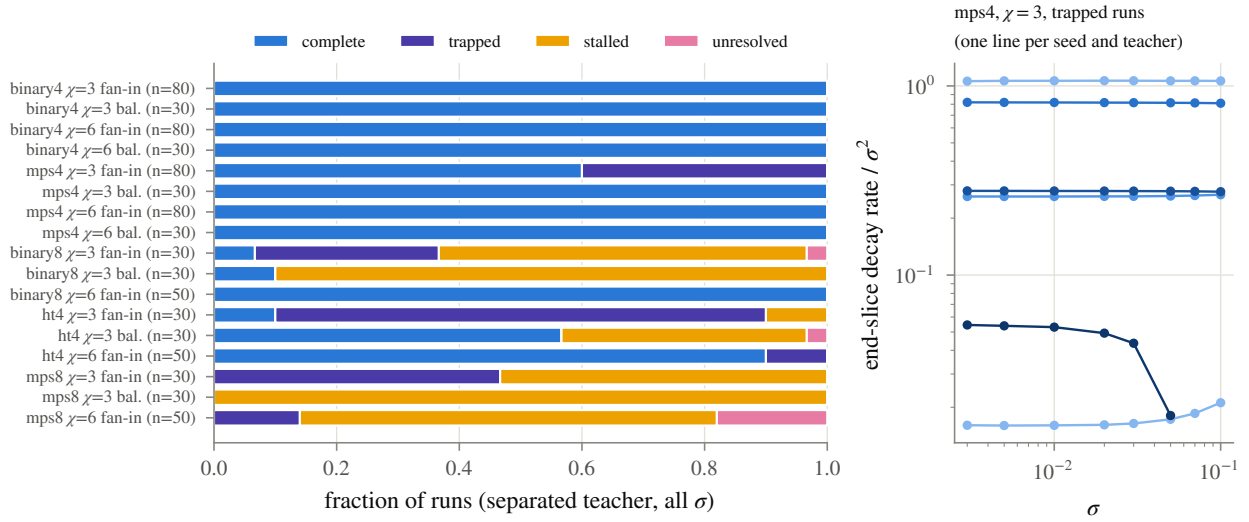


Figure 11: Odeco runs that do not complete. Left: fractions of run classes, separated teacher, all σ ($K = 7, 8$: $\sigma = 0.15$–0.3 at $\chi = 3$, 0.07–0.3 at $\chi = 6$); trapped fractions at $K = 7, 8$ are lower bounds. Right: decay rate/σ^2 of the route-end slices of the component never learned, MPS4, $\chi = 3$, one line per seed and teacher (upper four: separated, lower two: close), fitted from 1.5 times the last event (about $\pm 4\%$ bias from the post-event transient) to a slice of 10^{-9}, above the floating-point floor. Takeaway: no trap at $K = 3$, fixed by the initialization at $K = 4$, frequent at $K = 7, 8$ unless every bond has spare channels.

not in the escapes: across an escape a dormant amplitude $|\langle W, U_r \rangle|$ changes by a median factor 2.1–3.0 at $K = 3$ (it grows: the head start of Proposition A.25), 0.64–0.74 at $K = 4$ and 0.12–0.75 at $K = 7, 8$, and falls by more than 100 times in at most 1.3% ($K = 4$), 2% ($K = 7, 8$, $\chi = 6$) and 8.3% ($\chi = 3$; 11% for the later escapes on MPS8 with the separated teacher) of the escapes.

Stalled runs: slow where tested. empirical (prediction derived) new A stalled route with live ends should escape after about the plateau of a balanced route at its present amplitude, $b^{-(K-2)}/((K-2)s_r)$ with $b = (s_r|a_r|)^{1/K}$. Continued from their final state, 16 of 23 stalled or unresolved runs escaped, at 0.24–2.19 times this estimate (all 8 with the shortest estimates, 6 of 7 further runs of a seeded random sample, 2 of 8 MPS8 runs at $\chi = 6$); 2 did not escape within three times it and 5 ran out of wall-clock time or failed. The estimate is not a bound (4 escapes came later). For the stalled runs not continued (91 of 106 at $K = 7, 8$ with $\chi = 3$, 35 of 43 MPS8 runs at $\chi = 6$), "a long plateau, not a dead route" is a conjecture: their dormant slices are pinned at 0.01–0.04 σ on binary8 and balanced MPS8, but were still decaying in 6 of 13 balanced HT4 runs and 17 of 59 fan-in MPS8 runs.

Pattern teachers: dead routes at $k \geq 4$. empirical new Gradient flow can stop at a learned saddle for good. A run is *stalled* when, after a component has been learned, every gradient entry stays below 10^{-12} for 5,000 consecutive evaluations; the loss then sits at a partial fit (for two components $L = s_2^2/2$). Under fan-in initialization at $\chi = 2$, 398 of 5,682 two-component runs stall, all with $k \geq 4$; none of the 5,112 runs with $k \leq 3$ (two-component and robustness sweeps) does. By k, both protocols pooled: 130 of 768 runs ($k = 4$), 62/816 ($k = 5$), 133/360 (6), 29/118 (7), 44/60 (8). The rate depends on the tree more than on k: for $k = 5$, 7 of 624 binary and HT runs stall against 55 of 192 MPS runs, and stalls concentrate on MPS routes, especially those ending at the ends of the chain. The stalls checked are dead, not slow: in 22 (case, σ) pairs rerun with a 60 times longer patience (170 runs), all 37 flagged fan-in and saddle-start runs show no escape up to $t \approx 10^{10}$,

Table 7: Stalled runs among two-component runs with $k \geq 4$: small initialization, $d = 2$, the same 24 (tree, D) cases and σ grids in every column. Fan-in initialization at $\chi = 2$ and $\chi = 4$; balanced initialization ($Q_e = 0$, same W_0) at $\chi = 2$. Bottom rows: split by whether at least one end of the route sits on a bond below the cap (3) or both ends sit on bonds capped at d (MPS chain ends, HT leaves).

k	fan-in, $\chi = 2$	balanced, $\chi = 2$	fan-in, $\chi = 4$
4	67/384	2/384	18/384
5	20/384	5/384	0/384
6	58/180	5/180	59/180
7	8/30	0/30	12/30
8	16/30	4/30	17/27
all	169/1,008	16/1,008	106/1,005
route with an uncapped end (13 cases)	79	–	0
both route ends capped (11 cases)	90	–	106

with route-end amplitudes at roundoff (10^{-17}–10^{-13}). The pilot's code found the same collapse to the rank-one saddle.

Balanced initialization and spare channels (Table 7). Balanced initialization cuts the stalls tenfold, from 169 to 16 of 1,008 on the same cases, all on MPS routes, and none of the checked ones is dead: of 10 rerun with longer patience, 6 escaped very late (after $2 \cdot 10^9$–$5 \cdot 10^{10}$, about 10^5 times the usual plateau), 3 are pinned (route ends at $3 \cdot 10^{-10}$, $6 \cdot 10^{-9}$, $4 \cdot 10^{-8}$ at $t = 10^{11}$–$5 \cdot 10^{11}$) and one was in the middle of a late escape when its budget ended. Spare bond channels do *not* remove most stalls: $\chi = 4$ gives 106 of 1,005. They remove them on routes with at least one uncapped end (79 at $\chi = 2$, none at $\chi = 4$; all 13 such cases escape), but on routes with both ends capped the stalls rise from 90 to 106, in 7 of the 8 such cases that stall at all (they fall only for MPS4 $\{0, 3\}$, from 30 to 6 of 48). midroot6 and MPS6, the same graph with different roots, share exponents but not stall rates ($D = \{0, 5\}$ at $\chi = 4$: 30 and 18 of 30): the root changes the fan-in statistics of Q_e. The odeco runs are consistent with this: $\chi = 6$ removes the trap on binary8, where every bond has twice the teacher rank, but not on HT4 and MPS8, whose route ends sit on bonds capped at $d = 4$, one channel above it.

Effect on the exponents. Stalls depend on σ: in 20 of 33 fan-in rows with stalls a stalling seed escapes at other σ, so it enters the fits with fewer points. Restricting each row to seeds that never stall leaves the exponent unchanged in 27 of 46 rows with stalls and moves it by 0.1 or more in 9 fan-in rows (by up to 1.38; four of them keep only 1–3 seeds) and 3 other rows. This accounts for the off-target saddle-start $k = 4$ and HT8 rows of Table 6.

Relation to the conservation-law trap. In the reduced model of Proposition A.28 a route end decays iff $c = Q_{\nu\nu} + \|y_{\text{route}}\|^2 < 0$ (an *unfavourable* end), and a route dies for good only if all its ends are unfavourable. The odeco runs show the mechanism's signature where it was measured ($K = 4$): decay rates $\propto \sigma^2$, constant per seed and equal at both route ends, and no trap from balanced initialization. The pattern-teacher runs test the criterion itself. Evaluated once, at the start of the plateau ($a_1 = 0.99$), in the 122 fan-in and saddle-start runs of the stall diagnosis, this joint criterion is strongly associated with death (30 of 55 runs with both ends unfavourable died, 7 of 67 with a favourable end) but is neither sufficient (25 predicted-dead runs escaped) nor necessary: seven dead runs had a live end at the plateau start, which by the theory's own refutation criterion

("a dead stall with a live route end") counts against the mechanism as stated, unless c changes sign later in the plateau (y_{route} is not conserved; not checked). Further conflicts with the theory remain open: (i) exactly balanced initialization is expected to remove the trap, its route ends not decaying; it leaves no dead route in either experiment (consistent), but on the pattern-teacher MPS8 routes the ends shrink to 10^{-10}–10^{-8} and escapes are delayed by 10^5 or more (inconsistent; the balanced odeco MPS8 runs stall instead with route slices near $0.03\,\sigma$); (ii) the heuristic frequencies (about half of the two-ended HT routes dead under fan-in initialization, Proposition A.28; fewer stalls on MPS with $\chi = d$) are contradicted: 0 of 144 small-initialization HT4 runs with $k = 5$ stall, against 30 of 48 on MPS4 $\{0, 3\}$ and 17 of 48 on MPS6 $\{0, 3\}$ (the odeco HT4 routes, with four leaf ends, trap in at least 24 of 30 runs at $\chi = 3$); (iii) spare channels avoided stalls in the runs of Section A.8, here only on routes with an uncapped end, and the rise on capped routes has no established mechanism.

Steep MPS routes. empirical (mechanism open) One row stays steep at small σ: MPS8 $\{1, 5\}$ ($k = 5$) from the saddle, $p = 3.67 \pm 0.20$ at $\sigma \in [0.012, 0.03]$ (median 3.83; 4 fitted seeds, 0.4 decades, 10 of 24 runs stalled), with a matching Hessian exponent (3.74 projected, 3.75 raw). Its fitted seeds escape at every σ, so it meets a refutation criterion of the theory (a per-seed exponent off by more than 0.2 at $\sigma \leq 0.05$ in runs without stall or sign correction), on thin evidence. In the steep MPS rows the route-end amplitudes at the plateau midpoint scale as σ^α with $\alpha = 1.24$–1.91 and shrink during the plateau (median ratio of midpoint to first event 0.53–0.78, against 0.85–1.03 in on-target rows); but $\alpha > 1$ also occurs in on-target rows (binary8 $\{0, 7\}$, balanced: $\alpha = 1.34$, $p = 2.99$), so the steep exponents are associated with shrinking route ends, not shown to be caused by them. Neither the logarithmic correction expected if small ends caught up through the conservation laws (per seed it fits much worse than a free power law: median residual of $\ln \Delta t_2$ 0.25–0.62 against 0.05–0.12) nor the pilot's conjecture $\sigma^{-(2k-6)}$ (both ends pinned, $\alpha = 2$; measured α is seed-dependent, between 1 and 2) describes the data.

4.7 Dependence on each parameter

[Summary table: initialization scale, number of cores and tree shape, bond dimension, physical dimension, teacher spectrum, imbalance, learning rate; effect on plateau levels, durations and order; status of each statement.]

5 Other regimes

This section looks at the regimes of Section 3 other than saddle-to-saddle: lazy training and its failure below the cut dimension, the rich window between the two boundaries, runs that stall, and the limit of large bond dimension under three parameterizations; then at how discrete steps, gradient noise and finite samples (Section 5.5) and the optimizers used in practice (Section 5.6) modify the gradient-flow picture. Settings of the first four subsections: Sections B.4 and 3.1.

5.1 Lazy regime

At large σ the centred model trains like its linearization at θ_0: the function moves along the range of the tangent kernel $\Theta = JDJ^\top$, which hardly changes. Lazy training is known in general known (Jacot et al., 2018; Chizat et al., 2019; Lee et al., 2019); what the tree adds is the kernel. At initialization $\mathbb{E}\Theta$ is a multiple of the identity at every bond dimension, but the rank of Θ is the dimension of the manifold of tensors with the given bond ranks, which reaches d^N only when every bond is full (Proposition A.46) proven new (on every tree).

Lazy only with full bonds. derived empirical known (that a linearized model fits only the projection of the target on its tangent space: Chizat et al., 2019, Thm 2.5; Ghorbani et al., 2019, Thm 2) With a bond below its cut dimension the linearized flow fits only the projection of W^* on the range of Θ and stops at $\frac{1}{2}\|P_\perp W^*\|_F^2$, on binary4 a fraction $(1-\chi/d^2)^2$ of $L(0)$ on average (0.44, 0.31, 0.11 at $\chi = 3, 4, 6$; Section A.8). In addition the centred target $W^* + W_0$ is unrealizable at every σ, and gradient flow ends on its best approximation (Figure 12(a)): at small σ this floor rises like σ^{2K} (mean $(1 - \dim \mathcal{M}_\chi/d^N)\,\sigma^{2K}/\|W^*\|_F^2$, the seed median about 0.7 times it, (61)), and at large σ it saturates at the lazy residual: at $\sigma = 40$ the final loss of each of the 18 centred runs at $\chi = 3, 4, 6$ equals its residual computed at initialization to 0.13%, with $\delta_\Theta^{\text{probe}} < 10^{-3}$. Once the floor exceeds the convergence threshold (for every seed from $\sigma = 1.0, 1.0, 1.6$) the runs are stalled by the rule of Table 3, though their dynamics are kernel-like at large σ. A lazy regime therefore needs every bond at its cut dimension (binary4: $\chi \geq 9$), which grows like $d^{N/2}$ for the middle bonds.

Anisotropic, ill-conditioned kernels. derived empirical new (for TTNs; the mean kernels of MPS and TT models: Guo and Draper, 2021; Wesel and Batselier, 2025) A single initialization has an isotropic kernel only if every bond is far above its cut dimension; at a capped full bond the cores next to it are square Gaussian matrices and $\lambda_{\min}(\Theta)/\bar{\lambda} \sim 10^{-4}$ (Section A.8). The lazy fit then converges in a kernel time of order $(\bar{\lambda}/\lambda_{\min})\ln(1/\text{tol})$. On the binary tree with $N = 8$, $d = 2$, the square full bonds ($\chi = 16$) are not lazy within the step limit: every run from $\sigma = 2.3$ stops there at the same seed-dependent loss, $L/L(0) = 0.05$–0.44, for every σ. Over-parameterized twice (bonds 8 above two sites, 32 above four), the runs are lazy in 2 of 5 at $\sigma = 1.42$ and in 4 of 5 or 6 at each σ from 2.47 to 7.6 (the others stopped at the step limit at $L/L(0) = 0.0014$–0.0054 with $\delta_\Theta^{\text{probe}} < 0.02$); the kernel boundary is 1.45 (probe kernel, median of four seeds). At practical χ the lazy regime is thus kernel regression with a random, tree-dependent and anisotropic kernel, not with its mean.

Lazy by the kernel is not small motion. derived Under the standard parameterization the root of binary4 carries a share $\chi^2/(2d^2 + \chi^2)$ of the kernel, and its own kernel block does not depend on it: a lazy run can be a trained readout over fixed random features, with the root moving by more than 100% while $\delta_\Theta < 0.1$ (Section A.8). Per-core changes should be reported next to δ_Θ (Section 5.4).

5.2 Rich, non-incremental regime

Between the two boundaries the runs converge and move their kernel ($\delta_\Theta^{\text{probe}}$ of the binary4 rich runs: 0.10 to 10^7, median 68), but their events overlap: some $\rho_i \leq 3$.

What sets the crossover. derived (aligned initialization) empirical (random) The plateaus between events grow like $\sigma^{-(K-2)}$ as $\sigma \to 0$ while the event widths stay $O(1)$, so the separation ratios fall through 3 as σ grows; for aligned initialization Proposition A.48 gives the boundary exactly, with the threshold exponent of Gissin et al. (2020) known. From random initialization the crossover is a seed property: on binary4 the per-seed boundaries at one χ span a factor 1.9–16 for $\chi \leq 12$ and 1.4–2.7 for $\chi \geq 16$ (six seeds). In σ the rich window spans a factor 18–23 on binary4 at large χ and 2.7 on the binary tree with $N = 8$ (Section 3.2). If both boundaries have finite limits in the function-space scale $\alpha = \sigma^K$, the window shrinks to a band of width $O(1/K)$ in $\ln\sigma$ conjecture.

Transients. empirical During an escape, components not yet learned rise transiently (for matrix-like pairs the peak falls like $\sigma^{2(1-\varrho)}$, conjecture, Section A.8). In 116 of the 9,165 runs such a transient

crossed $\frac{1}{2}$ before the component was learned (14 more cross $\frac{1}{2}$ repeatedly only in a component never learned); with events dated at the first crossing, 15 runs would be rich instead of saddle-to-saddle. Events are therefore dated by the final rise (Section 2.7).

Lazy, then rich. derived ($N = 4$) empirical known (analogues: mixed lazy and active dynamics, Tu et al., 2024; lazy-then-rich transitions, Lyu et al., 2024) The two-copy model is realizable at a deficient bond when 2χ reaches the teacher's bond rank. At large σ it first fits the tangent part lazily and sits at the lazy residual, then escapes by rotating its bond subspaces, after a gradient-flow time $\propto \sigma^2$, a kernel time $\propto \sigma^{2K}$ (Section A.8). On binary4 at $\chi = 3$ (six seeds) the runs converge (rich) for $\sigma = 1.0$–4.0, end at $L/L(0) = 0.005$–0.05 at $\sigma = 6.3$, at 0.13–0.44 at $\sigma = 10$, and at the lazy residual from $\sigma = 16$ on: the escape leaves any fixed budget as σ grows. A light root with a deficient bond behaves the same way, lazy and then rich (Sections A.8 and 3.4).

Merged first escapes. empirical new In the two-copy model at small σ the two copies start identical, so their near-origin escapes coincide: each copy learns part of both of the first two components at once, and the first two events merge. On binary4 at $\chi = 18$, 24, 36 the smallest σ (0.010) is labelled saddle-to-saddle in 5, 4 and 5 of six seeds and the next (0.016) in 4, 3 and 2, but the crossing $\min_i \rho_i = 5$ exists in only 2, 3 and 0 seeds (the largest $\min_i \rho_i$ at these σ is 7.1, 14.4, 4.8), so the sweep fails condition (iii) of the saddle-to-saddle signature. Adam's runs from small initialization are rich, non-incremental in the same sense (Section 5.6).

5.3 Stalled runs

A run is stalled when its loss is still above $10^{-3}L(0)$ at the end of its budget (Table 3). In the maps of Section 3 this has four causes: the floor of an unrealizable centred target (Section 5.1); a plateau longer than the budget, since a route through k cores takes a time $\propto \sigma^{-(k-2)}$ (Section 4.4); a dead route, whose end cores have decayed to roundoff under the conservation laws (the trap of Proposition A.28); and the step limit, met by unfinished lazy fits at square bonds and by some deep-tree runs (for example on MPS with $N = 8$ at $\sigma = 0.1$ after $2 \cdot 10^4$ steps, at $t = 1.4 \cdot 10^9$ of a budget of $5.7 \cdot 10^9$). Only the first is specific to the centred model.

Bond dimension equal to the teacher rank. empirical confirming the prediction of Section A.8 (mechanism: conjecture, heuristic) With $\chi = R$ the last component must use the last free channel of every bond, through a route of all K cores, and fan-in initialization makes the root bonds child-heavy ($\mathbb{E}Q_e = \sigma^2(1 - 1/\chi)I \succ 0$), so the route ends decay at a rate $\propto \sigma^2$ while the escape needs a time $\propto \sigma^{-(K-2)}$ (Section 4.6); for $K \geq 5$ the route can die first. The stall test (Table 8; $d = 3$, $\chi = 3$, odeco teacher $(3, 2, 1)$, $\sigma = 0.3$, 0.45, 0.6, six seeds, budgets $6 \cdot 10^4$–$1.6 \cdot 10^7$ time units) confirms it: most runs on the binary tree with $N = 8$ ($K = 7$) and on the MPS with $N = 8$ ($K = 8$; every fan-in run) never learn the last component, in the plain as in the centred model, so W_0 is not the cause. From fan-in initialization the stalled route is dead: its third bond singular value ends at roundoff (below 10^{-12}; in 28 of 32 stalled plain runs). Gauge-balanced initialization (the same W_0 re-gauged to $Q_e = 0$) removes the dead routes but not the stall: the route still decays, its third singular value is $1.4 \cdot 10^{-5}$ (median) at the end, and 12 of 18 runs do not escape. This is the trap of Section 4.6 in the regime maps; that section gives the odeco and pattern-teacher evidence, the criterion for a dead route and the conflicts that remain open.

Stalls across the number of cores. empirical In the plain-model $\sigma \times K$ sweeps (Section B.4), 546 of the 795 stalled runs end with a dead route, 212 with a route still decaying and 37 on a long

Table 8: Stall test at χ equal to the teacher rank: $d = 3$, $\chi = 3$, odeco teacher $(3, 2, 1)$, $\sigma \in \{0.3, 0.45, 0.6\}$, six seeds; "balanced": the same W_0 with $Q_e = 0$ on every bond. Stall types from the third bond singular value of W at the end, minimized over edges: dead below 10^{-12}, decaying up to 10^{-4} (no stalled run is above); the centred runs did not record the spectra of W. Source: `stall_test_summary.csv`.

model	tree (K)	initialization	converged	stalled: dead route	decaying	not typed
plain	binary8 (7)	fan-in	4/18	12	2	–
		balanced	6/18	0	12	–
	MPS8 (8)	fan-in	0/18	16	2	–
centred	binary8 (7)	fan-in	4/18	–	–	14
		balanced	8/18	–	–	10
	MPS8 (8)	fan-in	0/18	–	–	18
		balanced	3/12	–	–	9

plateau (by the R-th bond singular value at the end, with the thresholds of Table 8 and 10^{-4}). Spare channels help where every route end has one: on the binary tree with $N = 8$ at $d = 3$, $\chi = 6$ no run stalls at $\sigma \leq 0.63$ (30 runs), but on the MPS with $N = 8$ every run stalls, its end bonds being capped at $d = 3$, the teacher rank; at $d = 2$ the MPS with full bonds still stall in up to 3 of 6 seeds. With the centred model at $\chi = 4$ the stalls below the boundary sit at a partial fit and are fixed by the initialization (Section 3.3). At practical budgets stalls thus occupy part of the saddle-to-saddle side of every deep tree whose route ends lack spare channels, as on the pattern-teacher routes of Section 4.6.

Long plateaus are not barren plateaus. derived From small initialization the gradient is $O(\sigma^{K-1})$ and the first plateau lasts $\propto \sigma^{-(K-2)}$ (Section 4.3): deep trees at small σ stall as if on a barren plateau. The cause differs: the gradient is small because the initialization is, not because it concentrates around zero over random initializations of fixed norm, and with the fan-in normalization at $\sigma = 1$ the initial function has unit norm on every tree (Proposition 2.2). For tensor networks, barren plateaus are known for global but not local losses of unitary-embedded MPS known (Liu et al., 2022) and are absent from the energy optimization of isometric MPS, TTNS and MERA known (Miao and Barthel, 2024; Barthel and Miao, 2025).

5.4 Large bond dimension: NTK versus μP

As χ grows at fixed σ, the scales of the cores and their learning-rate multipliers decide whether the features still move. We compare three parameterizations, written as initial scales and multipliers η_v (Section 2.4, variant (e)): the *standard* one (SP: fan-in initialization, $\eta_v = 1$); *NTK* ($\eta_v = 1/F_v$, so that every core contributes equally to the expected kernel); and *μP-TTN*, the spectral condition of Yang et al. (2024) applied with the injective norm of each core, since a core only ever acts on product vectors: fan-in initialization except at the root, which is scaled by $(\sum_\ell m_\ell)^{-1/2} = (2\chi)^{-1/2}$ on binary4 (m_ℓ its child-leg dimensions), and $\eta_v = \chi_v/F_v$ (Section A.8) derived new (the spectral condition is known; factor-wise rules for structured layers: Qiu et al., 2024; Potapczynski et al., 2024).

Predictions. derived (binary4, centred, isotropic limit $\chi \gg d^2$, fixed $\xi = \|W^*\|_F/\alpha$) In each parameterization the first- and second-order constants c_1, c_2 of the law (59) carry the same power of χ, so the kernel change scales as χ^{-3} (SP), χ^{-1} (NTK) and χ^0 (μP-TTN); the parameters move as in Table 9 (Section A.8; constants in `notes/theory_regimes.md`, R4 and R12). Under SP the root, with fan-in χ^2, carries almost all of the kernel and the model becomes linear in the root over fixed random features: lazy at any fixed σ as χ grows, centred or not. Under NTK the cores move

Table 9: Exponents p of χ^p at large bond dimension: binary4, $d = 3$, teacher $(3, 2, 1)$, centred model, $\sigma = 1.5$, $\chi = 16$–128, six seeds; least-squares slopes over the 24 runs with standard errors from the scatter; in brackets the prediction of Section A.8. δ_Θ: tensor kernel; δ_θ: all parameters; root and leaf cores: relative Frobenius change (the spectral change of the μP-TTN root is predicted $O(1)$). Source: `large_chi_slopes_tensor.csv`.

	δ_Θ	δ_θ	root	leaf cores
SP	-3.02 ± 0.06 (-3)	-1.63 ± 0.04 $(-\frac{3}{2})$	-1.02 ± 0.03 (-1)	-2.96 ± 0.07 $(-\frac{5}{2})$
NTK	-0.79 ± 0.04 (-1)	-0.52 ± 0.03 $(-\frac{1}{2})$	-0.93 ± 0.02 (-1)	-0.53 ± 0.03 $(-\frac{1}{2})$
μP-TTN	$+0.085 \pm 0.024$ (0)	-0.064 ± 0.018 (0)	-0.50 ± 0.01 $(-\frac{1}{2})$	-0.065 ± 0.018 (0)

by $O(\chi^{-1/2})$ or less, also lazy in the limit, as for the NTK of MPS known (Guo and Draper, 2021). Under μP-TTN every core's update stays of order one.

Measurements. empirical confirming the predictions, except for the SP leaf cores (6.5 standard errors steeper) and, at 3.3 standard errors, the SP parameter change. Binary4, $d = 3$, teacher $(3, 2, 1)$, centred, $\sigma = 1.5$ ($\xi = 1.11$), $\chi = 16$–128 (four values, 0.9 decade), six seeds; tensor kernel (Figure 12(b), (c); Table 9). Under μP-TTN the kernel change approaches the predicted level, 5.1, 5.7, 6.1, 6.2 at $\chi = 16$, 32, 64, 128 against 6.2–6.3: feature learning survives. Under NTK the fit over $\chi = 16$–128 is pulled by $\chi = 16$–32, where δ_Θ is still 0.4–0.6; between $\chi = 64$ and 128 the slope is -1.01, with levels 0.250 and 0.124 against 0.240 and 0.120 from the second-order law with the NTK constants. Under SP the kernel change falls like $\chi^{-3.02\pm0.06}$, while at $\chi = 128$ the root still moves by 9% and the leaf cores by $2 \cdot 10^{-4}$ (medians; Figure 12(c)). At $\sigma = 1$ ($\xi = 3.74$, further from the lazy limit) the slopes are shallower (SP -2.58 ± 0.06, NTK -0.62 ± 0.02, μP-TTN $+0.089 \pm 0.024$): the exponents hold as $\xi \to 0$. The plain model gives the same exponents within the errors ($\sigma = 1.5$: -3.06 ± 0.07, -0.83 ± 0.04, $+0.095 \pm 0.024$): with full, over-parameterized bonds, centring does not change the scaling. Scope: binary4 only; runs on the binary tree with $N = 8$ are incomplete and not interpreted. The factor-wise rule of structured layers, applied to a TTN, makes the initial kernel of that tree vanish like χ^{-2} and slows the initial fit about linearly in χ (Section A.8). The μP-TTN limit itself is not derived; on binary4 it may be a two-population mean field with a pairwise coupling through the root conjecture.

5.5 Discrete steps, gradient noise and finite samples

Gradient flow is the limit of gradient descent (GD) as the step $\eta \to 0$ and of online SGD as $\eta/B \to 0$, and the population loss is the limit of the empirical loss as $n \to \infty$. How much of Section 4 survives finite values? GD keeps gradient flow's saddle sequence up to a sharp stability limit, with event times shifted by a computable $O(\eta)$; gradient noise never shortens a plateau, but it lengthens later ones and can trap the dynamics at a saddle; and from a finite training set small initialization recovers the teacher with its bond ranks, while the lazy limit fails as dimension counting predicts.

Setup (details in Section B.6). Whitened Gaussian inputs, odeco teacher with weights $(3, 2, 1)$, $d = \chi = 3$, fan-in initialization (seeds 0–4); binary trees with $N = 4$ ($K = 3$) and $N = 8$ ($K = 7$), MPS with $N = 8$ ($K = 8$). GD and online SGD (fresh inputs at every step) run on the clock $t = \eta \times$ (steps) and are compared with gradient flow from the same initial point (DOP853, tolerances 10^{-10} and 10^{-14}). Events are persistent, as in Section 5.6: the last upward crossing of $a_r = \frac{1}{2}$.

The stability limit. proven known (deep linear networks: Mulayoff and Michaeli, 2020; Ghosh et al., 2025) new (general trees) At the balanced global minimum the Hessian $J^\top J$ has largest eigenvalue

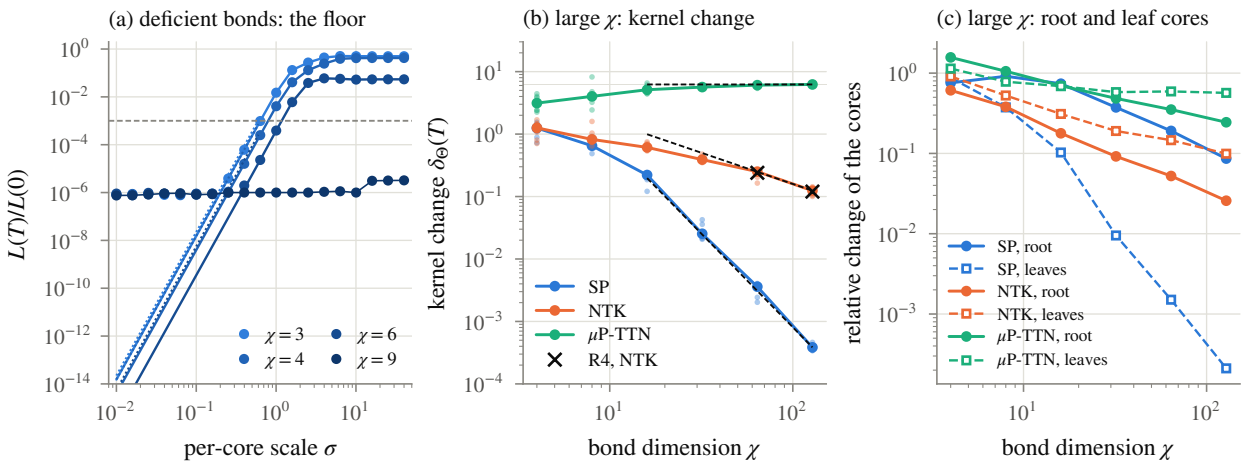


Figure 12: Lazy regime and large bond dimension; binary tree with $N = 4$, $d = 3$, teacher $(3, 2, 1)$, centred model, six seeds. (a) Final loss against σ (dots: medians) for $\chi = 3$, 4, 6 (deficient bonds) and $\chi = 9$ (full); solid: the Eckart–Young minimum of $W^* + W_0$ across the cut, median over seeds; dotted: 0.7 times the small-σ mean (61); dashed: the convergence threshold 10^{-3}. Runs stop at 10^{-6}; with deficient bonds they end on the minimum once it exceeds that, and are stalled above the threshold. (b) Kernel change (tensor kernel) against χ at $\sigma = 1.5$: seeds (dots), medians (lines), predicted slopes -3, -1, 0 (dashed, through the $\chi = 128$ medians) and the predicted NTK levels (crosses). (c) Relative change of the root (solid) and leaf cores (dashed), medians. Takeaway: below the cut dimension the large-σ end is a floor, not a lazy fit; at large χ only μP-TTN keeps the kernel and the leaf cores moving.

$\lambda_{\text{bal}} = K s_1^{2-2/K}$, and every other aligned minimum is sharper (Proposition A.43; with $\chi_e = R$ every other minimum, derived), so GD can converge to a global minimum only if

$$\eta < \eta_c = \frac{2}{K\, s_1^{2-2/K}} \tag{24}$$

(0.1541, 0.0435 and 0.0366 for the three trees). Along the saddle-to-saddle path the sharpness is $O(\sigma^{K-2})$ on the first plateau and $K \max_r s_r^{2-2/K}$ over the components learned so far derived (Section A.7): a step above η_c is felt only once component 1 is learned, and there is no catapult at the start (the condition $\eta\lambda_{\max}(H) > 2$ of Lewkowycz et al. (2020) fails near the origin). Measured empirical on binary $N = 4$ ($\sigma = 0.003$–0.3, five seeds, up to 18 steps from $0.01\eta_c$ to $3\eta_c$; Figure 13a): all 195 runs with $\eta \le 0.95\eta_c$ are GF-like and all 80 at 1.05–$1.3\eta_c$ oscillate (an independent code placed the edge between $0.99\eta_c$ and $1.01\eta_c$). Below η_c, GD learned the components in gradient flow's order in all 765 runs, including 40 more seeds on binary $N = 4$ and both deep trees at $\sigma = 0.9$; a larger step only enlarges transient excursions of near-tied components, which cross $\frac{1}{2}$ and fall back (at $\sigma = 0.1$ in 0.09 ± 0.04 of 45 initializations at $0.1\eta_c$ and 0.18 ± 0.06 at $0.9\eta_c$).

Below the limit, GD is gradient flow on a shifted clock. derived new (for TTNs) known (the modified flow: Barrett and Dherin, 2021) To first order in η, GD follows the modified flow $\dot\theta = -(I + \frac{\eta}{2}H)\nabla L$, whose critical points are those of L: the saddles are gradient flow's, and each event moves by $\eta c_e + O(\eta^2)$, with c_e computed without free parameter from the variational equation along the gradient-flow path (Proposition A.44 and Section B.6). Measured empirical: over 526 comparisons at $\eta \le 0.1\eta_c$ the shift agrees with ηc_e within 0.3%, 1% and 5% in 66%, 90% and 97% of the cases, all 202 first events within 1%; the 15 misses beyond 5% are later events, worst where $c_e \approx 0$ after a near-tied race and the $O(\eta^2)$ term dominates. Richardson extrapolation to $\eta \to 0$ recovers c_e to a

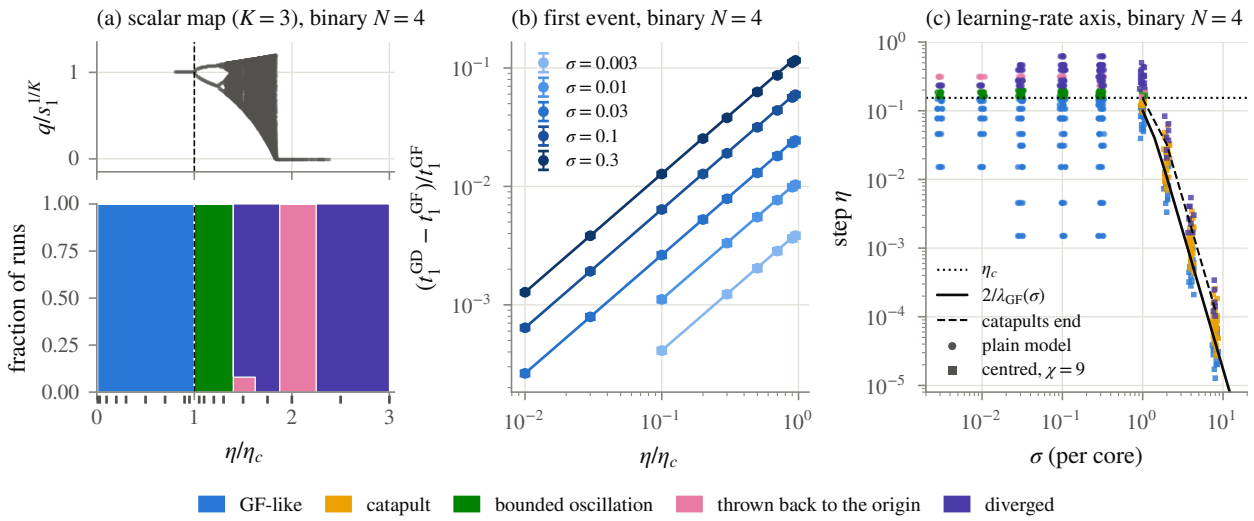


Figure 13: Gradient descent against gradient flow: binary tree, $N = 4$ ($K = 3$), $d = \chi = 3$, odeco teacher $(3, 2, 1)$, five seeds. (a) Top: attractor of the balanced scalar map $x \mapsto x + hx^{K-1}(1 - x^K)$, $x = q/s_1^{1/K}$, $h = \eta s_1^{2-2/K}$ (points at 0: returns to the origin). Bottom: outcome classes at the 18 steps marked below the axis, pooled over $\sigma = 0.003$–0.3 (15–25 runs per step). GF-like: monotone loss ending at gradient flow's loss (within 5% plus $10^{-3}L_0$); catapult: the same after a rise of the loss; thrown back: loss above $0.9L_0$ after the first event; diverged: above $10^6 L_0$. (b) Relative error of the first event time, mean ± s.e.; lines: $\eta c_1/t_1^{GF}$, no fit. (c) All runs in the (σ, η) plane (plain model with $\chi = 3$ and 9, $\sigma \le 0.3$; centred, $\chi = 9$, $\sigma \ge 1$): dotted, η_c; solid, $2/\lambda_{GF}$ (mean over seeds); dashed, the largest step at which a run still reaches gradient flow's loss.

median relative $1.2 \cdot 10^{-5}$ (57 events). Since c_1 grows like $\ln(1/\sigma)$ while $t_1 \propto \sigma^{-(K-2)}$, the relative error of t_1 vanishes as $\sigma \to 0$: at $0.9\eta_c$ it is 0.111, 0.056, 0.023, 0.0099 and 0.0036 for $\sigma = 0.3$, 0.1, 0.03, 0.01, 0.003 (five seeds; Figure 13b). At small initialization every stable step thus reproduces gradient flow, the more closely the smaller σ, as GD does in Section 5.6.

Above the limit: edge of stability. derived known (the scalar map and its two-cycle: Ghosh et al., 2025, Thm 1, MacDonald et al., 2025, Thm 5.3; edge of stability: Cohen et al., 2021) For $\eta_c < \eta < 2/\lambda_2(H)$ only the radial mode of component 1 is unstable, and GD on it is the balanced scalar map of Figure 13a (Section A.7): a two-cycle from η_c, period four from $1.245\eta_c$, chaos from $1.32\eta_c$, then returns to the origin interleaved with divergence ($K = 3$). Measured empirical on binary $N = 4$: component 1 oscillates while components 2 and 3 are still learned, and the time-averaged excess loss equals s_1^2 times the map's mean loss within 0.9% at 1.05–$1.3\eta_c$ (for example 0.4496 against 0.4496 at $1.2\eta_c$, 25 runs; (54) is its small-excess limit). The map's bounded chaotic window does not survive: the network diverges at 1.5–$1.75\eta_c$ (38 of 40 runs), is thrown back to the origin saddle and relearns its components again and again at $2\eta_c$ (25 of 25), and diverges from $2.5\eta_c$. The deep trees, started at $\sigma = 0.9$ where the gradient-flow states are 10–35% sharper than λ_{bal}, already oscillate or catapult at $0.9\eta_c$.

Larger initialization: catapults. empirical known (catapults: Lewkowycz et al., 2020; flatter, more balanced minima in scalar networks: Kreisler et al., 2023, Thms 3.2–3.3; limit sharpness at most $2/\eta$ in two-factor factorization: Wang et al., 2022, Thm 3.2; balancing while oscillating beyond the edge: Ghosh et al., 2025, Prop. 2) At larger σ the minimum that gradient flow reaches is sharper, and $2/\lambda_{GF}$, with λ_{GF} the largest Hessian eigenvalue there, replaces η_c. We follow it into the lazy regime with the

centred model (Section 2.4) on binary $N = 4$ at $\chi = 9$, the cut dimension, where all 45 gradient-flow references converge: $\lambda_{\mathrm{GF}} \propto \sigma^{3.965\pm0.014}$ over $\sigma = 1.41$–16 (five seeds), close to the scale $\sigma^{2(K-1)} = \sigma^4$ of the tangent kernel at initialization (Proposition A.46), so $2/\lambda_{\mathrm{GF}}$ equals η_c for $\sigma \le 0.3$, $0.65\eta_c$ at $\sigma = 1$ and $2.7 \cdot 10^{-4}\eta_c$ at $\sigma = 8$ (Figure 13c). With steps of 0.5–4 times each run's own $2/\lambda_{\mathrm{GF}}$, all runs up to 0.95 times are GF-like at every σ, and monotone GD ends between 0.95 and 1.2 times. Beyond, small initialization ($\sigma = 0.1$, 0.3) oscillates and then diverges, since the balanced minimum is already the flattest; from $\sigma = 1$ on GD catapults: the loss spikes (by up to $2 \cdot 10^5 L_0$ at $\sigma = 8$) and then reaches gradient flow's final loss (86 of 90 runs below $10^{-6}L_0$) at a flatter minimum, of sharpness 0.58–1.00 times $2/\eta$ (per run), for steps up to 1.5 ($\sigma = 1$) and 3 ($\sigma = 2$–8) times $2/\lambda_{\mathrm{GF}}$; at 4 times all runs diverge. For $\sigma \ge 2$ the window thus ends between $6/\lambda_{\mathrm{GF}}$ and $8/\lambda_{\mathrm{GF}}$, 1.5–2 times the $4/\lambda_0$ of the two-layer model of Lewkowycz et al. (2020). Imbalanced small initializations ($\sigma = 0.1$ with the root core at scale 2, 4 or 6) reproduce the known selection: the 104 of 150 runs that converge end 0.05–3% below $\min(\lambda_{\mathrm{GF}}, 2/\eta)$.

SGD: the first plateau keeps its law. derived new (for TTNs; framework: Li et al., 2017; Ziyin et al., 2024) The per-sample gradient carries the Jacobian $J = \partial W/\partial\theta$, which vanishes at the origin like the drift, so noise cannot drive the first escape and its exponent $K - 2$ is unchanged. On an aligned mode it only delays and spreads the escape, by (56): the mean delay is $O(\eta(1 + \kappa_{\mathrm{n}}/B)\ln(1/\sigma))$ and the standard deviation $(\kappa_{\mathrm{n}} t_1^{\mathrm{GF}} \eta/B)^{1/2}$, with $\kappa_{\mathrm{n}} \approx 3^N$ set by the kurtosis of the product features. This is consistent with the experiments of Corlouer et al. (2026) on deep linear networks (their Fig. 2: order kept, SGD slower). Measured empirical with online SGD on binary $N = 4$ (1780 runs; $B = 1$–256; five initializations with ten noise seeds each): no setting shortened the first plateau (Figure 14a). At $B = 1$ the mean delays are $+0.24 \pm 0.10$, -0.01 ± 0.12 and $+0.13 \pm 0.07$ at $\sigma = 0.03$, 0.1, 0.3 (s.e. across initializations; escape times 24.2, 7.2, 2.3), against +0.17, +0.12, +0.08 predicted: consistent, but resolved only at $\sigma = 0.03$. The spread scales as $(\eta/B)^p$ with $p = 0.43$–0.52 (five sweeps, s.e. 0.01–0.05), against $\frac{1}{2}$.

SGD: later plateaus are lengthened, and can trap. conjecture (heuristic law) known (mechanism: stochastic collapse, Chen et al., 2023) Once a component is learned, a route of $k \ge 3$ cores (Section 4.4) has no curvature, but its entries are coupled to the learned cores by zero-mean multiplicative noise of order one, which drives their log-amplitude down at a rate $c_\gamma \eta/B$. Proposition A.45 turns this into

$$\frac{T_{\mathrm{SGD}}}{T_{\mathrm{GF}}} = \frac{-\ln(1-x)}{x}, \qquad x = c_\gamma \frac{\eta}{B} T_{\mathrm{GF}}, \tag{25}$$

for a $k = 3$ plateau of gradient-flow length T_{GF}, and trapping for $x \ge 1$. Test empirical on binary $N = 4$ from the aligned route (component 1 learned, components 2 and 3 at per-core amplitude $b_0 = 0.002$–0.01, so that $T_{\mathrm{GF}} = 50$–250 is exact; $\eta/B = 3.1 \cdot 10^{-5}$–$2.5 \cdot 10^{-4}$; 144 runs, six noise seeds per point; Figure 14b): every escaped run was slower than gradient flow, and all 30 runs with $z = (\eta/B)T_{\mathrm{GF}} \ge 0.025$ were trapped. One fitted $c_\gamma = 32.6$ (the commuting-pair estimate of Section A.7 gives 26 for this teacher) yields $\chi^2 = 379$ for 18 degrees of freedom, against 535 for a pole $1/(1-x)$ and 929 for $1 + x/2$, and $x = 1$ separates escaped from trapped conditions in 23 of 24 cases; but the effective c_γ grows from 30 at small x to 45 near the trap, so the law holds to about 10% only for $T_{\mathrm{SGD}}/T_{\mathrm{GF}} \lesssim 1.5$. Only η/B matters: $B = 16$ and 64 agree within 1.8 combined s.e. at eight matched points, although $B = 64$ is higher at all three escaped points with $b_0 = 0.003$. From random initialization the second plateau is lengthened by $19.8 \pm 5.1\%$ at $\sigma = 0.03$, $B = 1$ (the 16 of 30 runs that kept component 1) against 17.7% from (25) with no further fit, but by $10.8 \pm 0.6\%$ against 4.4% at $\sigma = 0.1$ (Figure 14a). Consequence derived: at fixed η/B a later plateau grows

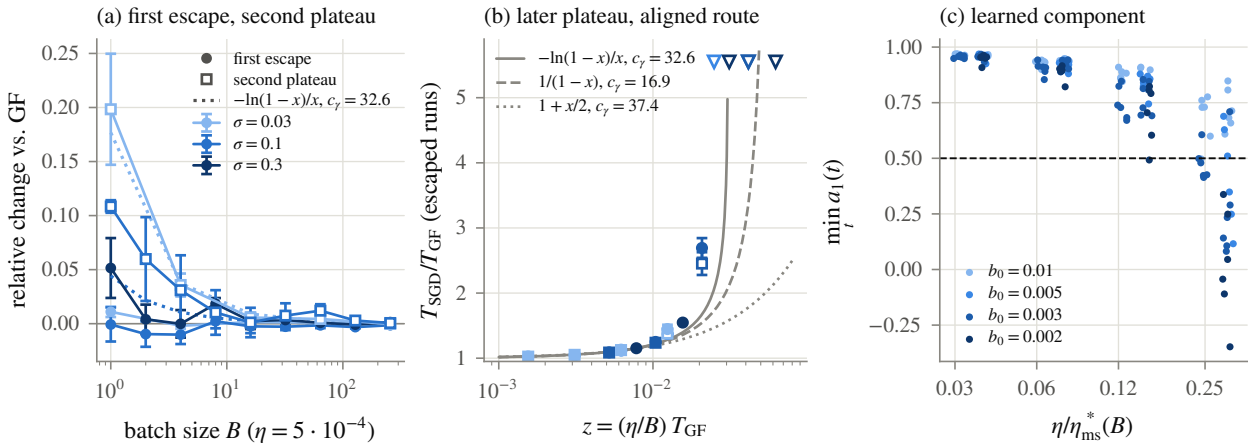


Figure 14: Online SGD, binary tree with $N = 4$ ($K = 3$), $d = \chi = 3$, odeco teacher $(3, 2, 1)$. (a) Change against gradient flow at $\eta = 5 \cdot 10^{-4}$ ($0.47\eta^*_{\text{ms}}$ at $B = 1$): first escape (filled; five initializations with ten noise seeds each) and second plateau (open; three initializations, runs that kept component 1), mean ± s.e. across initializations; dotted: (25) with c_γ from (b). (b) Second plateau from the aligned route, mean ± s.e. over the escaped runs of six noise seeds; $b_0 = 0.01$, 0.005, 0.003, 0.002 from light to dark ($T_{\text{GF}} = 50$–250); circles $B = 64$, squares $B = 16$; triangles: all runs trapped; grey: three laws, one fitted c_γ each. (c) Minimum of a_1 during each run of (b).

faster than $\sigma^{-(k-2)}$ and becomes infinite at a finite σ; on the aligned route at $\eta/B = 1.25 \cdot 10^{-4}$, $T_{\text{SGD}}/T_{\text{GF}} = 1.13$, 1.37 and 2.5–2.7 for $b_0 = 0.01$, 0.005, 0.003, and all runs are trapped at 0.002: SGD stalls where gradient flow is only slow. A strict ($k = 2$) route still escapes if $|\lambda_-| > c_\gamma \eta/B$ derived: on the MPS with $N = 4$ and a pattern teacher $(3, 0.3)$ whose second component needs $k = 2$, SGD at $B = 1$ learned it in 5 of 5 runs, $8 \pm 5\%$ later than gradient flow empirical. No SGD run escaped a plateau faster than gradient flow.

SGD: stability limit and forgetting. derived new (the reduction to the pattern basis) known (the criteria: Wu et al., 2018; Chemnitz and Engel, 2025; exact mean square: Mulayoff and Michaeli, 2024, Thm 5) At the minimum the noise vanishes, and the exact mean-square limit (57) on binary $N = 4$ is $\eta^*_{\text{ms}} = 0.00107$, 0.0161 and 0.112 for $B = 1$, 16, 256, against $\eta_c = 0.154$: per sample, small-batch SGD is about 150 times less stable than GD, because Gaussian product features have kurtosis 3^N. Started 10^{-5} from the minimum, SGD converged in 5 of 5 runs up to $1.5\eta^*_{\text{ms}}$ and diverged in 9 of 10 at $2.5\eta^*_{\text{ms}}$ ($B = 1$, 16) empirical, as the computed almost-sure threshold, about $1.9\eta^*_{\text{ms}}$, predicts. Well below it, noise on long plateaus erases learned components, which are then relearned empirical: in the random-initialization runs the first learned component fell back below $\frac{1}{2}$ in 20–66% of the runs at $0.47\eta^*_{\text{ms}}$ (most in the longest runs), 4% at $0.38\eta^*_{\text{ms}}$, 1 of 50 at $0.24\eta^*_{\text{ms}}$ and none below; these are also the only runs whose order differs from gradient flow's (10) or is unresolved (19). On the aligned-route plateaus (up to 1000 time units; Figure 14c), a_1 fell below $\frac{1}{2}$ in 0 of 72 runs at $\eta \le 0.076\eta^*_{\text{ms}}$, 1 of 36 at 0.12–$0.15\eta^*_{\text{ms}}$ and 20 of 36 at 0.25–$0.30\eta^*_{\text{ms}}$, so the safe fraction $0.25\eta^*_{\text{ms}}$ seen with another teacher (Section A.7) does not carry over. Near its limit SGD breaks the saddle sequence rather than compressing it.

Finite samples: which interpolant? empirical new (for TTN regression) Binary tree with $N = 8$, $d = 2$ ($W \in \mathbb{R}^{256}$), odeco teacher $(3, 2)$ of bond rank 2 (its rank-2 TTNs form a manifold of dimension 28), n Gaussian inputs with noise-free labels, five seeds; GD on the empirical loss (step $1/\lambda_{\max}$, $6 \cdot 10^4$ steps); students with $\chi = 4$, whose tangent space at a random point has dimension 112

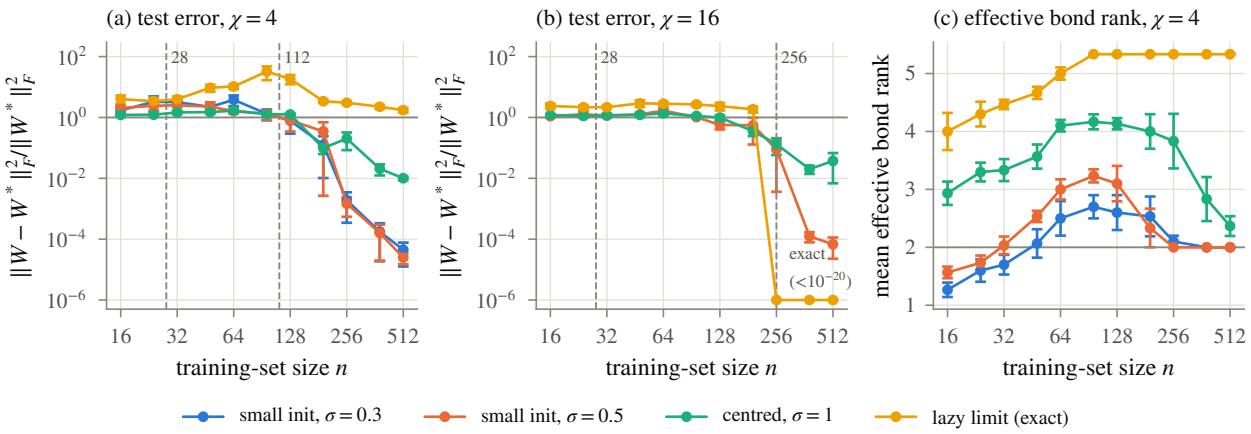


Figure 15: Finite training sets: binary tree, $N = 8$ ($K = 7$), $d = 2$, odeco teacher $(3, 2)$; mean ± s.e. over five seeds (data and initialization). (a), (b) Test error against n for $\chi = 4$ and 16 (lazy limit: kernel regression with the initial tangent kernel, values below 10^{-20} drawn at 10^{-6}); dashed: the teacher-manifold dimension 28, the tangent dimension 112 at $\chi = 4$, $d^N = 256$; grey: the zero predictor. (c) Effective bond rank (bond singular values above 0.2, a tenth of the smallest teacher weight, averaged over the six edges), $\chi = 4$.

(Proposition A.46), and $\chi = 16$, where all bonds are full; test error $\|W - W^*\|_F^2 / \|W^*\|_F^2$ (Figure 15). Small initialization ($\sigma = 0.3$, 0.5) recovers the teacher with its bond ranks: test error below 10^{-2} in 5 of 5 seeds from $n = 256$ at $\chi = 4$ (about $2 \cdot 10^{-3}$ at 256 and $2 \cdot 10^{-4}$ at 384, limited by unfinished training) and from $n = 384$ at $\chi = 16$, with mean effective bond rank 2.0–2.1. For $n \leq 96$ it fits the training data (to 1.3% of the initial loss) with a wrong tensor in 89 of 90 runs (mean test error 1.0–3.8). The exact lazy limit, the minimum-norm solution of the model linearized at initialization, fails as dimension counting predicts derived: at $\chi = 4$ the teacher lies outside the tangent space, and the error is 1.7–32 at every n, worse than predicting zero; at $\chi = 16$ it is exact only from $n = d^N = 256$. The centred model at $\sigma = 1$, which is not lazy (its parameters move by 12–69%), reaches 0.01–0.04 at $n = 384$–512 with bond rank 2.4–2.9; at $\sigma = 1.5$ and 2 it had not converged after $3 \cdot 10^5$ steps and is not ranked. With $N = 6$, $d = 3$ ($W \in \mathbb{R}^{729}$, manifold dimension 48), small initialization recovers the teacher in 5 of 5 seeds at $n = 384$, while the $\chi = 4$ lazy limit is worse than predicting zero at every $n \leq 384$. The learned bond ranks are ordered small initialization < centred < lazy limit at every n, except at $\chi = 16$ for $n \geq 256$, where the lazy limit is exact. A bias of small initialization towards low (hierarchical) rank is known for tensor factorizations known (Razin et al., 2021, 2022; tensor-train completion, empirically: Milanesi et al., 2021; matrices: Arora et al., 2019b), and a linearized model can only fit the projection of the target on its tangent space known (Chizat et al., 2019, Thm 2.5; Ghorbani et al., 2019, Thm 2; lazy training of a tensor decomposition needs $\Omega(d^{N-1})$ components for a rank-one target: Wang et al., 2020, Thm 1). Scope: one teacher per setting, noise-free labels, GD with capped steps, and no monotone trend in σ within small initialization. The threshold of small initialization, 200–250 samples or 5–9 times the teacher-manifold dimension in both settings, is unexplained.

When does gradient flow describe practical training? derived (from the results above; one teacher, trees with $N \leq 8$) GD follows gradient flow whenever $\eta < 2/\lambda_{\mathrm{GF}}(\sigma)$, with event times off by ηc_e and, from small initialization, the same saddle sequence; this limit is η_c at small initialization and falls like $\sigma^{-2(K-1)}$ on the lazy side, where steps up to 1.5–3 times larger still reach gradient flow's loss through a catapult. SGD needs in addition $\eta < \eta^*_{\mathrm{ms}}(B)$ (about $10^{-3}B$ here), $c_\gamma(\eta/B)T \ll 1$ for every later plateau of length T, and η a small fraction of η^*_{ms} when learned components must

survive long plateaus ($0.1\eta^*_{\mathrm{ms}}$ was safe over 1000 time units). Because plateaus grow like $\sigma^{-(k-2)}$, at fixed η/B SGD eventually stalls or forgets where gradient flow is only slow.

5.6 Optimizers used in practice

Tensor-network models are trained with Adam (Kingma and Ba, 2015), with ALS and DMRG sweeps (White, 1992; Holtz et al., 2012a) (in machine learning often with a gradient step on the merged two-site core instead of an exact local solve, Stoudenmire and Schwab, 2016) or with Riemannian methods (Novikov et al., 2018; Steinlechner, 2016), which use the geometry of fixed-rank tensor manifolds (Holtz et al., 2012b; Uschmajew and Vandereycken, 2013). Which features of the gradient-flow picture of Section 4 survive them? Table 10 collects the answers, Figure 16 shows them from one initial point, and the evidence follows.

Setup (details in Section B.7). Exact population loss, whitened inputs, double precision. Binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$: all methods start from the same 42 fan-in initializations (seven $\sigma \in [0.002, 0.128]$, six seeds). Deep trees, $N = 8$: binary ($K = 7$) and MPS ($K = 8$), with the same teacher and with a pattern teacher E8 whose weaker components need $k = 6$ and $k = 2$ cores on the MPS (Proposition 2.1); $\sigma \in \{0.3, 0.8\}$, plus seven $\sigma \in [0.25, 0.8]$ (binary) and eight $\sigma \in [0.01, 0.3]$ (MPS); six seeds; gauge-balanced initializations, and fan-in ones for the gauge-dependent methods (fan-in gradient flow stalls on the odeco MPS, Proposition A.28). Methods as listed in Table 10, with GD and heavy ball ($\beta = 0.9$) at effective steps $\eta/(1-\beta) \le 0.27\eta_c$ (Section A.7) and Adam's $\epsilon = 10^{-8}$ unless stated. Clocks: t for gradient flow (LSODA, events by root finding), ηn for GD, $\eta n/(1-\beta)$ for heavy ball, steps for the other gradient methods, sweeps for ALS and DMRG; exponents p ($T \propto \sigma^{-p}$) are per-seed fits, mean ± s.e.

GD and heavy ball reproduce gradient flow. empirical new (for TTNs) GD with $\eta = 0.13\eta_c$ on binary $N = 4$ has gradient flow's exponents to within 0.003 and its order from all 42 initial points; its first-escape delay grows by 0.0199 ± 0.0000 per e-fold of $1/\sigma$, the asymptotic slope $\eta(K-1)/2 = 0.020$ that follows from Proposition A.44. Heavy ball's delay grows $19.0 = (1+\beta)/(1-\beta)$ times faster (conjecture: the momentum analogue), hence $p = 0.970 \pm 0.002$. On the deep trees ($\sigma = 0.8$) GD and heavy ball escape after 1.001–1.006 and 1.01–1.11 times gradient flow's time.

Adam: same order, but the first escape forgets σ. derived new (for TTNs; in aligned deep matrix factorization the sign-descent escape time is likewise bounded as the initialization vanishes: Sabanayagam and Lucey, 2026, Thm 2 and Prop. 6) With $\epsilon = 0$ Adam's step $-\eta\hat{m}_n/\sqrt{\hat{v}_n}$ is invariant under a common rescaling of past gradients known (Kingma and Ba, 2015), and near the origin ∇L is homogeneous of degree $K-1$; so the iterates from $c\theta_0$ with step $c\eta$ are c times those from θ_0 with step η, Adam sees σ only through σ/η, and the number n_1 of steps to the first escape stays bounded as $\sigma \to 0$ (in the sign-flow limit $\eta \to 0$ at fixed β_1, β_2 known (Ma et al., 2022; Adam as sign descent: Balles and Hennig, 2018), ηn_1 tends to a constant). Measured empirical: $p = 0.009$–0.013 on binary $N = 4$ for four learning rates, RMSProp alike (Figure 17); $p = 0.049 \pm 0.022$ and 0.145 ± 0.033 ($\eta = 10^{-3}$, 10^{-2}; small but not zero) on binary $N = 8$, where gradient flow has 5.00 ± 0.02 ($\sigma \le 0.4$, prediction 5; fit computed from the committed runs in the review of this experiment). The selection survives empirical: Adam and RMSProp pick gradient flow's first component in 94–100% of the resolved deep-tree runs and keep E8's locality order on the MPS (the weaker $k = 2$ component before the stronger $k = 6$ one, in 4–5 of 6 seeds, as gradient flow). On binary $N = 4$ the first escape is as wide as its own time, so 79–100% of the runs fail $\rho_i > 3$ and are rich, non-incremental by Table 3.

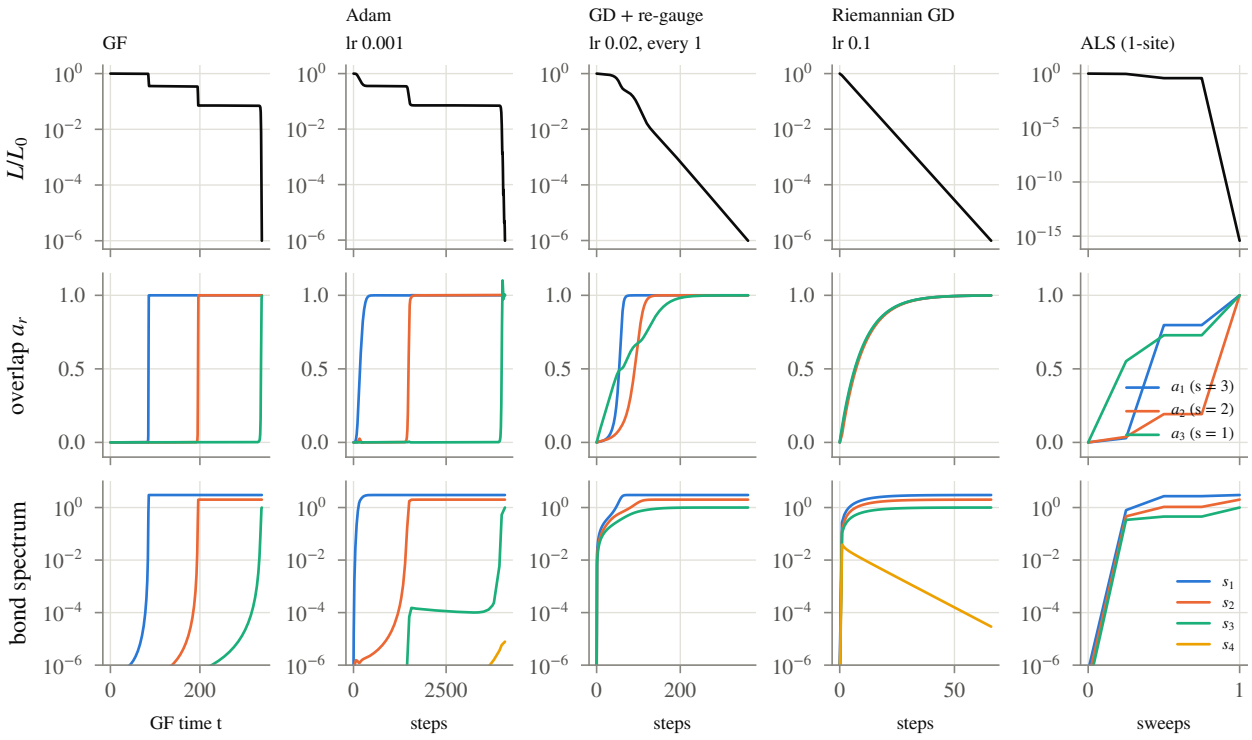


Figure 16: Five optimizers from one initial point: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, fan-in initialization, $\sigma = 0.008$, seed 0. Columns: gradient flow, Adam ($\eta = 10^{-3}$), GD re-gauged at every step ($\eta = 0.02$), Riemannian GD ($\eta = 0.1$), one-site ALS, each on its own clock. Rows: normalized loss, overlaps a_r, bond singular values of the cut $\{1, 2\}|\{3, 4\}$. Gradient flow and Adam learn one component at a time in the same order, with plateaus at the partial-fit levels $L_1/L_0 = 0.36$ and $L_2/L_0 = 0.07$; the other three have none. This is the only one of the 60 ALS and DMRG runs in which no two components cross $a_r = \frac{1}{2}$ in the same micro-step.

Adam's ϵ brings the gradient-flow law back. derived new (for TTNs; concurrently, the same law in aligned deep matrix factorization: Sabanayagam and Lucey, 2026, Prop. 7) Where $|\partial_j L| \ll \epsilon$ Adam is gradient flow with $t = \eta n/\epsilon$: from a small σ it crawls along the gradient-flow path until the gradient reaches ϵ, then escapes as in the sign regime,

$$n_1 \simeq \frac{\epsilon}{\eta} T_{\mathrm{GF}}(\sigma) + n_{\mathrm{sign}}(\eta), \tag{26}$$

so the law $T_{\mathrm{GF}} \propto \sigma^{-(K-2)}$ returns, with prefactor ϵ/η, below the crossover $\sigma_\times$ where $\epsilon T_{\mathrm{GF}} = \eta n_{\mathrm{sign}}$. Test empirical on the odeco MPS with $N = 8$ (eight $\sigma \in [0.01, 0.3]$, six seeds, $\eta = 10^{-3}$, 10^{-2}): per seed, from that seed's exact gradient-flow escape at $\sigma = 0.3$ times $(0.3/\sigma)^6$ and its own escape with $\epsilon = 10^{-30}$, with no fitted parameter, measured/predicted is 0.9989–1.0003 ($\eta = 10^{-3}$, $\sigma = 0.02$–0.03) and 1.0009–1.0062 ($\eta = 10^{-2}$, $\sigma = 0.015$–0.02) for every run that escaped within $2 \cdot 10^5$ steps, and (26) predicts which runs do in all 60 cases with $\sigma \leq 0.05$; near $\sigma_\times \approx 0.05$–$0.1$ the deviations are systematic (-5% to $+17\%$). The crossover depends on η and on the tree: on binary $N = 4$, (26) puts it near 10^{-8} (derived, not tested), far below the $2 \cdot 10^{-4}$ at which the initial gradient falls under ϵ. Only $K = 8$ was tested.

Later plateaus and minima under Adam. empirical Later gaps depend weakly on σ on binary $N = 4$ ($p = 0.0$–0.4). On binary $N = 8$ they grow as σ decreases, up to $8 \cdot 10^4$ steps

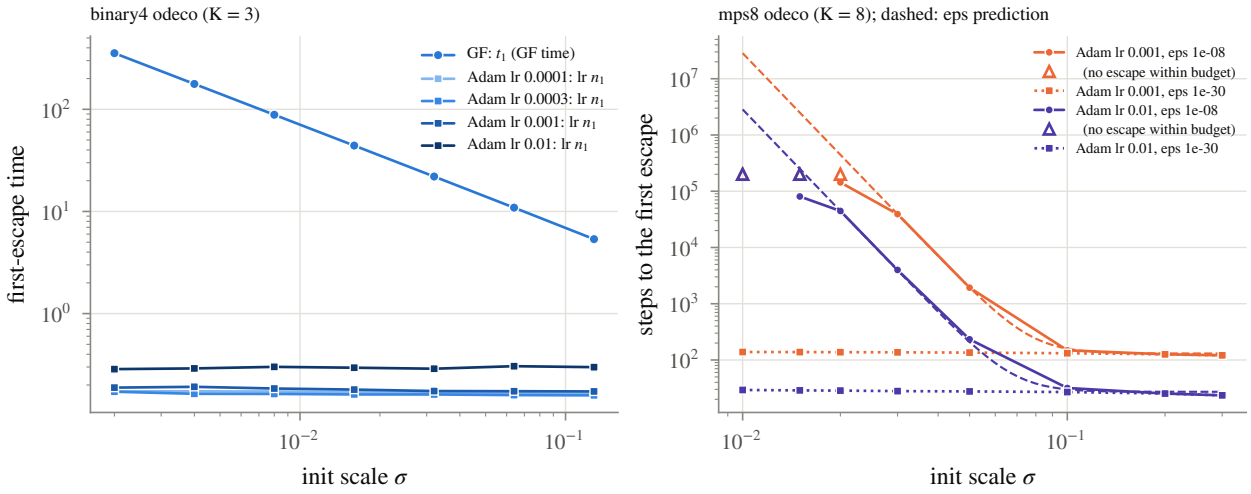


Figure 17: Adam's first escape. Left: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$, odeco teacher $(3, 2, 1)$, fan-in initialization; medians over six seeds of gradient flow's first-escape time t_1 and of Adam's ηn_1 ($\epsilon = 10^{-8}$). Gradient flow follows σ^{-1}; Adam does not depend on σ, and $\eta n_1 \to 0.17$ as $\eta \to 0$ (0.30, 0.18, 0.16, 0.17 for $\eta = 10^{-2}$ to 10^{-4}). Right: MPS, $N = 8$ ($K = 8$), same teacher, balanced initialization; median steps to the first escape over the runs that escaped within $2 \cdot 10^5$ steps, for $\epsilon = 10^{-8}$ (solid) and 10^{-30} (dotted); triangles, drawn at the budget, mark σ with runs that did not escape; dashed: (26) with median inputs. Without ϵ the escape takes 121–139 ($\eta = 10^{-3}$) and 23.5–29.3 steps ($\eta = 10^{-2}$) at every σ; with $\epsilon = 10^{-8}$ the law σ^{-6} returns below $\sigma \approx 0.05$–0.1.

at $\sigma = 0.25$, but per-seed local slopes range from -0.4 to 5.5, so over half a decade no law is established (gradient flow: 4.72 and 4.85, prediction 5); with $\sigma \le 0.4$, 83–100% of these runs are S2S. Adam and RMSProp break the conservation of Q_e and end at imbalanced minima (balance index $\|Q_e\|_F / (\|G_{v,(e)} G_{v,(e)}^\top\|_F + \|G_{p,(e)} G_{p,(e)}^\top\|_F) = 0.21$–$0.96$), 1.01–6.1 times sharper than the balanced value (53), the more so the larger the step; gradient flow, GD and heavy ball end within 0.2% of it.

No plateaus: sweeps and Riemannian GD. proven new (elementary) In the canonical form centred at core v the environment of v is an isometry, so the least-squares core is $\nabla_{G_v} \langle W, W^* \rangle$, a function of the other cores' frames, which are Q factors of QR decompositions and do not change when a core is multiplied by a positive number: every ALS and DMRG iterate is independent of σ. Riemannian GD, $W \leftarrow \mathrm{HSVD}_\chi(W - \eta P_{T(W)}(W - W^*))$ with $P_{T(W)}$ the projector onto the tangent space of the fixed-rank manifold, maps W_0 at $\eta = 1$ to $\mathrm{HSVD}_\chi(P_{T(W_0)} W^*)$, and that tangent space is the same at cW_0: it is exactly σ-free too. Measured empirical: ALS and DMRG fit the teacher within one sweep, with two components crossing $a_r = \frac{1}{2}$ in the same micro-step in 59 of 60 runs: no resolved order. Riemannian GD needs 2–7 steps; it fits all components in one step on binary $N = 4$ but stages them on binary $N = 8$, the weakest strictly last in 5 of 6 seeds for both teachers (3 and 2 of 6 on the MPS).

Re-gauged and Gram-preconditioned GD. Re-gauging moves all of $\|W\|_F \sim \sigma^K$ into the root, whose environment becomes an isometry, so the root fits the teacher's projection onto the current frames in a time $O(1)$ for any σ derived; on deep trees random frames keep only a fraction $\prod_e \chi / d^{n_e}$ of each component over the root's edges ($2.4 \cdot 10^{-3}$ on binary $N = 8$, 0.2 on $N = 4$), so they must rotate and learning is sequential again, with σ-free event times (empirical, $|p| \le 0.02$). Gram preconditioning moves each core towards its own least-squares solution, $\sim \sigma^{-(K-1)}$ times

Table 10: What survives which optimizer (Section 5.6). Binary $N = 4$ ($K = 3$): seven σ over 1.8 decades; binary $N = 8$ ($K = 7$) and MPS $N = 8$ ($K = 8$): two to eight σ over at most half a decade (1.5 decades for Adam's ϵ); six seeds each. "Gauge": the group (16). p: exponent of the first-escape time, $T \propto \sigma^{-p}$. Order: agreement with gradient flow from the same initial point, or counts over independent seeds for methods that give one trajectory per seed. Cost: median steps (sweeps, micro-steps) to $L < 10^{-3}L_0$, binary $N = 4$ at $\sigma = 0.008$, deep trees at $\sigma = 0.3$.

method	update equivariant under	plateaus	first escape vs. σ	learning order	cost
gradient flow	orthogonal maps of θ; conserves Q_e	saddle-to-saddle (S2S)	$p = 1.007$ ($K = 3$), 5.00 ($K = 7$, $\sigma \leq 0.4$); predicted $K - 2$	reference	$3.6 \cdot 10^3$–$1.1 \cdot 10^4$ evaluations
GD, heavy ball (steps $\leq 0.27\eta_c$)	as gradient flow; Q_e drifts	as gradient flow	$p = 1.005$, 0.970 ($K = 3$); delay $\propto \ln(1/\sigma)$	as gradient flow (100%)	T_{GF}/η ($1.7 \cdot 10^4$, $K = 3$)
Adam, RMSProp	coordinate permutations, sign flips; gradient rescaling ($\epsilon = 0$)	sequential; rich ($K = 3$), S2S ($K = 7$, $\sigma \leq 0.4$)	σ-free, $p = 0.00$–0.15; below $\sigma_\times$: (26)	first component as gradient flow: 94–100%; full order 87–98% ($K = 3$)	$1.7 \cdot 10^2$–$2.1 \cdot 10^5$
re-gauged GD	gauge (cores reset to the root-centred canonical form)	none ($K = 3$); σ-free ($K = 7$, 8)	σ-free, $p = 0.00$	largest first: 6/6 ($K = 3$), 2/6 and 5/6 ($K = 7$)	$2 \cdot 10^2$–$2 \cdot 10^4$; odeco MPS: budget
Riemannian GD ($\eta = 1$)	gauge (acts on W)	none	σ-free (exact)	one step ($K = 3$); weakest last: 5/6 ($K = 7$)	2–7, each a dense Jacobian
Gram-precond. GD	gauge, up to damping (per-core Gauss–Newton blocks)	overshoot, then return	not σ-free: $p = 0.65$ ($\eta = 0.01$); deep: 147–632 vs. 26–35 steps	mostly unresolved	$1.5 \cdot 10^2$–$4.3 \cdot 10^3$; odeco MPS: 9/24 converge
ALS, DMRG	gauge; invariant under per-core rescaling	none	σ-free (exact, proven)	none resolved (59/60 ties)	≤ 1 sweep (3–11 micro-steps)

larger than the core, so its first step overshoots the loss by $\sim \sigma^{-2K(K-1)}$ (derived; up to $4 \cdot 10^{34}$); it has no flat plateau, but its time depends on σ (empirical, Table 10).

Finite samples. empirical With $n = 45$ and 60 samples on binary $N = 4$ (four data draws, six initializations, $\sigma \in \{0.01, 0.1\}$, paired comparisons), gradient flow and Adam reach the same population loss: at $n = 60$ every run that met the training tolerance recovered the teacher; at $n = 45$ none did, and Adam was better in 8–13 of 21–23 pairs per setting. Only a threshold count differs (a fourth bond singular value above $0.1\, s^{e,*}_{\min}$: Adam with $\eta = 10^{-2}$ in 35 of 44 runs, gradient flow in 5); the entropy rank does not.

Interpretation: which invariance removes which feature. Two properties of an update decide what survives (Table 10, second column). The first is how its step scales with the gradient, which near the origin is $O(\sigma^{K-1})$ while the parameters must move by their own size: a step proportional to the gradient needs a time $\sigma^{-(K-2)}$, a step invariant under its rescaling (Adam) does not, and an absolute scale in the preconditioner, such as Adam's ϵ, brings the law back derived; Adam still follows the gradient's sign, which one component dominates, so learning stays sequential

and ordered (conjecture). The second is the metric. Under gradient flow W moves through $JJ^\top$ ($J = \partial W/\partial\theta$), which vanishes like $\sigma^{2(K-1)}$ at the origin, whereas Riemannian GD (in continuous time the Gauss–Newton flow $\dot\theta = J^+(W^* - W)$) and the exact block minimizations of ALS and DMRG move W at a rate of order one. These methods are equivariant under the whole gauge group (16), including the rescaling of a core against its neighbour, and see W_0 only through its frames, so they keep neither plateaus (derived) nor σ (proven); plateaus at singular points of a parametrization, and the better behaviour of the natural gradient there, are known for multilayer perceptrons known (Amari et al., 2006). Equivariance alone is not enough: Gram preconditioning is gauge-equivariant up to damping derived but moves all cores at once from their current values. Balance and the flattest minimum (Proposition A.43) rest on the conserved Q_e, the charges of the gauge symmetry under Euclidean gradient flow (Proposition 2.5); their measured relative drift is 10^{-11}, 10^{-5} and $3 \cdot 10^{-4}$ under gradient flow, GD and heavy ball (momentum flows conserve fewer quantities than gradient flow, with time- and velocity-dependent laws, Marcotte et al., 2024), while Adam, equivariant only under coordinate permutations and sign flips, breaks them at order one empirical. A law for Adam's later plateaus is open (Section 7).

6 Input covariance and teacher structure

Sections 4 and 5 use whitened inputs and orthogonally decomposable or pattern teachers. Here one complication is added at a time (covariance, correlations between sites, overlapping components, random TTN teachers, components sharing a core), and we ask whether the saddle-to-saddle picture changes qualitatively, quantitatively or not at all, and whether the baseline theory predicts the change after a substitution. An aligned covariance only reweights the components (Section 6.1). Anything that makes the components non-orthogonal in the metric of the loss changes the law of the escape from the rank-one saddle, and the tree decides how: strict and logarithmic in $1/\sigma$ on an MPS, of higher order and $\propto \sigma^{-1}$ on binary and HT trees (Section 6.2, the central result). Plateaus sit at critical points of the bond-rank-constrained loss, blends rather than partial sums of components (Sections 6.3 to 6.6); binary students learn bottom-up (Section 6.7); and two smaller components that share a core can destabilize a learned larger one (Section 6.8).

Setup (details in Section B.5). Gradient flow on the exact population loss (LSODA, tolerances 10^{-9} and 10^{-14}), fan-in initialization, $\chi = 4$ unless stated. The *generic teachers* have $d = 2$: (a) CP teachers with $u_1^{(i)} = e_1$ and $u_2^{(i)} = ce_1 + (1-c^2)^{1/2}e_2$ at every site (overlap c, weights $(1, 0.7)$); (b) an odeco teacher (weights $(1, 0.8)$ on e_1, e_2) under a misaligned covariance $\Sigma_i = R(\theta)\,\mathrm{diag}(\lambda_1, \lambda_2)R(\theta)^\top$, with the power-law spectrum of Section 2.3 (exponent β); (c) random binary-tree and MPS teachers with bond ranks 2. Students are MPS, binary and HT trees with $N = 4$ and 8. Protocols: random initialization at scale σ, or the saddle start of Section 4.4 with noise scale ε (the exact balanced bond-rank-one critical point of lowest loss, zero-padded, plus fan-in noise of scale ε). An escape is an onset event, the first time some edge has $s_2^e \geq f\, s_2^{e,*}$ (the teacher's value), $f = 0.05$ (0.03 and 0.1 as checks); the edge that fires is the *realized edge*. Fits are per seed; *asymptotic* fits use the three smallest σ or the five smallest ε; log and power laws are compared in log space; $\pm$ is the standard error over 5–10 seeds, statistical only, so agreements are cited at the 1–2% level. The float64-floor scan flags only the odeco baselines at $\sigma \leq 3 \cdot 10^{-3}$ (Figure 19).

6.1 Covariance aligned with the teacher: a reweighting

Prediction. Let the teacher be odeco with factors that are covariance eigenvectors, $\Sigma_i u_r^{(i)} = \lambda_{r,i} u_r^{(i)}$, and $\Lambda_r = \prod_i \lambda_{r,i}$. Near the origin the flow sees only the escape tensor $\Sigma W^* = \sum_r \Lambda_r s_r U_r$, so the *race* (which component escapes first, and when) is that of the whitened problem with weights $\Lambda_r s_r$; the *landscape* (critical points, plateau levels $\frac{1}{2}\sum_{r\notin S} \Lambda_r s_r^2$, saddle types) is that of $\Sigma^{1/2} W^*$, with weights $\Lambda_r^{1/2} s_r$ derived (Propositions A.39 and A.41). From an aligned initialization each component follows its whitened trajectory on the clock $\Lambda_r t$ proven known (Saxe-type modes: Saxe et al., 2014; depth two: Gidel et al., 2019; new: every tree) (Proposition A.40). When the two orderings disagree, a learned component a of binary4 is unstable toward b (a rotation through a strict saddle) iff $\Lambda_b s_b^2 > \Lambda_a s_a^2$; on MPS4 the rotation needs all four cores proven new (Proposition A.42).

Experiments. empirical $N = 4$, $d = \chi = 4$, $\beta = 0.25$, so the high-variance eigenvector e_1 has twice the Λ of e_2. Teachers T1 $(1, 0.8)$ and T2 $(1, 0.6)$ put the smaller component on e_1: Λs favours it in both, Λs^2 only in T1. Forty seeds, binary4 at $\sigma = 0.02$–0.1, MPS4 at 0.05–0.2, four variants from the same initialization: (Σ, s), $(I, \Lambda s)$, $(I, \Lambda^{1/2} s)$, and the last from the mapped initialization (same landscape and starting point as (Σ, s), whitened flow).

- *Clocks.* From aligned initialization half-times match the clock $\Lambda_r t$ to 10^{-9}; the naive substitution $s_r \to \Lambda_r s_r$ misses the widths by up to 47%.
- *Race.* Paired (Σ, s) and $(I, \Lambda s)$ runs learn the same component first in all MPS4 and 95–100% of the binary4 pairs, with first-escape times within 1.3%. The covariance reverses the order: in T1 the 0.8 component comes first in 90–98% (binary4) and 78% (MPS4) of the runs. The mapped-initialization control picks a different first component in 10–15% (MPS4) and 12–53% (binary4) of the seeds and escapes 11–23% later: the race is set by the flow metric, not by the landscape.
- *Exponents.* $t_1 \propto \sigma^{-1.010 \pm 0.002}$ (binary4) and $\sigma^{-1.999 \pm 0.001}$ (MPS4); $t_1 \propto (\Lambda s)^p$ with per-seed $p = -1.000$ to -1.003 within each eigenvector direction ($\Lambda = 0.008$–33, ten seeds).
- *Landscape.* On binary4 a learned component is replaced only if the other has the larger Λs^2: no swap after a landscape-correct first component in any of the 24 settings, while $P(\text{swap} \mid 0.6 \text{ first}) = 0.33, 0.68, 0.94$ at $\sigma = 0.1, 0.05, 0.02$ in T2. The Hessian at the saddle equals the formula to 10^{-15} in the 6 strict binary4 cases; MPS4 has no negative eigenvalue (12 of 12) and no swap in 960 runs. Plateau levels are $\frac{1}{2}\sum_{r\notin S} \Lambda_r s_r^2$ (median relative error 10^{-8}–10^{-3}).

The change is quantitative only and is predicted after two different substitutions new (for TTNs; in two-layer linear networks the split is implicit in Gidel et al., 2019, Thm 2, whose Corollary 2 states no ordering condition).

6.2 Generic teachers: the tree decides the saddle type

Prediction. derived new (Section A.5, "Generic teachers"; Proposition A.36 and Corollary A.37) After the first escape the flow sits near a bond-rank-one critical point $W_1 = \bar{s} \bigotimes_i w_i$. A bond can gain rank through a two-core route if both of its cores have other legs of dimension above its rank. For odeco and pattern teachers the route's residual vanishes by orthogonality, which gives the $\sigma^{-(k-2)}$ laws of Section 4.4; for a generic teacher it does not. On an MPS every core carries a site, so W_1 is a strict saddle, with Hessian eigenvalues $-\bar{s}^{(N-2)/N} \sigma_m(M_e)$, $m \geq 2$, each $\chi_e - 1$ times, that is $\sum_e (d-1)(\chi_e - 1)$ negative ones. It is left in a time $\gamma_e^{-1} \ln(1/\varepsilon) + O(1)$, $\gamma_e = \bar{s}^{(N-2)/N} \sigma_2(M_e)$,

and from random initialization with slope $h_e = (1 - \kappa_e)/\gamma_e$ in $\ln(1/\sigma)$, $\kappa_e = \sigma_2(M_e)/\bar{s}$ (the head start). On binary and HT trees the internal cores carry no site, so the cheapest route has $k = 3$: no negative eigenvalue and an escape $\propto \varepsilon^{-1}$ (or σ^{-1}). The landscape bound of Furman et al. (2026, Lemmas D.7–D.9) allows descent of order K at a rank-one point of an MPS; for generic teachers it is 2.

Experiments. Thirteen generic teachers (Table 11), each on the MPS, binary and HT student with the same N; five seeds; saddle start with ε from 10^{-8} (MPS) or 10^{-5} (binary, HT) to $5 \cdot 10^{-2}$, and random initialization for the six teachers with $N = 4$ on MPS4 ($\sigma = 3 \cdot 10^{-4}$–0.3) and binary4 (10^{-4}–0.1).

- *Saddle type.* derived and computed, new: all 13 MPS students are strict, with 5 (MPS4) and 17 (MPS8) negative eigenvalues, the count above; none of the 26 binary and HT students has an eigenvalue below -10^{-8}.
- *Escape from the saddle.* empirical new (Figure 18a,b) On MPS the log law wins in 65 of 65 seeds, and the slope equals $1/\gamma_e$ of the realized edge within 0.4% for all 13 teachers (changing f from 0.03 to 0.1 moves it by at most 0.4%); where the realized edge is not the fastest (two random teachers with $N = 8$) the slope follows it, not $1/|\lambda_{\min}|$. Fits over all ε deviate by up to 11% (pre-asymptotic points; a "slow crossover" read into them is withdrawn). On binary and HT, $t_{\text{esc}} \propto \varepsilon^{-p}$ with $p = 0.9998$–1.0005 in 23 of the 24 cases (1.0036 ± 0.0006 in the last, over a wider window); the log law wins in no seed. (The misaligned teacher with $N = 8$, $\theta = 20°$, timed as below, gives 1.07 and 1.05 over 1.3 decades, 1.03 and 1.02 over its three smallest ε.)
- *Random initialization, $N = 4$.* empirical new (Figure 18c,d) On MPS4 the log law holds, with slopes within 1.3% of h_e for the five teachers with a prediction: 33.95 ± 0.18 (34.14), 5.674 ± 0.059 (5.682), 12.005 ± 0.071 (12.014), 1.038 ± 0.005 (1.051; 2.78 without the head start), 1.401 ± 0.013 (1.386). The misaligned teacher, which has no prediction under the preconditioned flow, gives 52.46 ± 0.10, below the bound $1/|\lambda_{\min}| = 52.94$. Binary4 gives σ^{-p}, $p = 1.001$–1.005. The curvature at the plateau is independent of σ on MPS4 (-0.1510 against -0.1509 at the exact saddle, $c = 0.45$) and falls $\propto \sigma$ on binary4. The first escape keeps its $\sigma^{-(K-2)}$ law: per-seed exponents 1.998–2.014 (MPS4) and 1.002–1.017 (binary4) over all nine σ, 2.0000 and 1.0003 over the three smallest.
- *A failure mode of onset events.* empirical new (methodological) The misaligned teacher with $N = 8$, $\theta = 20°$ escapes at the predicted rate to an intermediate saddle with $s_2^e = 3.7$–$3.9 \cdot 10^{-4}$ on every bond, nine times below the threshold, so the f-rule times the next escape (a slope 24 times the prediction). Timed at the intermediate level, the slope is 1979.6 ± 1.8 against $1/\gamma_e = 1974.8$. A threshold relative to the teacher's spectrum misses escapes that end below it.

Scope: the escape from the rank-one saddle (the first rank increase), $N = 4$ and 8, $d = 2$; random initialization for $N = 4$ only. Later saddles on binary and HT students were not timed; there two-core routes can open on bottom edges at higher ranks, never for the root bipartition (Section A.5). Correlated inputs were never run on HT (Section 6.4).

6.3 Misaligned covariance

The critical points, critical values and saddle types under a product covariance are those of the whitened problem with teacher $\Sigma^{1/2}W^*$; escape rates are not, because the flow is preconditioned proven new (for TTNs) (Proposition A.39). With misaligned factors $\Sigma^{1/2}W^*$ is a non-orthogonal CP

Table 11: Escape from the exact bond-rank-one saddle, 13 generic teachers ($d = 2$, $\chi = 4$, five seeds, asymptotic per-seed fits, mean ± s.e.). MPS: $\lambda_{\min}$ at the saddle, $1/\gamma_e$ of the realized edge, and the slope A of t_{esc} against $\ln(1/\varepsilon)$, $\varepsilon = 10^{-8}$–10^{-4} (its deviation from $1/\gamma_e$). Binary, HT: $t_{\text{esc}} \propto \varepsilon^{-p}$, $\varepsilon = 10^{-5}$–10^{-3}. †: intermediate-level event, $\varepsilon = 10^{-6}$–$2 \cdot 10^{-3}$ (MPS), 10^{-3}–$2 \cdot 10^{-2}$ (binary, HT). ‡: up to $\varepsilon = 2 \cdot 10^{-2}$. §: two realized edges, $1/\gamma_e = 5.47$, 5.27.

N	teacher	$\lambda_{\min}$	$1/\gamma_e$	A (MPS)	p (binary)	p (HT)
4	CP, $c = 0.2$	-0.0285	35.14	35.156 ± 0.013 (+0.05%)	1.0001 ± 0.0001	1.0005 ± 0.0003
4	CP, $c = 0.45$	-0.151	6.627	6.642 ± 0.005 (+0.22%)	1.0001 ± 0.0001	1.0005 ± 0.0001
4	misaligned, 20°	-0.0189	52.94	53.04 ± 0.06 (+0.18%)	1.0001 ± 0.0001	1.0002 ± 0.0004
4	random binary (302)	-0.0764	13.09	13.12 ± 0.01 (+0.18%)	1.0002 ± 0.0001	1.0001 ± 0.0002
4	random binary (303)	-0.360	2.779	2.786 ± 0.002 (+0.27%)	1.0001 ± 0.0001	1.0001 ± 0.0001
4	random MPS (301)	-0.376	2.658	2.662 ± 0.001 (+0.15%)	1.0000 ± 0.0001	1.0002 ± 0.0001
8	CP, $c = 0.45$	-0.0048	209.70	209.60 ± 0.05 (-0.05%)	1.0001 ± 0.0001	$1.0036 \pm 0.0006^{\ddagger}$
8	CP, $c = 0.7$	-0.0574	17.41	17.45 ± 0.02 (+0.24%)	1.0001 ± 0.0001	0.9998 ± 0.0001
8	misaligned, 45°	-0.0174	57.39	57.49 ± 0.03 (+0.16%)	1.0003 ± 0.00004	0.9999 ± 0.0001
8	misaligned, 20°	-0.00051	1974.8	1979.6 ± 1.8 (+0.24%)†	$1.071 \pm 0.004^{\dagger}$	$1.048 \pm 0.012^{\dagger}$
8	random binary (302)	-0.371	2.698	2.708 ± 0.002 (+0.36%)	0.9999 ± 0.0001	0.9999 ± 0.0002
8	random binary (303)	-0.190	5.42§	5.43 ± 0.04 (+0.08%)	1.0001 ± 0.0001	1.0001 ± 0.0002
8	random MPS (303)	-0.164	6.181	6.202 ± 0.007 (+0.34%)	1.0003 ± 0.0001	0.9999 ± 0.0001

teacher, with factors $\propto \Sigma_i^{1/2} u_r^{(i)}$, while the escape is driven by ΣW^*, with factors $\propto \Sigma_i u_r^{(i)}$: the state must rotate after the escape, and misalignment alone can reverse the order derived new (Section A.6). Experiments empirical ($d = \chi = 4$, weights $(1, 0.8)$, $\beta \in \{0.5, 1\}$, $\theta = 0$–$45°$, $\sigma = 0.05$ on binary4 and 0.1 on MPS4, 20 seeds; an MPS4 sweep with $\beta = 1$, $\theta \in \{0, 5, 20\}°$, nine σ from $3 \cdot 10^{-4}$ to 0.3, ten seeds):

- The first plateau is the predicted rank-one critical point of the Σ-loss: median relative level errors 10^{-6}–10^{-3}, median $|\cos| \geq 0.998$ with the critical tensor for $\theta \leq 30°$, over the resolved plateaus (20–50% of the MPS4 runs per θ have none at $\sigma = 0.1$).
- Early alignment with the race tensor (the best bond-rank-one approximation of ΣW^*), then realignment to the saddle known (in linear networks, binary4 being one: Atanasov et al., 2022): in 75–100% of the binary4 runs at $\theta \geq 10°$, but on MPS4 only at 45°.
- Second escape on MPS4 (Figure 19b) new: $\sigma^{-2.000}$ at $\theta = 0$; at 20° the log law (10 of 10 seeds) with slope 12.14 ± 0.03, below the bound $1/|\lambda_{\min}| = 12.31$ of its σ-independent plateau curvature; at 5° the log law (10 of 10) with 194.9 ± 1.2, approaching its bound 193.2 from above (not yet asymptotic). The slope itself is not predicted: the head start under the preconditioned flow was not derived.

6.4 Cross-site correlations

In the one-factor model $x^{(i)} = (1 - \rho)^{1/2} z_i + \rho^{1/2} z_0$ (independent standard Gaussians; ρ is the c of Section A.6) every site is whitened and two sites have covariance ρI. The second-moment matrix is not a product, so no whitening substitution exists; the loss is computed with the dense matrix. The residual acquires *pairing* tensors such as $(x^{(0)} \cdot x^{(1)})\, x_r^{(2)} x_r^{(3)}$, of bond rank one on binary4 but d on MPS4 derived. On MPS4 the escape from the product saddle passes through binary4's first plateau point, strict on MPS4 but of higher order on binary4, so the second plateau grows by $\ln 2\, [(1 - \kappa_c)/\lambda_2 + 1/\lambda'']$ per halving of σ: 2.128 at $\rho = 0.4$, 14.39 at $\rho = 0.2$ ($d = 3$, weights $(1, 0.6)$) derived new (Section A.6); binary4 stays at σ^{-1}. Experiments empirical:

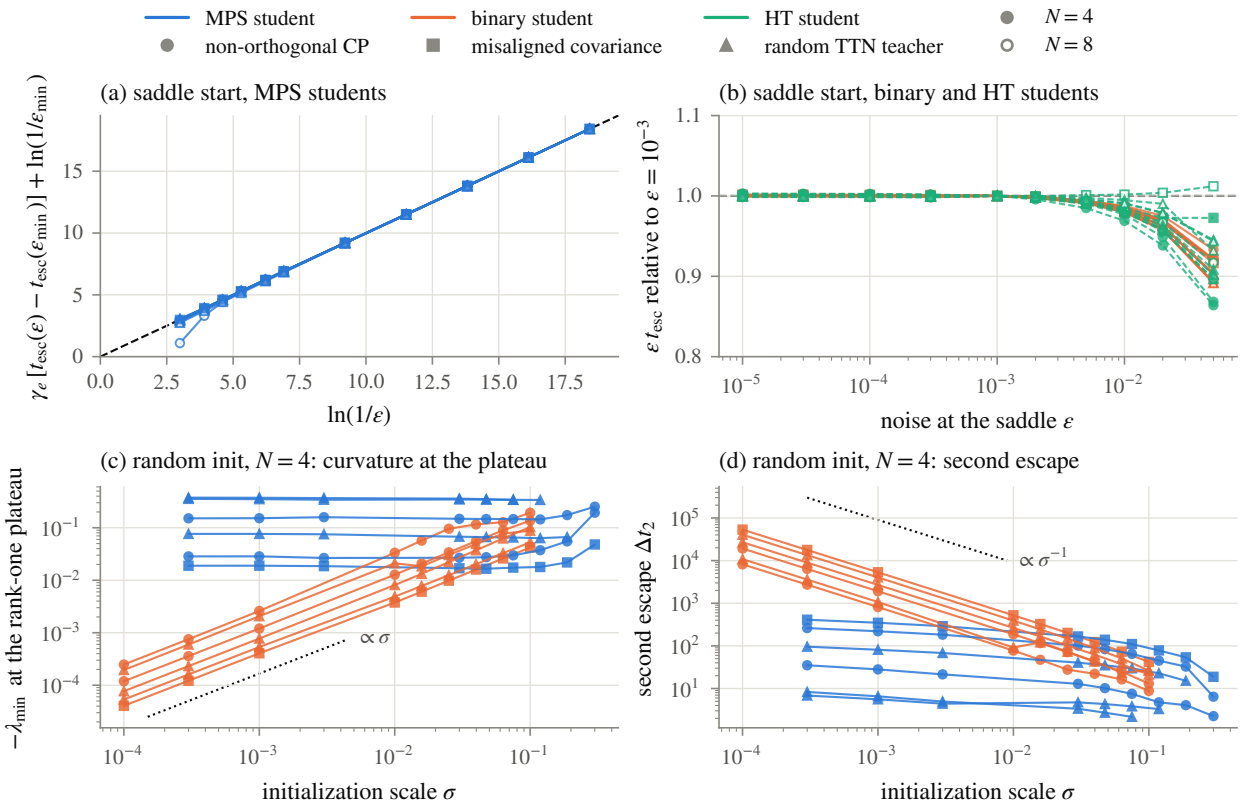


Figure 18: One generic teacher, three trees. (a) Saddle start, MPS students, 13 teachers: escape time rescaled by the predicted slope (mean over five seeds, γ_e of the realized edge) against $\ln(1/\varepsilon)$; dashed: the prediction. (b) Saddle start, binary and HT students, the same teachers except the misaligned one with $N = 8$, $\theta = 20°$: $\varepsilon\, t_{\rm esc}$ relative to its value at $\varepsilon = 10^{-3}$ (per seed, then geometric mean); flat means $t_{\rm esc} \propto \varepsilon^{-1}$. (c), (d) Random initialization, the six teachers with $N = 4$ on MPS4 and binary4 (runs on the major rank-one saddle with a resolved plateau, geometric means over seeds): $-\lambda_{\rm min}$ of the gauge-projected Hessian at the plateau, and the second escape Δt_2. Takeaway: the same teachers give a strict saddle with a logarithmic escape at the predicted rate on the MPS, and a higher-order saddle ($\lambda_{\rm min} \propto \sigma$, escape $\propto \varepsilon^{-1}$ or σ^{-1}) on binary and HT trees.

- binary4 ($d = \chi = 4$, weights $(1, 0.7)$, $\sigma = 0.05$, 20 seeds): the first plateau falls from $0.33\,L(0)$ at $\rho = 0$ to 0.29, 0.20, 0.14, $0.10\,L(0)$ at $\rho = 0.2$–0.5, always at the predicted critical value (median relative error $\leq 5 \cdot 10^{-4}$): a blend with pairing terms, no longer a partial fit.
- Second escape new (Figure 19c; $d = 3$, six σ from 0.0016 to 0.05, five seeds, events on the singular values across 01|23): on MPS4 the log law in 4 of 4 seeds at every $\rho > 0$ (seed 2 sits on the minor saddle and is excluded); increments per halving over the three smallest σ 2.133 ± 0.003 at $\rho = 0.4$ (predicted 2.128) and 14.67 ± 0.06 at $\rho = 0.2$ (14.39; +2%, still falling), and 113.9 ± 0.7, 4.616 ± 0.010 at $\rho = 0.1$, 0.3 (no prediction computed); $\sigma^{-2.000}$ at $\rho = 0$. Binary4: σ^{-p}, $p = 1.005$–1.011. This repeats the theory's own check with more seeds: a replication, not an independent test.
- The MPS4 plateaus separate only logarithmically: separation ratio $2.6 \to 11$ at $\rho = 0.4$ from $\sigma = 0.05$ to 0.0016, against 14–19 $\to$ 139–185 on binary4.
- Online SGD (binary4, $\rho = 0.3$, $B \geq 256$, three seeds) crosses the same loss levels within 4% of gradient flow.

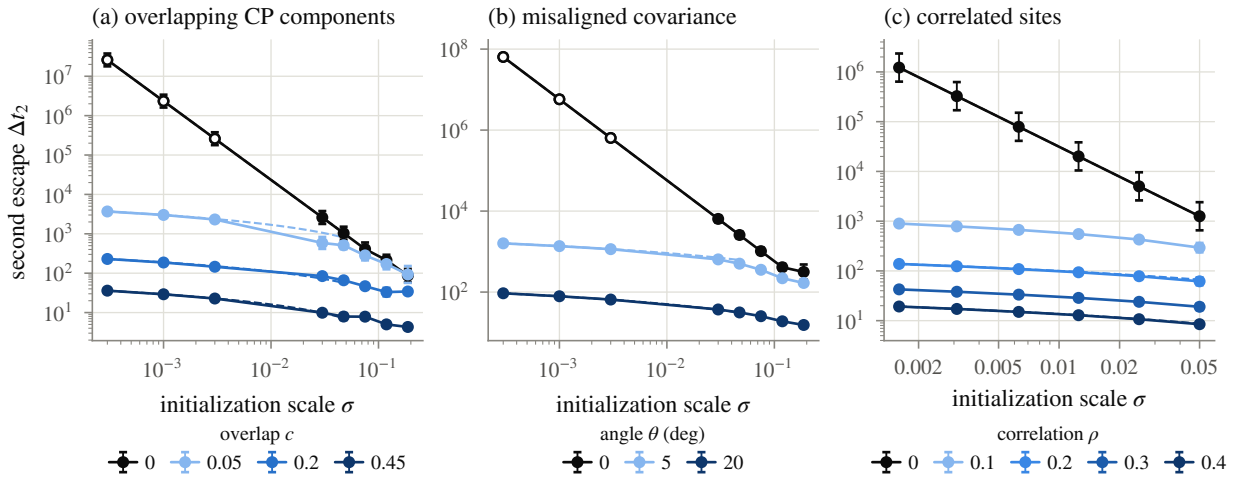


Figure 19: MPS4 from random initialization: the second escape Δt_2 (onset events; runs on the major rank-one saddle with a resolved plateau; geometric mean over seeds with its standard error) when the teacher's components are made non-orthogonal in three ways. (a) CP overlap c ($d = 4$, weights $(1, 0.7)$, ten seeds); dashed: $h_e \ln(1/\sigma)$ (Corollary A.37), anchored at the smallest σ since the intercept is not predicted. (b) Misaligned covariance at angle θ ($d = 4$, ten seeds); dashed: slope $1/|\lambda_{\min}|$, an upper bound. (c) Correlated sites ($d = 3$, four seeds); dashed: the two-stage law (Section A.6; not computed for $\rho = 0.1$, 0.3). Black: the orthogonal baseline, dashed $\propto \sigma^{-2}$; open markers: runs flagged by the float64-floor scan (dormant overlap 10^{-21}–10^{-12}); without them the baseline exponents are 2.008 ± 0.002 ($c = 0$) and 2.000 ± 0.002 ($\theta = 0$). Takeaway: any non-orthogonality in the metric of the loss turns the σ^{-2} second plateau into a logarithmic one, with the predicted slope where there is a prediction.

HT students and $N > 4$ were not run with correlated inputs.

6.5 Non-orthogonal CP teachers

The rank-one critical points are those of $g(x) = \langle \bigotimes_i x_i, W^* \rangle$ on unit rank-one tensors proven. With two components of equal weight and overlap c at every site, an MPS has two rank-one maxima (one near each component) iff $c < (N-2)/N$ and one blended maximum above proven (among symmetric points); unequal weights move the merger to a fold derived; on binary4 the rank-one point is the top singular pair of the (01|23) matricization, a blend for every $c > 0$ new (elementary) (Proposition A.35). The plateau is never a component. Experiments empirical ($d = \chi = 4$, weights $(1, s_2)$, $s_2 \in \{0.7, 1\}$, $c = 0$–0.8, $\sigma = 0.05$ on binary4 and 0.1 on MPS4, 20 seeds; an MPS4 sweep with $s_2 = 0.7$, $c \in \{0, 0.05, 0.2, 0.45\}$, nine σ from $3 \cdot 10^{-4}$ to 0.3, ten seeds):

- MPS4 has two rank-one minima for $c \leq 0.45$ and one for $c \geq 0.55$ ($s_2 = 1$), two for $c \leq 0.35$ and one for $c \geq 0.45$ ($s_2 = 0.7$); binary4 one. First plateaus sit at these blends (median relative level errors $5 \cdot 10^{-5}$–$2 \cdot 10^{-3}$ on binary4, up to 3% on barely resolved MPS4 plateaus; some binary4 runs with $s_2 = 1$ sit at the second singular pair, a strict saddle), with levels falling as c grows.
- Second escape on MPS4 (Figure 19a) new: $\sigma^{-2.00}$ at $c = 0$; for $c > 0$ the log law (9 of 9, 8 of 8, 8 of 8 seeds at $c = 0.45$, 0.2, 0.05) with slopes 5.686 ± 0.016 ($h_e = 5.682$), 34.40 ± 0.20 (34.14) and 595 ± 11 (570; not yet asymptotic, as expected near the crossover overlap $c_\times = O(\sigma)$ of Section A.5, a conjecture), and an σ-independent plateau curvature.

On binary4 the escape $\propto \sigma^{-1}$ was measured for $c = 0.2$ and 0.45 only (Section 6.2).

6.6 Random TTN teachers: plateaus are rank-constrained critical points

With bond ranks fixed, the critical points are the fixed points of one-site alternating least squares proven known (in substance: Eldén and Savas, 2011; Holtz et al., 2012a) (Proposition A.33); the hierarchical SVD is generically not one proven (Proposition A.34). The rank-one plateau should approach a local best rank-one approximation as σ^α, $\alpha = \min(\alpha_r, \alpha_a)$ (conjecture: a heuristic crossing argument), and an MPS then gains rank bond by bond, first on $e^* = \arg\max_e \sigma_2(M_e)$ with slope h_{e^*} derived. Experiments empirical new (MPS4 student, $d = 3$, $\chi = 4$; five random MPS4 teachers with bond ranks $(2, 3, 2)$ and $\|W^*\|_F = 1$; eight σ from 10^{-3} to 0.2; six seeds):

- The rank-one plateau converges to the bond-rank-one critical point: per seed its distance falls as σ^α, $\alpha = 0.72$, 0.83, 0.41, 0.93, 0.80 (± 0.01–0.03), 0.07–0.15 above the heuristic (0.64, 0.68, 0.34, 0.80, 0.72) for every teacher. For three teachers it is separated from every SVD-type truncation (hierarchical SVD, TT-SVD in both directions): at $\sigma = 10^{-3}$ it lies 10^{-4}–$2.5 \cdot 10^{-3}$ from the critical point and $1.5 \cdot 10^{-3}$–0.028 from the nearest truncation (relative to $\|W^*\|_F$). For the other two teachers, and at the higher-rank plateaus, the critical point lies within 0.0032 of an SVD-type truncation and the plateaus do not discriminate between them. (In CP factorization, small initialization follows a rank-one trajectory, Razin et al., 2021.)
- Ranks grow one bond at a time, e.g. $(1, 1, 1) \to (2, 1, 1) \to (2, 2, 1) \to (2, 3, 1) \to (2, 3, 2)$. The first increase after the rank-one plateau follows the log law in 26 of 26 seeds, with single-route slopes within 0.3–1.6% of h_{e^*} (e.g. 5.188 ± 0.008 against 5.175), always on e^* at $\sigma = 10^{-3}$; later increases were not timed. Gradient descent on tensor-train completion shows a plateau followed by abrupt rank growth (Milanesi et al., 2021).

6.7 Teacher and student on different trees

A student bond must reach the rank of the teacher's matricization across it, bounded by products of bond dimensions over cuts of the teacher's tree proven known (standard min-cut bound; cf. Grasedyck and Hackbusch, 2011, Lemmas 7–8) (Section A.5): a binary8 student of an MPS8 teacher of bond rank r needs rank $\min(d^2, r^2)$ on the bonds of sites $\{2, 3\}$ and $\{4, 5\}$. The routes give *bottom-up order* derived new: the root bipartition $\{0123\}|\{4567\}$ (the teacher's middle bond) has a three-core route only after both halves have bond rank above one below it, so it grows only after a bottom bond on each half, whatever its teacher Schmidt value. Experiments empirical new (binary8 student, $\chi = 4$, twelve random MPS8 teachers with bond ranks 2, $\sigma = 0.3$, 0.4, 0.5, eight seeds): in the 7 teachers whose root cut has a larger teacher Schmidt value than both bottom bonds of a half, so that a weight order disagrees, bottom-up order holds in 163 of 168 runs (56 of 56 at $\sigma = 0.3$), including 117 of 120 for five teachers whose root cut is the strongest of all by 5.5–13%; the 5 exceptions grow the root after a bottom bond on one half only. Over all 12 teachers: 249 of 256. "Largest teacher Schmidt value first" fails, and "a sibling pair first" is no rule (8–100% of the runs by teacher). Scope: $N = 8$, $\sigma = 0.3$–0.5. (An MPS8 student of binary8 teachers at $\sigma = 0.45$ grows the predicted bond first in 2 of 24 runs; with $\ln(1/\sigma) = 0.8$ this is no test of the asymptotic race of Proposition A.38.)

6.8 Star teachers: forgetting a learned component

A learned component j without a two-core route to the rest is a strict saddle iff $\lambda_{\max}(\Gamma) > s_j$, Γ the weighted adjacency matrix of the piggyback graph, and then $\lambda_{\min} = -q_j^{K-2}(\lambda_{\max}(\Gamma) - s_j)$ proven new (for general trees; deep linear networks: Achour et al., 2024, Thm 7) (Proposition A.22). The *star* has $q = 0000$ ($s_q = 2$) and $r = 1010$, $r' = 1001$ (weight w each, $d = 2$), which share the piggyback node: $\lambda_{\max}(\Gamma) = \sqrt{2}\, w$, so "$q$ learned" (θ_q) is strict iff $w > \sqrt{2}$, although both residual components

Table 12: Input covariance and teacher structure: what changes, and what predicts it. "Escape": the escape from the rank-one saddle.

complication	what changed (empirical)	predicted by
aligned covariance	quantitative only: order and times follow $\Lambda_r s_r$, plateaus and swaps $\Lambda_r s_r^2$; exponents unchanged	whitened problems with Λs (race) and $\Lambda^{1/2} s$ (landscape)
generic teachers	escape strict and logarithmic on MPS ($\leq 0.4\%$ from the saddle, $\leq 1.3\%$ from random init); $\propto \varepsilon^{-1}$, σ^{-1} on binary and HT	two-core routes and their rates
misaligned covariance	plateaus at critical points of the Σ-loss; early alignment (binary4); MPS4 escape logarithmic	$\Sigma^{1/2} W^*$; slope only bounded
cross-site correlations	binary4: first saddle with pairing terms; MPS4: logarithmic two-stage escape	two-stage law (replication)
overlapping CP components	blended plateaus; MPS4 rank-one maxima merge at $c = 1/2$; logarithmic escape	rank-one landscape; h_e within 1% ($c \geq 0.2$)
random TTN teachers	plateaus at rank-constrained critical points (3 of 5 teachers separated from SVD truncations); ranks grow bond by bond	critical points; h_{e^*} within 1.6%
teacher on another tree	bottom-up order on binary8 (163 of 168 discriminating runs)	routes
star teachers	larger learned component left iff $\lambda_{\max}(\Gamma) > s_q$; log time on MPS4, polynomial on HT4	piggyback graph; rates within 0.2%
real data (MNIST)	as for a generic teacher: binary first route a sibling pair, gap $A/\sigma + B$; MPS later routes logarithmic (6 of 9 edges); plateaus creep; the pixel order sets the final loss, not through locality	route counting with $T = \mathbb{E}[y\Phi]$ for W^*; route amplitudes (first route; ranking of orders, pre-registered)

are smaller and each needs three or more cores. Experiments ($\chi = 3$; MPS4, binary4, HT4; saddle start at θ_q with $\varepsilon = 10^{-6}$–10^{-2}, five seeds; random initialization at $w = 1.2$, 1.6, ten seeds):

- *Hessian* proven and computed: the formula holds to 10^{-15} on MPS4, binary4, HT4 and a binary8 analogue (-0.3716, -0.3310, -0.4311, -0.4311 at $w = 1.6$); no negative eigenvalue for $w \leq 1.40$. This repeats the theory's check; binary4 is a deep linear network, where it is known.
- *Threshold* empirical new (dynamics): at $\varepsilon = 10^{-6}$ no seed leaves θ_q at $w = 1.4138$ ($\lambda_{\max}(\Gamma)/s_q = 0.9998$) and all do at 1.4146 (1.0003), on MPS4 and binary4; on HT4 the step lies between 1.40 and 1.42 for $\varepsilon \leq 10^{-3}$. At $\varepsilon = 10^{-2}$ it moves to 1.43–1.44: near the threshold the strict escape is slow and the polynomial route that adds r wins at moderate noise.
- *One or two strict stages* derived (from the computed Hessians) and empirical new: the slope of the escape time against $\ln(1/\varepsilon)$ is $1/|\lambda_q|$ within 0.21% where the event times one strict stage ($w \geq 1.48$ on all trees, every w on binary4). Near the threshold on MPS4 the flow stops at a blended rank-one point that keeps $a_q = 0.98$, 0.93 ($w = 1.42$, 1.44) and is itself strict: the measured 130.0 ± 0.2 and 29.39 ± 0.02 equal $1/|\lambda_q| + 1/|\lambda_b| = 129.9$ and 29.3 (one stage: 86.4, 19.4). On HT4 that point has no negative eigenvalue, and leaving it takes a time $\propto \varepsilon^{-1}$ ($p = 0.995$, 0.998; at $w = 1.6$, $a_q < \frac{1}{2}$ after $3.4 \cdot 10^3$, $3.3 \cdot 10^4$, $3.3 \cdot 10^6$ for $\varepsilon = 10^{-3}$, 10^{-4}, 10^{-6}). A star forgets q in logarithmic time on MPS4 and binary4 but polynomially on HT4: the tree contrast of Section 6.2 again.
- *Random initialization* empirical new ($w = 1.6$; $\sigma = 0.01$–0.1 on MPS4, 0.005–0.05 on binary4): the flow never sits at θ_q (at most 0.13 time units at its loss level, one seed 1.7; up to 538 at "q forgotten"). The initialization direction sets the class: q learned last (binary4 7 of 10 seeds, MPS4 1 of 10), a_q rising to 0.78–1.00 and decaying (3, 4 of 10), or a blended start with

$a_q \leq 0.68$ (MPS4, 5 of 10). The decay deepens steeply as σ decreases, not as a power law; q is relearned last. Sub-critical stars ($w = 1.2$) keep q ($\min a_q \geq 0.992$).

Withdrawn: the laws $\min a_q \propto \sigma^{5.7\pm0.3}$ and a relearning time $\propto \sigma^{-1.4}$ from random initialization; they pooled two classes of runs, dropped the points where a_q crosses zero, and describe runs that never visit θ_q.

6.9 Real data

Do the teacher–student results survive real inputs, with a non-product input law and no teacher? The empirical loss is $L = \frac{1}{2}\langle W, CW\rangle - \langle T, W\rangle + \text{const}$, with the second-moment operator C of the features, which is not a product, and $T = \mathbb{E}[y\Phi]$ (averages over the training set). Two predictions need no product law derived. Near the origin $L \approx L(0) - \langle T, W\rangle$, so the first escape is that of Section 4.3 with $W^* \to T$ (Proposition A.41). At a bond-rank-one critical point the routes that can open depend only on bond ranks and site dimensions (Section A.5, "Generic teachers"): on a binary tree, whose internal cores carry no pixel, the cheapest routes join two sibling bottom cores ($k = 3$: a higher-order saddle, a plateau $\propto \sigma^{-1}$); on an MPS any adjacent pair of cores can open one ($k = 2$: strict, $\propto \ln(1/\sigma)$). These are the generic-teacher predictions of Section 6.2 with T for W^*. The data decide which of the cheapest routes goes first, and the assignment of pixels to sites decides which pixel interactions lie on short routes.

Setup (details in Section B.8). MNIST 3 ($y = +1$) against 5 ($y = -1$): 4000 training and 1600 test images, cropped to 24×24 and block-averaged to 4×4 pixels ($N = 16$), with the features $\phi(x) = (\cos\frac{\pi x}{2}, \sin\frac{\pi x}{2})$ of Stoudenmire and Schwab (2016); square loss ($L(0) \approx \frac{1}{2}$), fan-in initialization, $\chi = 4$. Gradient flow runs through the first plateau to the time t_a at which the loss has fallen by 1%, then on the clock $\tau = t - t_a$ up to $\tau = 10^4$. Trees: binary16 ($K = 15$; two pixels per bottom core, none on the internal cores) and MPS16 ($K = 16$). Pixel orders: *quadtree* (Morton order; bottom cores carry horizontally adjacent pixel pairs, level-2 subtrees 2×2 blocks), *snake* (boustrophedon, for the MPS) and random permutations (*scrambled*). Main sweep: binary quadtree, snake MPS and three scrambled binary orders, seven σ from 0.05 to 0.5, seeds 0–4. Events are loss drops (peaks of $-\mathrm{d}L/\mathrm{d}t$; gaps on the τ clock) and rank events (a ratio s_k^e/s_1^e of data-weighted Schmidt values crossing 0.05). Runs of one (order, seed) at different σ mostly reach the same rank-one critical point, so counts are per (order, seed) or per *unit* (order, seed, critical point), never per run; $\pm$ is the standard error over seeds.

What survives.

- *First plateau: an identity, not evidence.* $t_a\sigma^{K-2}$ is constant per seed to four digits or better (exponents 13.000 ± 0.000 on binary16, 14.000 ± 0.000 on MPS16). This holds for any input law, since as $\sigma \to 0$ the flow from σZ is the flow from Z on a clock slowed by $\sigma^{-(K-2)}$ proven known (the exponent: Jacot et al., 2022; Kumar and Haupt, 2025a): it checks the integrator and the clock, not the theory.
- *The first binary route is a sibling pair* empirical new. After the escape, 96 of the 124 binary states lie within 0.05 (relative) of a rank-one product, the others mostly at $\sigma \geq 0.3$. In all 20 (order, seed) pairs the first two edges to gain a singular value belong to two sibling bottom cores, the predicted $k = 3$ route, and in all 96 rank-one states they cross the threshold together; the 11 binary runs of the round-1 review (nine orders, $\sigma = 0.1$) agree.

- *Route amplitudes pick the route* empirical new. At the rank-one state the *route amplitude* μ_{uw} of two bottom cores is the norm of the residual's correlation between their unused feature directions, weighted by the other cores (63); it sets the rate of the route through them up to factors common to all routes with the same number $k_{uw} \in \{3, 5, 7\}$ of cores (Definition A.17). The sibling pair with the largest μ_{uw} rises first in 91 of the 96 rank-one states, in about 80–85% of the critical-point units (22 of 26 with our grouping of states, 19–20 of 23–24 with others; chance $\frac{1}{4}$), and in 11 of 11 review runs (10 of them new orders or seeds). Degree comes before amplitude: in 95 of the 96 states the largest of the 28 amplitudes lies on a route with $k = 5$ or 7, yet a sibling pair rises first (Remark A.27).
- *The $k = 3$ gap is $A/\sigma + B$, not logarithmic* empirical new (Figure 20). Over the 11 series (one per order and seed) whose first route, one sibling pair, rises inside the second loss drop at every σ (at least four values over at least half a decade), the gap between the first two drops fits $A/\sigma + B$, the $k = 3$ law (23) with its $O(1)$ correction (Proposition A.25), with relative rms residual at most 0.060 and $B < 0$ in all 11; the log law of a two-core route leaves 0.11–0.30. A free exponent (1.22 ± 0.03, range 1.06–1.36) mostly measures the offset and depends on the σ range. The verdicts hold for all 124 binary runs, each point attributed to the drop that contains the route's rank crossing (91 points kept, ten seed series). All of these use raw features; the one whitened series that meets the same rule has $B = +10.8$.
- *Saddle types* empirical new, from the σ-scaling of $\lambda_{\min}(H_\perp)$ at plateau midpoints (Section 2.7). Binary16 at the rank-one plateau ($L \approx 0.27$): $|\lambda_{\min}| \propto \sigma^p$ with $p = 0.87$–0.89 on the three seeds with five or six σ (0.6–0.8 decades) and 1.15 on a seed with three (mean 0.95 ± 0.07): far from 0, so higher order, but below the 1 of $k = 3$ (Proposition A.26) at these σ. MPS16 at $L \approx 0.23$ (five seeds): $\lambda_{\min} = -0.054$ to -0.079, exponent 0.047 ± 0.016, a strict saddle. That MPS plateau comes after the first route has risen and has two or three negative directions: it is the saddle of later two-core routes.
- *Logarithmic later routes on the MPS* empirical new. The first rank increase is on one edge in all 25 runs where it is resolved (in 10 of 35 the threshold is met at $\tau = 0$), within two time units of the first drop's peak: it merges with the escape and has no measurable gap. Of the later edges, timed by their first rise after the first drop, edges 3, 4 and 6–9 prefer the log law (28 of 29 seed fits; relative rms 0.03–0.11 against 0.17–0.36), edge 4 with slope 14.7 ± 1.5 per unit of $\ln(1/\sigma)$, comparable to $1/|\lambda_{\min}| = 13$–19. Edges 1 and 2 prefer a power law and edge 0 is mixed, which rank-one counting ($k = 2$ on every MPS edge) does not explain.
- *Plateaus creep* empirical: between drops the loss falls by a median 0.76 of the next drop (56 intervals in 20 binary runs, $\sigma = 0.1$). With drops dated by rate peaks the S2S signature holds on binary trees (quadtree: median smallest separation ratio 2.8 at $\sigma = 0.5$, 24 at 0.05), not on the MPS.

Pixel order: a pre-registered test. In the main sweep the quadtree ends lowest: at $\sigma \leq 0.14$ a scrambled order ends below the quadtree of the same seed in 1 of 50 runs (at $\sigma = 0.1$, $L(10^4) = 0.098 \pm 0.001$ against 0.19–0.23 for the three scrambled orders), and it puts most of $\sum \mu_{uw}^2$ on routes with $k \leq 5$ (0.89 ± 0.01 at $\sigma = 0.05$; scrambled orders 0.17 ± 0.07) empirical. In the review runs, though, the equally local transposed quadtree (vertical pairs) ended above six random orders at seed 0. Before running any new order we registered a test of the share (`notes/real_data.md`, section 9; the registration and the predictor values were committed before the runs), reported here as registered.

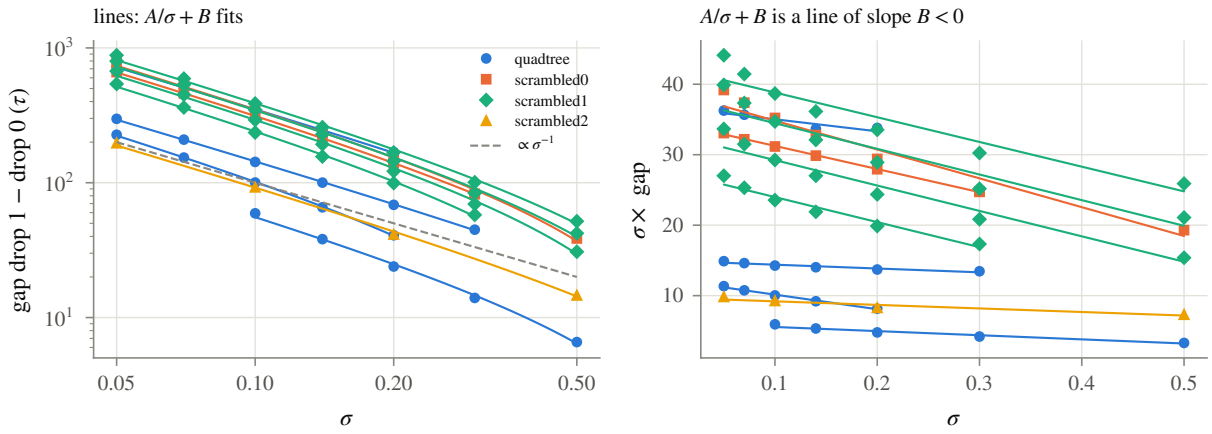


Figure 20: First route after the escape on binary16 (MNIST 3 against 5, $\chi = 4$): gap between the first two loss drops (τ clock) against σ, for the 11 series (one per order and seed; colour: order) whose first route, one sibling pair, rises inside the second drop at every σ. Left: log–log axes; lines: $A/\sigma + B$ fitted with relative residuals; dashed: $\propto \sigma^{-1}$. Right: σ times the gap, in which $A/\sigma + B$ is a line of slope B. Takeaway: every series follows the $k = 3$ law with a negative $O(1)$ offset; a logarithmic law fails.

- *Design.* Hypothesis: across pixel orders of binary16 at $\sigma = 0.1$, the share S of $\sum_{uw} \mu_{uw}^2$ on routes with $k_{uw} \leq 5$, evaluated at the best rank-one fit of the labels (no training), predicts (O1) the training loss at $\tau = 10^4$ and (O2) $\log_{10}$ of the time to $L \leq 0.25$: Spearman $\rho < 0$ with a 95% bootstrap interval that excludes 0, for each outcome, over all orders and within the new random orders. Orders: 20 new random ones and 7 structured ones (quadtree; quadtreeT, the Morton order of the transposed image; row-major; column-major; snake; a Hilbert curve; mortonrev, bit-reversed Morton keys, which put every sibling pair two rows apart). Seeds 0–2: 81 runs, 74 new (the outcomes of quadtree, quadtreeT and row-major were seen before registration); outcomes averaged over seeds; O2 censored at $\tau = 10^4$ (4 of 81 seeds).
- *Result* empirical (pre-registered) new (Figure 21). O1: $\rho = -0.60$ $[-0.82, -0.22]$ over the 27 orders (permutation $p = 0.001$) and -0.55 $[-0.85, -0.09]$ within the 20 random ones ($p = 0.015$): *supported*. O2: -0.30 $[-0.59, +0.06]$ and -0.26 $[-0.63, +0.23]$: *not supported*. The registered secondary analyses agree (O1 without the three orders seen before registration: -0.51 $[-0.80, -0.08]$; with the round-0 scrambled orders added: -0.60). Orders with the same pixel pairs share S (24 distinct values over 27 orders), so the within-random result is the cleaner test. Also registered: the rank-one search on the data does not always find the rank-one state that gradient flow reaches (quadtree: a fit at loss 0.310, while four of five runs plateau at 0.27), so S is a noisy proxy.
- *Deviations.* None in the design or the analysis. The verification found two procedural points missing from the experiment's own list: the run configuration and analysis script were committed four minutes after the runs started (before any outcome), and an interim summary of 72 of the 81 runs (all 60 random-order runs) was written, and probably analysed, before the last runs finished.

The pixel order thus sets how low a binary TTN ends at fixed training time, and a route-amplitude share computed from the data before training ranks the orders by that loss, but not by their early speed. Locality in the image is neither necessary nor sufficient: the three orders with horizontal pixel pairs (quadtree, snake, row-major; $S = 0.989$) rank 1, 2 and 4 of 27 by O1, but the equally local orders with vertical or mixed pairs (quadtreeT, column-major, Hilbert) rank 13–21; the transposed

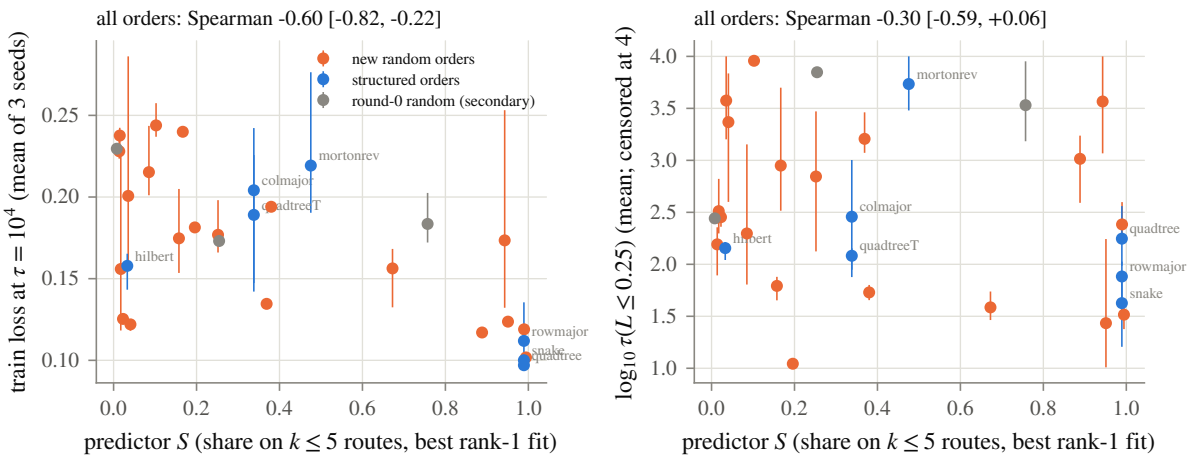


Figure 21: Pre-registered order test: 27 pixel orders of binary16 (orange: 20 new random orders; blue: 7 structured orders, labelled), $\sigma = 0.1$, seeds 0–2 (points: mean; bars: range over seeds); grey: the three round-0 scrambled orders (secondary analysis only). Horizontal axis: the predictor S, computed from the data before training. Left: training loss at $\tau = 10^4$ (O1). Right: $\log_{10}$ of the time to $L \leq 0.25$ (O2), censored at 4. Titles: Spearman ρ over the 27 orders, 95% bootstrap intervals. Takeaway: S ranks the orders by their final loss, not by their early speed; orders with the same pixel pairs share S, and local orders with vertical or mixed pairs rank only 13–21 of 27.

quadtree ends at twice the loss of the quadtree (0.189 against 0.097; the quadtree's outcome, the lowest, was seen before registration); and a random order with $S = 0.994$ ranks third. Tree tensor networks for images are often built to follow the pixel geometry (e.g. Cheng et al., 2019). Our explanation is a conjecture: an order helps by placing label-relevant pixel interactions on routes that escape within the horizon ($k = 5$, plateaus $\propto \sigma^{-3}$) rather than on routes with $k = 7$ ($\propto \sigma^{-5}$). *Exploratory, not registered:* the same share at the state each run actually reaches ranks O1 more strongly ($\rho = -0.83$ $[-0.91, -0.66]$; -0.85 $[-0.95, -0.60]$ within random orders) and O2 no better empirical.

Narrowed or withdrawn after review (`notes/reviews/real_data_round1.md`).

- *"The MPS is fastest"* holds only on the clock τ started at the first escape (MNIST, snake order; at $\sigma = 0.1$ the MPS reaches $L \leq 0.2$ after 56 time units, the quadtree after 333) empirical. On the absolute clock the first escape dominates ($t_a \geq 10^5$), $t_a \propto \sigma^{-14}$ on MPS16 against σ^{-13} on binary16 proven, and the MPS escapes later than the quadtree in 4 of 5 seeds at $\sigma = 0.1$; on Fashion-MNIST it is the slowest tree to $L \leq 0.25$ at $\sigma \leq 0.14$ (but the first to reach $L \leq 0.12$). Only snake chains were run, so tree shape and pixel order are confounded, and that the logarithmic routes make the MPS fast on the τ clock is a conjecture.
- *Locality.* "A small-initialization dynamical effect", the round-0 contrast of 88% against 21% of the interaction on short routes (three random orders from the low tail, pooled over σ) and "gone at $\sigma = 0.5$" are withdrawn; the order effect above remains.
- *Exponents.* Withdrawn: the round-0 gap law $\sigma^{-1.01 \pm 0.14}$ (an attribution bug), "later routes have higher degree" (late events not identified by route), and the first-route exponents of the Fashion-MNIST ($\sigma^{-0.92}$ to $\sigma^{-0.97}$) and $\chi = 8$ sweeps (1.17 ± 0.06, 0.97 ± 0.11), which are in no committed output; the $\chi = 8$ comparison at $\sigma = 0.5$ is confounded by 23 runs stopped at the wall-clock limit. Kept empirical: a sibling pair first at $\chi = 8$ (14 of 15 runs per binary order)

and on Fashion-MNIST (first two rising edges, 15 and 14 of 15, rising together in only 3 and 2), where a strict saddle ($\lambda_{\min} \propto \sigma^{0.03\pm0.09}$) separates two rank-one states and the quadtree has no advantage.

- *Regimes.* The centred sweep ($\sigma = 0.5$–4, quadtree) gives no regime map: "rich" rests on $\sigma = 1.4$ alone, the S2S–rich boundary moves over $\sigma = 0.07$–1 with the separation threshold, the lazy runs are unconverged at their horizon, and "the lazy floor lies well above what feature learning reaches" is withdrawn (the small-initialization runs end below the floor for four of five seeds). Kept empirical: at $\sigma \geq 2$ the runs follow the linearized model, whose tangent kernel has rank $304 = \sum_v F_v\chi_v - \sum_e \chi_e^2$ of 528 parameters (Proposition A.46(c)).

Scope. The scaling claims rest on one task at 4×4 pixels, gradient flow and 4000 fixed images; 8×8 images (12 runs) show the same first routes but allow no fits, and whitened site features do not make the dynamics cleaner. The horizon $\tau = 10^4$ cuts the escapes along routes with $k \geq 5$ at $\sigma < 0.1$, so final losses are losses at fixed training time. Events are read from loss rates and spectral thresholds, so late events are only partly identified by route.

6.10 Summary

Only the aligned covariance leaves the qualitative picture unchanged (Table 12). Every other complication breaks the orthogonality of the teacher's components in the metric of the loss, and then the student's tree decides whether the escape from the rank-one saddle is logarithmic (MPS) or polynomial (binary, HT). The baseline theory predicts each change after the substitution in the table, except the slope under a misaligned covariance from random initialization, which is only bounded. On MNIST the same tree contrast appears with the data's escape tensor T in place of W^* (sibling-pair routes with $A/\sigma + B$ gaps on binary trees, logarithmic later routes on the MPS), while the effect of the pixel order is only ranked, by a data-only route-amplitude share (pre-registered; Section 6.9).

7 Open questions

[Skeleton. Questions left open by this work, each with what would settle it.]

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A Derivations

This appendix derives the theoretical statements of the main text not proved in Section 2.

Conventions. Notation is that of Section 2. Unless stated otherwise inputs are whitened ($\Sigma = I$), gradient flow has unit multipliers ($\eta_v = 1$), so that $L = \frac{1}{2}\|W - W^*\|_F^2$, and the teacher is a CP teacher (4). For two components j, r of a pattern teacher, $D_{jr} = \{i : \pi_j(i) \neq \pi_r(i)\}$ is the set of sites where they differ, $N(D_{jr})$ the set of (*terminal*) cores carrying a site of D_{jr}, $m_{jr} = |N(D_{jr})|$, and $\mathrm{St}(N(D_{jr}))$ the smallest subtree of $\mathcal{T}$ containing $N(D_{jr})$ (its Steiner subtree; compare Proposition 2.1(b)). At a learned saddle the learned component is j, with per-core amplitude $q_j = s_j^{1/K}$ (Proposition 2.4), and r is a component still to be learned. A ratio of two teacher weights, smaller over larger, is written ϱ. $\sigma_k(M)$ is the k-th singular value of a matrix M; σ is the initialization scale. "LO" means leading order in the small quantities of each statement; $\hat{\theta} = \theta/\|\theta\|$. A statement that combines a proven identity with a derived consequence carries both tags. Arguments the theory notes call *heuristic* (assumptions not controlled) are tagged conjecture, or empirical with the words "heuristic mechanism" where experiments support them. "Checked" refers to the numerical checks of the theory notes (`notes/theory_*.md`), which give all values; only the key ones are quoted here.

A.1 Conservation laws, gauge and invariant subspaces

Besides Proposition 2.5, three facts are used below.

Lemma A.1 (Equal norm shift). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Let $L(\theta) = \mathcal{L}(W(\theta))$ for a differentiable $\mathcal{L}$. Under gradient flow (15) (no mask), $\eta_v^{-1}\frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = -2\langle \nabla_W \mathcal{L}, W\rangle$ for every core v, the same for all cores. For the population loss (6) the right-hand side is $2(\langle W, W^*\rangle_\Sigma - \|W\|_\Sigma^2)$; for the leading-order flow $\dot{\theta} = \nabla g$ of Section A.2 it is $2g$.*

Proof. W is linear in G_v, so $\langle G_v, \nabla_{G_v} L\rangle = \langle \nabla_W \mathcal{L}, W\rangle$, and $\frac{\mathrm{d}}{\mathrm{d}t}\|G_v\|_F^2 = -2\eta_v\langle G_v, \nabla_{G_v} L\rangle$. □

The lemma gives the common shift $y(t)$ in $\|G_v(t)\|_F^2 = \|G_v(0)\|_F^2 + y(t)$. Along the leading-order flow $\dot{g} = \|\nabla g\|^2 \geq 0$, so $y = 2\int_0^t g$ is convex: *all cores shrink, by the same amount, and only while $g < 0$*, i.e. only if $\langle W_0, W^*\rangle < 0$. The identity holds for any block of variables in which g is multilinear, e.g. the reduced cores of Proposition A.11 (checked).

Lemma A.2 (Unused bond channels are invariant). proven new (elementary) *Let $e = e_v$ with $p = p(v)$, and let $\mathcal{B}$ be a set of channels of e such that every entry of G_v and of G_p whose e-index lies in $\mathcal{B}$ vanishes. Then, for every loss of the form $\mathcal{L}(W(\theta))$ (any input law, population or empirical), the gradient vanishes on these entries, so gradient flow, gradient descent and SGD keep them at zero. In particular, a critical point of the problem with bond dimensions $r_e \leq \chi_e$, padded with zeros, is a critical point of the problem with bond dimensions χ_e.*

Proof. The derivative of W with respect to an entry of G_v with e-index $b \in \mathcal{B}$ contracts the other cores with G_p at e-index b, a vanishing slice; likewise for G_p. □

Lemma A.3 (Conservation laws under discrete steps). proven known (two-layer factorization: Du et al., 2018, App. B.1; the $O(\eta^2)$ change in deep linear networks: Arora et al., 2019a, App. D.2.3; scalar form: Hofmann, 2026, Prop. 14) *With the gradients $g_v = \nabla_{G_{v,(e)}} L$, $g_p = \nabla_{G_{p,(e)}} L$ matricized as in (17), one step $\theta \mapsto \theta - \eta\nabla L$ (unit multipliers) changes Q_e by exactly $Q_e^+ - Q_e = \eta^2(g_v g_v^\top - g_p g_p^\top)$; likewise for SGD. For one aligned component (Proposition 2.4), $Q = q_{v,1}^2 - q_{p,1}^2$ obeys $Q^+ = Q(1 - \eta^2\Delta)$, $\Delta = (s_1 - \Pi_1)^2\Pi_1^2/(q_{v,1}^2 q_{p,1}^2)$: a step contracts the imbalance iff $0 < \eta^2\Delta < 2$.*

Proof. With $X = G_{v,(e)}$, $Y = G_{p,(e)}$, the proof of Proposition 2.5 gives $g_v X^\top = Y g_p^\top =: C$ for every loss that depends on θ through W; so the terms linear in η cancel in $(X - \eta g_v)(X - \eta g_v)^\top - (Y - \eta g_p)(Y - \eta g_p)^\top$. For the aligned component $g_v = -(s_1 - \Pi_1)\Pi_1/q_{v,1}$, and likewise for p. □

The change is negligible on plateaus and large during oscillations at large steps (Section A.7). At initialization with a common scale, $\mathbb{E}\, Q_e(0) = \sigma^2(1 - \chi_p/\chi_v)I$ (Section 2.6; χ_p is the parent's own bond dimension, $\chi_{\text{root}} = 1$): edges into the root are child-heavy ($\mathbb{E}\, Q_e \succ 0$), and HT leaf edges with $\chi_v = d = 2$ below $\chi_p = 3$ are parent-heavy ($\mathbb{E}\, Q_e = -\frac{1}{2}\sigma^2 I$).

A.2 Escape from the origin

Setting. Whitened inputs, a CP teacher, and gradient flow from the fan-in initialization $\theta_0 = \sigma Z$ (Equation (11); Z is the initialization at $\sigma = 1$); Section A.6 extends this to product covariances by $W^* \to \Sigma W^*$. Write

$$L(\theta) = \tfrac{1}{2}\|W^*\|_F^2 - g(\theta) + \tfrac{1}{2}\|W(\theta)\|_F^2, \qquad g(\theta) := \langle W(\theta),\, W^* \rangle. \tag{27}$$

g is homogeneous of degree K and linear in each core, while $\|W\|_F^2$ has degree $2K$. The *leading-order (LO) flow* $\dot\theta = \nabla g(\theta)$ drops a term of relative size $\|W\|_F/\|W^*\|_F = O(\sigma^K)$ on the first plateau. Let t^* be its blow-up time and $\nabla_{\mathbb{S}} g(\hat\theta) = \nabla g(\hat\theta) - K g(\hat\theta)\hat\theta$ the gradient of g on the unit sphere.

Proposition A.4 (Spherical–radial decomposition). proven known (directional convergence: Kumar and Haupt, 2025a) *Along the LO flow, with the time change* $\mathrm{d}\tau = \|\theta\|^{K-2}\mathrm{d}t$,

$$\frac{\mathrm{d}\hat\theta}{\mathrm{d}\tau} = \nabla_{\mathbb{S}} g(\hat\theta), \qquad \frac{\mathrm{d}\ln\|\theta\|}{\mathrm{d}\tau} = K\, g(\hat\theta). \tag{28}$$

The direction follows the spherical gradient flow of g, independent of $\|\theta\|$ and σ, and $Q_e(\hat\theta(\tau)) = Q_e(\theta_0)/\|\theta(\tau)\|^2$.

Proof. By Euler's identity and the homogeneity of ∇g (degree $K-1$), $\frac{\mathrm{d}}{\mathrm{d}t}\|\theta\| = K\|\theta\|^{K-1} g(\hat\theta)$ and $\frac{\mathrm{d}}{\mathrm{d}t}\hat\theta = \|\theta\|^{K-2}\nabla_{\mathbb{S}} g(\hat\theta)$; Q_e is quadratic in θ. □

As $\sigma \to 0$ the escape needs $\|\theta\|$ to grow by a factor of order $1/\sigma$ (τ of order $\ln(1/\sigma)$), so the escape direction is $\lim_{\tau\to\infty}\hat\theta(\tau)$, a KKT point of g on the sphere (Kumar and Haupt, 2025a, Thm 1, which applies verbatim to TTNs with $K \geq 3$). The conservation laws bind only while $\|\theta\| \approx \|\theta_0\|$ ($\tau = O(1)$), the phase that decides the escape time and the KKT point reached; they do not change the balanced form of the limit (Proposition A.9).

Corollary A.5 (Blow-up time). proven known (the identity is the activation law of alternating gradient flows, Kunin et al., 2025, Eqs. 2–3, written in spherical time with homogeneity degree K and utility g; the exponent $K-2$ in analogues: Jacot et al., 2022; Gissin et al., 2020) new (its per-initialization form for TTNs) *With* $G(\tau) = \int_0^\tau g(\hat\theta)\,\mathrm{d}\tau'$,

$$t^* = \|\theta_0\|^{-(K-2)} \int_0^\infty e^{-(K-2)KG(\tau)}\,\mathrm{d}\tau. \tag{29}$$

For $\theta_0 = \sigma Z$, $t^* = \sigma^{-(K-2)}\,\hat t_1(Z)/s_1$, where $\hat t_1(Z) := s_1\, t^*|_{\sigma=1}$ depends on the initialization direction and on W^*/s_1 but not on σ; the exponent $K-2$ counts every core.

Proof. By (28), $\|\theta(\tau)\| = \|\theta_0\| e^{KG(\tau)}$ and $\mathrm{d}t = \|\theta\|^{-(K-2)}\mathrm{d}\tau$. The LO field is homogeneous of degree $K-1$, so $\theta_\sigma(t) = \sigma\,\theta_1(\sigma^{K-2}t)$; replacing W^* by cW^* divides all times by c. □

There is no separate alignment time: the whole integral is alignment ($g(\hat\theta)$ starts near 0).

Proposition A.6 (From the blow-up to the first event). derived new (for TTNs) *Assume that at the end of the LO phase the direction $\hat{\theta}$ has converged to the balanced KKT point of the first component learned, of weight s. Then, as $\sigma \to 0$,*

$$t_1 = t^* + \delta_K \, s^{-(2-2/K)} + o(1), \qquad \delta_K = -\frac{2^{(K-2)/K}}{K-2} + \int_0^{2^{-1/K}} \frac{b \, \mathrm{d}b}{1-b^K}, \tag{30}$$

with $\delta_3 = -0.850$, $\delta_4 = -0.266$, $\delta_5 = -0.047$, $\delta_6 = 0.072$, $\delta_7 = 0.148$ and $\delta_8 = 0.201$. Hence $t_1 \sigma^{K-2} s_1 / \hat{t}_1(Z) = 1 + O(\sigma^{K-2})$ for every initialization.

Derivation. Once $\|\theta\| = O(1)$ the remaining time does not depend on σ. Along a balanced KKT point the flow is the balanced mode $\dot{q} = q^{K-1}(s - q^K)$ of Proposition 2.4, whose half-time is $q_0^{-(K-2)}/((K-2)s) + \delta_K s^{-(2-2/K)} + O(q_0^2)$ (write $b = q s^{-1/K}$ and $1/(b^{K-1}(1-b^K)) = b^{-(K-1)} + b/(1-b^K)$); the first term is the LO blow-up time. □

Checked per initialization (within 0.8% of gradient flow for $K = 3, 4$, $\sigma \le 0.03$); for $K = 6, 7$ at $\sigma = 0.15$–0.5 the direction has not converged (heuristically), so a 1% test needs $\sigma^{K-2} \le 10^{-3}$.

Remark A.7 (Per-initialization prediction). proven at LO. Integrate in τ the state (u, ℓ, T) from $u(0) = \theta_0/\|\theta_0\|$, $\ell(0) = T(0) = 0$: $\mathrm{d}u/\mathrm{d}\tau = \nabla g(u) - K g(u) u$ (renormalizing u), $\mathrm{d}\ell/\mathrm{d}\tau = K g(u)$, $\mathrm{d}T/\mathrm{d}\tau = e^{-(K-2)\ell}$, up to $\ell = 40/(K-2) + 5$, and add the tail $e^{-(K-2)\ell}/((K-2)K g(u))$. Then $t^* = \|\theta_0\|^{-(K-2)} T$ for the actual teacher (any number of components), and the dominant component of the final u is the first one learned as $\sigma \to 0$: a parameter-free prediction of t_1 (tested in Section 4.3).

Proposition A.8 (Maximum of g on the sphere). proven new (for TTNs; deep linear networks, where the injective norm is the spectral norm: Jacot et al., 2022, App. C.1.1) *Group the sites by the core that carries them (cores without sites form no group), and let $\|W^*\|_{\mathrm{inj}} = \max\{\langle W^*, \bigotimes_v Y_v \rangle : \|Y_v\|_F = 1\}$ be the injective norm of W^* seen as a tensor with one leg per group. Then*

$$g(\theta) \le \|W^*\|_{\mathrm{inj}} \prod_v \|G_v\|_F \le K^{-K/2} \|W^*\|_{\mathrm{inj}} \|\theta\|^K, \tag{31}$$

with equality on the unit sphere at the balanced bond-rank-one configurations $G_v = K^{-1/2} Y_v \otimes \varepsilon_1 \otimes \cdots \otimes \varepsilon_1$ built from a maximizing (Y_v). For odeco teachers $\|W^*\|_{\mathrm{inj}} = s_1$ for every grouping; in general the grouping matters (for $W^* = (I/\sqrt{d}) \otimes (I/\sqrt{d})$ on the site pairs $(0,1)$, $(2,3)$ the unit-sphere maximum is $3^{-3/2}$ on binary $N = 4$ and $4^{-2}/d$ on the MPS).

Proof. First inequality, by induction on the number of cores: for a leaf ℓ with parent p and the SVD $G_\ell = \sum_k \mu_k Y_k \otimes z_k$ across (physical legs | bond leg), $g = \sum_k \mu_k \langle W'_k, W^*[Y_k] \rangle$, with W'_k the network without ℓ and G_p contracted with z_k on the ℓ-leg, and $W^*[Y_k]$ the contraction of W^* with Y_k on the sites of ℓ. Since $\|W^*[Y_k]\|_{\mathrm{inj}} \le \|W^*\|_{\mathrm{inj}}$, induction bounds the sum by $\|W^*\|_{\mathrm{inj}} \prod_{v \ne \ell, p} \|G_v\|_F \sum_k \mu_k \|G_p \times_\ell z_k\|_F$, and Cauchy–Schwarz with $\sum_k \|G_p \times_\ell z_k\|_F^2 \le \|G_p\|_F^2$ closes the step. The second inequality is AM–GM. □

Proposition A.9 (KKT points are balanced). proven new (TTN form; linear chains: Kumar and Haupt, 2025a, Lemma 5; it also follows from the proof of Furman et al., 2026, Prop. F.10) *Every KKT point θ^* of g on the unit sphere with $g^* := g(\theta^*) > 0$ has $Q_e = 0$ on every bond and $\|G_v\|_F^2 = 1/K$ for every core.*

Proof. At a KKT point $\nabla g(\theta^*) = \lambda \theta^*$, and Euler's identity gives $\lambda = K g^* > 0$. g is invariant under the gauge transformations (16), so $\langle \nabla g, \delta\theta \rangle = 0$ for every gauge direction $\delta\theta = (M G_{v,(e)}, -M^\top G_{p,(e)})$, i.e. $\lambda \operatorname{tr}\left((G_{v,(e)} G_{v,(e)}^\top - G_{p,(e)} G_{p,(e)}^\top) M\right) = 0$ for all M; hence $Q_e = 0$. Since g is linear in each core, $\langle G_v, \nabla_{G_v} g \rangle = g^*$, i.e. $\lambda \|G_v\|_F^2 = g^*$. □

Hence (checked to 10^{-15}) the near-origin escape direction is balanced, so of minimal norm in its fibre (Furman et al., 2026, Prop. F.10), and the $O(\sigma^2)$ initial imbalance does not affect it at LO.

Proposition A.10 (Second-order structure at a component). *derived* *new* (for general trees; for binary $N = 4$ the unstable direction toward the top component is Jacot et al., 2022, Lemma 15; strict and non-strict critical points of the loss: Achour et al., 2024) *For a pattern teacher with distinct weights let θ_j be the balanced bond-rank-one configuration of component j on the unit sphere (per-core amplitude $K^{-1/2}$, the pattern of j on the physical legs, channel 1 on every bond), so that $g(\theta_j) = s_j K^{-K/2}$. (a) If some component r has $m_{jr} = 1$, θ_j is not a critical point of g on the sphere. (b) Otherwise, assume that no two components r with $m_{jr} = 2$ share a terminal core. Each such r, with terminal cores v, w, contributes n_{jr} eigenvalues $K^{1-K/2}(s_r - s_j)$ to the Riemannian Hessian of g at θ_j, where $n_{jr} = \chi_e$ if v and w share the bond e and $n_{jr} = 1$ otherwise; all other eigenvalues are negative, except the $\sum_e(\chi_e - 1)$ gauge zero modes. So θ_j is a strict local maximum modulo gauge iff no $m_{jr} = 1$ and every r with $m_{jr} = 2$ has $s_r < s_j$; otherwise it has $\sum_{r:\, m_{jr}=2,\, s_r>s_j} n_{jr}$ unstable directions. Ties are decided at higher order; partners sharing a terminal core: Proposition A.22.*

Derivation. The Riemannian Hessian is $\nabla^2 g - s_j K^{1-K/2} I$. As g is linear in each core, $\nabla^2 g$ has only cross terms between cores $v \neq w$, which contract W^* with the pattern of j outside $S_v \cup S_w$: only components with $D_{jr} \subseteq S_v \cup S_w$ contribute. If $m_{jr} = 1$, $\nabla g(\theta_j)$ has a component along the pattern of r on the terminal core. Component j's own cross terms vanish on the gauge orbit and give $-2s_j K^{1-K/2}$ off it; uncoupled directions give $-s_j K^{1-K/2}$. For $m_{jr} = 2$, rotating G_v, G_w toward the pattern of r (channel c on the bond joining them if adjacent, channel 1 elsewhere) gives the cross term $s_r K^{1-K/2}$ for every c if v, w are adjacent and for $c = 1$ only otherwise; each surviving channel gives the block $K^{1-K/2}\left(\begin{smallmatrix} -s_j & s_r \\ s_r & -s_j \end{smallmatrix}\right)$, with eigenvalues $K^{1-K/2}(\pm s_r - s_j)$. □

Checked on 19 pattern-teacher cases and 7 odeco settings. For odeco teachers m_{jr} is the number of cores carrying sites (N on MPS and HT trees, $3, 4$ on binary $N = 6, 8$, and 2 on binary $N = 4$, whose bottom cores are not adjacent, $n = 1$): every odeco component is a local maximum on MPS, HT and binary $N \geq 6$ trees (*tensor-like*: one basin per component), only the largest on binary $N = 4$ (*matrix-like*); pattern teachers whose components differ on exactly two cores are matrix-like on every tree. Spherical flows from 200 random initializations (odeco teachers, five trees) ended at component configurations *empirical*; other local maxima for non-odeco teachers: Section A.5.

Proposition A.11 (Only the teacher slices move). *proven* *new* (for TTNs) *For an odeco teacher $\nabla_{G_v} g = \sum_r s_r (\bigotimes_{i \in S_v} u_r^{(i)}) \otimes E_v^{(r)}$, $E_v^{(r)}$ the contraction of all other cores with $u_r^{(i)}$ on their sites. Under the LO flow the part of G_v orthogonal on its physical legs to every $\bigotimes_{i \in S_v} u_r^{(i)}$ is therefore frozen; the moving variables are the reduced cores $H_v^{(r)}$ (G_v contracted with the $u_r^{(i)}$ on its physical legs; a core without sites is shared by all components), and $g = \sum_r s_r \mathcal{W}_{\mathrm{red}}(H^{(r)})$, $\mathcal{W}_{\mathrm{red}}$ the scalar contraction of the reduced tree. Under (11) the $H_v^{(r)}$ have i.i.d. $\mathcal{N}(0, \sigma^2/F_v)$ entries, independent across r (a fan-in scalar TTN with per-core scale $\sigma d^{-|S_v|/2}$); on an MPS the components have disjoint moving variables and evolve independently at LO.*

Proof. The physical legs of $\nabla_{G_v} g$ lie in the teacher span, so g depends on G_v only through the $H_v^{(r)}$, which are i.i.d. Gaussian as orthonormal contractions of i.i.d. Gaussians. □

The same holds for pattern teachers (one slice per distinct pattern on each core). Most of $\|\theta_0\|$ is frozen: $\mathbb{E}\|H_v^{(r)}\|_F^2 = \sigma^2 \chi_v d^{-|S_v|}$, against $\mathbb{E}\|G_v\|_F^2 = \sigma^2 \chi_v$.

Corollary A.12 (The Kumar–Haupt time is a lower bound). *proven* *(bounds)* *empirical* *(sizes)* *new* (the lower bound and its sizes; t_{KH} itself is the escape time of Kumar and Haupt, 2025b, text before

Thm 11) *Let the teacher be odeco and let r_{m} be the norm of the moving variables at initialization. Then, for every initialization,*

$$t^* \;\geq\; \frac{K^{K/2-1}}{(K-2)s_1}\, r_{\mathrm{m}}^{-(K-2)} \;\geq\; \frac{K^{K/2-1}}{(K-2)s_1}\, \|\theta_0\|^{-(K-2)} =: t_{\mathrm{KH}}, \tag{32}$$

with equality iff the initial direction already maximizes g; $t_{\mathrm{KH}} = 1/(K(K-2)g_*\|\theta_0\|^{K-2})$, $g_* = s_1 K^{-K/2}$, is, up to $O(1)$, the escape time of Kumar and Haupt (2025b, text before Thm 11) from an initialization along a maximizer. Under fan-in initialization the median of t^*/t_{KH} ranges from 2.5–5.2 (binary, $N=4$, $d=2$–4) to 182–12378 (MPS, $N=8$), growing with d, χ and K.

Proof. By Proposition A.8, $g \leq g_*$ on the unit sphere, so $\ln\|\theta(\tau)\| \leq \ln\|\theta_0\| + Kg_*\tau$ and (29) gives $t^* \geq t_{\mathrm{KH}}$. The moving variables follow the LO flow of the same g restricted to them, so the same argument with their norm $r_{\mathrm{m}} \leq \|\theta_0\|$ gives the first inequality. □

Proposition A.13 (Escape along an imbalanced product path). derived known (the trichotomy: Rawal and DeWeese, 2026, Thm 4.2; new: the TTN reduction and the constants) *If the escape follows a rank-one path whose per-core amplitudes $x_v > 0$ obey $\dot{x}_v = s\prod_{w\neq v} x_w$, then the $x_v^2 - x_w^2$ are conserved and, with $x_v^2 = c_v^2 + y$,*

$$\hat{t} = \frac{1}{2s}\int_0^\infty \prod_{v=1}^{K} (y + c_v^2)^{-1/2}\mathrm{d}y. \tag{33}$$

Let m internal cores start at amplitude γ and the other $K-m$ (the ends) at $c_e \gg \gamma$. As $\gamma \to 0$: for $m=0$ and equal amplitudes c, $\hat{t} = c^{-(K-2)}/((K-2)s)$ (integrating the mode equation of Saxe et al., 2014, eq. 15); for $m=1$ and ends at c, $\hat{t} \to \frac{1}{2}\mathrm{B}(\frac{1}{2}, \frac{K-2}{2})c^{-(K-2)}/s$ (B is Euler's beta function), a finite limit; for $m=2$, $\hat{t} = \ln(1/\gamma)/(s\prod_e c_e) + O(1)$; for $m \geq 3$, $\hat{t} \simeq \gamma^{-(m-2)}/((m-2)s\prod_e c_e)$.

The integral follows from Lemma A.1 ($\dot{y} = 2s\prod_w (y + c_w^2)^{1/2}$). Binary $N=4$ and MPS $N=3$ have $m=1$, MPS $N=4$ $m=2$ (only qualitatively: its internal amplitude is not small), binary $N=8$ and HT $N=4$ $m=3$, the MPS $m = N-2$. The escape time is set by the internal path amplitudes, while $\|\theta_0\|$ is dominated by frozen slices and by internal cores whose norm is spread over $\sim \chi^2$ directions: hence t_{KH} fails most at large K. That the end directions stay frozen is empirical (median cosine 0.92–1.00; a minority rotate strongly). Given it, on binary $N=4$ (reduced network $g = s_1 a^\top C b$, $a, b \in \mathbb{R}^{\chi_b}$, $\chi_b = \min(\chi, d^2)$) the median is $t_1 \approx \pi d/(2s_1\sigma\sqrt{\chi_b})$ for $d \leq \chi_b \leq d^2$, with $\mathrm{sd}\ln\hat{t}_1 \approx 1/(2\sqrt{\chi_b})$ derived new (0.80–1.00 of measured medians empirical).

Learning order is decided in the spherical phase by the basin of the initial direction.

Proposition A.14 (Race on trees whose cores all carry sites). proven *(at LO)* new *Let the teacher be odeco and the tree an MPS. By Proposition A.11 the components evolve independently at LO, and component r blows up at $\sigma^{-(K-2)}\hat{t}_{1,r}/s_r$, where the $\hat{t}_{1,r}$ are i.i.d. copies of the prefactor of Corollary A.5 for a single component of unit weight. The first component learned is $\arg\min_r \hat{t}_{1,r}/s_r$, and for two components with $\varrho = s_2/s_1 < 1$*

$$P(\textit{larger first}) = P\Big(\frac{\hat{t}_{1,2}}{\hat{t}_{1,1}} > \varrho\Big) = \frac{1}{2} + \frac{1}{2}P\Big(\Big|\ln\frac{\hat{t}_{1,2}}{\hat{t}_{1,1}}\Big| < \ln\frac{1}{\varrho}\Big), \tag{34}$$

which does not depend on σ. In the spherical flow the basin of component r is exactly the set of initial directions with $\hat{t}_{1,r}/s_r < \hat{t}_{1,r'}/s_{r'}$ for all $r' \neq r$.

Proof. The components are independent at LO and the $\hat{t}_{1,r}$ are exchangeable. □

The race agreed with gradient flow on 100 of 100 initializations (MPS, $N = 4$). empirical: $P(\text{larger first}) \approx F_{\mathcal{N}}(\ln(1/\varrho)/S)$ within ± 0.05 in all tensor-like settings computed ($F_{\mathcal{N}}$ the standard normal distribution function, S the standard deviation of $\ln(\hat{t}_{1,2}/\hat{t}_{1,1})$), and $\operatorname{sd} \ln \hat{t}_1 \approx c\sqrt{K-2}$ on MPS ($c = 0.27$–0.49). Hence conjecture: $P(\text{larger first}) \to \frac{1}{2}$ as $K \to \infty$ at fixed ϱ, d, χ, and $P \to 1$ needs $d \to \infty$ as well as large χ (compare the largest-first (deflation) order of over-parameterized orthogonal tensor decomposition, proven for a modified gradient flow and observed for plain gradient flow, Ge et al., 2021). When some cores carry no site (binary $N = 8$, HT) the race predicts the first component on 94–100% of initializations empirical.

Proposition A.15 (Matrix-like pairs)**.** *Let the teacher have two components, the smaller j and the larger r, with $m_{jr} = 2$, and let $n = n_{jr}$ and $\varrho = s_j/s_r$. (a)* derived known (binary $N = 4$, the limit $P \to 0$: Jacot et al., 2022, Thm 2; their probability-one proof for depth three assumes a full-rank gradient at the origin) *θ_j is a saddle of the spherical flow with n unstable directions while θ_r is a local maximum (Proposition A.10), so $P(\text{smaller first}) \to 0$ as $\sigma \to 0$. (b)* conjecture *(heuristic)* new *As $\sigma \to 0$,*

$$P(\text{smaller first}) \propto \sigma^{n(1-\varrho)/\varrho}. \tag{35}$$

Heuristic argument for (b). A trajectory that escapes near θ_j passes close to it, where the deviations along the n unstable directions grow in τ at rate $K^{1-K/2}(s_r - s_j)$ while $\mathrm{d}\ln\|\theta\|/\mathrm{d}\tau = K^{1-K/2}s_j$, i.e. like $(\|\theta\|/\|\theta_0\|)^{(1-\varrho)/\varrho}$. As $\|\theta\|/\|\theta_0\| \sim 1/\sigma$ at the escape, all n initial deviations must be below $\sim \sigma^{(1-\varrho)/\varrho}$, which for a smooth density has probability $\propto \sigma^{n(1-\varrho)/\varrho}$. □

Checks of (b): in an LO emulation (MPS, $N = 4$, $n = 1, \ldots, 4$) the slopes $\mathrm{d}\ln P/\mathrm{d}\ln\sigma$ are ordered as n and are 0.67–1.03 of $n(1-\varrho)/\varrho$, steepening toward it as $\sigma \to 0$. For gradient flow (binary $N = 4$, $\varrho = 0.7$) the estimated $P(\text{smaller first}) = 0.155, 0.064, 0.024$ at $\sigma = 0.1, 0.01, 0.001$ (emulation probability times gradient flow's agreement rate) has local slopes 0.38, 0.43 against 0.43 empirical; whether gradient flow has the LO exponent is open. A wrong order is corrected through a strict saddle (Remark A.24); tests in Section 4.5.

A.3 Saddles after the first escape

Setting and assumptions. After the first escape, gradient flow passes near a *learned saddle* θ_S representing the partial fit $W_S = \sum_{j \in S} s_j U_j$. We assume: (A1) whitened inputs; (A2) a pattern teacher with distinct patterns (orthonormal U_r); (A3) θ_S is a balanced ($Q_e = 0$) critical point with bond ranks $r_e = \operatorname{rank}(W_S)^{\langle e \rangle}$, padded with zeros to χ_e (Lemma A.2 lifts criticality from r_e to χ_e; with one learned component θ_S is critical iff no residual has $k = 1$, Proposition A.21(c); small initialization gives $Q_e = O(\sigma^2)$); (A4) $\chi_e \geq r_e + 1$ on every bond whose rank must grow; (A5) leading order in the entries that vanish at θ_S (of size σ_{eff} on the plateau). In a basis of each bond with the learned channels first, entries whose bond indices are all learned are *old*, the others *new*; $c_e = \chi_e - r_e$. Let $\mathcal{R} = W^* - W_S$ and $L_S = \frac{1}{2}\|\mathcal{R}\|_F^2$. With one learned component j, balance forces equal core norms: each core is $q_j = s_j^{1/K}$ times the pattern of j on its sites and channel 1 on its bonds proven.

Expansion around the saddle. For a set $\mathcal{J}$ of cores and perturbations $\delta = (\delta_v)_v$, let $W_{\mathcal{J}}[\delta]$ be the tensor with the cores in $\mathcal{J}$ replaced by the δ_v and the others at θ_S. Multilinearity gives exactly $W(\theta_S + \delta) = \sum_{\mathcal{J}} W_{\mathcal{J}}[\delta]$, $W_\emptyset = W_S$, and

$$L(\theta_S + \delta) = L_S - \sum_{\mathcal{J} \neq \emptyset} \langle \mathcal{R}, W_{\mathcal{J}}[\delta] \rangle + \tfrac{1}{2} \Big\| \sum_{\mathcal{J} \neq \emptyset} W_{\mathcal{J}}[\delta] \Big\|_F^2. \tag{36}$$

Lemma A.16 (Routes). proven new *(i) If δ_v has only new entries, $W_{\{v\}}[\delta_v] = 0$: new entries carry no Gauss–Newton cost at second order. (ii) For new entries, $W_{\mathcal{J}}[\delta] \neq 0$ requires every new index of every δ_v to sit on a bond inside $\mathcal{J}$ and to be matched there; so every bond leaving $\mathcal{J}$ carries a learned index, and every connected component of $\mathcal{J}$ has at least two cores. For a connected set $\mathcal{C}$ (a* route*), $W_{\mathcal{C}}[\delta] = \mathcal{W}_{\mathcal{C}}(\delta) \otimes \bigotimes_{f\in\partial\mathcal{C}} T_f$, where $\mathcal{W}_{\mathcal{C}}(\delta)$ is the small TTN formed by the δ_v (internal bond dimensions c_e), $\partial\mathcal{C}$ the set of bonds leaving $\mathcal{C}$, and T_f the contraction of the learned network beyond f (a map from the r_f learned indices of f to tensors on the sites H_f beyond f). Every proper subset $\mathcal{C}' \subsetneq \mathcal{C}$ has $W_{\mathcal{C}'}[\delta] = 0$:* a route pays no Gauss–Newton cost at the order of its gain.

Proof. A new index of δ_v meets, in the neighbouring core, a perturbation or the zero block of a learned core (Lemma A.2), and in a proper subset of a route some new index meets a zero block. □

Disconnected sets $\mathcal{J}$ also contribute, possibly below the route degree (MPS $N = 5$, $D = \{0, 4\}$: degree 4), but each connected component $\mathcal{J}_i$ pays the Gauss–Newton cost $\frac{1}{2}\|W_{\mathcal{J}_i}\|_F^2$ at the order of the gain; for one learned component and two blocks the quadratic form is positive semidefinite iff $s_j \geq s_r$, the condition of Proposition A.22 derived. In general (one learned component) whether disconnected sets descend before the route is decided by the generalized piggyback operator of Section A.4 proven.

Definition A.17 (Local residual). For a route $\mathcal{C}$, $\mathcal{R}_{\mathcal{C}} := \langle \mathcal{R}, \bigotimes_{f\in\partial\mathcal{C}} T_f \rangle$ (contraction over all sites beyond $\partial\mathcal{C}$) is a tensor with one mode per core of $\mathcal{C}$, made of its physical legs and its learned boundary indices; then $\langle \mathcal{R}, W_{\mathcal{C}}[\delta] \rangle = \langle \mathcal{R}_{\mathcal{C}}, \mathcal{W}_{\mathcal{C}}(\delta) \rangle$.

Proposition A.18 (Escape degree). derived known (mechanism: Kumar and Haupt, 2026, Thm 8 (a lower bound on the plateau, for active weights that start near a local minimum of the restricted loss with a Łojasiewicz inequality), and the accumulated utility of Kunin et al., 2025; new: the degree is set by the tree) *The new entries first lower L at order*

$$k^* = \min\{|\mathcal{C}| : \ \mathcal{C} \text{ a route}, \ \mathcal{R}_{\mathcal{C}} \neq 0\}. \tag{37}$$

At LO (old entries frozen at θ_S) the new entries of a minimizing route follow the gradient flow of a k^*-core TTN that fits the target $\mathcal{R}_{\mathcal{C}}$ from small initialization: the escape from the saddle is the first-escape problem of Section A.2 for the route, with $K \to k^*$ and $W^* \to \mathcal{R}_{\mathcal{C}}$. This needs the old entries to sit at a local minimum of the fixed-rank problem; it fails at the strict saddles of Proposition A.22, where the old entries move first. It also needs the generalized piggyback operator below the learned weight (Theorem A.30); otherwise, from $k^* = 5$ on, a smaller cluster of cores plus old rotations can escape first (Theorem A.31).

Proposition A.19 (Steiner identity). proven *(identity and upper bound on the order; order $= k^*$ under the condition of Theorem A.30)* new (the route is the Steiner subtree of the residual's sites, which can be a proper subtree of a connected component of rank-deficient bonds; descent along a whole such component (model rank below target rank): Furman et al., 2026, Lemma D.8; with several such components the smallest gives the least suppressed escape: their Fig. 4) *Let $S = \{j\}$. For every route $\mathcal{C}$,*

$$\mathcal{R}_{\mathcal{C}} = q_j^{K-|\mathcal{C}|} \sum_{r:\, N(D_{jr})\subseteq\mathcal{C}} s_r \, U_r|_{\mathcal{C}}, \tag{38}$$

where $U_r|_{\mathcal{C}}$ is U_r restricted to the sites of $\mathcal{C}$. Hence $k^* = \min_r |\mathrm{St}(N(D_{jr}))|$; the route of component r has degree $k = |\mathrm{St}(N(D_{jr}))|$ and coupling

$$c_r = s_r \, q_j^{K-k} = s_r \, s_j^{(K-k)/K}. \tag{39}$$

Table 13: Route degree $k = |\mathrm{St}(N(D))|$ (Proposition A.19) for a two-component pattern teacher whose components differ on the sites D (numbered from 0), on the trees of Table 1; $m = |N(D)| = 2$ unless shown. With the larger component learned first, $k = 1$: no plateau; $k = 2$: strict saddle, plateau $\propto \ln(1/\sigma)$; $k \geq 3$: higher-order saddle, plateau $\propto \sigma^{-(k-2)}$.

D ($N = 4$)	binary $K = 3$	MPS $K = 4$	HT $K = 7$	D ($N = 8$)	binary $K = 7$	MPS $K = 8$	HT $K = 15$
$\{0,1\}$	1 ($m = 1$)	2	3	$\{0,1\}$	1 ($m = 1$)	2	3
$\{1,2\}$	3	2	5	$\{1,2\}$	3	2	5
$\{0,2\}$	3	3	5	$\{3,4\}$	5	2	7
$\{0,3\}$	3	4	5	$\{0,4\}$	5	5	7
$\{0,1,2,3\}$	3	4 ($m = 4$)	7 ($m = 4$)	$\{2,5\}$	5	4	7
				$\{1,6\}$	5	6	7
				$\{0,7\}$	5	8	7
				$\{0,\dots,3\}$	3	4 ($m = 4$)	7 ($m = 4$)
				all	7 ($m = 4$)	8 ($m = 8$)	15 ($m = 8$)

The bonds whose rank must grow to add r are exactly the bonds of $\mathrm{St}(N(D_{jr}))$, so k is one plus their number, and $k = 1$ iff D_{jr} lies on a single core. With route entries proportional to t, $L - L_S = -c_r t^k + O(t^{2k})$: the saddle has order at most k^*. No curve descends faster if the generalized piggyback operator stays below s_j (Theorem A.30; always for $k^* = 2$, for $k^* \in \{3,4\}$ with a positive definite fixed-rank block, and for two-component teachers in the correct order); otherwise the order can be smaller, from $k^* = 5$ on (Theorem A.31 and Corollary A.32).

Proof. Beyond a boundary bond f the learned network is $T_f = q_j^{n_f} U_j|_{H_f}$ (n_f cores, sites H_f, $\sum_f n_f = K - |\mathcal{C}|$). As $\mathcal{R} = \sum_{r \neq j} s_r U_r$ and U_r contracted with T_f gives $q_j^{n_f} \prod_{i \in H_f} \langle u_r^{(i)}, u_j^{(i)} \rangle$ ($q_j^{n_f}$ if $\pi_r = \pi_j$ on H_f, else 0), U_r survives iff $N(D_{jr}) \subseteq \mathcal{C}$, and a connected $\mathcal{C}$ containing $N(D_{jr})$ contains its Steiner subtree. The rank statement is Proposition 2.1(b). For the descent curve let δ_v be t times the pattern of r on S_v, with a new channel on the bonds inside $\mathcal{C}$ and channel 1 on $\partial\mathcal{C}$: then $W_{\mathcal{C}}[\delta] = t^k q_j^{K-k} U_r$, every other $W_{\mathcal{J}}[\delta]$ vanishes (Lemma A.16), and $L - L_S = -c_r t^k + \frac{1}{2} q_j^{2(K-k)} t^{2k}$. □

Table 13 lists k for two-component teachers. For Tucker decompositions, Frandsen and Ge (2022, proof of Lemma 7) exhibit, at balanced points, descent of order 2, 3 and 4 when 2, 3 and 4 cores must move (no lower bounds).

Lemma A.20 (Several learned components). proven new *Let E_+ be the set of bonds e with* $\mathrm{rank}(W_S + U_r)^{\langle e \rangle} > \mathrm{rank}(W_S)^{\langle e \rangle}$. *(i) E_+ is connected. (ii) If $E_+ = \emptyset$, one core can produce U_r: $\nabla L(\theta_S) \neq 0$.*

Proof. (i) If the restriction of U_r to one side of a bond lies outside the learned column space, so does its restriction to any larger side. (ii) A walk from any core toward a branch on which U_r is new cannot return, so it ends at a core all of whose branches are old, which alone can produce U_r. □

With several learned components, adding r exactly needs $k_{\mathrm{ex}} = |E_+| + 1$ cores, and $k^* \leq k_{\mathrm{ex}}$ derived, strictly when the learned space of a bond holds a superposition of learned patterns that overlaps $U_r|_{H_f}$ without containing it: a cheaper route then first learns the projection, adding a plateau at the best fit of an intermediate rank profile (*partial learning*, Remark A.27). When $k^* = k_{\mathrm{ex}}$ the coupling is $c_r = s_r \prod_{f \in \partial\mathcal{C}} \|T_f^\top U_r|_{H_f}\|$ in the balanced gauge derived.

Proposition A.21 (Hessian at a balanced learned saddle). proven *(at θ_S)* new (the block structure, the rate and its multiplicity; that a two-core descent curve gives a strict saddle follows from Furman et al.,

2026, Lemmas D.8–D.9, though they do not state it; deep linear networks: Achour et al., 2024) *(a) The Hessian at θ_S is block diagonal between old and new entries; the old (*fixed-rank*) block is the Hessian of the problem with bond ranks fixed at r_e (Gauss–Newton term plus the piggyback couplings of Proposition A.22). (b) The new (*rank-increasing*) block vanishes except on two-core routes: a route $\{u, v\}$ through the bond e contributes $\left(\begin{smallmatrix} 0 & -B \\ -B^\top & 0 \end{smallmatrix}\right)$, $B = \mathcal{R}_{\{u,v\}}$, for each of the c_e new channels, i.e. eigenvalues $\pm\sigma_k(B)$ with multiplicity c_e; with one learned component the negative one is $-\lambda$,*

$$\lambda = \sigma_{\max}(\mathcal{R}_{\{u,v\}}) = s_r\, s_j^{(K-2)/K}. \tag{40}$$

(c) Hence θ_S is a strict saddle iff the fixed-rank block has a negative eigenvalue or a route with $k^* = 2$ exists. If $k^* = 1$ it is not critical (one learned component: $\|\nabla L(\theta_S)\| = s_r q_j^{K-1}$); if $k^* \geq 3$ and the fixed-rank block is positive semidefinite, the Hessian has no negative eigenvalue: a higher-order saddle. (d) The gauge orbit through θ_S has dimension $\sum_e[\chi_e^2 - (\chi_e - r_e)^2]$, and the Hessian kernel has dimension $\sum_e r_e^2 + \dim\theta_{\text{new}} - 2\sum_{k^*=2 \text{ routes}} c_e$ ($\dim\theta_{\text{new}}$: the number of new entries).

Proof sketch. An old–new cross term of $\langle \mathcal{R}, W_{\mathcal{J}} \rangle$ needs an old index to meet a new one, which gives zero; the Gauss–Newton cross term vanishes by Lemma A.16(i). On the new entries the only second-order terms are $-\langle \mathcal{R}, W_{\{u,v\}}[\delta] \rangle$ for adjacent pairs, $\mathcal{R}_{\{u,v\}} = q_j^{K-2} \sum_r s_r U_r|_{\{u,v\}}$ for one learned component (38). Gauge generators on new–new blocks fix θ_S. □

Checked in 21 (tree, D) cases (λ to six digits); λ is also the escape rate of a $k = 2$ route.

Proposition A.22 (Piggyback block and piggyback graph)**.** proven new (the graph criterion on general trees; deep linear networks: Achour et al., 2024, Thm 7; odeco components are the isolated local maximizers of $W^*(u, u, u)$ on the unit ball in the symmetric third-order case: Anandkumar et al., 2014, Thm. 4.2) *Let $S = \{j\}$. A* piggyback direction *keeps the learned bond index and rotates the physical vectors of one terminal core u toward a site pattern w on S_u. (i) For one residual r with $m_{jr} = 2$ and terminal cores u, v the rotation block has diagonal $q_j^{2(K-1)}$, off-diagonal $-s_r q_j^{K-2}$ and eigenvalues $q_j^{K-2}(s_j \mp s_r)$; for $m_{jr} \geq 3$ it is $q_j^{2(K-1)} I \succ 0$. (ii) In general it is*

$$H_{\text{rot}} = q_j^{K-2}(s_j I - \Gamma), \tag{41}$$

where Γ is the weighted adjacency matrix of the piggyback graph*: its nodes are the pairs (u, w) of a terminal core and a rotated site pattern, with one edge between $(u, \pi_r|_u)$ and $(v, \pi_r|_v)$, of weight s_r, for every residual component r with $N(D_{jr}) = \{u, v\}$. Hence $\lambda_{\min}(H_{\text{rot}}) = q_j^{K-2}(s_j - \lambda_{\max}(\Gamma))$, and θ_j is a strict saddle iff $\lambda_{\max}(\Gamma) > s_j$ or a route with $k^* = 2$ exists.*

Proof. Rotating core u by ε_u toward w creates a mixed tensor (pattern w on S_u, j elsewhere) of norm $q_j^{K-1}\varepsilon_u$; those of different nodes are orthonormal, so the Gauss–Newton part is diagonal ($q_j^{2(K-1)} = s_j q_j^{K-2}$). A residual r with terminals $\{u, v\}$ couples two rotations through $-\langle \mathcal{R}, W_{\{u,v\}} \rangle = -s_r q_j^{K-2} \varepsilon_u \varepsilon_v$; residuals with $m_{jr} \geq 3$ enter at order m_{jr}; the radial part is $\succeq 0$. □

With one learned component j and one residual r the saddle is thus not critical for $k = 1$, strict for $k = 2$ ($\lambda_{\min} = -s_r q_j^{K-2}$), of higher order for $k \geq 3$ if $m \geq 3$ or $s_j > s_r$, and strict via the piggyback block ($\lambda_{\min} = -(s_r - s_j) q_j^{K-2}$) for $k \geq 3$, $m = 2$, $s_j < s_r$ proven. A *star* (residuals $r_1, \ldots, r_n$ sharing one node) has $\lambda_{\max}(\Gamma) = (\sum_i s_{r_i}^2)^{1/2}$ and a triangle of equal weights s has $2s$, so θ_j can be strict although every residual is smaller than s_j and every route has $k^* \geq 3$ (checked to six digits). Dynamics empirical: for $j = 0000$ (weight 2) and $r = 1010$, $r' = 1001$ (weight 1.6 each) on MPS $N = 4$ ($\lambda_{\max}(\Gamma) = 2.263$), gradient flow passes near j in 4 of 5 seeds and in all 5 forgets it, learns r, r' and relearns j; with weights 1.2 ($\lambda_{\max}(\Gamma) = 1.70$) it plateaus at j.

Remark A.23 (Binary tree with $N = 4$). proven known (Saxe et al., 2014; Jacot et al., 2022; Achour et al., 2024) With the bottom cores as matrices $\tilde{A}, \tilde{B} \in \mathbb{R}^{d^2 \times \chi}$ (sites $(0,1)$ and $(2,3)$ as rows), $W^{\langle e \rangle} = \tilde{A}\, G_{\text{root}} \tilde{B}^\top$ on both bonds, and L is exactly the loss of a depth-3 linear network fitting the $d^2 \times d^2$ matrix $(W^*)^{\langle e \rangle}$. Its critical points are SVD truncations onto subsets of singular pairs (Achour et al., 2024, Prop. 1); from small initialization the flow leaves the origin along the top singular pair (Jacot et al., 2022, Thm 2), and that the later saddles are the successive truncations is their Conjecture 3. Critical points that select non-top singular values are strict, balanced ones that select the top values are not (Achour et al., 2024, Thm 7, for full-rank targets with distinct singular values; the computation above covers ours).

Remark A.24 (After a wrong first escape). derived new Two-component teachers, smaller component j learned first; ε and b are the piggyback and route amplitudes when the plateau starts. (1) $k \geq 3$, $m = 2$ (matrix-like pair), if the rotation beats the direct add route (always as $\sigma \to 0$): the learned bond index rotates from j to r after $\Delta t \approx \ln(1/\varepsilon)/((s_r - s_j) s_j^{(K-2)/K})$, the loss plateaus at $\frac{1}{2} s_j^2$ with $a_j \approx 0$, and j is relearned through its route after a time $\propto \sigma_{\text{eff}}^{-(k-2)}$, σ_{eff} set by the largest new-index seed. (2) $k = 2$: r is added directly, at the larger rate $s_r s_j^{(K-2)/K}$, and j is kept. (3) $m \geq 3$ (tensor-like pair): no rotation; r is added through its route after $\Delta t \approx b^{-(k-2)}/((k-2) s_r s_j^{(K-k)/K})$, and a_j stays at 1. This needs $N(D_{jr})$ to have no split into two sets of degree below $k/2$; with one (from $k = 5$ on, e.g. HT with $N = 4$ and $D =$ all sites) the order is $2\ell^* < k$ (Corollary A.32), and at small enough σ a cluster escape is expected to make j forgotten and relearned (derived, Section A.4). So a correction through a strict saddle exists only for matrix-like pairs, where the wrong order becomes rare as $\sigma \to 0$ (Proposition A.15). Checked on binary and MPS $N = 4$, before the random streams of teacher and initialization were separated.

Escape times. Let the route of component r have $k = |\mathcal{C}|$ cores and amplitudes b_v (its new entries: the pattern of r on the physical legs, a new channel on internal bonds, channel 1 on $\partial\mathcal{C}$). With the old entries frozen, one learned component and $A := q_j^{K-k}$, (36) and Lemma A.16 give

$$\dot{b}_v = A \prod_{u \in \mathcal{C} \setminus v} b_u \Big(s_r - A \prod_{u \in \mathcal{C}} b_u \Big), \qquad a_r = \frac{A}{s_r} \prod_{u \in \mathcal{C}} b_u, \tag{42}$$

a depth-k mode with target s_r/A that conserves the differences $b_u^2 - b_v^2$.

Proposition A.25 (Escape times with head start)**.** derived known (dormant amplitudes carried across phases: Kunin et al., 2025, Eqs. 2–3 and Sec. 2.2, and, in diagonal linear networks, Pesme and Flammarion, 2023, Algorithm 1; the $k = 2$ head start (weaker directions are $O(\sigma^{1 - s_r/s_j})$ once the stronger one is learned, two-layer linear networks): Zhang et al., 2026, Thm 4; the trichotomy in k: Rawal and DeWeese, 2026, Thm 4.2; $k = K$: the decoupled modes of Saxe et al., 2014) new (only the closed forms for TTN routes, in particular the effective amplitude for $3 \leq k < K$) *Let the initialization be aligned: every core of the learned component j and of the route starts at amplitude σ, and the hanging cores are aligned with the shared pattern. Let $\Delta t_2 = t(a_r = \frac{1}{2}) - t(a_j = \frac{1}{2})$. As $\sigma \to 0$,*

$$k = 2: \quad \Delta t_2 = \frac{(1 - s_r/s_j) \ln(1/\sigma) + O(1)}{s_r\, s_j^{(K-2)/K}}, \tag{43}$$

$$k \geq 3: \quad \Delta t_2 = \frac{\sigma^{-(k-2)}\, (1 - s_r/s_j)}{(k-2)\, s_r\, s_j^{(K-k)/K}} (1 + o(1)), \tag{44}$$

and $\Delta t_2 = O(1)$ (no plateau) for $k = 1$. For $k \geq 3$ the route enters the plateau with the effective amplitude $\sigma_{\text{eff}} = \sigma(1 - s_r/s_j)^{-1/(k-2)}$; for $k = K$ it never sees j, and r escapes at $\sigma^{-(K-2)}/((K-2) s_r)$.

Derivation. The route exists from the start, with the then-small learned cores in its hanging part, so it grows during the rise of j with the coupling $c(t) = s_r q_j(t)^{K-k}$, where $\dot{q}_j \simeq s_j q_j^{K-1}$ while q_j is small. For $k = 2$, $\ln b$ grows by $\int c\,\mathrm{d}t = \int s_r q_j^{-1} \mathrm{d}q_j/s_j = (s_r/s_j)\ln(1/\sigma) + O(1)$, so the route enters the plateau at $\ln b_{\text{eff}} = -(1 - s_r/s_j)\ln(1/\sigma) + O(1)$ and needs a further $\ln(1/b_{\text{eff}})/\lambda$, with λ from (40). For $k \geq 3$ the balanced route obeys $\frac{\mathrm{d}}{\mathrm{d}t} b^{-(k-2)} = -(k-2)c(t)$, so at the end of the rise of j $b_{\text{eff}}^{-(k-2)} = \sigma^{-(k-2)} - (k-2)\int c\,\mathrm{d}t = \sigma^{-(k-2)}(1 - s_r/s_j) + O(1)$, and the plateau lasts $b_{\text{eff}}^{-(k-2)}/((k-2)s_r s_j^{(K-k)/K})$; as $q_j^{K-k}\mathrm{d}t = q_j^{1-k}\mathrm{d}q_j/s_j$, this head start accrues while $q_j \approx \sigma$. □

The factor $1 - s_r/s_j$ is the head start (without it the $k = 2$ slope would be $1/\lambda$; Section 4.4). Checked to 10^{-4} with aligned initialization; with fan-in initialization the $k = 2$ slope is 1.066 against 1.065, per-seed exponents -0.97 to -1.02 ($k = 3$) and -1.99 to -2.00 ($k = 4$). An unbalanced route with positive product has $\Delta t = \int_0^\infty \mathrm{d}\beta/(2c\prod_{v\in\mathcal{C}}(b_v^2 + \beta)^{1/2})$ derived (as in Proposition A.13); a negative product lengthens the plateau. With random initialization the prefactors are set by the route amplitudes at each plateau start (seed-to-seed factors above 20 for $k \geq 4$) empirical.

Proposition A.26 (Signatures of the route degree)**.** derived known (the exponent $2 - 2/k$ of a balanced product of k factors: Arora et al., 2019b; Razin et al., 2022; new: its use to read k off TTN plateaus) *(i) Per seed (fixed initialization direction, σ varied), $\Delta t_2 \propto \sigma^{-(k-2)}$ for $k \geq 3$. (ii) On a balanced route,* (42) *gives $\dot{a}_r = kA^{2/k} s_r^{2-2/k} a_r^{2-2/k}(1 - a_r)$, so the local growth exponent is*

$$\gamma(a_r) = \frac{\mathrm{d}\ln \dot{a}_r}{\mathrm{d}\ln a_r} = 2 - \frac{2}{k} - \frac{a_r}{1 - a_r}, \tag{45}$$

that is $1, \frac{4}{3}, \frac{3}{2}, \frac{8}{5}$ for $k = 2, \ldots, 5$ as $a_r \to 0$. The route is balanced only once $b^2 \gg |b_u^2 - b_v^2| = O(\sigma^2)$, so k is read in the window $a_r \in [10^{-4}, 10^{-2}]$ (a smaller γ at smaller a_r indicates an unbalanced route). (iii) At a plateau point (route amplitudes small but nonzero) each active route contributes one negative Hessian direction modulo gauge, with

$$\lambda_{\min} \simeq -s_r s_j^{(K-2)/K} \quad (k = 2), \qquad \lambda_{\min} \simeq -(k-1)\, s_r A\, b^{k-2} = O(\sigma_{\text{eff}}^{k-2}) \quad (k \geq 3), \tag{46}$$

where $b = (s_r a_r/A)^{1/k}$ (for $k \geq 3$ on a balanced route). The count of one negative eigenvalue per route holds for $k \leq 3$ only: at $k = 4$ a second negative eigenvalue of the same order appears empirical (heuristic mechanism: new-index pairs that connect to learned structure add indefinite terms of order σ^2, the same order as the route eigenvalue).

In (iii) the route block of the Hessian is $-s_r A b^{k-2}(\mathbf{1}\mathbf{1}^\top - I)$, whose $k - 1$ positive directions are gauge. Checked on MPS $N = 4$ (γ within 0.01; $|\lambda_{\min}|/\lambda_{\max} \propto \sigma^0, \sigma^1, \sigma^2$ for $k = 2, 3, 4$; (46) within a factor 1.4 for $k = 3$, 3.5 for $k = 4$), so the saddle type is read from the σ-scaling of $\lambda_{\min}$ or from γ.

Remark A.27 (The whole sequence). derived at LO for each step; conjecture (heuristic) as an algorithm. new (1) The first escape ends near a critical point of the bond-rank-one problem; it is a single component j only if θ_j is a local maximum of the spherical problem (Proposition A.10, with the piggyback graph when several partners share a core); a star with $\lambda_{\max}(\Gamma) > s_j$ sends the flow past j toward a superposition. (2) At a later saddle whose fixed-rank block has a negative eigenvalue (wrong order, stars) the next event is a rotation of the learned structure or a $k = 2$ route, whichever is more unstable, in time $\propto \ln(1/\sigma)$ (Proposition A.18 does not apply); otherwise it is the route of smallest k^* (logarithmic beats power law, smaller powers beat larger ones), ties being decided by the coupling and head start $c\, b_{\text{eff}}^{k-2}$, dominated for $k \geq 3$ by the random route amplitudes, provided the generalized piggyback operator stays below the learned weight; if a level $\ell^* < k^*/2$ exceeds it, the order $2\ell^*$ competes with k^* (Corollary A.32; a leading-order race). (3) The escape raises the

bond rank by one on the bonds inside the route, and the next saddle is the best approximation of W^* with the new rank profile (for pattern teachers without superpositions, the partial sum of the learned components), whose loss is the plateau level. Since k^* depends on the tree, *locality beats weight* (teacher B4, Section 4.4). The number of cores that must change at once plays the role of the leap of Abbe et al. (2023) (the number of new support coordinates that SGD must align with at once, which sets a runtime $\tilde{\Theta}(d^{\max(L,2)})$ in the input dimension d for a leap L in their conjecture; Furman et al., 2026, Remark D.10, draw the same analogy for the order of their descent curves); specific to the TTN is that it is set by the tree, and that the plateau scales with the initialization, $\sigma^{-(k-2)}$, not with the dimension.

Proposition A.28 (Conservation-law trap)**.** derived *(mechanism, in a reduced model)* new (the route-end condition; in the landscape, a point that is not of minimum norm, with dead leaf columns and live core entries, can be a spurious local minimum: Furman et al., 2026, Thm G.1) *Let a route end at a leaf core, on the bond e (child v, parent p); let ν be the new channel used by the route, $Q_{\nu\nu} = \nu^\top Q_e \nu$, x the leaf's new column, and split the parent's slice along ν into its route entry y_{route} and the* visible *entries y_{vis}, which connect through the other side to learned structure. Conservation gives $\|x\|^2 - \|y_{\text{vis}}\|^2 - \|y_{\text{route}}\|^2 = Q_{\nu\nu}$. On the plateau the Gauss–Newton term makes x and y_{vis} decay together, at a rate of order $|Q_{\nu\nu}| = O(\sigma^2)$, so the leaf column decays to zero iff the end is* unfavourable*, $Q_{\nu\nu} + \|y_{\text{route}}\|^2 < 0$. The route's drive on x is proportional to the product of the other route amplitudes, so while another end is favourable it pins x at a positive value and the route escapes, slowly. In the reduced model a permanent stall ($x \to 0$ down to roundoff) therefore requires every end of the route to be unfavourable.*

The dead state is linearly stable if $\gamma_1\gamma_2 > \kappa^2$, with decay rates $\gamma_i \sim |Q_i| \sim \sigma^2$ and mutual drive $\kappa \sim \sigma^{k-2}$ (conjecture, heuristic): stable as $\sigma \to 0$ for $k \geq 5$, never for $k = 3$. On HT leaf bonds ($\chi_v = d = 2$ below a parent bond of dimension 3) fan-in initialization gives $Q_{\nu\nu} = \sigma^2(X_2/2 - X_6/4)$ (X_n independent chi-squared, n degrees of freedom; ν independent of the initialization), so $\mathbb{E}Q_{\nu\nu} = -\sigma^2/2$; then $P(Q_{\nu\nu} > 0) \approx 0.30$ and about half of the two-ended routes have both ends unfavourable (conjecture, heuristic). Data empirical (HT $N = 4$, teacher B4, $\sigma = 0.2$, nine fan-in initializations): all 6 runs with all ends unfavourable stalled with dead leaves, the 2 with one favourable end had the other end pinned; balanced initialization ($Q_e = 0$, same W_0) kept the routes alive (3 of 3, before the random streams were separated). conjecture: balanced initialization always avoids the trap (a property of unbalanced initialization, not of the landscape; in another model, four-layer matrix factorization of a target with equal singular values, exactly balanced initialization fails to converge with probability at least $\frac{1}{2}$, Luo et al., 2025, Thm 2), and for $k \geq 5$ the stalled fraction tends to P(all ends unfavourable) as $\sigma \to 0$. The locality experiments contradict parts of this picture (dead runs with a live route end, no stall in 144 fan-in HT4 runs with $k = 5$; Section 4.6).

A.4 Saddle order: lower bound and sharpness

Proposition A.19 gives a descent curve of order k^* at a learned saddle, an upper bound on the saddle order (as Lemma D.8 of Furman et al., 2026 does for a whole connected component of rank-deficient bonds). Here we ask when no curve descends faster. Lower bounds on saddle orders are known in special cases only: in deep linear networks the strict/non-strict classification (Achour et al., 2024, Thm 7) (order 2 versus more than 2), and for one parity saddle of a binary tree (Furman et al., 2026, Prop. 5.3).

Setting. One learned component j: θ_S represents $W_S = s_j U_j$ (balanced or not), $\mathcal{R} = W^* - W_S$ is any residual with $\langle U_j, \mathcal{R}\rangle = 0$ (a pattern teacher only where stated), inputs are whitened. The

saddle order of θ_S is the largest p with $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^p$ near θ_S; measuring the distance to the local gauge orbit instead gives the same p, because the gauge acts linearly and preserves L. Let P_v project onto the physical vector of core v in U_j (the pattern of j on S_v), $P_v^\perp = I - P_v$, and for $B \subseteq \mathcal{V}$ let $\Pi_B = \bigotimes_{v\in B} P_v^\perp \otimes \bigotimes_{v\notin B} P_v$. These 2^K *sector* projectors are orthogonal and sum to the identity, and the sector $\mathcal{H}_B = \mathrm{range}(\Pi_B)$ satisfies $\mathcal{H}_{B_1} \otimes \mathcal{H}_{B_2} \cong \mathcal{H}_{B_1 \sqcup B_2}$ for disjoint sets. θ_S is critical iff $\Pi_{\{v\}}\mathcal{R} = 0$ for every core v. The *degree* of a set is $\deg B = |\mathrm{St}(B)|$, the number of cores of its Steiner subtree, and $k^* = \min\{\deg B : \Pi_B\mathcal{R} \neq 0\}$ (equal to (37) for a pattern teacher, where $\Pi_B\mathcal{R} = \sum_{r:N(D_{jr})=B} s_r U_r$). Let $\mathcal{L}_\ell = \{B \neq \emptyset : \deg B \leq \ell\}$.

Definition A.29 (Generalized piggyback operator)**.** $\Gamma^{(\ell)}$ is the symmetric operator on $\bigoplus_{B\in\mathcal{L}_\ell} \mathcal{H}_B$ whose block (B_1, B_2) is the bilinear form $(x, y) \mapsto \langle \Pi_{B_1\sqcup B_2}\mathcal{R},\, x \otimes y\rangle$ if $B_1 \cap B_2 = \emptyset$ and zero otherwise; $\lambda_\ell = \lambda_{\max}(\Gamma^{(\ell)})$, nondecreasing in ℓ. $\Gamma^{(1)}$ is the piggyback matrix Γ of Proposition A.22. For a pattern teacher with all $s_r > 0$, $\Gamma^{(\ell)}$ is, in the pattern basis, the weighted adjacency matrix of the *generalized piggyback graph*: its nodes are pairs (B, w) of a set $B \in \mathcal{L}_\ell$ and a pattern w on its sites that differs from j on every core of B, and every residual r adds an edge of weight s_r between $(B_1, \pi_r|_{B_1})$ and $(B_2, \pi_r|_{B_2})$ for every split $N(D_{jr}) = B_1 \sqcup B_2$ with $B_1, B_2 \in \mathcal{L}_\ell$. A node with $|B| \geq 2$ is reached by new entries on $\mathrm{St}(B)$, a single core by an old rotation (Proposition A.22).

Theorem A.30 (Lower bound)**.** proven new (lower bounds were known for deep linear networks, Achour et al., 2024, Thm 7 (order 2 or more), and for one parity saddle, Furman et al., 2026, Prop. 5.3) *Let θ_S be critical. (a) If $\lambda_{\lceil k^*/2\rceil - 1} < s_j$, there are $C, r_0 > 0$ with $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^{k^*}$ for $\|\theta - \theta_S\| < r_0$; with the route of Proposition A.19 ($\chi_e \geq 2$ on the bonds of one minimal Steiner subtree) the saddle order is exactly k^*. (b) (Level version.) If $2 \leq \ell \leq \lceil k^*/2\rceil$ and $\lambda_{\ell-1} < s_j$, then $L(\theta) - L_S \geq -C\|\theta - \theta_S\|^{\min(k^*, 2\ell)}$ near θ_S.*

Theorem A.31 (Sharpness)**.** proven new *Let the teacher be a pattern teacher with all $s_r > 0$, let $\ell < k^*/2$, and let X be a set of nodes of the generalized piggyback graph on $\mathcal{L}_\ell$ whose adjacency matrix has spectral radius $\rho_X > s_j$. Assume the index condition: on every bond e, $\chi_e - 1$ is at least the number of nodes $(B, w) \in X$ with $|B| \geq 2$ and e a bond of $\mathrm{St}(B)$. Then there is a curve with $\|\theta(t) - \theta_S\| \leq t$ and $L(\theta(t)) - L_S \leq -ct^{2\ell}$: the saddle order is at most $2\ell < k^*$.*

Proof sketch of Theorem A.30. (Full proof: `notes/theory_saddle_order.md`, sections 3.1-3.5.) Let $\omega = \langle U_j, W\rangle$ ($= s_j$ at θ_S) and $\mu_B = \Pi_B W/\omega \in \mathcal{H}_B$. Sector orthogonality gives the exact identity

$$L - L_S = \tfrac{1}{2}(\omega - s_j)^2 + \sum_{B\neq\emptyset} \Big[\tfrac{\omega^2}{2}\|\mu_B\|^2 - \omega\langle \Pi_B\mathcal{R},\, \mu_B\rangle\Big]. \tag{47}$$

Families $(f_B)_{B\subseteq\mathcal{V}}$, $f_B \in \mathcal{H}_B$, form a commutative algebra under $(f * g)_B = \sum_{B_1\sqcup B_2=B} f_{B_1} \otimes g_{B_2}$; the cumulants $\kappa = \log \mu$ are given by the set-partition moment-cumulant formulas. *Locality of cumulants:* for $|B| \geq 2$, $\|\kappa_B\| \leq C_0 \prod_{v\in\mathrm{St}(B)} \|\nu_v\|$, ν_v the new entries of core v. Indeed, if $\nu_v = 0$ the contraction factorizes at v, μ is a $*$-product of factors supported on $\{v\}$ and on the branches at v, its logarithm is a sum of such factors, and none contains B; analyticity turns this vanishing into the product bound. Now fix ℓ, call a set *low* if its degree is at most $\ell - 1$ and *high* otherwise, and let $\varepsilon = \|\theta - \theta_S\|$. High cumulants are $O(\varepsilon^\ell)$, low sets carry no residual ($\ell - 1 < k^*$), and the low part of the Gauss–Newton sum is at least $(1 - C\varepsilon)\|\kappa_{\mathrm{low}}\|^2$. In the gain, the terms with two low cumulants sum exactly to $\frac{1}{2}\langle \kappa_{\mathrm{low}}, \Gamma^{(\ell-1)}\kappa_{\mathrm{low}}\rangle$; every other term contains a single cumulant on a set of degree at least k^* ($O(\varepsilon^{k^*})$), or two high cumulants ($O(\varepsilon^{2\ell})$, after Young's inequality for one low and one high), or is ε times the quadratic form. So $L - L_S \geq \frac{\omega}{2}(s_j - \lambda_{\ell-1} - O(\varepsilon))\|\kappa_{\mathrm{low}}\|^2 - O(\varepsilon^{\min(k^*, 2\ell)})$, which gives (b); (a) is $\ell = \lceil k^*/2\rceil$. Balance, pattern structure, bond dimensions and the gauge are not used. □

Proof sketch of Theorem A.31. (Full proof: section 4 of the note.) Take a Perron vector $x \geq 0$, $\|x\| = 1$, of the adjacency matrix on X. For $0 < \eta \leq 1$, rotate each single core v of a node $(\{v\}, w)$ by $\eta x_{v,w}$ toward w (old entries), and give each node $n = (B, w)$ with $|B| \geq 2$ a *polymer*: new entries of size $q_j(\eta x_n)^{1/|\mathrm{St}(B)|}$ on the cores of $\mathrm{St}(B)$, with the pattern w on B, the pattern of j elsewhere, and a spare index on the bonds of $\mathrm{St}(B)$ that no other node of X uses there (the index condition). A nonzero contraction term then uses whole polymers with disjoint Steiner subtrees, so $\omega = s_j$ exactly, the Gauss–Newton term at order η^2 is $\frac{s_j^2}{2}\|x\|^2$, and the gain at order η^2 comes exactly from the edges of the graph, $\frac{s_j}{2} x^\top A_X x = \frac{s_j}{2}\rho_X$. Hence $L - L_S = -\frac{s_j}{2}(\rho_X - s_j)\eta^2 + O(\eta^3)$ while $\|\theta - \theta_S\| = O(\eta^{1/\ell})$. □

Corollary A.32 (When the saddle order is k^*)**.** proven new *With one learned component j and the route of Proposition A.19: (i) $k^* = 2$: order 2. (ii) $k^* \in \{3, 4\}$: then $\Gamma^{(\lceil k^*/2 \rceil - 1)} = \Gamma$, so the order is k^* if $\lambda_{\max}(\Gamma) < s_j$, i.e. if the fixed-rank block is positive definite modulo gauge, and 2 (strict saddle) if $\lambda_{\max}(\Gamma) > s_j$; at equality it is 3 for $k^* = 3$ (positive semidefinite Hessian) and open for $k^* = 4$. (iii) Two-component teachers (one residual r): every node of the graph has at most one edge, so $\lambda_\ell \in \{0, s_r\}$. In the correct order ($s_r < s_j$) the order is k^* on every tree. In the wrong order it is k^* if no split $N(D_{jr}) = B_1 \sqcup B_2$ has both degrees below $k^*/2$, and otherwise exactly $2\ell^* < k^*$, ℓ^* the smallest $\max(\deg B_1, \deg B_2)$ over such splits (lower bound by Theorem A.30(b) at level ℓ^*, upper bound by Theorem A.31). (iv) In general, for a pattern teacher with the index condition, the order is $\min(k^*, 2\ell^*)$, ℓ^* the first level below $k^*/2$ with $\lambda_{\ell^*} > s_j$, provided $\lambda_{\ell^*-1} < s_j$.*

So the order can drop below k^* only for $k^* \geq 5$ (given a positive definite fixed-rank block; the boundary case $k^* = 4$, $\lambda_{\max}(\Gamma) = s_j$, is open), and only through a residual with at least three terminal cores, two of which lie in a set of degree below $k^*/2$. Two examples on the HT tree with $N = 4$ ($K = 7$) and wrong order: three-site D have $k^* = 6$ and no low split (two leaves already have degree $3 = k^*/2$), so the order is 6; $D = \{0, 1, 2, 3\}$ has the split into the two leaf pairs (degrees 3, 3), so the order is 6, not $k^* = 7$. The binary tree with $N = 8$ and $D =$ all sites is the same case.

Counterexample with a positive semidefinite Hessian. proven (order) empirical (numerics) MPS with $N = 6$, $d = \chi = 2$, $j = 000000$ ($s_j = 1$), residuals $r_1 = 110001$ and $r_2 = 110010$ with weight 0.8 each. The route of r_2 has $k^* = 5$ (that of r_1 has 6); the gradient vanishes and the Hessian is positive semidefinite (eigenvalues 0, 1, 6), since no residual has two terminal cores and $\Gamma^{(1)} = 0$. At level 2 the node $(\{0, 1\}, 11)$ is joined to $(\{5\}, 1)$ through r_1 and to $(\{4\}, 1)$ through r_2, a star with $\lambda_2 = 0.8\sqrt{2} = 1.131 > 1$: by Theorem A.30(b) and Theorem A.31 the order is exactly $4 < k^*$. The descent combines new entries of size t on the bond between cores 0 and 1, which point at the *non-residual* pattern 110000, with old rotations of cores 4 and 5 of size t^2. Measured minimum over the sphere of radius t: $-0.0350\, t^4$ (slope 4.000), against $-(2 \cdot 0.8^2 - 1)/8 = -0.0350$; with weights 0.6 ($\lambda_2 = 0.85$) the order is 5 ($-0.0107\, t^5$, the route of r_2). Single wrong-order residuals give counterexamples to "order $= k$" for two-component teachers: $r = 110001$ with weight 1.5 on the same MPS ($k^* = 6$, order 4) and $r = 1111$ with weight 1.5 on the HT tree with $N = 4$ ($k^* = 7$, order 6, $-t^6/432$). The orders predicted by Theorems A.30 and A.31 were measured on 32 instances ($k^* = 2$ to 7, MPS, binary and HT trees, pattern and random residuals), 12 of them with independent code, with slopes within 0.003 of the prediction (one slowly converging case). Dynamics derived: where Theorem A.31 applies at level ℓ, the old rotations follow the cluster adiabatically and the cluster obeys $\dot{b} = c\, b^{2\ell-1}$, so the escape takes $\propto \sigma^{-(2\ell-2)}$ instead of $\sigma^{-(k^*-2)}$ once it beats the route. On the counterexample, gradient flow from θ_S plus a perturbation of size σ escapes in $\propto \sigma^{-2}$ (local exponents 1.98 to 2.04 at the smallest σ, 5 seeds) along 110000, then forgets j and relearns

it later; from small initialization only some seeds visit θ_S, and in those the σ^{-2} law sets in at a seed-dependent σ empirical.

A.5 Teachers that are not orthogonally decomposable

For a general teacher the saddles are no longer partial sums of components. $\mathcal{M}_{\mathbf{r}}$ is the manifold of tensors with bond ranks exactly $\mathbf{r} = (r_e)_e$ (Holtz et al., 2012b; Uschmajew and Vandereycken, 2013). For $W \in \mathcal{M}_{\mathbf{r}}$ and $e = e_v$, $p = p(v)$, U_e and V_e are the column and row spaces of $W^{\langle e \rangle}$, the environment spaces of e are $E_e^- = (\bigotimes_{c \in \mathrm{ch}(v)} U_c) \otimes \mathbb{R}^{d_v}$ and $E_e^+ = (\bigotimes_{c \in \mathrm{ch}(p) \setminus \{v\}} U_c) \otimes \mathbb{R}^{d_p} \otimes V_p$ ($d_v = \prod_{i \in S_v} d_i$; no V_p at the root), and $\widetilde{W}_e^* := P_{E_e^-} (W^*)^{\langle e \rangle} P_{E_e^+}$ is the *environment-projected teacher*.

Proposition A.33 (Critical points of the rank-constrained problem)**.** proven known (in substance, for the Tucker format: the first-order optimality conditions of best multilinear-rank approximation, Eldén and Savas, 2011 (HOOI, De Lathauwer et al., 2000, iterates their dominant-subspace version), and the fixed points of one-site ALS, Holtz et al., 2012a; tangent spaces: Uschmajew and Vandereycken, 2013) *$W \in \mathcal{M}_{\mathbf{r}}$ is a critical point of L restricted to $\mathcal{M}_{\mathbf{r}}$ iff, for every bond e,*

$$W^{\langle e \rangle} = P_{U_e} \widetilde{W}_e^* = \widetilde{W}_e^* P_{V_e}, \tag{48}$$

i.e. iff U_e is spanned by r_e left singular vectors of $\widetilde{W}_e^*$ (not necessarily the top ones) and $W^{\langle e \rangle}$ is the corresponding SVD truncation, i.e. iff W is a fixed point of one-site alternating least squares. Padded with zeros it is critical for every $\chi_e \geq r_e$ (Lemma A.2).

Proof sketch. At full bond ranks the differential with respect to G_v spans $\mathcal{V}_v = E_e^- \otimes V_e$ ($e = e_v$), so W is critical iff $P_{\mathcal{V}_v} W^* = W$ for every v, i.e. $\widetilde{W}_e^* P_{V_e} = W^{\langle e \rangle}$ at v and $P_{U_e} \widetilde{W}_e^* = W^{\langle e \rangle}$ at p. If $P_U A = A P_V = X$ has rank $\dim U$, $P_U A A^\top = X X^\top$ is symmetric: U is a left singular space of A. □

Proposition A.34 (When is the hierarchical SVD critical?)**.** proven *(a, b)* derived *(c)* new (the ε^4 gap; the HSVD is Grasedyck, 2010) *Let W_H be the HSVD truncation of W^* at ranks $\mathbf{r}$. (a) W_H is critical iff its frames satisfy (48), which generic teachers violate (parts discarded at other bonds leak into each bond's environment). (b) If the best rank-r_e approximation of W^* across every bond e is one and the same tensor W (consistent truncations, e.g. an odeco teacher truncated to the same components on every bond, or binary $N = 4$), then $W_H = W$ is critical and the global minimizer over bond ranks at most $\mathbf{r}$. (c) If $W^* = W + \varepsilon E$ with W consistent and E generic, the nearby critical point W_c has $\|W_H - W_c\|_F = O(\varepsilon^2)$ and $L(W_H) - L(W_c) = \frac{1}{2}\|W_H - W_c\|_F^2 (1 + O(\varepsilon)) = O(\varepsilon^4)$.*

For (b), $\|W^* - X\|_F \geq \|W^* - W\|_F$ for every X with one bond rank at most r_e; for (c), both sets of frames are first-order rotations of those of W with the same driving terms, $W^* - W_c$ is normal to $\mathcal{M}_{\mathbf{r}}$ at W_c and $W_H - W_c$ tangent up to curvature (checked for $\varepsilon \leq 0.2$).

The rank-one plateau. derived new The origin plateau ends along a KKT direction of g, and the global maximum of g has bond rank one (Proposition A.8). That local maxima do too, and that the flow reaches one, are conjectures (true at every rank-one plateau computed); then the next plateau is a critical point of the bond-rank-one problem, not an HSVD truncation. If every core carries at most one site (MPS, HT) it is a local best rank-one approximation $\bar{s} \bigotimes_i w_i$, $\bar{s} = \langle W^*, \bigotimes_i w_i \rangle$, at level $\frac{1}{2}(\|W^*\|_F^2 - \bar{s}^2)$; on binary $N = 4$ it is the top singular pair of the matricization. In the cases computed it lies 6–12% below the HSVD truncation empirical, which is not general.

Proposition A.35 (Two non-orthogonal components). proven *(stationarity; threshold among symmetric critical points)* derived *(fold)* new (elementary) *Let $W^* = s_1 U_1 + s_2 U_2$ with N sites, $s_1 \geq s_2$ and overlaps $\langle u_1^{(i)}, u_2^{(i)} \rangle = \cos\phi_i \in [0, 1)$, at most one of them zero. Every rank-one critical point has w_i in the plane of $u_1^{(i)}, u_2^{(i)}$, at angle ψ_i from $u_1^{(i)}$, with value $s_1 \prod_i \cos\psi_i + s_2 \prod_i \cos(\phi_i - \psi_i)$. For equal weights and overlaps $\cos\phi$, the symmetric point $\psi_i = \phi/2$ has Hessian eigenvalues $-2c^{N-2}[c^2 - (N-1)(1-c^2)]$ along $(1, \ldots, 1)$ and $-2c^{N-2}$ on its complement ($c = \cos\frac{\phi}{2}$, unit weight), so there are two local maxima, one near each component, iff $\cos\phi < (N-2)/N$, and one merged maximum otherwise (asymmetric maxima excluded numerically). For $\varrho = s_2/s_1 < 1$ the maximum near U_2 disappears at a fold, $\cos\phi_{\text{fold}} \approx \frac{N-2}{N-1}(\frac{\varrho}{N-1})^{1/(N-2)}$ for small ϱ (1–13% below the exact values). The rank-one plateau is never one of the components.*

Proposition A.36 (Rates at the rank-one saddle of an MPS). derived *(rates)* proven *(bound)* new *Let the student be an MPS ($K = N$, core k carries site k) at the balanced, zero-padded rank-one critical point $W_1 = \bar{s} \bigotimes_i w_i$, a local maximum of the rank-one problem, and let $M_e = W^* \times_{i \notin \{k,k+1\}} w_i$ be the $d \times d$ matrix of the bond e between cores k and $k+1$. Then (w_k, w_{k+1}) is the top singular pair of M_e (singular value $\bar{s}$), the route block of e is $\bar{s}^{(N-2)/N}(M_e - \bar{s}\, w_k w_{k+1}^\top)$, with Hessian eigenvalues $-\bar{s}^{(N-2)/N}\sigma_m(M_e)$, $m = 2, \ldots, d$ (each $\chi_e - 1$ times), and the escape rate of bond e is*

$$\gamma_e = \bar{s}^{(N-2)/N}\, \sigma_2(M_e), \qquad \text{with} \quad \sigma_m(M_e) \leq \sigma_m((W^*)^{\langle e \rangle}) \tag{49}$$

by interlacing (M_e compresses $(W^*)^{\langle e \rangle}$ by isometries).

For two components with $d = 2$, $\sigma_2(M_e) = \det M_e/\bar{s}$ with $\det M_e = s_1 s_2 \sin\phi_k \sin\phi_{k+1} a_e b_e$, $a_e = \prod_{i \neq k,k+1} \cos\psi_i$, $b_e = \prod_{i \neq k,k+1} \cos(\phi_i - \psi_i)$: *the bond that grows first is the one whose two sites differ most while the other sites agree most*, not the one with the largest teacher Schmidt value (checked to four digits).

Corollary A.37 (Second escape). derived known (the degree-two case of the activation rule of alternating gradient flows, with carried-over amplitudes: Kunin et al., 2025, Eq. 3) new (the TTN rates: Proposition A.25 with $s_j \to \bar{s}$ and $s_r \to \sigma_2(M_e)$) *Assume $\sigma \to 0$, balanced cores and route projections $\propto \sigma$ at initialization, and let $\kappa_e = \sigma_2(M_e)/\bar{s}$. During the first escape the route of bond e grows by the factor $\sigma^{-\kappa_e}$, so the first bond to gain rank is $e^* = \arg\min_e h_e = \arg\max_e \sigma_2(M_e)$, and*

$$t_{e^*} - t_1 = h_{e^*} \ln(1/\sigma) + O(1), \qquad h_e = \frac{1 - \kappa_e}{\gamma_e}. \tag{50}$$

Here t_e is when σ_2 across e first exceeds $f \in [0.03, 0.1]$ times the teacher's (slopes checked to 1%).

Proposition A.38 (Later escapes on an MPS: an accumulated-utility race). derived *(as $\sigma \to 0$, under (A-carry))* known (the rule: alternating gradient flows, Kunin et al., 2025; new: the TTN routes and their rates) *Give every pending two-core route ρ the log-amplitude $\ell_\rho = \ln x_\rho / \ln(1/\sigma) \in [-1, 0]$ and use the clock $\vartheta = (t - t_1)/\ln(1/\sigma)$. Then $\ell_\rho(0) = -1 + \kappa_\rho$; while the flow sits at the saddle S_n, $\mathrm{d}\ell_\rho/\mathrm{d}\vartheta = \gamma_\rho(S_n)$, the rate of the route at the exact critical point S_n; ρ escapes when $\ell_\rho = 0$, and the flow moves in time $O(1)$ to S_{n+1}, with the rank of its bond raised by one. (A-carry): at each transition a pending route's amplitude projects onto its new unstable direction with a σ-independent factor $p \neq 0$.*

Event times are $t_e - t_1 = \vartheta_e \ln(1/\sigma) + C_e + o(1)$ with initialization-dependent offsets C_e (up to about 50 time units), so near-tied later escapes come in a seed-dependent order at accessible σ (conjecture, heuristic). Unlike greedy low-rank learning (Li et al., 2021), the race keeps memory across saddles. Checked (two components, overlaps $(0.2, 0.5, 0.7, 0.3)$): predicted bond order 0, 2, 1 with slopes 6.823, 12.178, 14.467, confirmed to 0.5% at $\sigma \leq 10^{-6}$ in 7 of 8 seeds.

Generic teachers: dependence on the tree. derived new At any critical point the local residual of an adjacent pair $\{u, v\}$ through the bond e annihilates the learned subspaces on both sides, so it lives on complements of dimensions $\mathsf{d}_u - r_e$, $\mathsf{d}_v - r_e$ (d_u: the dimension of the legs of u other than e), where a generic teacher makes it nonzero: *a route with $k^* = 2$ (strict saddle, logarithmic escape) exists whenever some adjacent pair has $\mathsf{d}_u, \mathsf{d}_v > r_e$* (and $\chi_e > r_e$). (i) On an MPS every core has a physical leg, so a generic teacher gives one polynomial plateau (at the origin) and then only logarithmic ones, with ranks growing bond by bond (consistent with the single long plateau followed by an abrupt, nearly simultaneous rise of the leading unfolding singular values seen with gradient descent on tensor-train completion, Milanesi et al., 2021, Fig. 3(b)). (ii) On binary and HT trees the internal cores carry no site and, at the rank-one saddle, have rank-one outer legs, so the cheapest routes join two sibling bottom cores through their parent ($k = 3$): escapes from the rank-one saddle are polynomial even for generic teachers, and the root bipartition grows only after both halves (bottom-up order); two-core routes can open later on bottom edges, never for the root bipartition. (iii) Odeco and pattern teachers suppress two-core routes by orthogonality, so power-law later plateaus need components orthogonal on the relevant site blocks; for nearly orthogonal ones (overlaps below a crossover $c_\times = O(\sigma)$ up to logarithms) they persist at accessible σ (conjecture, heuristic). Checked on one random CP teacher: strict saddles with logarithmic gaps on MPS $N = 4$, $\lambda_{\min} \propto \sigma$ and gaps $\propto \sigma^{-1}$ on the binary tree.

Teacher on another tree. proven known (standard (min-cut bound); cf. Grasedyck and Hackbusch, 2011, Lemmas 7–8) A student bond e must reach $\operatorname{rank}(W^*)^{\langle e \rangle}$, which for a TTN teacher is at most the minimum, over cuts of the teacher's tree that separate the two sides of e, of the product of the cut dimensions; a binary student with $N = 8$ of an MPS teacher with bond dimension r, for example, needs rank $\min(d^2, r^2)$ on the bonds of sites $\{2, 3\}$, $\{4, 5\}$, i.e. about r^2 separate saddles. A binary student with $N = 4$ is the deep linear network of Remark A.23 for any teacher: plateaus at $\frac{1}{2}\sum_{i>m} \sigma_i((W^*)^{\langle (01\mid 23) \rangle})^2$, escapes of degree three ($\propto \sigma^{-1}$), as many plateaus as the rank of the matricization. One teacher thus has different plateau levels and exponents on an MPS and on a binary tree derived new (tests: Sections 6.5 to 6.7).

A.6 Input covariance

Let $\Sigma = \Sigma_1 \otimes \cdots \otimes \Sigma_N$ (cross-site correlations: end of this section), and let Ξ multiply the physical leg of each site i of every core by $\Sigma_i^{1/2}$; Ξ is linear, symmetric and invertible, and $W(\Xi\theta) = \Sigma^{1/2} W(\theta)$.

Proposition A.39 (A product covariance is a preconditioned whitened problem)**.** proven new (for TTNs) *(a) $L(\theta) = \tilde{L}(\Xi\theta)$ with $\tilde{L}(\tilde{\theta}) = \frac{1}{2}\|W(\tilde{\theta}) - \Sigma^{1/2} W^*\|_F^2$. (b) In the variables $\tilde{\theta} = \Xi\theta$ gradient flow reads $\dot{\tilde{\theta}} = -\Xi^2 \nabla \tilde{L}(\tilde{\theta})$ from the anisotropic initialization $\Xi\theta_0$: a whitened problem with teacher $\Sigma^{1/2} W^*$ and the preconditioner Σ_i on every physical leg. (c) $\nabla L(\theta) = \Xi \nabla \tilde{L}(\Xi\theta)$ and $\nabla^2 L(\theta) = \Xi\, \nabla^2 \tilde{L}(\Xi\theta)\, \Xi$ at every θ. Critical points, critical values, Morse indices and saddle types are therefore those of the whitened problem with teacher $\Sigma^{1/2} W^*$ (Sylvester's law of inertia), but escape rates are eigenvalues of $\Xi^2 \nabla^2 \tilde{L}$, not of $\nabla^2 \tilde{L}$. (d) The gauge group acts on bond legs only, so Proposition 2.5 holds unchanged in θ: small initialization is balanced in θ, not in $\tilde{\theta}$. (e) Plain whitened gradient flow is recovered iff $\Xi^2 \propto I$, i.e. iff every $\Sigma_i = \kappa_i I$ and $\prod_{i \in S_v} \kappa_i$ is the same for every core (equal to 1 on binary and HT trees, whose internal cores have factor 1); for $\Sigma_i = \kappa_i I$ time is rescaled by $\prod_i \kappa_i$.*

Proof. (a) is $W(\Xi\theta) = \Sigma^{1/2} W(\theta)$; (b) and (c) are the chain rule through the symmetric map Ξ (checked to $2 \cdot 10^{-8}$; without the preconditioner the flow deviates by 4–100%). □

Proposition A.40 (Aligned covariance: exact clocks). proven known (Saxe-type modes: Saxe et al., 2014; depth two: Gidel et al., 2019; new: every tree) *Let the teacher be odeco with factors that are eigenvectors of the site covariances, $\Sigma_i u_r^{(i)} = \lambda_{r,i} u_r^{(i)}$, and let $\Lambda_r = \prod_i \lambda_{r,i}$. On the aligned family of Proposition 2.4: (a) $L = \frac{1}{2}\sum_r \Lambda_r(\Pi_r - s_r)^2$; (b) the family is invariant and $\dot{q}_{v,r} = \Lambda_r(s_r - \Pi_r)\prod_{u\neq v} q_{u,r}$, so each component evolves as in the whitened problem with its own clock, $\Pi_r(t;\Sigma) = \Pi_r(\Lambda_r t; I)$, the same for every core; (c) from a balanced start q_0, $t_{1/2,r} = q_0^{-(K-2)}/((K-2)\Lambda_r s_r) + \delta_K/(\Lambda_r s_r^{2-2/K}) + O(q_0^2)$ (Proposition A.6); (d) with the set S learned the loss is $\frac{1}{2}\sum_{r\notin S}\Lambda_r s_r^2$, and the events of j and then r are separated by $(1 - \frac{\Lambda_r s_r}{\Lambda_j s_j})/((K-2)\Lambda_r s_r q_0^{K-2})$.*

Proof. $\langle U_r, \Sigma U_{r'}\rangle = \delta_{rr'}\Lambda_r$, so $\Sigma(W - W^*) = \sum_r \Lambda_r(\Pi_r - s_r)U_r$; argue as for Proposition 2.4. □

Three orderings coexist derived new: by escape speed $\Lambda_r s_r$, by loss drop $\frac{1}{2}\Lambda_r s_r^2$, and the whitened one by s_r. With $s_1 > s_2$ and aligned initialization, component 2 escapes first iff $\Lambda_2/\Lambda_1 > s_1/s_2$ (LO in q_0); in the *mismatch window* $s_1/s_2 < \Lambda_2/\Lambda_1 < (s_1/s_2)^2$ the faster component is the smaller loss drop, so the first plateau lies above the reduced-rank-regression optimum: the visited saddles are not the reduced-rank solutions of Gidel et al. (2019), whose Corollary 2 states no ordering condition. With power-law spectra $\lambda_k \propto k^{-\beta}$ and component r on eigenvector k_r at every site, $\Lambda_2/\Lambda_1 = (k_1/k_2)^{\beta N}$, so the pairing $k_r = r$ reinforces the whitened order.

Proposition A.41 (Random initialization: what replaces the teacher). derived new (TTN form of the covariance-weighted first escape of Jacot et al., 2022; Atanasov et al., 2022; Rawal and DeWeese, 2026) *Near the origin $L = L_0 - \langle W, \Sigma W^*\rangle + \frac{1}{2}\|W\|_\Sigma^2$, and the fan-in initialization law of θ does not depend on Σ. Hence: (a) the escape from the origin (direction, time and learning order of Section A.2) is that of the whitened problem with the* escape tensor *$\Sigma W^* = \mathbb{E}[f^*(x)\Phi(x)]$ in place of W^*; for aligned covariance $s_r \to \Lambda_r s_r$, and for a single (or dominant) aligned component $\mathrm{d}\ln t_1/\mathrm{d}\ln\Lambda = -1$ exactly, for every K, tree and split of Λ over the sites; (b) the landscape (critical points, plateau levels, saddle types) is that of $\Sigma^{1/2}W^*$ (Proposition A.39); for aligned covariance $s_r \to \Lambda_r^{1/2} s_r$; (c) the problems (Σ, W^*) and $(I, \Sigma W^*)$ have the same escape at LO; for an aligned component their half-times, each relative to its own target, differ by the $O(1)$ offset $-\delta_K s_r^{-(2-2/K)}(\Lambda_r^{-(2-2/K)} - \Lambda_r^{-1})$.*

Without the preconditioner the exponent would be wrong ($-1/2$ with isotropic initialization). Checked: exponent -1.000 (binary, MPS, HT), offset -0.2218 against -0.2219. The order stays random (Proposition A.14), at rate ratio $\Lambda_2 s_2/(\Lambda_1 s_1)$.

Proposition A.42 (Piggyback block under aligned covariance). proven *(at the saddle)* new *Let j be learned (balanced, $q_j = s_j^{1/K}$) and let r differ from j on $D = D_{jr}$, with two terminal cores c_1, c_2. With $\mu_x = \prod_{i\in D\cap S_{c_1}}\lambda_{x,i}$, $\nu_x = \prod_{i\in D\cap S_{c_2}}\lambda_{x,i}$ for $x \in \{j, r\}$ and $\lambda_{\mathrm{sh}} = \prod_{i\notin D}\lambda_{j,i}$, the Hessian on the two rotations is*

$$H_{\mathrm{pb}} = \frac{\lambda_{\mathrm{sh}}}{s_j^{2/K}}\begin{pmatrix} s_j^2\mu_r\nu_j & -s_r s_j\mu_r\nu_r \\ -s_r s_j\mu_r\nu_r & s_j^2\mu_j\nu_r \end{pmatrix}. \tag{51}$$

It has a negative eigenvalue iff $\Lambda_r s_r^2 > \Lambda_j s_j^2$, for any split of the eigenvalues over the cores. If c_1, c_2 are adjacent ($k = 2$), its rate is below the add-route rate $b = \Lambda_r s_r s_j^{(K-2)/K}$, so with comparable seeds r is added directly: a rotation needs $m = 2$ and $k \geq 3$.

Proof. At the learned amplitude, up to constants, $L = \frac{1}{2}s_j^2\lambda_{\mathrm{sh}}(\mu_j\cos^2\varphi + \mu_r\sin^2\varphi)(\nu_j\cos^2\psi + \nu_r\sin^2\psi) - s_j(\Lambda_j s_j\cos\varphi\cos\psi + \Lambda_r s_r\sin\varphi\sin\psi)$ in the terminal-core rotations φ, ψ, so $\det H_{\mathrm{pb}} \propto \Lambda_j s_j^2 - \Lambda_r s_r^2$. For adjacent terminals the off-diagonal entry is $-b$ ($\lambda_{\mathrm{sh}}\mu_r\nu_r = \Lambda_r$), and a symmetric 2×2 matrix with positive diagonal h_1, h_2 and off-diagonal $-b$ has $\lambda_{\min} > -b$, since $(\frac{h_1+h_2}{2} + b)^2 - (\frac{h_1-h_2}{2})^2 - b^2 = h_1 h_2 + b(h_1 + h_2) > 0$. □

In the mismatch window, if the faster component wins the first escape derived: on binary $N = 4$ ($m = 2$, $k = 3$) the learned term rotates to the other component at bond rank one, the loss plateaus at $\frac{1}{2}\Lambda_2 s_2^2$, and the faster component is relearned later; on MPS $N = 4$ and every tree with $m \geq 3$ there is no rotation, and the loss plateaus at the higher level $\frac{1}{2}\Lambda_1 s_1^2$ (rotation rates checked).

Misaligned covariance. proven (identities and inequality) derived (order, to leading order in the cross overlaps) new For every CP teacher, exactly, $\Sigma W^* = \sum_r s_r \Lambda_r^{\mathrm{esc}} \widetilde{U}_r$ and $\Sigma^{1/2} W^* = \sum_r s_r (\Lambda_r^{\mathrm{loss}})^{1/2} \widehat{U}_r$, with $\widetilde{U}_r = \bigotimes_i \Sigma_i u_r^{(i)} / \|\Sigma_i u_r^{(i)}\|$, $\widehat{U}_r = \bigotimes_i \Sigma_i^{1/2} u_r^{(i)} / \|\Sigma_i^{1/2} u_r^{(i)}\|$, $\Lambda_r^{\mathrm{esc}} = \prod_i \|\Sigma_i u_r^{(i)}\|$ and $\Lambda_r^{\mathrm{loss}} = \prod_i \langle u_r^{(i)}, \Sigma_i u_r^{(i)} \rangle$; by Cauchy–Schwarz $\Lambda_r^{\mathrm{esc}} \geq \Lambda_r^{\mathrm{loss}}$, with equality iff every $u_r^{(i)}$ is an eigenvector of Σ_i. Components escape in the order of $s_r \Lambda_r^{\mathrm{esc}}$ along $\widetilde{U}_r$ and then realign to U_r without an extra plateau (the TTN form of the early alignment of Atanasov et al., 2022), while the loss drops are $\frac{1}{2} s_r^2 \Lambda_r^{\mathrm{loss}}$: misalignment alone can reverse the order. It also opens routes of degree below k^* derived, with couplings of order $\epsilon^{n_{\mathrm{out}}}$ at per-site misalignment ϵ (n_{out} sites outside the route).

Cross-site correlations. In the one-factor model $x^{(i)} = \sqrt{1-c}\, z_i + \sqrt{c}\, z_0$, with independent $z_0, \ldots, z_N \sim \mathcal{N}(0, I_d)$, every marginal is standard and two sites have covariance cI. (i) proven new (the model is the standard one-factor Gaussian model; new: its tree contraction) Given z_0 the sites are independent with second moments $M_i(z_0) = (1-c)I + c z_0 z_0^\top$, so $L = \mathbb{E}_{z_0}[\frac{1}{2}\|W - W^*\|^2_{\bigotimes_i M_i(z_0)}]$, a polynomial of degree $2N$ in z_0, is exact by Gauss–Hermite quadrature with $(N+1)^d$ product-metric tree contractions; a site-aligned component has the weight $\Lambda_{\mathrm{eff}}(c, N) = \sum_k \binom{N}{k} (1-c)^{N-k} c^k (2k-1)!!$. (ii) derived The first plateau survives for every $c < 1$ (the origin is a degree-K critical point for any input law; rank-constrained critical points remain critical, Lemma A.2). (iii) proven (rates at the saddle) new On an MPS with even N and a site-aligned odeco teacher, the correlations open strict two-core routes at the product saddle on every bond, with rate $\lambda_2 \approx s_1^{(N-2)/N} s_2 c^{N-2} ((N-3)!!)^2$ at small c; whitened power-law plateaus therefore become logarithmic once c exceeds a crossover $c^* = O(\sigma)$, up to logarithms (heuristic mechanism: empirical for $N = 4$; conjecture for $N \geq 6$, checked on one example with $N = 6$). (iv) derived On MPS $N = 4$ this escape passes through the first plateau point of the binary tree (with junk pairing components; strict on the MPS), and with spectral events the second plateau grows by $\ln 2\, [(1 - \kappa_c)/\lambda_2 + 1/\lambda'']$ per halving of σ, with λ'' the rate at that intermediate saddle and $\kappa_c \in (0, 1)$ the head start of the first escape (2.128 predicted, 2.13 measured, at $c = 0.4$).

A.7 Discrete steps and gradient noise

Odeco teacher with $s_1 > s_2 > \ldots$, whitened Gaussian inputs, $q_r = s_r^{1/K}$; gradient descent (GD) and online SGD as in Section 2.5, recorded at $t = \eta \times$ (number of steps).

Proposition A.43 (Hessian at aligned global minima: the pattern rule)**.** proven known (binary $N = 4$ is a depth-3 linear network: spectrum for square layers, Ghosh et al., 2025, Lemma 1; top eigenvalue at the flattest minimum, Mulayoff and Michaeli, 2020, Thm 1; new: general trees) *At a global minimum $H = J^\top J$, $J = \partial W / \partial \theta$. At an aligned minimum (Proposition 2.4 with $\Pi_r = s_r$), an elementary perturbation of core w changes W by a gain times a unit rank-one* pattern*: the chosen vectors at the sites of w, the teacher vector ℓ_c at every site below a child c whose bond index is ℓ_c, and ℓ_p at every site outside the subtree of w. The gain is $\prod_{u \neq w} q_{u,\ell(u)}$, $\ell(u)$ the label of the branch of w that contains u (zero if a label exceeds R). Distinct patterns are orthonormal, so*

$$\mathrm{spec}(H) \setminus \{0\} = \Big\{ \lambda_P = \sum_{w \in \mathrm{prod}(P)} \mathrm{gain}_w(P)^2 \ : \ P \text{ a pattern} \Big\}, \tag{52}$$

where $\mathrm{prod}(P)$ *is the set of cores that can produce* P*. The radial mode of component* r *(pattern* U_r*, produced by every core) has* $\lambda = s_r^2 \sum_v q_{v,r}^{-2} \geq K s_r^{2-2/K}$*, with equality iff the component is balanced. At the balanced minimum* $\lambda_P \leq |\mathrm{prod}(P)|\, q_1^{2(K-1)}$*, hence (nondegenerate if* $s_2 < s_1$*)*

$$\lambda_{\max}(H) = K\, s_1^{2-2/K}. \tag{53}$$

Consequences derived new (for TTNs; deep linear networks: Mulayoff and Michaeli, 2020, Thm 1): GD is linearly stable at the balanced minimum iff $\eta < \eta_c := 2/(K s_1^{2-2/K})$ (tested in Section 5.5). Imbalanced aligned minima are at least as sharp (AM–GM; proven; deep linear networks: Mulayoff and Michaeli, 2020), and with $\chi_e = R$ so is every gauge-equivalent minimum (derived; random GL gauges were at least 1.43 times sharper): for $\eta > \eta_c$ no global minimum is linearly stable (derived for $\chi_e = R$; for binary $N = 4$ with $\chi \geq d^2$ it follows from Mulayoff and Michaeli, 2020, Thm 1; conjecture otherwise). The minimum reached by gradient flow, and its sharpness λ_{GF}, are fixed by $Q_e(0)$ (a strongly imbalanced weaker component can carry the top mode); for $2/\lambda_{\mathrm{GF}} < \eta < \eta_c$ GD oscillates, the imbalance contracts (Lemma A.3), and GD converges to a flatter, more balanced minimum, with sharpness just below $2/\eta$ on binary $N = 4$ (empirical; known (scalar networks: Kreisler et al., 2023, Thms 3.2–3.3; two-factor factorization, limit sharpness at most $2/\eta$: Wang et al., 2022, Thm 3.2; beyond the edge, deep matrix factorization balances while oscillating: Ghosh et al., 2025, Prop. 2)). Hessians checked to $6 \cdot 10^{-15}$ on five trees.

Sharpness along the saddle-to-saddle path. derived new On the first plateau the Hessian is indefinite and $O(\sigma^{K-2})$. During the event of a balanced component of amplitude a its radial eigenvalue $(2K-1)a^{2K-2} - (K-1)s_r a^{K-2}$ rises to $K s_r^{2-2/K}$ without overshoot, and the $K-1$ transverse ones $(s_r - a^K)a^{K-2}$ stay below $K s_r^{2-2/K}$. At a later plateau that has learned the set S the radial mode of the largest learned component is an exact eigenvector: the sharpness is $K \max_{r \in S} s_r^{2-2/K}$, up to a relative excess $O(\sigma^4)$ from the $O(\sigma^2)$ imbalance. So at a fixed step η a plateau is linearly unstable iff it has learned a component with $s_r > s_c(\eta) = (2/(\eta K))^{K/(2K-2)}$, and GD first becomes unstable at that component's event.

Edge of stability. derived known (edge of stability: Cohen et al., 2021; the scalar map and its two-cycle, given implicitly: Ghosh et al., 2025, Thm 1; the amplitude below follows from the normal form of MacDonald et al., 2025, Thm 5.3 and Prop. B.3; $K = 2$: Hofmann, 2026) new (only the TTN form of the reduction and the closed-form time-averaged loss (54)) For $\eta_c < \eta < 2/\lambda_2(H)$ only the radial mode of component 1 is unstable, and GD restricted to it (exactly for aligned initialization) is the balanced scalar map $x \mapsto x + h x^{K-1}(1 - x^K)$, $x = a/s_1^{1/K}$ (a the per-core amplitude), $h = \eta s_1^{2-2/K}$, which doubles its period at $hK = 2$, i.e. at $\eta = \eta_c$. To third order around $x = 1$ the period-two orbit $1 \pm \delta_{\mathrm{p2}}$ has $\delta_{\mathrm{p2}}^2 \simeq 3(hK-2)/(4(K-1)(5K-4))$, so for $\eta_c < \eta < \min(2/\lambda_2, 1.245\eta_c)$ the time-averaged loss is

$$L_{\mathrm{EoS}} \simeq \frac{3K^2 s_1^2}{4(K-1)(5K-4)} \Big(\frac{\eta}{\eta_c} - 1 \Big). \tag{54}$$

Later events still occur and lower the loss. Checked (binary $N = 4$, $\eta = 1.05\eta_c$): late loss $1.45 \cdot 10^{-2}$ against $1.452 \cdot 10^{-2}$ (map) and $1.53 \cdot 10^{-2}$ (54); divergence at 1.5–$1.65\eta_c$.

Proposition A.44 (Shift of event times under gradient descent)**.** derived new (for TTNs; the modified flow is known: Barrett and Dherin, 2021) *To order* η*, GD follows the modified flow* $\dot\theta = -(I +$

$\frac{\eta}{2}H)\nabla L$. *Its component along the gradient-flow path reparametrizes time, and since* $\frac{\mathrm{d}}{\mathrm{d}t}\ln\|\nabla L\| = -\nabla L^\top H\nabla L/\|\nabla L\|^2$ *along gradient flow,*

$$t_{\mathrm{GD}} - t_{\mathrm{GF}} = \frac{\eta}{2}\ln\frac{\|\nabla L(t_{\mathrm{event}})\|}{\|\nabla L(0)\|} + (\textit{transverse part}) + O(\eta^2). \tag{55}$$

For aligned initialization the path is exact and the event of component r *moves by* $\frac{\eta}{2}\ln(f(a_1)/f(a_0))$, $f(a) = a^{K-1}(s_r - a^K)$, a_0, a_1 *the per-core amplitudes at initialization and at the event: plateau exponents and prefactors are those of gradient flow, and no saddle is skipped or reordered while the gradient-flow gaps exceed the differences of the shifts* $+O(\eta^2)$.

For generic initialization $\|\nabla L(0)\| \propto \sigma^{K-1}$, so $\Delta t_1/\eta \approx \frac{K-1}{2}\ln(1/\sigma) + \text{const}$ asymptotically (conjecture, heuristic). At a strict saddle the growth per step is $1 + \eta|\lambda_-|$ instead of $e^{\eta|\lambda_-|}$, so $t_{\mathrm{GD}}/t_{\mathrm{GF}} = \eta|\lambda_-|/\ln(1+\eta|\lambda_-|) \in [1, 1.82]$ for $\eta|\lambda_-| < 2$. Checked: aligned shifts within 1.3%, generic slopes 0.69–0.70 per halving of σ (predicted 0.69).

SGD on the first plateau. derived new (the TTN delay law; framework: stochastic differential equations for SGD, Li et al., 2017, 2019, and Type-II saddles, Ziyin et al., 2024; consistent with the experiments of Corlouer et al., 2026 (Fig. 2: order kept, SGD slower)) For online SGD with batch B the per-sample gradient is $\langle W - W^*, \Phi(x)\rangle J^\top\Phi(x)$; near the origin the gradient covariance is $O(s_1^2 3^N\|\theta\|^{2K-2})/B$ and the drift $O(s_1\|\theta\|^{K-1})$, so in the clock $\mathrm{d}\tau = \|\theta\|^{K-2}\mathrm{d}t$ the noise vanishes: noise cannot drive the first escape. For one aligned balanced component, $\mathrm{d}a = s_1 a^{K-1}\mathrm{d}t + \sqrt{\eta/B}\, s_1\kappa_{\mathrm{n}}^{1/2}a^{K-1}\mathrm{d}\mathcal{B}_t$ ($\mathcal{B}_t$ a standard Brownian motion, $\kappa_{\mathrm{n}} = 3^N - 1 + \sum_{r\geq 2} s_r^2/s_1^2$), whose hitting time is inverse Gaussian at leading order:

$$\mathbb{E}t_1^{\mathrm{SGD}} - t_1^{\mathrm{GF}} \simeq \frac{\eta}{2}(K-1)\Big(1 + \frac{\kappa_{\mathrm{n}}}{B}\Big)\ln\frac{a_{1/2}}{a_0} > 0, \qquad \mathrm{sd}(t_1^{\mathrm{SGD}}) \simeq \Big(\kappa_{\mathrm{n}}\frac{\eta}{B}t_1^{\mathrm{GF}}\Big)^{1/2}, \tag{56}$$

with $a_{1/2} = (s_1/2)^{1/K}$. Both corrections vanish relative to $t_1^{\mathrm{GF}} \propto a_0^{-(K-2)}$: the exponent is unchanged. Checked: within 10–15% for aligned, order of magnitude for generic initialization.

SGD at the minimum. derived (exact mean-square threshold) proven (the inequality) new (the reduction to the pattern basis; general criteria are known: Ziyin et al., 2024; exact mean square, Mulayoff and Michaeli, 2024, Thm 5, of which (57) is the evaluation in the pattern basis; one-dimensional, Wu et al., 2018; top Lyapunov exponent, Chemnitz and Engel, 2025) In the eigenbasis of (52) linearized SGD at the minimum acts on one coordinate per pattern, with the products $Z_P(x) = \prod_i (t_{P,i}\cdot x^{(i)})$ of its site vectors; the second-moment operator maps diagonal matrices to diagonal ones ($\mathbb{E}[Z_R^2 Z_P Z_Q] = 0$ for $P \neq Q$) and preserves the positive semidefinite cone, which gives the exact mean-square threshold

$$\eta_{\mathrm{ms}}^*(B) = \frac{2}{\lambda_{\max}(G_B)}, \qquad G_B = \Big(1 - \frac{1}{B}\Big)\Lambda_H + \frac{1}{B}\Lambda_H^{1/2}\mathcal{K}\Lambda_H^{1/2}, \qquad \mathcal{K}_{PQ} = \prod_i m(t_{P,i}, t_{Q,i}), \tag{57}$$

with $\Lambda_H = \mathrm{diag}(\lambda_P)$, $m = 3$ if the two site vectors coincide and 1 if they are orthogonal (Gaussian sites). Jensen's inequality gives $\eta_{\mathrm{as}} \geq \eta_{\mathrm{ms}}^*$ for the almost-sure divergence threshold; empirical: the ratios are about 2 ($N = 4$, $B \leq 16$) and 8–21 (binary $N = 8$), and SGD near the minimum diverges at η_{as}.

Proposition A.45 (SGD lengthens later plateaus)**.** *conjecture (heuristic law)* known (mechanism: stochastic collapse, Chen et al., 2023) *At a later saddle with a route of degree* $k \geq 3$ *the population*

Hessian vanishes on the route and the per-sample gradients vanish at zero route amplitude. Zero-mean multiplicative noise coupling the route entries to mixed learned entries gives the log-amplitude of the coherent route a drift $-\gamma_c$, $\gamma_c = c_\gamma \eta/B$; added to the route equation, $\dot{b} = s_r b^{k-1} - \gamma_c b$. For $k = 3$, with $T_{\mathrm{GF}} = 1/(s_r b_0)$,

$$\frac{T_{\mathrm{SGD}}}{T_{\mathrm{GF}}} = \frac{-\ln(1-x)}{x}, \qquad x = \gamma_c T_{\mathrm{GF}}, \tag{58}$$

and the route is trapped (decays) for $x \geq 1$; for general k escape needs roughly $(k-2)\gamma_c T_{\mathrm{GF}} < 1$. As b_0 scales with the initialization, at fixed η/B SGD stalls on a later degenerate plateau once σ is small enough; strict saddles escape if $|\lambda_-| > \gamma_c$.

Solving $\dot{w} = -s_r + \gamma_c w$ for $w = 1/b$ gives (58). Checked on binary $N = 4$ only (weights $(1, 0.6)$, Gaussian sites; 21 runs): delays 1.02–2 times T_{GF}, trapping for $x \geq 1$, never faster than gradient flow, $c_\gamma \approx 1.35$ empirical (1.08 from a commuting-pair estimate; about 33 for weights $(3, 2, 1)$, $d = \chi = 3$: c_γ depends on teacher and tree). Near the stability limit SGD breaks the saddle sequence instead empirical: with weights $(1, 0.6)$ no learned component was erased in 13 runs at $\eta \leq 0.25\eta^*_{\mathrm{ms}}$, but with $(3, 2, 1)$ one was erased (mostly relearned) in 20 of 36 longer runs at 0.25–$0.30\,\eta^*_{\mathrm{ms}}$ (Section 5.5); at 0.7–$1.2\,\eta^*_{\mathrm{ms}}$ components were repeatedly erased and relearned, or runs diverged.

A.8 Regime boundaries

Here $d_i = d$, and $\Theta = \sum_v \eta_v J_v J_v^\top$ on $\mathbb{R}^{d^N}$, $J_v = \partial W/\partial G_v$, is the tensor-space tangent kernel that drives the function-space dynamics of (15) (for whitened inputs the probe-set kernel of Table 2 is $\Phi(x_p)^\top \Theta \Phi(x_{p'})$). Recall $W^{\langle e_v \rangle} = T_v C_v^\top$ (Section 2.1); the initialization is (11) with scales σ_v, $\alpha = \prod_v \sigma_v$. A bond e is *full* if χ_e reaches its cut dimension $\min(d^{n_e}, d^{N-n_e})$ (n_e sites below e).

Proposition A.46 (Tangent kernel at initialization)**.** proven new (on every tree; for MPS and TT the bond-dimension-independent mean kernel is implicit in Guo and Draper, 2021; Wesel and Batselier, 2025; the rank in (c) is the manifold dimension of Uschmajew and Vandereycken, 2013) *(a) $J_v J_v^\top = \bigotimes_{c \in \mathrm{ch}(v)} T_c T_c^\top \otimes I_{d_v} \otimes C_v C_v^\top$ (legs in site order). (b) For all bond dimensions, $\mathbb{E}\,\Theta = \frac{\alpha^2}{d^N} \sum_v \frac{\eta_v F_v}{\sigma_v^2} I$ and $\mathbb{E}[W_0 \otimes W_0] = \frac{\alpha^2}{d^N} I$; with $\sigma_v = \sigma$ and $\eta_v = 1$ the kernel scale is $\bar{\lambda} := \mathbb{E}\,\mathrm{tr}\,\Theta/d^N = \sigma^{2(K-1)} \sum_v F_v/d^N$ (core v contributes the term F_v). (c) If the bond dimensions are consistent (each χ_e at most the product of the other leg dimensions of both of its cores) and the initialization is generic, $\mathrm{rank}\,\Theta = \sum_v F_v \chi_v - \sum_e \chi_e^2$ (capped at d^N), the dimension of the manifold $\mathcal{M}_\chi$ of tensors with bond ranks exactly χ_e (dense in the set with ranks at most χ_e; for inconsistent bond dimensions, first reduce each χ_e to its effective value); $\mathrm{rank}\,\Theta = d^N$ iff every bond is full.*

Proof sketch. (a) W is linear in G_v. (b) The cores are independent, T_c and C_v involve disjoint cores, and the initialization law is invariant under orthogonal maps of each physical leg (irreducible); the traces follow by induction from the leaves (Proposition 2.2). (c) $\mathrm{rank}\,\Theta$ is the dimension of the tangent space of $\mathcal{M}_\chi$ at W_0, parameters minus gauge (Uschmajew and Vandereycken, 2013, Thms 2 and 4). □

Consequences derived: with a bond that is not full, the linearized (lazy) dynamics fits only the projection of the target on the range of Θ and stops at $\frac{1}{2}\|P_\perp W^*\|_F^2$, with $\mathbb{E}\|P_\perp W^*\|_F^2/\|W^*\|_F^2 = (1 - \chi/d^2)^2$ for binary $N = 4$ (compare Chizat et al., 2019, Thm 2.5 and the remark after it, and Ghorbani et al., 2019, Thm 2). $\Theta \approx \bar{\lambda} I$ needs every χ_e far above its cut dimension; at capped full bonds $\lambda_{\min}(\Theta)/\bar{\lambda} \sim 10^{-4}$ empirical, so lazy runs need uncapped $\chi \geq 2d^2$ (Section 5.1); at practical χ the lazy regime follows a random, tree-dependent kernel.

Proposition A.47 (Lazy boundary of the centred model, $N = 4$). derived new (the TTN constants; the criterion is a threshold on the relative kernel change, as in Geiger et al., 2020, refining the a-priori relative scale of Chizat et al., 2019, Sec. 1.2; a coherent second-order kernel change is known in wide networks: Bordelon and Pehlevan, 2022) *Binary tree with $N = 4$, standard parameterization, centred model; let $\xi = \|W^*\|_F/\alpha$, assume $\chi \gg d^2$ and keep the leading order in d^2/χ. Then*

$$\delta_\Theta^2 \approx (c_1\xi)^2 + 2\zeta c_1 c_2 \xi^3 + (c_2\xi^2)^2, \qquad c_1 = \frac{4d^3\sqrt{d^2+1}}{\chi^3}, \quad c_2 = \frac{2d^4\sqrt{2d^2\nu+2}}{\chi^3}, \tag{59}$$

with $\nu = \sum_r s_r^4/(\sum_r s_r^2)^2$ and a seed-dependent correlation $\zeta \in [-1,1]$: a zero-mean first-order term, linear in W_0, and a coherent second-order term $\propto (W^*)^{\langle e\rangle}((W^*)^{\langle e\rangle})^\top$ (the bottom cores drift toward the singular directions of the target). The boundary $\delta_\Theta = \delta_*$ lies in $[\xi_c^+, \xi_c^-]$, $\xi_c^\pm = (\mp c_1 + \sqrt{c_1^2 + 4c_2\delta_*})/(2c_2)$, and for large χ ($c_1^2 \ll c_2\delta_*$)

$$\sigma_{\text{lazy}} = \Big(\frac{\|W^*\|_F^2 c_2}{\delta_*}\Big)^{1/6} \propto \chi^{-1/2} \tag{60}$$

($6.50\,\chi^{-1/2}$ for $d = 3$, weights $(3,2,1)$, $\delta_* = 0.1$), not χ^{-1} as the first-order term alone would give. In the two-copy model the first-order term cancels, $\delta_\Theta \approx \frac{1}{4}c_2\xi^2$, and $\sigma_{\text{lazy}}^{(2)} = 2^{-1/3}\sigma_{\text{lazy}}$.

Checked empirical: predicted $\xi_c \in [2.17, 2.81]$, $[6.67, 7.31]$ at $\chi = 32, 64$ against 2.25–2.82, 6.73–7.53 in gradient flow. The boundary moves with the threshold as $\sigma_{\text{lazy}} \propto \delta_*^{-1/(Ks)}$, $s = \mathrm{d}\ln\delta_\Theta/\mathrm{d}\ln\xi \in [1,2]$ (1.5–1.8 measured); the root can move by more than 100% while $\delta_\Theta < 0.1$.

Deficient bonds. derived new If some bond is not full, $W^* + W_0$ generically has more bond rank than the student can carry: the centred model is unrealizable at every σ. At small σ (first order in W_0 around a smooth point W^* of $\mathcal{M}_\chi$, i.e. $\chi_e = \operatorname{rank}(W^*)^{\langle e\rangle}$, with σ^K far below the smallest teacher Schmidt value) the best reachable point is $W^* + P_T W_0 + O(\|W_0\|_F^2)$ (P_T, $P_\perp$: tangent and normal projectors of $\mathcal{M}_\chi$ at W^*), so the loss floors at $\frac{1}{2}\|P_\perp W_0\|_F^2$, with mean, by Proposition A.46(b),

$$\frac{\mathbb{E}L_{\text{end}}}{L(0)} \approx \Big(1 - \frac{\dim\mathcal{M}_\chi}{d^N}\Big)\frac{\sigma^{2K}}{\|W^*\|_F^2}. \tag{61}$$

A typical run meets the convergence rule $L < 10^{-3}L(0)$ only below $\sigma_{\text{floor}} = [10^{-3}\|W^*\|_F^2/(1 - \dim\mathcal{M}_\chi/d^N)]^{1/(2K)}$ (0.56 for binary $N = 4$, 0.74 for $N = 8$; $d = \chi = 3$, weights $(3,2,1)$); runs above it are stalled (unrealizable). At large σ the floor tends to $\|P_\perp W^*\|_F^2/\|W^*\|_F^2$ ($P_\perp$ at W_0). In the two-copy model (realizable iff $2\chi_e \geq \operatorname{rank}(W^*)^{\langle e\rangle}$) a deficient bond gives, at large σ, a lazy plateau at that level and then a saddle-free escape by rotation of the bond subspaces: adiabatic elimination gives $t_{\text{esc}} = 2\vartheta\sigma^2/\|W^*\|_F^2$, $\vartheta = 0.37$–0.70, a kernel time $\propto \sigma^{2K}$ (derived for $N = 4$; conjecture, heuristic, for general K; compare Tu et al., 2024; Lyu et al., 2024).

Proposition A.48 (Saddle-to-saddle boundary for aligned initialization). proven *(given the exact aligned reduction)* known (mode equation: Saxe et al., 2014; threshold exponent $K/(K-2)$ and a product threshold that stays finite as $K \to \infty$, so that $a_{0c} \to 1$: Gissin et al., 2020, Thm 2, with the $K \to \infty$ mode solution in their Thm 1; new: the constants for the criterion of Section 2.7, including their $K \to \infty$ limits) *Let every component of an odeco teacher start at product amplitude $u_0 = q_0^K$, let $I_K(v_0, v_1) = \frac{1}{K}\int_{v_0}^{v_1} v^{-2+2/K}(1-v)^{-1}\mathrm{d}v$ and $\omega_K = I_K(0.1, 0.9)$. Then $t_r = s_r^{-(2-2/K)} I_K(u_0/s_r, \frac{1}{2})$ and $w_r = s_r^{-(2-2/K)}\omega_K$, so, with $\varrho_i = s_{i+1}/s_i$,*

$$\rho_i = \frac{I_K(u_0/s_{i+1}, \frac{1}{2}) - \varrho_i^{2-2/K} I_K(u_0/s_i, \frac{1}{2})}{\omega_K} \simeq \Big(\frac{u_0}{s_{i+1}}\Big)^{-(1-2/K)} \frac{1-\varrho_i}{(K-2)\,\omega_K} \quad (u_0 \to 0), \tag{62}$$

and the boundary $\rho_i = 3$ *lies at* $u_{0c} \simeq s_{i+1}[(1-\varrho_i)/(3(K-2)\omega_K)]^{K/(K-2)}$. *As* $K \to \infty$, $KI_K(v_0, v_1) \to \frac{1}{v_0} - \frac{1}{v_1} + \ln\frac{v_1}{v_0} + \ln\frac{1-v_0}{1-v_1}$ *and* $K\omega_K \to \frac{80}{9} + 2\ln 9 = 13.283$*:* u_{0c} *stays finite (*$1.306 \cdot 10^{-2}$ *for weights* $(3,2,1)$*), and* $a_{0c} = u_{0c}^{1/K} = 1 + \ln u_{0c}/K + O(K^{-2}) \to 1$.

Values: $\omega_3 = 2.009$, $\omega_5 = 1.612$, $\omega_7 = 1.318$; for weights $(3,2,1)$, $a_{0c} = 0.066$, 0.324, 0.480 at $K = 3, 5, 7$. Aligned gradient flow reproduces (62) to four decimals at $K = 3$ and 7.

Random initialization. derived (t_1) conjecture (later gaps; large K) empirical (boundary) By Corollary A.5 and Proposition A.6, $t_1 = \sigma^{-(K-2)}\hat{t}_1(Z)/s_1 + O(1)$. For teachers whose escape routes span the whole tree (odeco), later gaps scale the same way if the dormant amplitudes entering each plateau stay $\propto \sigma$ (exact for aligned initialization), so $\rho_i \approx R_i\sigma^{-(K-2)}$ with seed-dependent R_i; in general a later gap scales as $\sigma^{-(k-2)}$, k the route degree (Section A.3). The boundary is a seed distribution: median per-seed $\sigma_c = 0.27$–0.31 for binary $N = 4$, $d = \chi = 3$, weights $(3,2,1)$, and 0.71 for $N = 8$, $d = 2$, $\chi = 16$, weights $(3,2)$; as aligned initialization with per-core amplitude $\kappa_K\sigma$, $\kappa_3 \approx 0.22$ ($d = 3$) and $\kappa_7 \approx 0.69$ ($d = 2$). conjecture: $\kappa_K \to 1$, so $\sigma_c(K) \to a_{0c}(K) \to 1$; the data (all at $d = 2$ for large K) do not distinguish this from $\kappa_K \to d^{-1/2}$, i.e. $\sigma_c \to \sqrt{d}$ (Section 3.3). For matrix-like pairs the transient overlap of a component not yet learned peaks at a value $\propto \sigma^{2(1-\varrho)}$ (conjecture, heuristic), hence events are timed by their final rise. With χ equal to the teacher rank R and $K \geq 5$, the last component needs a route through all K cores, which the child-heavy root bonds of fan-in initialization (Section A.1) can kill before it escapes (conjecture, heuristic). empirical: it was not learned in 2 of 4 runs at $K = 5$ and 2 of 2 at $K = 7$ ($d = 3$); balanced initialization removed these stalls at $K = 5$ (leaving long plateaus) but not at $K = 7$; spare channels ($\chi \geq 2R$) avoided stalls in the runs made (one at $d = 3$; 26 at $d = 2$).

Large bond dimension. derived new (the spectral condition is Yang et al., 2024; factor-wise rules for structured layers: Qiu et al., 2024; Potapczynski et al., 2024) The input of a core is always a product of vectors, so its signals are controlled by its injective norm, not by the spectral norm of its matricization. The spectral condition with the injective norm keeps the fan-in initialization of every non-root core, multiplies the root by $(\sum_\ell m_\ell)^{-1/2}$ (m_ℓ: its child-leg dimensions) and sets $\eta_v \propto \chi_v/F_v$ ("μP-TTN"). At fixed ξ ($N = 4$) this gives $\delta_\Theta \propto \chi^0$, against χ^{-3} (standard) and χ^{-1} (NTK parameterization; compare Guo and Draper, 2021); the factor-wise rule makes the initial kernel of binary $N = 8$ vanish like χ^{-2} and slows the initial fit about linearly in χ empirical.

Imbalance. derived (aligned mode) new (compare two-layer linear networks: Kunin et al., 2024) With the root at scale $\lambda_{\text{root}}\sigma$ and the other cores at σ (fixed α), the root bonds have $\mathbb{E}Q_e = \sigma^2(1 - \lambda_{\text{root}}^2/\chi_e)I$, balanced at $\lambda_{\text{root}} = \sqrt{\chi_e}$ (fan-in initialization is root-light). A small root ($\lambda_{\text{root}} \ll 1$) makes training lazy when $s \ll \alpha/\lambda_{\text{root}}$ (s a teacher weight), with $\delta_\Theta \propto \lambda_{\text{root}}^2$, but only with full bonds (with a deficient bond: lazy, then rich). A large root freezes, leaving a degree-$(K-1)$ model, which does not make training lazy for $K \geq 3$ (unlike the two-layer case). Tests: Sections 3.2 to 3.4 and 5.4.

B Experimental details

[Skeleton. For every experiment: trees, dimensions, teachers, initialization, dynamics and tolerances, sweeps, seeds, time budgets, compute cost.]

B.1 Common settings

[Double precision; Dormand–Prince 5(4) with relative and absolute tolerances 10^{-8} and 10^{-12} unless stated; recording grid; conservation-drift threshold 10^{-6}; event detection; Hessian and kernel computations.]

B.2 Saddle-to-saddle experiments

Settings of Sections 4.1 to 4.3 and of the odeco runs of Section 4.6 (experiment s2s-core; the locality experiment is in Section B.3). Experiment folder `experiments/s2s_core/` (README with every command in order; YAML configurations in `configs/`); findings in `notes/s2s_core.md`; reviews in `notes/reviews/s2s_core_round1.md` and `notes/reviews/s2s_core_round2_verification.md`, whose errata E1–E21 are applied in Section 4. Numbers that are not single entries of a results table are computed by `report_numbers.py` into `results/report_numbers.csv`.
[Order experiment (Section 4.5): settings to be added.]

Problems. Whitened Gaussian inputs, exact population loss, double precision. Odeco teachers (`ttn.teachers.odeco_teacher`) with $R = 3$, $d = 4$ at every site and Haar-random site bases from teacher seed 1000, a stream independent of the initialization seeds; weights $(3, 2, 1)$ (separated) and $(1, 0.85, 0.7)$ (close). Trees binary4, MPS4, binary8, HT4 and MPS8 ($K = 3, 4, 7, 7, 8$); nominal bond dimension $\chi = 3$ or 6, capped by (3) (at $\chi = 6$: MPS4 bonds $(4, 6, 4)$, MPS8 end bonds and HT4 leaf bonds 4, all other bonds 6). Initializations: fan-in (11) with seeds 0–9; *balanced*, the same W_0 re-gauged to $Q_e = 0$ on every bond (`s2s.gauge_balance`); *aligned*, the same per-core amplitude on every core and component (`ttn.network.aligned_init`). The leading-order sweeps also use MPS, binary and HT trees with up to 16 sites (Table 14).

Dynamics. Gradient flow by SciPy's LSODA (`ttn.dynamics.integrate_gf`) with the exact Hessian as Jacobian, instead of the default integrator of Section 2.5: past the first escape the flow is stiff, and an explicit method needs a number of steps $\propto \sigma^{-(K-2)}$ per plateau (Radau, tried on one $K = 7$ run, did not finish in 25 minutes). Tolerances: relative 10^{-8} and absolute $10^{-11}\sigma$ (sweeps `main`, `first`, `balanced`), or 10^{-6} and $10^{-9}\sigma$ (`large`, `k78`, whose plateaus last up to 10^7); on binary8 at $\chi = 6$, $\sigma = 0.07$, the looser tolerances move the event times by $7 \cdot 10^{-7}$ relative. Aligned runs: Dormand–Prince 5(4) and LSODA with tolerances 10^{-10} and 10^{-14}. Events are last upward crossings of $a_r = \frac{1}{2}$ and widths final rises from 0.1 to 0.9 (Section 2.7), both refined by re-integrating the event window on linear grids until the rise holds at least 40 samples (`s2s.zoom_rise`); no duration is read off the logarithmic output grid. The plateau midpoint is the time of minimum parameter speed between two events; the parameters there are saved, and the Hessian, raw and gauge-projected (`ttn.diagnostics.project_out_gauge`), was recomputed from them after the library's projection fix (`recompute_hessians.py`). Bond spectra are recorded on the output grid and, in all but the first 332 complete runs of `main`, in the event windows. The conservation drift (per integration chunk, as in (18)) has median $9 \cdot 10^{-10}$ and exceeds 10^{-6} in 11 of 1,576 runs (at most $3.3 \cdot 10^{-6}$; ten MPS8 runs and one binary8 run).

Leading-order flow. $T(Z)$ and r_1 (Remark A.7) come from `s2s.correlation_flow_blowup`: the flow $\dot{\theta} = \nabla g$ from the initialization at $\sigma = 1$, by DOP853 with relative tolerance 10^{-10}, integrated in segments until the second-largest component amplitude $|\langle W, U_r \rangle|$ is below 10^{-6} of the largest and no longer growing relative to it, with a cap at amplitude 10^{30}; the remaining time of a balanced mode from the stop is added. The 40 binary4 close-teacher jobs and 2 separated binary4 jobs at $\chi = 6$ stopped at the cap with the direction still converging; T is converged regardless (the time

left beyond amplitude A is $\propto A^{-(K-2)/K}$), and an independent integration of the spherical flow agrees to $1.5 \cdot 10^{-11}$ (binary4) and $1.7 \cdot 10^{-10}$ (MPS4) relative on all 160 binary4 and MPS4 jobs (verification review). Round-1 values of T, from a stop at amplitude 10^6, were biased by up to $2.2 \cdot 10^{-3}$ relative (binary4, close teacher) and are superseded.

Sweeps and run counts. Table 14.

Table 14: Sweeps of the s2s-core experiment, one configuration file each in `experiments/s2s_core/configs/` except the last two rows. Seeds 0–9 unless stated; "sep." and "close": the two teachers.

sweep	content	runs
`aligned`	five trees, sep., $\chi = 3$, balanced aligned initialization; Dormand–Prince and LSODA	10
`main`	five trees, $\chi = 3, 6$, both teachers, fan-in; $K = 3, 4$: 8 values of σ in $[0.003, 0.1]$; $K = 7, 8$: 6 values in $[0.07, 0.4]$ at $\chi = 3$, plus 50 runs at $\sigma = 0.03$, 0.05 ($\chi = 3$) and 76 runs at $\chi = 6$ ($\sigma = 0.03$–0.1) made before these were dropped or moved to `large`, all kept	1,126
`large`	binary8, HT4, MPS8, $\chi = 6$, sep., fan-in, $\sigma \in \{0.07, 0.1, 0.15, 0.2, 0.3\}$, wall-clock limit 2400 s	150
`k78`	binary8, HT4, MPS8, $\chi = 3$, sep., fan-in and balanced, $\sigma \in \{0.15, 0.2, 0.3\}$, no amplitude stop	180
`balanced`	binary4, MPS4, $\chi = 3, 6$, sep., balanced, $\sigma \in \{0.003, 0.01, 0.03\}$	120
`first`, `first_k78`	first escape only, 8 values of σ: binary4, MPS4 ($\sigma = 0.002$–0.1; $\chi = 3, 6$, both teachers, fan-in and balanced); binary8, HT4, MPS8 ($\sigma = 0.01$–0.4; sep., fan-in, $\chi = 3, 6$, MPS8 at $\chi = 3$ only; binary8 also close and balanced at $\chi = 3$, 6–9 seeds); 231 (initialization, teacher) pairs	1,848
`lo_d`	leading-order flow only: binary4, MPS4, $d = 2, 3, 4$, $\chi = d^2$, one component of weight 1, 200 seeds	1,200
`lo_chi`	leading-order flow: the five trees, $\chi = 1$–16 (binary8 to 12) with one component of weight 1, and $\chi = 3, 6, 12$ with weights $(3, 2, 1)$; 100 seeds	5,400
`lo_K`	leading-order flow: MPS with $N = 3$–$8, 10, 12$, binary with $N = 4, 6, 8, 12, 16$, HT with $N = 3, 4, 6, 8$ ($K = 3$–15), $\chi = 3$, weights $(3, 2, 1)$, 50 seeds	850
continuation	stalled or unresolved runs continued from their final state (`continue_stalled.py`)	23
tolerance check	9 runs of `k78` rerun with tolerances 10^{-8}, $10^{-11}\sigma$ (`tolcheck.py`)	9

Budgets and run classes. Runs stop at $L \leq 10^{-9}L(0)$; at $t_{\max}$, 10^3 times the RMS-overlap escape time of the weakest component; at a wall-clock limit (300 s in `main`, 400 s in `balanced`, 2400 s in `large` and `k78`, so the number of unresolved runs depends on the machine) or a step limit per output chunk; and, except in the first ∼310 runs of `main`, when a component never learned has a route-end slice below $10^{-8}\sigma$ for a whole output chunk (a dead end, after which nothing is physics). `main` also stopped when every dormant amplitude $|\langle W, U_r\rangle|$ stayed below 10^{-15} for a chunk; at $K = 7, 8$ this was too strict (the amplitude is a product of K per-core amplitudes and can be 10^{-15} while each of them is 10^{-3}), so those runs are classed unresolved unless a slice is dead, and the cells were rerun without it in `k78`. Classes (`analyze_main.py`): *complete* (all three components learned); *trapped* (a component never learned has a route-end slice below $10^{-8}\sigma$ after the first escape); *stalled* (the time budget ended with every such slice above it); *unresolved* (wall-clock limit or the amplitude stop). The classes do not depend on the tolerance: the 9 runs of the tolerance check keep their class and order, and their event times change by at most $3.6 \cdot 10^{-6}$ relative. At $K = 7, 8$ the trapped fractions are lower bounds, since some stalled runs end with slices still decaying (`slice_trends.py`).

Fits. Exponents are fitted per seed, ln of the time against $\ln \sigma$ with at least four values of σ, on run sets classified by the saddle sequence (`analyze_main.fit_sets`): t_1 on the runs whose first

component is the seed's most frequent one; $t_2 - t_1$ on the runs whose first two components are the seed's most frequent pair, with $\rho_1 > 3$; $t_3 - t_2$ on complete runs in the seed's most frequent order with $\rho_1, \rho_2 > 3$. The Hessian slopes use the same sets. Each cell comes from one sweep (`large` at $K = 7, 8$ with $\chi = 6$, `main` otherwise); sweeps with different tolerances are never pooled. The cut $\rho_i > 3$ matters only for binary4 with the close teacher, where it leaves some seeds with four or five values of σ (about 0.8 decade); without it, overlapping and wrong-order runs are mixed in and the second gap gives 0.92 ± 0.11 at $\chi = 3$ and 1.12 ± 0.05 at $\chi = 6$ (verification review, E9). First-escape exponents are also given with the offset $\delta_K s_{r_1}^{-(2-2/K)}$ of (21) subtracted from t_1.

Continued runs. `continue_stalled.py` integrates a stalled or unresolved run further from its saved final state, with the escape time resolved like every event. Three samples: the 8 stalled `k78` runs with the shortest predicted remaining plateau (3 seeds); a seeded random sample (`--seed 1`) of 8 of the 74 `k78` runs (73 stalled, 1 trapped) whose estimate is at most 100 times the time already integrated, one of them also in the first sample; and 8 MPS8 runs of `large`, spread over the ratio of estimate to elapsed time (0.2–420). The budget multiplier of the third sample was not recorded, and the README command selects a different set (verification review, E13); its results are reported as measured.

Excluded and superseded runs. The scipy pilots made before the library existed, part of them before the teacher and initialization streams were separated, support no claim. Superseded after review round 1: T from the amplitude-10^6 stop (kept as `T_v1` in the raw records), gap fits on pooled run sets, Hessians with the earlier gauge projection, and decay rates fitted into the floating-point floor (now fitted above 10^{-9} absolute). No run was dropped: runs that do not complete are reported in their class. The raw records (3,417 files, about 220 MB; local copy `data/s2s_core/`) are not in the repository; Section C says how to regenerate them.

B.3 Locality of later escapes

Settings of Sections 4.4 and 4.6. Experiment folder `experiments/locality/` (README; one YAML configuration per sweep in `configs/`, written by `make_configs.py`); findings, and the changes made after two review rounds, in `notes/locality.md`, sections 7–10; reviews in `notes/reviews/locality_round1.md` and `notes/reviews/locality_round2_verification.md`.

Problems. Whitened Gaussian inputs, exact population loss, double precision, the library model (`ttn.dynamics.Problem`). Pattern teachers (`ttn.teachers.pattern_teacher`) with $d = 2$ and random orthonormal site bases, teacher seed 1000, from a stream independent of the initialization seeds. Two-component teachers have patterns 0^N and the indicator of D, weights $(1, 0.5)$, also $(1, 0.25)$, $(1, 0.75)$ and $(2, 1)$, on 60 (tree, D) pairs. Multi-component teachers (patterns as site values, weights in brackets): TA 0000 (1), 1100 (0.6), 1110 (0.35); TB as TA plus 1111 (0.2); N4 0000 (1), 1100 (0.6), 1111 (0.4); B4 0000 (1), 0110 (0.3), 1001 (0.6); E8 0^8 (1), $0^3 1^2 0^3$ (0.4), $010^4 1\,0$ (0.7); TC 0^8 (1), $1^2 0^6$ (0.6), $1^4 0^4$ (0.35), 1^8 (0.2); TD 0^8 (1), $0^3 1^2 0^3$ (0.6), $1^2 0^6$ (0.35). Bond dimension $\chi = 2$, except $\chi = 3$ for B4, N4 and E8, and $\chi \in \{4, 8\}$, $d = 4$ in the robustness sweep. Trees: MPS4, MPS6, MPS8 ($K = 4, 6, 8$), binary4, binary8 ($K = 3, 7$), HT4, HT8 ($K = 7, 15$), star3 ($K = 4$: a root without sites and three children carrying sites $\{0, 1\}$, $\{2, 3\}$, $\{4, 5\}$) and midroot6 ($K = 6$: the MPS6 chain rooted at the core of site 3).

Protocols and dynamics. *Small*: fan-in initialization (11). *Saddle*: the exact balanced saddle $s_1 U_1$ (`ttn.network.aligned_init`) plus a fan-in perturbation of per-core scale σ. *Balanced*: the

fan-in W_0 re-gauged so that $Q_e = 0$ on every bond. Gradient flow by SciPy's LSODA (`solve_ivp`) with the exact Hessian as Jacobian, relative and absolute tolerances 10^{-9} and 10^{-14}, instead of the default integrator of Section 2.5; Radau as fallback (fewer than 1% of runs). Event times are located by root finding on the dense output; event r is the last upward crossing of $a_r = \frac{1}{2}$, and a run with an overlap that crosses $\frac{1}{2}$ more than once is a *rotation*. Budgets: $t_{\max} = 10^{14}$ and a wall-clock limit of 1500 s per run (900 s for the $k \geq 6$ rows of the main sweep, 2400 s in the $k \geq 6$ extension and the stall diagnosis). A run is *stalled* when, after an overlap has passed $\frac{1}{2}$, every gradient entry stays below 10^{-12} for more than 5,000 consecutive evaluations (300,000 in the stall diagnosis, which also keeps the trajectory). The Hessian is computed by automatic differentiation at the plateau midpoint (minimum gradient norm between consecutive events), raw and with the gauge directions projected out (`ttn.diagnostics.project_out_gauge`, relative cutoff 10^{-3}). An earlier cutoff, 10^{-10}, kept the generators of nearly unused bond channels, whose normalized direction overlaps the escape route, and biased the $k = 2$ values to -0.44 to -0.46 instead of -0.5000; all Hessians were recomputed (`hess_recompute.py`). The fixed projection is still biased on the short small-initialization $k = 2$ plateaus at $\sigma \geq 10^{-3}$ (-0.411 and -0.502 against raw -0.437 and -0.581), so raw values are quoted there; raw and projected σ-exponents of $\lambda_{\min}$ differ by up to 0.11.

Sweeps. Table 15; initialization seeds 0–7 (0–4 for the $k \geq 6$ rows of the main sweep).

Table 15: Sweeps of the locality experiment (Sections 4.4 and 4.6), one configuration file each in `experiments/locality/configs/`; tables and fits use only the runs of these configurations.

sweep	content	runs
`twocomp`	60 (tree, D) pairs, small and saddle (HT8 saddle only), weights $(1, 0.5)$; 7 values of σ in $[10^{-4}, 0.1]$ for $k \leq 3$ (lower end 10^{-4} to $8 \cdot 10^{-3}$), 6 values in $[0.01, 0.2]$, $[0.03, 0.3]$, $[0.08, 0.5]$, $[0.15, 0.6]$, $[0.2, 0.6]$ for $k = 4, 5, 6, 7, 8$	5,682
`twocomp_balanced`	the 24 cases with $k \geq 4$ (small), balanced initialization	1,008
`twocomp_chi4`	the same 24 cases, fan-in, $\chi = 4$	1,005
`robust`	MPS4, binary4, HT4 (6 cases with $k \leq 4$): $(\chi, d) \in \{(4, 2), (2, 4), (4, 4), (8, 4)\}$, weights $(1, 0.25)$, $(1, 0.75)$; MPS4 $\{1, 2\}$ also from the saddle	1,840
`k5_extension`	MPS6 $\{1, 3, 5\}$, MPS8 $\{1, 5\}$, saddle and balanced, $\sigma \in \{0.012, 0.019, 0.03\}$	96
`khigh_extension`	binary8 $\{0, 2, 4\}$, HT4 $\{0, 1, 2\}$, HT4 all sites, MPS8 $\{1, 6\}$; balanced, 8 values of σ over one decade	256
`lambda2_K`	MPS4, MPS6, MPS8 with $k = 2$, weights $(2, 1)$, saddle, 7 values of σ in $[10^{-4}, 0.1]$	168
`multicomp`	TA, TB, N4, B4 on MPS4, binary4, HT4 (HT4 also balanced); E8 and TC on MPS8 and binary8; TD on MPS8; small, 6 values of σ	1,008
`multicomp_wide`	B4, N4, TA, TB on MPS4 (7–9 values of σ down to 10^{-5}); TC, TD on MPS8 from the saddle (7 values in $[10^{-4}, 0.1]$)	384
`stall_diagnosis`	22 (case, σ) pairs with stalls and escaping controls, longer patience	170
total		11,617

Analysis. Run classes: greedy, other order, rotation, stalled, timeout, incomplete, error. Fits as in Section 4.4 (per seed, at least three σ; rotations excluded; the log law is fitted on the residuals of $\ln \Delta t_2$, like the power law). Several components: plateaus detected by `ttn.diagnostics.find_plateaus` on $L(t)$ (linear time, separation ratio above 3), matched to a predicted level within 5%; laws of levels too short to detect from the time spent within 2% of the level. Bond events: effective ranks just after the event ($a_2 = 0.9$) against the plateau midpoint, threshold 0.05 times the teacher's singular value (0.03, 0.1 as sensitivity); rank profiles of several components in the middle of the time spent within 2% of each predicted level. Growth exponent: least squares of $\ln \dot{a}_2$ against $\ln a_2$ over $a_2 \in [10^{-4}, 10^{-2}]$.

Accuracy. The conservation drift $\max_e \|Q_e(T) - Q_e(0)\|_{\max}$ is at most 10^{-6} in every run (99.9% below $3.5 \cdot 10^{-8}$, median $1.2 \cdot 10^{-9}$), against Gram entries of order one after learning. Relative to the initial imbalance $|Q_e(0)| \sim \sigma^2$ it is at most $1.6 \cdot 10^{-4}$ in every run with $k \geq 4$, where the imbalance matters (Section 4.6), but up to 0.5 for $k \leq 3$ at $\sigma \leq 10^{-3}$, where σ^2 is below the absolute tolerance; those runs never stall, and their gaps agree with an independent integration to 10^{-7}–10^{-9}. LSODA (tolerances 10^{-9} and 10^{-10}), Radau and RK45 agree on a $k = 6$ gap to $3 \cdot 10^{-8}$. A reviewer's independent code (dense contraction, automatic differentiation, Radau at relative tolerance 10^{-10}) reproduced the gaps of 11 rows to 10^{-7}–10^{-9}, the gauge-projected $\lambda_{\min}$ to $7 \cdot 10^{-11}$ and the Steiner counts of all 60 (tree, D) pairs.

Excluded runs. One `lambda2_K` run (MPS8, seed 7, $\sigma = 10^{-4}$) was lost to the runner's start-up check, which compared two implementations of the velocity at the exact saddle without an absolute floor. Six slow runs did not finish within their time limit; they belong to rows reported as not measured ($k = 8$) or as spanning 0.6 decades (HT8, $k = 7$). Round-1 tables (rows on σ grids outside the committed configurations, rotation runs pooled, Hessians with the biased projection, a bond-rank statistic taken at the end of the run) are superseded. The raw run records and trajectories (every accepted integration step of every run, 8 GB) are not in the repository; Section C says how to regenerate them.

Local pilot. The pilot in `experiments/pilot_h2_locality/` (README; code `h2sim.py`, `run_sweep.py` and `analyze.py`) is an independent implementation of the same model, used as a cross-check in Section 4.4. $d = \chi = 2$; MPS4, binary4, MPS6, binary8; teacher seed 1000, weights $(1, 0.5)$; $D = \{0, 1\}$, $\{0, N-1\}$, all sites, $\{0, 2\}$, $\{0, 3\}$, $\{0, 4\}$; fan-in initialization with seeds 0–7 ($N = 4$; $\sigma \in [10^{-3}, 0.1]$ on MPS4, $[2 \cdot 10^{-4}, 0.05]$ on binary4) and 0–5 (MPS6, $\sigma \in [0.05, 0.3]$; binary8, $[0.07, 0.3]$): 784 runs, 19 of them rerun with a 100 times longer horizon, plus 340 runs from the saddle (seeds 0–3, five σ). Gradient flow by LSODA with the exact Hessian (relative tolerance 10^{-10}, absolute $10^{-13}\sigma$); events are first crossings of $a_r = \frac{1}{2}$, located by root finding; t_1 and t_2 agree with LSODA at 10^{-12}, Radau, BDF, DOP853 and RK45 to $1.8 \cdot 10^{-7}$ relative (47 comparisons). A run *collapses* when, at $t \geq 30\, t_1$, the largest second bond singular value is below 10^{-9} of its value after the first escape and still falling, or at roundoff; horizons $10^5 \sigma^{-(K-2)}$ ($N = 4$) and $10^4 \sigma^{-(K-2)}$. Fits per seed, mean ± s.d. over seeds; gap fits for $k \geq 3$ use runs with a distinct intermediate plateau. A first sweep, made before the teacher and initialization streams were separated, was discarded.

B.4 Regime-map experiments

Settings of Section 3 and Sections 5.1 to 5.4. Experiment folder `experiments/regime_map/` (README with the run order; YAML configurations in `configs/`); findings in `notes/regime_map.md`, whose section 3 (fix round 1) overrides its sections 0–2; review in `notes/reviews/regime_map_round1.md`; theory in `notes/theory_regimes.md` and Section A.8. Every number of Section 3 and of Sections 5.1 to 5.4 is an entry of the committed tables in `results/` (per-seed boundaries), `results/fix1/` (fix round 1) and `results/report/` (written from the raw runs by `report_tables.py`), or an aggregate of them in `results/report/numbers.csv` (`report_numbers.py`).

Problems. Whitened Gaussian inputs, exact population loss, double precision, the library model (`ttn.dynamics.Problem`). Odeco teachers with weights $(3, 2, 1)$ at $d = 3$ or $(3, 2)$ at $d = 2$, Haar-random site bases from teacher seed $1000 + s$ for initialization seed s (an independent stream); fan-in initialization (11). Bond dimensions: binary4 uncapped ($\chi = 3$–128); the other trees capped

by (3), except the variant of the binary tree with $N = 8$ with bonds 8 above two sites and 32 above four. Models: *centred*, target $W^* + W_0$ through the cancellation-free shifted teacher, or, for $\alpha < 10^{-2}$, through the direct loss $\frac{1}{2}\|W - W^* - W_0\|_F^2$ (cancellation error about $10^{-16}\alpha^2$; on two configurations both forms gave identical labels, event times to 10^{-4} and kernel changes to six digits); *plain*; *two-copy* (`antisymmetric_double` with the gradient mask of Remark 2.3). Imbalance: root scale $\lambda\sigma$ at fixed $\alpha = \lambda\sigma^3$. Large χ: the three parameterizations of Section 5.4 as initial scales and multipliers η_v.

Dynamics. Gradient flow (`ttn.dynamics.integrate_gf`) by Dormand–Prince 5(4) with tolerances 10^{-8} and 10^{-12}, or LSODA with the exact Hessian as Jacobian for long plateaus (the $\sigma \times K$ sweeps, the stall test, and $\sigma < 0.7$ in the sweeps of the binary tree with $N = 8$). A run stops when $L < 10^{-6} \cdot \frac{1}{2}\|W^*\|_F^2$, at the time budget $T = 50\, t_{\mathrm{esc}} + 10^4/\max(\bar{\lambda}, 1)$, with $t_{\mathrm{esc}} = (0.3\sigma)^{-(K-2)}/((K-2)s_{\min})$ the aligned escape time of the weakest component and $\bar{\lambda} = \operatorname{tr}\Theta(0)/d^N$, or at a step limit ($10^5$ steps by default; $3 \cdot 10^4$ or $2 \cdot 10^4$ in the $N = 8$ and $\sigma \times K$ sweeps, 10^4 for $K \geq 10$). Each final rise is re-integrated from the saved parameters on a 200-point linear grid (`run_regime.refine_events`); the event time is the last upward crossing of $a_r = \frac{1}{2}$ before a_r first reaches 0.9, the width the final rise from 0.1 to 0.9. Runs written before this rule are relabelled from their recorded log and fine rows (`analysis.relabel_last_crossing`). Deep in the saddle-to-saddle regime at $K \geq 7$ the widths are not resolved; separation ratios there are lower bounds, and no label depends on them.

Labels, kernels and boundaries. Labels of Table 3 with convergence measured against $\frac{1}{2}\|W^*\|_F^2$ (Remark 2.6), at the nine threshold pairs $\delta_* \in \{0.05, 0.1, 0.2\}$, $\rho \in \{2, 3, 5\}$; a run that ends on a floor below 10^{-3} counts as converged. $\delta_\Theta^{\mathrm{probe}}$ uses 128 Gaussian probe inputs (generator seed 12345) and the weights D (multipliers times mask); δ_Θ uses the tensor-space kernel $JDJ^\top$, computed in the run when $d^N \dim\theta \leq 2 \cdot 10^7$, and for earlier runs from their final parameters by `fix_round1.py` (`tensor`). Per-seed boundaries: `analysis.seed_rho_boundary` (log-log interpolation of the crossing of $\min_i \rho_i$), `seed_s2s_boundary` (label brackets; status "unresolved" when the next run is stalled) and `seed_kernel_boundary` (crossing of δ_* among converged runs); slopes against χ are pooled least-squares fits with standard errors from the scatter, checked by per-seed fits. Stall types come from the R-th bond singular value of W at the end, minimized over edges (`analysis.stall_type`); most centred runs recorded only the spectra of $W - W_0$ and are not typed.

Sweeps and run counts. Table 16; six seeds (0–5) everywhere.

Checks. 1,323 runs regenerated locally with the final driver from the committed configurations (the binary4 saddle-to-saddle side and lazy windows, large χ at $\sigma = 1.5$, $\sigma \times K$ runs above the brackets) give the labels of the remote runs in every case, with median relative differences of $\min_i \rho_i$ at most $2.5 \cdot 10^{-4}$, of $\delta_\Theta^{\mathrm{probe}}$ at most $2 \cdot 10^{-4}$ and of δ_Θ at most $7 \cdot 10^{-7}$ per sweep (`results/fix1/validation_local_vs_remote.csv`). The fixed analysis applied with the first-crossing rule reproduces the per-seed tables committed before the fix round to $5 \cdot 10^{-15}$. The conservation drift (18) has a median of 10^{-12}–10^{-9} per sweep and a maximum of $9.5 \cdot 10^{-7}$. Of the 2,736 runs of `sigma_chi_b4`, 317 stopped at the step limit, all at $\sigma \geq 4$: 283 are labelled stalled and 34 had converged.

Excluded and superseded runs. Pilot runs made before the teacher and initialization streams were separated were discarded. A first map of binary4 with SciPy's RK45 on a logarithmic output grid

Table 16: Sweeps of the regime-map experiment, one configuration each in `experiments/regime_map/configs/` (`stall_test` has two); runs after de-duplication, from `results/report/inventory.csv`.

sweep	content	runs
`sigma_chi_b4`	binary4, $d = 3$; centred, plain, two-copy; $\chi = 3, 4, 6, 9, 12, 18, 24, 36$; 19 values of $\sigma \in [0.01, 40]$	2,736
`chi_b4`	binary4; centred, χ = 9–128 (9 values), 22 values of $\sigma \in [0.004, 4]$; two-copy, χ = 16–128, $\sigma \in [0.25, 4]$	1,428
`sigma_chi_b8`	binary8, $d = 3$, centred and plain, $\chi = 3, 6, 24$, $\sigma \in [0.3, 4]$	227
`sigma_chi_b8_d2`	binary8, $d = 2$, centred, $\chi = 2, 4, 8, 16$ and bonds 8/32, $\sigma \in [0.3, 10]$	375
`sigma_K_centred_d2`	$d = 2$, centred, $\chi = 4$ on binary4, 8, 16 and MPS4, 5, 6, 8, 10, 12; full bonds on MPS6, binary8, MPS8; σ from $\max(10^{-8/K}, 0.02)$ to 2	827
`sigma_K_d2`, `sigma_K_full_d2`	the same trees, plain, $\chi = 2$ and 4; full bonds on MPS6, binary8, MPS8	1,293, 220
`sigma_K`	$d = 3$, plain, $\chi = 3$, 6, binary8 and 16, MPS8, 10, 12	302
`imbalance_b4`	binary4, centred; $\chi = 24$ and 3: 9 values of α times 10 of λ; $\chi = 16$, $\alpha = 3.375$, 9 values of λ	1,134
`large_chi_b4`	binary4, centred and plain, χ = 4–128, SP, NTK, μP-TTN, $\sigma = 1$, 1.5	432
`large_chi_b8`	binary8, $d = 2$, plain, χ = 4–32, $\sigma = 1$ (not interpreted)	71
`stall_test`	binary8 and MPS8, $d = 3$, $\chi = 3$, $\sigma = 0.3$, 0.45, 0.6; fan-in and balanced; plain and centred	120
total		9,165

(`b4_sigma_chi`) is superseded: its widths, read off that grid, were too coarse. Runs unfinished when the remote machine was stopped are missing from the tables (13 of 240 in `sigma_chi_b8`, 15 of 390 in `sigma_chi_b8_d2`, 85 of 912 in `sigma_K_centred_d2`, among them the full-bond binary8 and MPS8 runs at large σ and binary16 at $\sigma = 0.76$–0.92; the $\chi = 6$ variants of the stall test); every count above is of finished runs. Compute: one remote machine with 384 vCPUs for 6.2 hours (`notes/spend.md`). The raw records (one row and one trajectory per run, 1.2 GB; local copy `data/regime_map/`, with the fix-round reruns and the tensor-kernel cache in `data/regime_map_local/`) are not in the repository; Section C says how to regenerate them.

B.5 Covariance and teacher experiments

Settings of Section 6. Experiment folder `experiments/data_teacher/` (README with the run order; YAML configurations in `configs/`; the star and correlated-input sweeps are defined in `r2_star.py` and `r2_corr.py`); findings in `notes/data_teacher.md`; reviews in `notes/reviews/` (`data_teacher_round1.md`, `data_teacher_round2.md` and `data_teacher_round3_verification.md`, whose errata are applied in Section 6); theory in Sections A.5 and A.6.

Problems. Exact population loss, double precision, fan-in initialization (11); teacher seeds from a stream independent of the initialization seeds. Site covariances `ttn.losses.power_law_cov`(d, β) (eigenvalues $\propto k^{-\beta}$, trace d), rotated by θ in the (e_1, e_2) plane where misaligned.

- *Aligned covariance* ($N = 4$, $d = \chi = 4$): odeco teachers on covariance eigenvectors, T1 $(1, 0.8)$ and T2 $(1, 0.6)$ with the smaller component on e_1, $\beta = 0.25$; one-component teachers on e_1 or e_4 with $\beta = 0$–1.5 ($\Lambda = 0.008$–33); three components $(1, 0.7, 0.5)$ for the exact-clock check. Variants from the same initialization: (Σ, s), (I, s), $(I, \Lambda s)$, $(I, \Lambda^{1/2} s)$, and $(I, \Lambda^{1/2} s)$ from the mapped initialization (physical legs multiplied by $\Sigma_i^{1/2}$).

- *Round 1, generic teachers* ($N = 4$, $d = \chi = 4$): misaligned covariance with an odeco teacher on e_1, e_2 (weights $(1, 0.8)$), $\beta \in \{0.5, 1\}$, $\theta \in \{0, 10, 20, 30, 45\}^\circ$, also as the absorbed white problem from the mapped initialization; non-orthogonal CP teachers $u_2^{(i)} = ce_1 + (1 - c^2)^{1/2} e_2$, weights $(1, s_2)$; correlated inputs (one-factor model, odeco teacher on e_1, e_2, weights $(1, 0.7)$, $\rho = 0$–0.5), with the dense second-moment matrix from Isserlis' theorem (`correlated.py`, checked against Monte Carlo and the closed form for $N = 2$).
- *Random TTN teachers*: MPS4 teachers with $d = 3$, bond ranks $(2, 3, 2)$, bond index k weighted by 0.5^k, $\|W^*\|_F = 1$, teacher seeds 101–105, MPS4 student with $\chi = 4$; cross-tree: MPS8 teachers ($d = 2$, bond ranks 2, index weights 0.6^k, $\|W^*\|_F = 1$) on a binary8 student ($\chi = 4$), teacher seeds 201–203, then 203, 204, 209, 212, 218, and 491, 548, 643, 694, 846 (the five of seeds 200–999 whose root cut exceeds every bottom bond by the largest margins); binary8 teachers (seeds 201–203) on an MPS8 student.
- *Round 2, generic teachers* ($d = 2$, $\chi = 4$, Table 11): CP $c = 0.2$, 0.45 ($N = 4$) and 0.45, 0.7 ($N = 8$), weights $(1, 0.7)$; misaligned $\theta = 20^\circ$ ($\beta = 1$ at $N = 4$, $\beta = 2$ at $N = 8$) and 45° ($\beta = 2$, $N = 8$); random binary-tree teachers (seeds 302, 303) and random MPS teachers (seed 301 at $N = 4$, 303 at $N = 8$), bond ranks 2, bond index k weighted by q^k ($q = 1$ and 1.5), $\|W^*\|_F = 1$. Correlated inputs: $d = 3$, site-aligned odeco teacher $(1, 0.6)$, $\rho = 0$–0.4.
- *Stars* ($d = 2$, $\chi = 3$): $q = 0000$ (weight 2), $r = 1010$, $r' = 1001$ (weight w); the binary8 analogue has r, r' differing from $q = 0^8$ on sites $\{0, 2\}$ and $\{0, 4\}$.

Dynamics.

- Round-1 runs at a single σ (and the aligned-covariance sweeps): SciPy DOP853 on the flat parameter vector, tolerances 10^{-9} and 10^{-14}, 2000 logarithmic output times; checked against the library integrator (event times and losses to $7 \cdot 10^{-7}$), conservation drift at most $1.1 \cdot 10^{-9}$.
- σ sweeps: `ttn.dynamics.integrate_gf` (LSODA, tolerances 10^{-9} and 10^{-14}) in two stages: until $L < 0.98L(0)$, then on 3000 times geometric in $t - t_A$, with the recorder's fine rows; stopped at $L < 10^{-6}$–$10^{-4} L(0)$ or at $t_{\max}$ (up to 10^{10}).
- Saddle start: the lowest-loss bond-rank-one critical point (L-BFGS from 12 starts on the rank-constrained problem, balanced, zero-padded), stored in `results/r2_saddle_cores.pt`, since gauge copies with losses one ulp apart shift per-run times on another platform (not the laws); fan-in noise of scale ε; LSODA in chunks of doubling length until $\max_e s_2^e \geq 0.2$ or $t = 10^7$ (10^{11} for the small-ε extension; $\max_e s_2^e \geq 2 \cdot 10^{-3}$ for the re-timed misaligned teacher). Hessians at the exact saddle, raw and gauge-projected; saddle type by the sign of $\lambda_{\min}$ (threshold -10^{-6}); per-edge rates γ_e from the Hessian block of the two cores of the edge.
- Correlated inputs: SciPy LSODA with the exact Hessian as Jacobian, on the dense loss, in the two stages above. Online SGD check: fresh Gaussian batches, $B = 256$ and 4096, $\eta = 0.01$ and 0.002.

Events and fits.

- Random initialization: t_1 is the first time $L \leq \frac{1}{2}(L(0) + L_1)$, L_1 the level of the rank-one critical point visited (matched within 5% at the point of minimum speed; runs with no such point are *unresolved* and left out); Δt_2 is the first onset event after t_1 minus t_1. Onset event: $s_2^e \geq f\, s_2^{e,*}$ on some edge, $f \in \{0.03, 0.05, 0.1\}$ (and, from the saddle, at least 10 times its

start value); event times from merged log and fine rows (within 0.5% of fine-row times). The re-timed misaligned teacher uses $\max_e s_2^e \geq 1.8 \cdot 10^{-4}$, where $s_2^e(0) \leq 1.8 \cdot 10^{-5}$.

- Correlated inputs: events at half the final singular values across 01|23, final-rise widths; Δt_2 is the time between the events of s_2^e and s_1^e. Stars: θ_q is left when $a_q < 0.99$ (departure) or $a_q < 0.9$ before any $a_r > \frac{1}{2}$; the two rules agree for every run with $\varepsilon \leq 10^{-3}$.
- Fits per seed (`lawfit.py`): $t = A \ln(1/\sigma) + B$ and $t \propto \sigma^{-p}$, both scored in log space; asymptotic fits through the three smallest σ (random initialization) or the five smallest ε (saddle start). Errors are standard errors over seeds; per-seed fit errors are not propagated, and changing the window moves slopes by 1–3%. Increments per halving: per-seed regression of Δt_2 on $\log_2(1/\sigma)$ with the actual σ.
- Rank-one predictions: `constrained.py` (L-BFGS on the bond-rank-constrained loss from many starts) and `predict_rank1.py`; slopes h_e from `theory_log_slope.py` and the exact-saddle Hessians; heuristic exponents α from `p1_exponents.py`.

Sweeps and run counts. Table 17; about 9,100 runs in all.

Table 17: Sweeps of the experiment of Section 6: configuration files in `configs/` of the experiment, except where a script is named.

sweep	problem	σ or ε, seeds	runs
`e1_order`	aligned, T1, T2, four variants, binary4, MPS4	$\sigma = 0.02$–0.2 (3 per tree), 40	1920
`e1_scaling_sigma`, `_lambda`	aligned, t_1 against σ and Λ	6 σ; 13 teachers; 10	360 + 520
`e2_misaligned`, `e4a_nonorth`, `e3_correlated`	round 1, one σ per tree	0.05, 0.1; 20	800 + 560 + 240
`e4a_sigma`, `e2_sigma`	MPS4, $c \in \{0, 0.05, 0.2, 0.45\}$; $\theta \in \{0, 5, 20\}^\circ$	9 σ, $3 \cdot 10^{-4}$–0.3; 10	360 + 270
`e4b_ttn`	random MPS4 teachers	8 σ, 10^{-3}–0.2; 6	240
`e4c_cross`, `r2_cross`, `r2_cross_b`	cross-tree, $N = 8$	0.3–0.5; 8	48 + 120 + 120
`r2_random`	6 generic teachers, MPS4, binary4	9 σ per tree; 5	540
`r2_saddle` (+`_b`, `_ext`, `_m20`)	13 generic teachers, 6 trees, saddle start	$\varepsilon = 10^{-8}$–$5 \cdot 10^{-2}$; 5	1950 (+54 fine rows)
`r2_corr.py`	correlated, $d = 3$, MPS4, binary4	6 σ, 0.0016–0.05; 5	300
`r2_star.py`	stars: saddle start; random init	$\varepsilon = 10^{-6}$–10^{-2}, 15 w; 6 σ, 10	468 + 240
`e3_sgd_check.py`	online SGD, binary4, $\rho = 0.3$	$\sigma = 0.05$; 3	9

Checks and exclusions. A float64-floor scan (`check_float_floor.py`) takes the smallest dormant signal between the first escape and the next learning event (overlap envelope below 10^{-12}, or $\max_e s_2^e$ below 10^{-14} for TTN teachers): it flags only the odeco baselines ($c = 0$, $\theta = 0$) at $\sigma \leq 3 \cdot 10^{-3}$, whose exponents without the flagged runs are in `results/report_float_floor_refit.csv`. Excluded: the round-1 single-σ plateau durations (withdrawn), the pilot versions of `e4b` and `e4c`, seed 2 of MPS4 under correlated inputs (minor saddle at every ρ), and 14 HT8 runs at $\varepsilon \leq 3 \cdot 10^{-4}$ (CP, $c = 0.45$) stopped beyond $t = 10^8$. An independent verification recomputed the Hessians at 12 stored saddles with its own code and reproduced the committed saddle-start escape times on Windows to $4 \cdot 10^{-9}$ (MPS4) and $4 \cdot 10^{-5}$ (binary4). Raw outputs (about 10 GB, local copy

`data/data_teacher/`) are not in the repository; the summary tables in `results/` suffice to redraw every figure.

B.6 Discrete steps, gradient noise and finite samples

Settings of Section 5.5. Experiment folder `experiments/optim_samples/` (README with the script list; YAML configurations in `configs/`); findings in `notes/optim_samples.md`; the two reviews in `notes/reviews/` (`optim_samples_round1.md` and `optim_samples_round2_verification.md`, whose errata are applied in Section 5.5); theory in Section A.7.

Problems. Whitened Gaussian inputs, exact population loss (empirical loss in the finite-sample sweep), double precision. The GD and SGD loops of the experiment are its own (PyTorch), checked against `ttn.dynamics` (final losses agree to 10^{-14}, event times to $5 \cdot 10^{-4}$; `validate_library.py`). Teacher seed 101, independent of the initialization seeds.

- *Main problems*: binary trees with $N = 4$ ($K = 3$) and $N = 8$ ($K = 7$), MPS with $N = 8$ ($K = 8$); $d = 3$, $\chi = 3$; odeco teacher with weights $(3, 2, 1)$ and Haar-random site bases; $\eta_c = 0.1541$, 0.0435, 0.0366. Fan-in initialization (11); *imbalanced*: $\sigma = 0.1$ with the root core at scale 2, 4 or 6. Centred model (Section 2.4, variant (b)) with $\chi = 9$, the cut dimension of binary $N = 4$.
- *Aligned starts* (`ttn.network.aligned_init`): every per-core amplitude 0.05 (SGD sweep); component 1 at its balanced value $3^{1/3}$ and components 2 and 3 at b_0 (route test); the balanced minimum plus a Gaussian perturbation of size 10^{-5} per entry (SGD stability test).
- *Strict route*: MPS with $N = 4$, $d = \chi = 3$, pattern teacher with weights $(3, 0.3)$ and patterns 0000, 1100 (second component needs $k = 2$), $\sigma = 0.1$.
- *Finite samples*: binary $N = 8$, $d = 2$, odeco teacher $(3, 2)$ (bond rank 2; teacher-rank TTN with 52 parameters, manifold dimension 28), students with $\chi = 4$ (208 parameters, tangent dimension 112, the measured rank of the Jacobian) and $\chi = 16$ (832 parameters, all bonds full); binary $N = 6$ ($K = 5$), $d = 3$, teacher $(3, 2)$ (manifold dimension 48), $\chi = 4$ (tangent dimension 120). n inputs with data seed 500+seed and noise-free labels; seed s fixes both the data and the initialization.

Dynamics.

- *Gradient-flow references*: SciPy DOP853 with relative and absolute tolerances 10^{-10} and 10^{-14}, instead of the default integrator of Section 2.5; every upward crossing of $a_r \in \{0.1, 0.5, 0.9\}$ is located by the integrator's event finder; conservation drift (18) at most $2 \cdot 10^{-12}$. In the $\chi = 9$ sweep: LSODA (`ttn.dynamics.integrate_gf`, tolerances 10^{-8} and 10^{-12}, 3000 logged times), stopped at $L = 10^{-8}$; λ_{GF} is the top eigenvalue of the full Hessian at the end. Route test and strict route: LSODA as well.
- *GD*: η a multiple of η_c, or in the $\chi = 9$ sweep of each run's own $2/\lambda_{\mathrm{GF}}$, for a gradient-flow time T: 2800, 900, 300, 100, 40 for $\sigma = 0.003$, 0.01, 0.03, 0.1, 0.3 on binary $N = 4$, 150 (binary $N = 8$), 60 (MPS), 100 (imbalanced), and as long as the reference took to converge ($\chi = 9$). Loss and overlaps at every step; sharpness (full Hessian), conservation drift and bond ranks at 41 checkpoints; diverged: loss non-finite or above $10^6 L_0$.

- *Event shift*: c_e from the variational equation $\dot{\delta} = -H(\delta + \frac{1}{2}\nabla L)$, $\delta(0) = 0$, integrated along the reference path, $c_e = -\nabla a_r \cdot \delta / \dot{a}_r$ at the event (`predict_shift.py`); the Richardson check extrapolates the measured shifts at $0.01\eta_c$ and $0.03\eta_c$ linearly to $\eta = 0$.
- *Online SGD*: fresh Gaussian inputs every step, no label noise (except in the label-noise runs of the strict route, $\zeta \in \{10, 20, 30\}$), an independent generator for every (initialization, noise seed, B, η). Overlaps every 4 steps; runs continue to $1.3 \max(t_1^{\mathrm{GF}}, t_2^{\mathrm{GF}}) + 2$, capped at 150, 60, 40 time units for $\sigma = 0.03$, 0.1, 0.3. Route test: `ttn.dynamics.run_sgd` for up to $4T_{\mathrm{GF}}$, with T_{GF} from the one-dimensional aligned integral (checked against LSODA to 10^{-8}). Stability test: 25 time units; converged if the final loss is below the initial one, diverged if non-finite or above 10^3. Mean-square limits $\eta^*_{\mathrm{ms}}(B)$ from the dense second-moment operator (`sgd_stability.py`), equal to (57).
- *Finite samples*: GD on the empirical loss with step $1/\lambda_{\max}$ (power iteration on the training Hessian every 200 steps), the relative parameter change per step capped at 1% (near the origin $1/\lambda_{\max}$ is huge), $6 \cdot 10^4$ steps ($3 \cdot 10^5$ in the long centred runs); no run reached the training tolerance 10^{-8} (final training loss $2 \cdot 10^{-7}$–$2 \cdot 10^{-4}$ of its initial value for small initialization from $n = 256$). Lazy limit: minimum-norm least squares with the Jacobian at initialization (relative cutoff 10^{-12}), exact.
- Auxiliary computations: Hessians at balanced and gauged minima on five trees ($\chi = 3$, 5; `check_sharpness.py`); the scalar map for $K = 3, 4, 7, 8$, h/h_c from 0.8 to 3.2 in steps of 0.005, $6 \cdot 10^4$ iterations from $x = 0.3$, attractor from the last 2000 (`map1d.py`); Hessians at the second-plateau midpoint of the strict route ($\sigma = 0.01$–0.1, five seeds; `check_k2_saddle.py`).

Sweeps and run counts. Table 18; 4,165 runs in total, plus 150 exact lazy limits.

Events and classification.

- Overlap events, in the GD, gradient-flow and SGD sweeps and the route test: the last upward crossing of $a_r = \frac{1}{2}$, with $a_r \geq \frac{1}{2}$ at the end of the record (`common.record_events`, or the integrator's event finder for the DOP853 references); first crossings and the number of crossings are kept to count transients; widths are final rises (the first crossing of 0.9 minus the last crossing of 0.1 before it). The strict-route scripts (`collapse_test.py`, `check_k2_saddle.py`) use the first upward crossing of $\frac{1}{2}$ on the logged trajectory. No experiment of this section uses spectral events: bond spectra enter only through effective bond ranks (singular values above 0.1 times the smallest teacher weight, at checkpoints and at the end) and, in the route test, through the second Schmidt value as the route amplitude.
- GD outcome classes, checked in this order (`analyze_gd.py`): *diverged*; *GF-like*: final loss at most $1.05 L_{\mathrm{GF}}(T) + 10^{-3} L_0$, with $L_{\mathrm{GF}}(T)$ the reference at the same time, and no step raising the loss by more than $10^{-3} L_0$; *catapult*: the same final loss after such a rise; *thrown back to the origin*: not at that level and the loss back above $0.9 L_0$ after the first event; *bounded oscillation*: over the last 20% of the steps (at most 2000) the total variation of the loss exceeds ten times its net change and the tail reaches above gradient flow's level, or the end point is linearly unstable ($\eta\lambda_{\max}/2 > 1.001$); *other*: none of these (it did not occur in the runs shown). GF-like includes 39 of the 195 small-initialization runs that end, like their references, on the last plateau ($L/L_0 = 0.071$).
- SGD: the first escape is the first time any component reaches $\frac{1}{2}$, compared with gradient flow's; a run *forgot* if its first-learned component later falls below $\frac{1}{2}$. Runs are classified before

Table 18: Sweeps of the experiment of Section 5.5: configuration files in the folder `configs/` of the experiment, except the last four rows, which are defined in their scripts. Seeds 0–4 unless stated; GF: gradient-flow references.

sweep	problem, σ	steps, batches, sizes	runs
`gd_sweep`	binary $N = 4$, $\sigma = 0.003$–0.3; binary and MPS $N = 8$, $\sigma = 0.9$; imbalanced	η/η_c: 18 values in $[0.01, 3]$ (10 for $\sigma \le 0.01$; 10 in $[0.1, 1.05]$ imbalanced)	700 + 50 GF
`gd_order`	binary $N = 4$, $\sigma = 0.03$, 0.1, 0.3, seeds 5–44	$\eta/\eta_c = 0.1$, 0.3, 0.5, 0.9	480 + 120 GF
`gd_centred_chi9`	binary $N = 4$, $\chi = 9$: centred $\sigma = 1$–16 (GF), 1, 2, 4, 8 (GD); plain $\sigma = 0.1$, 0.3	$\eta\lambda_{\mathrm{GF}}/2$: 10 values in $[0.5, 4]$	300 + 75 GF
`sgd_sweep`	binary $N = 4$, $\sigma = 0.03$, 0.1, 0.3; aligned start	$B = 1$–256 at $\eta = 5 \cdot 10^{-4}$; $\eta = 10^{-4}$–$2 \cdot 10^{-3}$ at $B = 4$, 10^{-3}–$2 \cdot 10^{-2}$ at $B = 64$; 10 noise seeds per initialization (20 aligned)	1780
`p9_sgd.py`	binary $N = 4$, aligned route, $b_0 = 0.01$, 0.005, 0.003, 0.002	$\eta/B = 3.125 \cdot 10^{-5} \cdot 2^j$, $j = 0, \ldots, 3$, at $B = 64$ (and 16 for $b_0 = 0.01$, 0.003); 6 noise seeds	144
`sgd_limit_test.py`	binary $N = 4$, balanced minimum	$\eta/\eta^*_{\mathrm{ms}} \in [0.5, 3]$ (8 values), $B = 1$, 16	80
`collapse_test.py`, `check_k2_saddle.py`	MPS $N = 4$, strict route	SGD $B = 1$, $\eta = 5 \cdot 10^{-4}$, $\zeta \in \{0, 10, 20, 30\}$; GF at six $\sigma \in [0.01, 0.1]$	21 + 30
`samples_sweep`	binary $N = 8$, $d = 2$: small $\sigma = 0.3$, 0.5, centred $\sigma = 1$ ($\chi = 4$, 16), centred $\sigma = 1.5$, 2 ($\chi = 4$); binary $N = 6$, $d = 3$: small $\sigma = 0.5$	$n = 16$–512 (11 values; 5 for $\sigma \ge 1.5$, and 256, 512 with $3 \cdot 10^5$ steps; 8 values 32–384 for $N = 6$)	385

pooling; means ± s.e. are across initializations. Second plateaus use only initializations with a resolved gradient-flow plateau ($\rho_1 > 3$) and no near-tie (the eventually second component does not peak between 0.15 and 0.6 during the first event), and only runs that kept the first component and gradient flow's order. Spread exponents: per initialization, least squares of $\ln \mathrm{sd}(t_1)$ against $\ln(\eta/B)$. Collapse law: weighted least squares (weights $1/\mathrm{s.e.}^2$) of $T_{\mathrm{SGD}}/T_{\mathrm{GF}}$ over the 19 conditions in which every run escaped, one c_γ per law; *trapped*: no persistent event by $4T_{\mathrm{GF}}$.

- Finite samples: test error $\|W - W^*\|_F^2/\|W^*\|_F^2$; *recovered*: test error below 10^{-2} (thresholds 10^{-3} and 10^{-1} give the same picture); effective bond rank: singular values above 0.2, a tenth of the smallest teacher weight, averaged over the edges; the lazy limit depends only on the initial direction and is counted once per (setting, n, seed).

Excluded runs. Superseded after review round 1 and not analysed: the first GD and SGD sweeps, whose events were first crossings and whose SGD runs stopped right after the second crossing; the first route test, whose $B = 16$ and 64 runs shared noise streams (kept as `p9_sgd_round1.csv`); the $\chi = 3$ centred-model sweep, whose gradient-flow references stalled because $W^* + W_0$ has bond rank up to 6 there; and a first strict-route test on binary $N = 4$, where the pattern lies inside one core ($k = 1$). On MPS $N = 8$, seed 3 never learns component 1 within $T = 60$ under gradient flow and is left out of the oscillation-loss comparison; in the strict-route Hessians two runs at $\sigma = 0.01$ have no

resolved plateau midpoint. Raw outputs (about 420 MB, local copy `data/optim_samples/`) are not in the repository; the summary tables in `results/` suffice to redraw every figure (Section C).

B.7 Optimizers used in practice

Settings of Section 5.6. Experiment folder `experiments/optimizers/` (one YAML configuration per sweep in `configs/`); findings and claims withdrawn after review in `notes/optimizers.md`, sections 2, 3 and 5.

Problems. Whitened Gaussian inputs, exact population loss (empirical loss in the finite-sample sweep), double precision. Teacher seed 100; initialization seeds 0–5, from a stream independent of the teacher's.

- *binary4*: binary tree, $N = 4$ ($K = 3$), $d = 3$, $\chi = 4$; odeco teacher with weights $(3, 2, 1)$ and Haar-random site bases.
- *binary8*, *MPS8*: $N = 8$ ($K = 7$ and 8), the same odeco teacher, $d = 3$, $\chi = 4$.
- *E8* on binary8 and MPS8: pattern teacher with $d = 2$, $\chi = 3$ and patterns $\pi_0 = (1, \dots, 1)$ (weight 1), π_A equal to 2 at sites 4 and 5 (weight 0.4) and π_B equal to 2 at sites 2 and 7 (weight 0.7), 1 elsewhere. The cores that carry the differing sites span $k = 2$ (π_A) and $k = 6$ (π_B) cores on the MPS and $k = 5$ for both on the binary tree.

Initializations. Fan-in (11) with per-core scale σ; *balanced*: the same W_0 in the gauge with $Q_e = 0$ on every edge. All deep-tree baselines use the balanced version, because fan-in gradient flow converges by $t = 2 \cdot 10^6$ in none of the 12 MPS8 odeco runs (the conservation-law trap of Proposition A.28, verified at the route ends in 11 of the 12 stalls), against 10 of 12 (binary8 odeco), 11 of 12 (binary8 E8) and 11 of 12 (MPS8 E8) fan-in runs. Methods whose iterates depend on the gauge were run from both versions; re-gauged GD, Riemannian GD, ALS and DMRG depend only on W_0 and were run from the balanced one.

Dynamics. All methods stop at $L < 10^{-6} L_0$ (finite samples: empirical loss below 10^{-8} times its initial value) or at their budget.

- *Gradient flow*: SciPy LSODA with relative and absolute tolerances 10^{-9} and 10^{-14} and the exact Hessian as Jacobian (Radau as fallback), instead of the default integrator of Section 2.5; the crossings $a_r \in \{0.1, 0.5, 0.9\}$ are located by root finding, so event times are exact. Budgets: $t_{\max} = 8/\sigma$ (binary4), $2 \cdot 10^6$ (deep trees), $4 \cdot 10^6$ (binary8 scan), 10^6 (finite samples).
- *GD and heavy ball* (PyTorch): $\eta = 0.02$ and 0.002 with $\beta = 0.9$ on binary4 ($0.13\,\eta_c$ for both, counting heavy ball's effective step $\eta/(1-\beta)$); $\eta = 0.01$ and 0.001 on the deep trees (0.23–$0.27\,\eta_c$), at $\sigma = 0.8$ only, because gradient-flow plateaus at $\sigma = 0.3$ last 10^4–10^6 time units. Budgets $8/\sigma$ (binary4) and $2 \cdot 10^4$ (deep) gradient-flow time units, at most $5 \cdot 10^5$ steps.
- *Adam* (PyTorch, bias-corrected): $\beta_1 = 0.9$, $\beta_2 = 0.999$ (also 0.99 and 0.9 at $\eta = 10^{-3}$ on binary4), $\epsilon = 10^{-8}$ (10^{-30} in the ϵ sweep), $\eta \in \{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 10^{-2}\}$ on binary4 and $\{10^{-3}, 10^{-2}\}$ on the deep trees. *RMSProp*: $\alpha = 0.99$, $\epsilon = 10^{-8}$, $\eta \in \{10^{-4}, 10^{-3}\}$. Budgets 10^5–$3 \cdot 10^5$ (binary4), $3 \cdot 10^5$ (deep), $5 \cdot 10^5$ (binary8 scan) and $2 \cdot 10^5$ (ϵ sweep) steps.
- *Re-gauged GD*: $\eta = 0.02$ (binary4) and 0.01 (deep); QR sweeps from the leaves to the root bring the cores to the root-centred canonical form every 1 or 50 steps; $5 \cdot 10^4$, 10^5 and $2 \cdot 10^5$ (binary8 scan) steps.

- *Riemannian GD*: $\eta \in \{0.1, 1\}$; tangent projection by least squares with the dense Jacobian of $W(\theta)$ (relative cutoff 10^{-12}), so only small d^N are feasible; leaves-to-root HSVD retraction; 2000 steps.
- *Gram-preconditioned GD*: $\eta \in \{0.01, 0.1, 0.5\}$; each Kronecker factor of a core's Hessian block is damped by 10^{-3} times its mean eigenvalue (scale-free) or, at $\eta = 0.1$, by an absolute floor 10^{-4} or 10^{-2}; no divergence guard, because the first step overshoots by up to $10^{34} L_0$ before the loss returns; $5 \cdot 10^4$ (binary4; $2 \cdot 10^5$ with a floor) and 10^5 (deep) steps.
- *ALS* (one site) and *DMRG* (two sites, then SVD truncation to at most χ values with relative cutoff 10^{-12}): exact least squares; the orthogonality centre moves by QR along a depth-first tour of the tree; at most 50 sweeps.

Sweeps and run counts. Table 19 lists the sweeps; six initialization seeds per setting, 2529 runs in total.

Table 19: Sweeps of the optimizer experiment (Section 5.6), one configuration file per sweep in `experiments/optimizers/configs/`.

sweep	problem	σ	methods	runs
`scaling_b4`	binary4, fan-in	$0.002 \cdot 2^j$, $j = 0, \ldots, 6$	19 settings: gradient flow, GD, heavy ball, Adam (6), RMSProp (2), re-gauged GD (2), Riemannian GD (2), Gram (2), ALS, DMRG	798
`precond_b4`	binary4, fan-in	as above	Gram: 3 step sizes with scale-free damping, 2 absolute floors	210
`deep`	binary8, MPS8; odeco, E8; balanced and fan-in	0.3, 0.8	gradient flow, GD and heavy ball ($\sigma = 0.8$), Adam (2), RMSProp, Gram; from the balanced W_0 only: re-gauged GD (2), Riemannian GD (2), ALS, DMRG	864
`later_b8`	binary8 odeco, balanced	0.25, 0.3, 0.35, 0.4, 0.5, 0.6, 0.8	gradient flow, Adam (2; every step recorded for the first 2000 steps), re-gauged GD	168
`eps_mps8`	MPS8 odeco, balanced	0.01, 0.015, 0.02, 0.03, 0.05, 0.1, 0.2, 0.3	Adam: $\eta \in \{10^{-3}, 10^{-2}\}$, $\epsilon \in \{10^{-8}, 10^{-30}\}$	192
`implicit_bias2_b4`	binary4, $n \in \{45, 60\}$, data seeds 200–203	0.01, 0.1	gradient flow, Adam (2); pairs share draw and initialization	288
`adam_probe`	binary4, binary8, MPS8 E8	single runs	Adam and RMSProp internal state (illustration only)	9

Events and classification.

- Event r is the last upward crossing of $a_r = \frac{1}{2}$ after which a_r stays at least $\frac{1}{2}$; transient crossings are not events (Section 2.7). Width: final rise from 0.1 to 0.9. Saddle-to-saddle: every separation ratio $\rho_i > 3$ (Table 3).
- Discrete methods record loss and overlaps at every step for the first 2000 steps, then every $\max(1, n_{\text{steps}}/(5 \cdot 10^4))$ steps. A first escape or gap shorter than two recording intervals is unresolved and excluded from fits (and counted in the result tables). Two components that cross $\frac{1}{2}$ in the same step or micro-step have no resolved order. For Gram-preconditioned GD, events are defined only after the loss has returned below L_0; a component already above $\frac{1}{2}$ at that point has no resolved event.

- Exponents: per seed, a least-squares fit of $\ln T$ against $\ln \sigma$ over the σ at which the seed follows its most common learning order (for plateau durations, not for times to reach a loss level); mean $\pm$ standard error over seeds. A logarithmic law fitted in the same space serves as comparison (gradient flow on binary4: root-mean-square residual of $\ln T$ 0.005 for the power law, 0.62 for the logarithmic law).
- Orders are compared with gradient flow only where both are resolved, for the same problem, σ and seed. Re-gauged GD, Riemannian GD, ALS and DMRG give one trajectory per seed at every σ, so their orders are counted over the six independent seeds (at the smallest σ).
- ϵ *test* (26): $T_{\mathrm{GF}}(\sigma)$ is the seed's exact gradient-flow first-escape time at $\sigma = 0.3$ (`deep` sweep, same balanced initial point up to scale) times $(0.3/\sigma)^6$, and n_{sign} the median over σ of the same seed's escape with $\epsilon = 10^{-30}$ at the same η. Runs without an escape within $2 \cdot 10^5$ steps enter only through the check that the prediction exceeds the budget.
- Convergence for cost and time-to-loss tables: $L < 10^{-3} L_0$. Runs that hit their budget are reported as budget-limited, never as stalls. RMSProp hovers above $10^{-6} L_0$ on the deep trees within $3 \cdot 10^5$ steps; its events are used, its convergence flags are not.

Excluded runs. Runs replaced after review round 1 are kept in `superseded/` subfolders of the raw data and ignored by the analysis: gradient-flow runs whose event times were interpolated on an output grid (errors up to 1.7%), Gram-preconditioned runs stopped by a $10^6 L_0$ loss guard after one step, binary8 Adam runs recorded only every 10 steps, and the first finite-sample sweep (`implicit_bias_b4`, 288 runs, one data draw per initialization). The pilots (`pilot_calib`, `pilot_trap`) support no claim. Raw trajectories (one file per run, 2.2 GB) are not in the repository; Section C says how to regenerate them.

B.8 Real data

Settings of Section 6.9. Experiment folder `experiments/real_data/` (README with every command in order; YAML configurations in `configs/`); findings in `notes/real_data.md`, whose section 8 (fix round) supersedes its sections 4–7 and whose section 9 holds the pre-registered order test; review in `notes/reviews/real_data_round1.md`.

Data and model. MNIST (torchvision mirror) and Fashion-MNIST, fetched and downsampled by `download.py`: images cropped to the central 24×24 pixels and block-averaged to 4×4 (6×6 blocks) or 8×8 (3×3 blocks), intensities in $[0, 1]$. Tasks: MNIST 3 ($y = +1$) against 5 ($y = -1$); Fashion-MNIST sneaker (7) against ankle boot (9), chosen before any run for a visible shape difference at 4×4. Balanced subsamples of 4000 training and 1600 test images, data seed 0. Site features $\phi(x) = (\cos \frac{\pi x}{2}, \sin \frac{\pi x}{2})$ (*raw*), or per site $A_i \phi$ with $A_i \mathbb{E}[\phi\phi^\top] A_i^\top = I$ (*white*; uncentred second moment, eigenvalues floored at 10^{-3} of the largest), which keeps the function class. Model $f(x) = \langle W, \bigotimes_i \phi(x_i) \rangle$ and loss $\frac{1}{2n} \sum_j (f(x_j) - y_j)^2$; in the regime sweep the centred model trains $f - f_0$. Trees from `data.make_tree`: binary16 ($K = 15$) and MPS16 ($K = 16$), binary64 ($K = 63$) and MPS64 ($K = 64$) for 8×8 images; $\chi = 4$ (8 in one sweep, where the bottom bonds are capped at 4 by their two-pixel cut). Pixel orders (`data.pixel_order`; site s carries pixel order$[s]$, pixels numbered row by row): quadtree (Morton key, column bit lowest), snake (boustrophedon rows), scrambled$k$ (random permutation from `numpy.random.default_rng`$(k + 12345)$), and for the order test quadtreeT (Morton order of the transposed image), rowmajor, colmajor, hilbert (Hilbert curve) and mortonrev (site s carries the pixel whose Morton key is s with its bits reversed). Fan-in

initialization (11) at per-core scale σ with initialization seeds 0–4, independent of the data seed. Double precision, one thread per process.

Dynamics (`run.py`). Phase 1: Dormand–Prince 5(4) with relative and absolute tolerances 10^{-8} and 10^{-14}, the clock re-anchored ($t \to 0$, offset kept) whenever an accepted step is shorter than $10^{-9}t$, so that escapes at $t \sim 10^{15}$–10^{20} are resolved; it stops at the first accepted step at which the loss has fallen by 1% of $L(0)$ (time t_a). Phase 2: LSODA with the same tolerances on the clock $\tau = t - t_a$, with the Gauss–Newton part of the Hessian (closed-form per-sample Jacobians) as Newton matrix, up to $\tau = 10^4$ (3000 for 8×8 images; $\min(3000, 10^4$ kernel units) in the regime sweep, a kernel unit being $128/\operatorname{tr}\Theta$ at t_a). Records: 30 logarithmic times per decade, 1000 linear times (600 for 8×8) and the library's fine rows, merged for every event time. Conservation drift (cumulative, relative to the largest squared core norm) at most $4.6 \cdot 10^{-8}$, median $1.8 \cdot 10^{-9}$ over the 234 main runs. Two cases rerun with tolerances 10^{-10} and 10^{-16} (`tol_check`) agree in t_a to 7 digits, in the final loss to 6, and in later drop times to 0.1%; the first drop ($\tau \approx 10$–15) moves by 0.2–0.8 time units, below every gap reported.

Events and analyses.

- *Loss drops* (`analysis.py`): peaks of $-\mathrm{d}L/\mathrm{d}t$ that stand out by a factor 3 against the minima separating them from higher peaks and lower the loss by at least 1% of $L(0)$; time: the peak; width: its full width at half maximum. Widths by progress (from 10% to 90% of a drop) are not used, since the loss creeps between drops and such widths measure the creep. Separation ratios and the S2S signature as in Section 2.7, with these drops.
- *Data-weighted bond spectra* (`data.function_spectra`): across edge e the network computes $f(x) = \sum_a u_a(x)v_a(x)$, with messages u_a from below the edge and v_a from above it; the Schmidt values are the square roots of the eigenvalues of $G_u^{1/2}G_vG_u^{1/2}$, where G_u and G_v are the Gram matrices of the two messages on the training set. They are gauge invariant and equal the Frobenius spectra under a product law with identity second moments (checked to $2 \cdot 10^{-12}$). Frobenius spectra are not used: after the escape $\|W\|_F$ grows to 10^6–10^7 along directions the data barely see.
- *Rank events*: s_k^e/s_1^e crossing 0.05 (log-interpolated); an edge belongs to a drop if it crosses inside the drop's half-maximum window widened by one width on each side. Crossings already met at $\tau = 0$ are flagged as unresolved.
- *First-route gaps* (`scaling.py`): gap = drop 1 − drop 0 on the τ clock; a route's drop is the one whose widened window contains its rank crossing, and points where it rose between drops, merged with drop 0, or where another route rose first are excluded with their reason. Clean series: one (order, seed) whose first route, one sibling pair, rises inside drop 1 at every σ, with at least four σ over at least half a decade (11 series); full sweep: 91 of the 124 binary points kept (16 between drops, 2 merged, 15 another route first), 10 seed series. Fits per seed: power law by least squares in $\ln g$, log law and $A/\sigma + B$ by weighted least squares with weights $1/g$; all compared by the relative rms residual.
- *MPS edges*: time of each edge's first rise after the first drop, per seed over the seven σ, both laws fitted with relative residuals (`scaling.py`, `edge_scaling`).
- *Route amplitudes* (`route_amplitudes.py`). A binary TTN of bond rank one computes $f = c\prod_v q_v$, $q_v = g_v \cdot \Phi_v$, over its bottom cores v, Φ_v the features of the two pixels of v. For a

pair (u, w) of bottom cores, with the residual $r_j = f(x_j) - y_j$ and $h_{uw} = c \prod_{v \neq u,w} q_v$,

$$\mu_{uw} = \|P_u M_{uw} P_w\|_2, \qquad M_{uw} = \frac{1}{n} \sum_j r_j \, h_{uw}(x_j) \, \Phi_u(x_j) \Phi_w(x_j)^\top, \tag{63}$$

with $P_u = I - g_u g_u^\top / \|g_u\|^2$; k_{uw} is the number of cores of the tree path from u to w. Evaluated for all 28 pairs at the plateau midpoint (minimum of $\|\dot\theta\|$ between the first two drops, reached by re-integrating from the end of phase 1). States farther than 0.05 (relative) from a rank-one product are dropped (28 of 124). Units (`analysis.route_units`): rank-one states of the same (order, seed) whose normalized patterns $\mu_{uw}^2 / \sum \mu^2$ differ by less than 0.2 in total variation and whose plateau losses differ by less than 0.01; 96 states form 26 units in the 20 (order, seed) pairs.

- *Hessians* (`plateau_hessians.py`): exact Hessian (dense residual tensor) at the plateau midpoint (minimum $\|\dot\theta\|$ between the first two drops), raw and gauge-projected (Section 2.7); exponents per seed over the runs at the common plateau level ($L \in [0.265, 0.275]$ binary, $[0.225, 0.235]$ MPS; $[0.31, 0.345]$ on Fashion-MNIST, pooled).
- *Regime sweep* (`regime_analysis.py`): kernel on the first 128 test images; labels by Table 3 without the convergence rule, which no $\chi = 4$ network meets on 4000 labels (lazy if $\delta_\Theta(T) < 0.1$; S2S if at least two drops, all $\rho > 3$; rich otherwise); the linearized model at $\theta(t_a)$ in closed form, with its floor $L_{\text{lin}}(\infty)$.
- *Order test* (`order_predictor.py`, `order_test.py`; registered analysis): the best rank-one fit $f = \prod_v g_v \cdot \Phi_v$ of the labels from 32 starts (start i: unit g_v from `default_rng`$(1000 + i)$, 50 steps of higher-order power iteration on $\frac{1}{n} \sum_j y_j \bigotimes_v \Phi_v(x_j)$, rescaling to the least-squares amplitude, then alternating least squares until the relative loss change is below 10^{-12} or 2000 sweeps), the lowest-loss fit balanced, S from (63) at that state; outcomes O1 (mean over seeds of $L(10^4)$) and O2 (mean of $\log_{10} \tau(L \leq 0.25)$ on the running minimum, censored at 10^4); Spearman ρ over orders, percentile bootstrap intervals (10,000 resamples of orders) and two-sided permutation p-values (10,000 permutations of S).

Sweeps and run counts. Table 20; 507 runs of the experiment's own configurations, plus the round-1 reviewers' runs with the same code (among them 11 binary runs in nine orders and three MPS chains in other orders, $\sigma = 0.1$, seeds 0 and 1; local copy `data/review_real_data/`), which enter only where the text says "review runs". The main, regime, Fashion-MNIST, $\chi = 8$ and 8×8 sweeps ran on one 64-core machine (about 60 processes, as the runs are bound by memory bandwidth); the fix-round reruns and the order test ran locally (the order test: 8 processes, 70 minutes).

Checks and exclusions. The fix-round reruns reproduce the committed main-sweep values: t_a to $7 \cdot 10^{-15}$ (relative) and the final loss to 5 digits for the quadtree at $\sigma = 0.1$, and the timing of the first MPS rise within 0.07 time units (median 0.01). The round-1 review reproduced the first-plateau times to about 10^{-15}, the gaps within 1.3% and the Hessians and route amplitudes within 4%, and an independent reimplementation agreed. Excluded: the first-route points of other routes (above) and the 28 route-amplitude states that are not near rank one. The round-0 claims superseded by the fix round are listed in `notes/real_data.md`, section 8.13. Raw outputs (about 0.8 GB, plus 0.2 GB of local reruns; local copies `data/real_data/` and `data/real_data_local/`) are not in the repository; the JSON summaries in `results/` suffice to redraw the figures of Section 6.9.

Table 20: Sweeps of the experiment of Section 6.9: configuration files in the folder `configs/` of the experiment. Raw features and $\chi = 4$ unless stated; scrambled orders are random pixel orders on binary16.

configuration	problem	σ, seeds	runs
`main` (+`main_extra`)	MNIST 3/5, 4×4: quadtree, snake MPS, scrambled0, scrambled1, scrambled2 (raw); quadtree, MPS, scrambled0 (white)	7 values 0.05–0.5 (scrambled1: 4, and 3 more in `main_extra`; scrambled2: 4; white: 5); 0–4	220 + 14
`regime`	centred model, quadtree	0.5, 0.7, 1, 1.4, 2, 2.8, 4; 0–4	35
`fashion`	Fashion-MNIST 7/9: quadtree, snake MPS, scrambled0	0.05, 0.14, 0.5; 0–4	45
`chi8`, `chi8b`	$\chi = 8$: quadtree, snake MPS, scrambled0 (one sweep, two runners)	0.05, 0.14, 0.5; 0–4	45
`img8`	8×8 images ($N = 64$): quadtree, snake MPS, scrambled0; $\tau \leq 3000$	0.1, 0.3; 0, 1	12
`tol_check`	tolerances 10^{-10}, 10^{-16}: quadtree, MPS	0.1; 0	2
`fix1_local`	reruns: four binary orders to $\tau = 10^4$; MPS phase 1	0.1; 0–4	25
`fix1_mps`	reruns: snake MPS, $\tau \leq 1000$	as `main`; 0–4	35
`extra_orders`	order test: 27 orders on binary16 (7 runs reused)	0.1; 0–2	74 (+7)

C Reproducibility

[Skeleton. How every number and figure in this report is regenerated.]

C.1 Software and data

[Repository layout, environment and where raw outputs live.]

C.2 Figures

[Table: figure, experiment folder, script and command, config, raw data needed.]

Table 21: Figures of the report: the command that draws each one (from the repository root) and the sweeps whose raw runs it reads. Figures are copied into `report/figures/` by `report/collect_figures.sh` from the list in `report/figures.txt`.

figure	command	raw data (sweeps; Section B)
2	`uv run python experiments/regime_map/plot_report.py` draws this figure (`report_map`) and Figure 12 from committed tables in `results/`, `results/fix1/` and `results/report/`; the last are written by `report_tables.py --raw data/regime_map --tensor-cache data/regime_map_local/ tensor_kernel` (README)	`sigma_chi_b4`, `chi_b4`, `sigma_K_centred_d2`, `imbalance_b4` (Table 16); local copy `data/regime_map/`

continued on the next page

Table 21: Figures of the report (continued).

figure	command	raw data (sweeps; Section B)
3	`uv run python experiments/s2s_core/plot_report.py --raw-main data/s2s_core/main --raw-large data/s2s_core/large --out experiments/s2s_core/results` draws this figure (`report_runs`) and Figure 6; scripts below are in `experiments/s2s_core/`, run with `uv run python` and `--out experiments/s2s_core/results`; full order in the experiment's README	`main`, `large` (one run each; Table 14); local copy `data/s2s_core/`
4	`plot_aligned.py --raw data/s2s_core/aligned` (`fig_a_aligned`)	`aligned`
5	`plot_scaling.py` (`fig_c_first_escape_chi3_fanin`) from `first_runs.csv` (`analyze_first.py`); Table 5 is `main_exponents.csv`, written by `recompute_hessians.py` and `analyze_main.py`	`first`, `first_k78`; table: `main`, `large`
6	`plot_report.py` as above (`report_prefactor`), from `lo_d.csv`, `lo_chi_table.csv`, `lo_K_table.csv` (`lo_sweep.py`, `plot_lo.py`)	`lo_d`, `lo_chi`, `lo_K`
7, 8	`bash experiments/locality/finalize.sh R` writes every figure and table of Sections 4.4 and 4.6 into directory `R` (`LOCALITY_EAGER=1` without a C++ compiler). Single steps, with scripts in `experiments/locality/` run by `uv run python`: these two by `make_figures.py $O/twocomp --config configs/twocomp.yaml --hessians R/hessians.csv --out R/twocomp`	`twocomp` (Table 15); `$O` is `LOCALITY_OUT`, local copy `data/locality/`
9	`plot_trees.py $O/multicomp --teacher TA --n 4 --sigma 0.03 --seed 0 --levels R/levels_runs.csv --out R/trees_TA.png`	`multicomp`, `multicomp_wide` (levels by `levels.py`)
10	`plot_bonds.py $O/twocomp --cases mps8:25 binary8:34 ht8:12s --sigma 0.05 --out R/bond_spectra.png`	`twocomp` (saved trajectories)
11	`uv run python experiments/s2s_core/plot_dormant.py --out experiments/s2s_core/results` (`fig_g_dormant`) from `main_runs.csv` and `dormant_decay.csv` (`analyze_main.py`, `analyze_dormant.py`)	`main`, `large`, `k78`, `balanced` (Table 14)
12	as Figure 2 (`report_other`)	`sigma_chi_b4`, `large_chi_b4`

continued on the next page

Table 21: Figures of the report (continued).

figure	command	raw data (sweeps; Section B)
13	`uv run python experiments/optim_samples/plot_report.py --out-dir experiments/optim_samples/results` draws this figure and the next two from committed tables, here `gd_sweep.csv`, `map1d_points.csv` and `centred_chi9_gf.csv` (written from the raw runs by `analyze_gd.py`, `map1d.py`, `analyze_centred.py`; order of all steps in the experiment's README)	`gd_sweep`, `gd_centred_chi9` (Table 18); local copy `data/optim_samples/`
14	as above; tables `sgd_summary.csv`, `sgd_second_plateau.csv` (`analyze_sgd.py`), `p9_sgd.csv` (`p9_sgd.py`), `p9_conditions.csv` and `p9_fit.csv` (`plot_p9.py`)	`sgd_sweep`, `p9_sgd.py`
15	as above; table `samples_summary.csv` (`analyze_samples.py`)	`samples_sweep`
16	`uv run python -m experiments.optimizers.make_figures --results-dir experiments/optimizers/results b4`	`scaling_b4` (Table 19)
17	as above, with target `eps`	`scaling_b4`, `eps_mps8`, and `deep` for the gradient-flow reference times
18	`uv run python experiments/data_teacher/plot_report.py` draws this figure and the next from committed tables, here `r2_saddle_runs_all.csv`, `r2_saddle_fits.csv` (`analyze_r2_saddle.py`) and `r2_random_v2_runs.csv` (`analyze_law_v2.py r2_random`); it also writes the tables `report_first_escape.csv`, `report_lambda_scaling.csv` and `report_float_floor_refit.csv` quoted in Section 6 (order of all steps in the experiment's README)	`r2_saddle` (with `_b`, `_ext`, `_m20`), `r2_random` (Table 17); local copy `data/data_teacher/`
19	as above; tables `e4a_sigma_v2_runs.csv`, `e2_sigma_v2_runs.csv`, `e2_sigma_v2_summary.csv` (`analyze_law_v2.py`), `r2_corr_runs.csv`, `r2_corr_fits.csv` (`analyze_r2_corr.py`), `float_floor_check.csv` (`check_float_floor.py`)	`e4a_sigma`, `e2_sigma`, `r2_corr.py`

continued on the next page

Table 21: Figures of the report (continued).

figure	command	raw data (sweeps; Section B)
20	`uv run python experiments/real_data/make_figures.py --json experiments/real_data/results experiments/real_data/results` draws this figure (`first_route_gaps`) and the next from committed JSON only, here `fix1_analysis.json` (`fix1_analysis.py`, from `events_table.json` and the run summaries of `summarize.py`; order of all steps in the experiment's README)	`main`, `fix1_local`, `fix1_mps` (Table 20), review runs; local copies `data/real_data/`, `data/real_data_local/`
21	as above; table `order_test.json` (`order_test.py`, with the predictor `extra_orders_predictor.json` written by `order_predictor.py` before the runs)	`extra_orders`; `main` for the secondary analysis

C.3 Building the report

The report is built by `report/build.sh` (pdflatex, BibTeX, then pdflatex until the cross-references are stable), after `report/collect_figures.sh` has copied the figures listed in `report/figures.txt` from `experiments/*/results/` into `report/figures/`. `report/README.md` describes both scripts.